HL7 FHIR R4 — Chapter 3: FHIR Exchange Module (sections 3 – 3.8)

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3 FHIR Exchange Module

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Work Group FHIR Infrastructure | Standards Status: Informative

3.0 FHIR Exchange Module

3.0.1 Introduction / Index

FHIR is designed as an interface specification - it specifies the content of the data exchanged between healthcare applications, and how the exchange is implemented and managed. FHIR defines the following methods for exchanging data between systems:

- RESTful API
- Messaging
- Documents
- Services
- Database / Persistent Storage

Each of these approaches can be used to exchange information, and each has its own strengths and weaknesses and applicability. Note that applications are allowed to use any other method to exchange resources; the methods described in this specification are the common methods that are used enough to justify the effort to describe or standardize their use.

3.0.1.1 RESTful API

Most implementers focus on RESTful API. This is a client/server API designed to follow the principles of RESTful design for Create, Read, Update and Delete operations, along with Search and Execute (Operations) support.

The RESTful API is a general-purpose interface that can be used to push and pull data between systems. Which is appropriate depends on architecture and deployment considerations. The RESTful API also supports Asynchronous Use and GraphQL.

3.0.1.2 Messaging

In addition to the RESTful API, a messaging exchange framework is documented, which supports exchange between systems by sending routed messages from system to system. This exchange can be implemented on the RESTful API or using some other messaging technology.

Implementers should note that the messaging framework is not provided to fill any functional deficiency in the RESTful API (or vice versa), these frameworks are provided to allow implementers to choose how to exchange content based on their own architectural and deployment considerations. Messaging may be more suitable for exchange between disparate organizations with low levels of process integration and/or trust.

3.0.1.3 Documents

This specification also defines a document based exchange framework, where content to be exchanged is wrapped by a Composition that provides the context of the content, and that has a fixed presentation for a human reader. The document framework is provided to help with computer-assisted human to human communication uses - which are not uncommon in healthcare.

Typically, exchanging documents is associated with exchanging clinical information across clinical governance borders, while data-based exchange using the RESTful API is appropriate within where there are well established clinical governance arrangements.

3.0.1.4 Services / SOA

In addition, this specification describes the use of FHIR in a services framework(e.g. a SOA). Note that any use of any of the above approaches in production is a 'service' by some or many definitions. The services description provides context regarding the use of FHIR (and particularly the RESTful API) in a wider enterprise architecture.

3.0.1.5 Database / Persistent Store

Another way to make use of the resources defined by FHIR is to store them natively in a database or persistent store, where different applications or modules write and read the resources as part of their implementation. Using resources in this fashion is described here.

3.0.2 Security and Privacy

All forms of data exchange should be appropriately secured. This requires the following:

• Only the parties to the exchange can access the communication

• The parties are authenticated and authorized as required

• Access control always checks that the only the appropriate data is exchanged

• Appropriate patient consent has been obtained for the exchange

This subject is described further in the Security Page.

With regard to the RESTful API, implementers should always consider the Smart App Launch protocol as part of the overall secure API approach

3.0.3 Developmental Roadmap

RESTful API | This is stable and currently being balloted as Normative. No breaking changes are expected. No significant development is planned, but HL7 will continue to respond to user experience

Messaging | Messaging has only be implemented in a few projects; some of the infrastructure has not yet been used in production. It is not clear whether significant development will be needed or appropriate

Documents | Documents have mostly only been used in prototype projects, though there is considerable impetus around implementation at this time. No significant development is planned, but HL7 will continue to respond to user experience

Services | At this point in time, it's not clear whether further work is required or appropriate in terms of service orientated architecture / enterprise integration. HL7 will continue to monitor implementer experience and feedback

Database / Persistent Storage | This is a new area with considerable action at this time, and many production implementations, though RDF itself is not getting much use. At this time, HL7 is monitoring implementer experience and feedback to see whether additional standardization is required

Trial-Use Note:

Note to balloters: There's some interest in standardizing the use of ProtocolBuffers directly in the specification itself (basis). Ballot comments are welcome.

3.1 RESTful API

URL: https://hl7.org/fhir/R4/http.html

3.1.0 RESTful API

FHIR Infrastructure Work Group | Maturity Level: Normative | Standards Status: Normative

| This page has been approved as part of an ANSI standard. See the Infrastructure Package for further details.

FHIR is described as a 'RESTful' specification based on common industry level use of the term REST. In practice, FHIR only supports Level 2 of the REST Maturity model as part of the core specification, though full Level 3 conformance is possible through the use of extensions. Because FHIR is a standard, it relies on the standardization of resource structures and interfaces. This may be considered a violation of REST principles but is key to ensuring consistent interoperability across diverse systems.

Each "resource type" has the same set of interactions defined that can be used to manage the resources in a highly granular fashion. Applications claiming conformance to this framework claim to be conformant to "RESTful FHIR" (see Conformance).

Note that in this RESTful framework, transactions are performed directly on the server resource using an HTTP request/response. The API does not directly address authentication, authorization, and audit collection - for further information, see the Security Page. All the interactions are all described for synchronous use, and an Asynchronous use pattern is also defined.

The API describes the FHIR resources as a set of operations (known as "interactions") on resources where individual resource instances are managed in collections by their type. Servers can choose which of these interactions are made available and which resource types they support. Servers SHALL provide a Capability Statement that specifies which interactions and resources are supported.

In addition to a number of General Considerations this page defines the following interactions:

Instance Level Interactions
read | Read the current state of the resource
vread | Read the state of a specific version of the resource
update | Update an existing resource by its id (or create it if it is new)
patch | Update an existing resource by posting a set of changes to it
delete | Delete a resource
history | Retrieve the change history for a particular resource
Type Level Interactions
create | Create a new resource with a server assigned id
search | Search the resource type based on some filter criteria
history | Retrieve the change history for a particular resource type
Whole System Interactions
capabilities | Get a capability statement for the system
batch/transaction | Update, create or delete a set of resources in a single interaction
history | Retrieve the change history for all resources
search | Search across all resource types based on some filter criteria

In addition to these interactions, there is an operations framework, which includes endpoints for validation, messaging and Documents. Also, implementers can use GraphQL.

3.1.0.1 General Considerations

Note: Where the FHIR specification does not specify behavior with regards to HTTP capabilities (such as OPTIONS), implementers cannot expect greater consistency than is mandated in the underlying HTTP protocol.

3.1.0.1.1 Style Guide

The interactions on this page are defined like this:

```
VERB [base]/[type]/[id] {?_format=[mime-type]}
```

• The first word is the HTTP verb used for the interaction

• Content surrounded by [] is mandatory, and will be replaced by the string literal identified. Possible insertion values: base: The Service Base URL mime-type: The Mime Type type: The name of a resource type (e.g. "Patient") id: The Logical Id of a resource vid: The Version Id of a resource compartment: The name of a compartment parameters: URL parameters as defined for the particular interaction

• Content surrounded by {} is optional

Implementations constructing URLs using these patterns SHOULD conform to RFC 3986 Section 6 Appendix A which requires percent-encoding for a number of characters that occasionally appear in the URLs (mainly in search parameters).

This specification uses the underscore as a prefix to disambiguate reserved names from other names in 3 cases:

• To differentiate system wide history and search interactions from interactions on Resource Types

• To differentiate search, history and similar interactions from instances of a resource type

• To differentiate search parameters defined for all resources from those defined for specific resource types

In addition, the character $ is used as a prefix to operation names that are RPC-like additions to the base API defined either by this specification or by implementers.

3.1.0.1.2 Service Base URL

The Service Base URL is the address where all of the resources defined by this interface are found. The Service Base URL takes the form of

```
http{s}://server{/path}
```

The path portion is optional, and does not include a trailing slash. Each resource type defined in this specification has a manager (or "entity set") that lives at the address /[type] where the [type] is the name of the resource type. For instance, the resource manager for the type Patient will live at:

```
https://server/path/Patient
```

All the logical interactions are defined relative to the service root URL. This means that if the address of any one FHIR resource on a system is known, the address of other resources may be determined.

Note: All URLs (and ids that form part of the URL) defined by this specification are case sensitive. Clients SHOULD encode URLs using UTF-8, and servers SHOULD decode them assuming they are UTF-8 (for background, see here).

Note that a server may use a path of the form http://server/...[xx]... where the [xx] is some variable portion that identifies a particular instantiation of the FHIR API. Typically, the variable id identifies a patient or a user, and the underlying information is completely compartmented by the logical identity associated with [xx]. In this case, the FHIR API presents a patient or user centric view of a record, where authentication/authorization is explicitly granted to the URL, on the grounds that some identifiable user is associated with the logical identity. It is not necessary to explicitly embed the patient id in the URL - implementations can associate a FHIR end-point with a particular patient or provider by using an OAuth login. See Compartments for the logical underpinning.

Identity

Systems often need to compare two URLs to determine whether they refer to the same underlying object or not. For the purposes of this specification, the following rules apply:

• The query part of the URL (anything after ?) is ignored

• The comparison of the document portion of the URL (i.e. not the server/port) is case sensitive

• The protocols http: and https: SHALL NOT be used to refer to different underlying objects

• If a port is specified, then the ports must be identical or the objects are different (due to the prevalence of port mapping and/or interface engines running on different ports). Ports should only be explicit when they have explicit meaning to the server

For example: http://myserver.com/Patient/1 and https://myserver.com/Patient/1 refer to the same underlying object, while http://myserver.com:81/Patient/1 is a distinct entity from either of the above. This does not mean that the two addresses need to be treated the same, or that a server must serve both addresses, or

that the content from the two addresses must be identical, but just that if these two addresses have the same identity, and if both are served, they must both represent the same underlying object. Systems are not required to check that this is true. Note: the identity comparison for protocols other than http:/https: is undefined.

3.1.0.1.3 Resource Metadata and Versioning

Each resource has an associated set of resource metadata elements. These map to the HTTP request and response using the following fields:

Metadata Item | Where found in HTTP
Logical Id (.id) | The Id is represented explicitly in the URL
Version Id (.meta.versionId) | The Version Id is represented in the ETag header
Last modified (.meta.lastUpdated) | HTTP Last-Modified header

Note that the Version Id is considered a "weak" ETag and ETag headers should be prefixed with W/ and enclosed in quotes, for example:

```
ETag: W/"3141"
```

3.1.0.1.4 Security

Using HTTPS is optional, but all production exchange of healthcare data SHOULD use SSL and additional security as appropriate. See HTTP Security for further information. Most operations will require user authentication, and all operations that do so are subject to RBAC and/or ABAC , and some operations may depend on appropriate consent being granted.

Note: to support browser-based client applications, servers SHOULD implement cross-origin resource sharing (CORS) for the interactions documented here.

See the HTTP Security guidance for further information about both security generally and the use of CORS.

3.1.0.1.5 HTTP Status Codes

This specification makes rules about the use of specific HTTP status codes in particular circumstances where the status codes SHALL map to particular states correctly, and only where the correct status code is not obvious. Other HTTP status codes may be used for other states as appropriate, and this particularly includes various authentication related status codes and redirects. Authentication redirects should not be interpreted to change the location of the resource itself (a common web programming error).

FHIR defines an OperationOutcome resource that can be used to convey specific detailed processable error information. For some combinations of interactions and specific return codes, an OperationOutcome is required to be returned as the content of the response. The OperationOutcome may be returned with any HTTP 4xx or 5xx response, but this is not required - many of these errors may be generated by generic server frameworks underlying a FHIR server.

3.1.0.1.6 Managing Return Content

If the server has a default timezone, it SHOULD return the default timezone in the HTTP Response headers using the "Date" header.

In the interests of managing band-width, this specification allows clients to specify what kind of content to return for resources.

3.1.0.1.7 conditional read

Clients may use the If-Modified-Since, or If-None-Match HTTP header on a read request. If so, they SHALL accept either a 304 Not Modified as a valid status code on the response (which means that the content is unchanged since that date) or full content (either the content has changed, or the server does not support conditional request).

Servers can return 304 Not Modified where content is unchanged because the If-Modified-Since date-time or the If-None-Match ETag was specified, or they can return the full content as normal. This optimisation is relevant in reducing bandwidth for caching purposes and servers are encouraged but not required to support this. If servers don't support conditional read, they just return the full content.

3.1.0.1.8 create/update/patch/transaction

These interactions are performed using POST, PUT or PATCH, and it may be appropriate for a server to return either only a status code, or also return the entire resource that is the outcome of the create or update (which may be different to that provided by the client). In the case of transactions this means returning a Bundle with just the Bundle.entry.response populated for each entry, and not the Bundle.entry.resource values.

The client can indicate whether the entire resource is returned using the HTTP return preference :

```
Prefer: return=minimal
Prefer: return=representation
Prefer: return=OperationOutcome
```

The first of these asks to return no body. The second asks to return the full resource. The third asks the server to return an OperationOutcome resource containing hints and warnings about the operation rather than the full resource. Servers SHOULD honor this header. In the absence of the header, servers may choose whether

to return the full resource or not (but not the OperationOutcome; that should only be returned if explicitly requested). Note that this setting only applies to successful interactions. In case of failure, servers SHOULD always return a body that contains an OperationOutcome resource.

See also the Asynchronous use pattern for another use of the Prefer header.

3.1.0.1.9 Content Types and encodings

The formal MIME-type for FHIR resources is application/fhir+xml or application/fhir+json. The correct mime type SHALL be used by clients and servers:

• XML: application/fhir+xml

• JSON: application/fhir+json

• RDF: application/fhir+turtle (only the Turtle format is supported)

Servers SHALL support server-driven content negotiation as described in section 12 of the HTTP specification.

Implementation Notes:

• The content type application/x-www-form-urlencoded (Specification) is also accepted for posting search requests.

• If a client provides a generic mime type in the Accept header (application/xml, text/json, or application/json), the server SHOULD respond with the requested mime type, using the XML or JSON formats described in this specification as the best representation for the named mime type (except for binary - see the note on the Binary resource).

• Note: between FHIR DSTU2 and STU3, the correct mime type was changed from application/xml+fhir and application/json+fhir to application/fhir+xml and application/fhir+json. Servers MAY also support the older mime types, and are encouraged to do so to smooth the transition process.

• 406 Not Acceptable is the appropriate response when the Accept header requests a format that the server does not support, and 415 Unsupported Media Type when the client posts a format that is not supported to the server.

UTF-8 encoding SHALL be used for the mime type application/fhir. This MAY be specified as a MIME type parameter to the application/fhir mime type, but is not required.

3.1.0.1.10 FHIR Version Parameter

This specification defines the MIME-type parameter fhirVersion as a parameter to indicate which version of the FHIR release a resource is based on:

```
Accept: application/fhir+json; fhirVersion=4.0
```

The value of this parameter is the publication and major version number for the specification:

FHIR R1 (DSTU 1) | 0.0
FHIR R2 (DSTU 2) | 1.0
FHIR R3 (STU3, or just R3) | 3.0
FHIR R4 (this version) | 4.0 (once published)

Intermediate balloted releases may also be encountered occasionally - see publications directory . Versions from before the publication of the first DSTU (which is 0.0) are not supported.

This parameter can be used anywhere that the FHIR Mime type is used. When used in an HTTP request, the version parameter may be used on either the Content-Type header, or the Accept header, or both, and applies to the entire interaction (the behavior of the interactions as described on ths page, the search parameters and functionality, and the accompanying conformance resources). It is an error for a client to attempt to use two different versions in the same interaction. For further information about specifying FHIR version, see Managing FHIR Versions.

3.1.0.1.11 General parameters

The following parameters are defined for use with all of the interactions defined on this page:

_format | Override the HTTP content negotiation - see immediately below
_pretty | Ask for a pretty printed response for human convenience - see below
_summary | Ask for a predefined short form of the resource in response - see Search Summary
_elements | Ask for a particular set of elements to be returned - see Search Elements

Note that the impact of _elements is not defined for interactions other than search where the response is a bundle that contains more than one type of resource.

_format

In order to support various implementation limitations, servers SHOULD support the optional _format parameter to specify alternative response formats by their MIME-types. This parameter allows a client to override the accept header value when it is unable to set it correctly due to internal limitations (e.g. XSLT usage). For the _format parameter, the values xml, text/xml, application/xml, and application/fhir+xml SHALL be interpreted to mean the XML format, the codes json, application/json and application/fhir+json SHALL be interpreted to mean the JSON format, and the codes ttl, application/fhir+turtle, and text/turtle SHALL be interpreted to mean the Turtle RDF format. In addition, the values html and text/html are allowed.

Note: the _format parameter does not override the Content-Type header for the type of the body of a POST request. If neither the accept header nor the _format parameter are specified, the MIME-type of the content returned by the server is undefined and may vary.

_pretty

Clients that wish to request for pretty-printed resources (either in JSON or XML) can use the _pretty parameter:

```
GET [base]/Patient/example?_pretty=true
```

Value values are true and false. Since pretty printed or not makes no difference to the content, this is only of interest for development tools, and servers MAY choose to support this parameter.

3.1.0.1.12 Support for Versions

Servers that support this API SHOULD provide full version support - that is, populate and track versionId correctly, support vread, and implement version aware updates. Supporting versions like this allows for related systems to track the correct version of information, and to keep integrity in clinical records. However, many current operational systems do not do this, and cannot easily be re-engineered to do so.

For this reason, servers are allowed to not provide versioning support and this API does not enforce that versioning is supported. Clients may elect to only interact with servers that do provide full versioning support. Systems declare their support for versioning in their Capability Statements, where they can indicate one of three levels for versioning support:

• no-version: Versioning and meta.version is not supported (server) or used (client)

• versioned: Versioning and meta.version is supported (server) or used (client)

• versioned-update: Versioning and meta.version is supported, and version aware updates are used - Version ID must be correct for updates (server) or will be specified (If-match header) for updates (client)

Servers that do not support versioning SHALL ensure that Resource.meta.versionId is not present on resources they return, and SHALL update the value of Resource.meta.lastUpdated correctly.

3.1.0.2 read

The read interaction accesses the current contents of a resource. The interaction is performed by an HTTP GET command as shown:

```
GET [base]/[type]/[id] {?_format=[mime-type]}
```

This returns a single instance with the content specified for the resource type. This url may be accessed by a browser. The possible values for the Logical Id ("id") itself are described in the id type. The returned resource SHALL have an id element with a value that is the [id]. Servers SHOULD return an ETag header with the versionId of the resource (if versioning is supported) and a Last-Modified header.

Note: Unknown resources and deleted resources are treated differently on a read: a GET for a deleted resource returns a 410 status code, whereas a GET for an unknown resource returns 404. Systems that do not track deleted records will treat deleted records as an unknown resource. Since deleted resources may be brought back to life, servers MAY include an ETag on the error response when reading a deleted record to allow version contention management when a resource is brought back to life.

In addition, the search parameter _summary can be used when reading a resource:

```
GET [base]/[type]/[id] {?_summary=text}
```

This requests that only a subset of the resource content be returned, as specified in the _summary parameter, which can have the values true, false, text, count and data. Note that a resource that only contains a subset of the data is not suitable for use as a base to update the resource, and might not be suitable for other uses. The same applies to the _elements parameter - both that it should be supported, and the subset implications. Servers SHOULD define a Resource.meta.tag with the SUBSETTED as a Simple Tag to explicitly mark such resources.

A HEAD request can also be used - see below.

3.1.0.3 vread

The vread interaction performs a version specific read of the resource. The interaction is performed by an HTTP GET command as shown:

```
GET [base]/[type]/[id]/_history/[vid] {?_format=[mime-type]}
```

This returns a single instance with the content specified for the resource type for that version of the resource. The returned resource SHALL have an id element with a value that is the [id], and a meta.versionId element with a value of [vid]. Servers SHOULD return an ETag header with the versionId (if versioning is supported) and a Last-Modified header.

The Version Id ("vid") is an opaque identifier that conforms to the same format requirements as a Logical Id. The id may have been found by performing a history interaction (see below), by recording the version id from a content location returned from a read or from a version specific reference in a content model. If the version referred to is actually one where the resource was deleted, the server should return a 410 status code.

Servers are encouraged to support a version specific retrieval of the current version of the resource even if they do not provide access to previous versions. If a request is made for a previous version of a resource, and the server does not support accessing previous versions (either generally, or for this particular resource), it should return a 404 Not Found error, with an operation outcome explaining that history is not supported for the underlying resource type or instance.

A HEAD request can also be used - see below.

3.1.0.4 update

The update interaction creates a new current version for an existing resource or creates an initial version if no resource already exists for the given id. The update interaction is performed by an HTTP PUT command as shown:

```
PUT [base]/[type]/[id] {?_format=[mime-type]}
```

The request body SHALL be a Resource with an id element that has an identical value to the [id] in the URL. If no id element is provided, or the id disagrees with the id in the URL, the server SHALL respond with an HTTP 400 error code, and SHOULD provide an OperationOutcome identifying the issue. If the request body includes a meta, the server SHALL ignore the provided versionId and lastUpdated values. If the server supports versions, it SHALL populate the meta.versionId and meta.lastUpdated with the new correct values. Servers are allowed to review and alter the other metadata values, but SHOULD refrain from doing so (see metadata description for further information). Note that there is no support for updating past versions - see notes on the history interaction.

A server SHOULD accept the resource as submitted when it accepts the update, and return the same content when it is subsequently read. However systems might not be able to do this; see the note on transactional integrity for discussion. Also, see Variations between Submitted data and Retrieved data for additional discussion around update behavior. Note that update generally updates the whole content of the resource. For partial updates, see patch below.

If the interaction is successful, the server SHALL return either a 200 OK HTTP status code if the resource was updated, or a 201 Created status code if the resource was created (or brought back to life/re-created), with a Last-Modified header, and an ETag header which contains the new versionId of the resource. If the resource was created (i.e. the interaction resulted in a 201 Created), the server SHOULD return a Location header (this is for HTTP conformance; it's not otherwise needed). The body of response is as described in Managing Return Content.

Note: Servers MAY choose to preserve XML comments, instructions, and formatting or JSON whitespace when accepting updates, but are not required to do so. The impact of this on digital signatures may need to be considered.

3.1.0.4.1 Update as Create

Servers MAY choose to allow clients to PUT a resource to a location that does not yet exist on the server - effectively, allowing the client to define the id of the resource. Whether a server allows this is a deployment choice based on the nature of its relationships with the clients. While many servers will not allow clients to define their ids, there are several reasons why it may be necessary in some configurations:

• client is reproducing an existing data model on the server, and needs to keep original ids in order to retain ongoing integrity

• client is a server doing push based pub/sub (this is a special case of the first reason)

• multiple clients doing push in the context of agreed data model shared across multiple servers where ids are shared across servers

Alternatively, clients may be sharing an agreed identification model (e.g. key server, scoped identifiers, or UUIDs) where clashes do not arise. Note that this use of update has security implications.

Servers can choose whether or not to support client defined ids, and indicate such to the clients using CapabilityStatement.rest.resource.updateCreate.

3.1.0.4.2 Rejecting Updates

Servers are permitted to reject update interactions because of integrity concerns or other business rules, and return HTTP status codes accordingly (usually a 422). Note that there are potential security issues relating to how rejections are handled. See the security page for more information.

Common HTTP Status codes returned on FHIR-related errors (in addition to normal HTTP errors related to security, header and content type negotiation issues):

• 400 Bad Request - resource could not be parsed or failed basic FHIR validation rules (or multiple matches were found for conditional criteria)

• 401 Not Authorized - authorization is required for the interaction that was attempted

• 404 Not Found - resource type not supported, or not a FHIR end-point

• 405 Method Not allowed - the resource did not exist prior to the update, and the server does not allow client defined ids

• 409/412 - version conflict management - see below

• 422 Unprocessable Entity - the proposed resource violated applicable FHIR profiles or server business rules

Any of these errors SHOULD be accompanied by an OperationOutcome resource providing additional detail concerning the issue. In general, if an instance fails the constraints documented in the CapabilityStatement then the response should be a 400, whereas if the instance fails other non-externally described business rules, the response would be a 422 error. However, there's no expectation that servers will tightly adhere to this differentiation (nor is it clear that it makes much difference whether they do or not). In practice, servers may also return 5xx errors in these cases without being deemed non-conformant.

For additional information on how systems may behave when processing updates, refer to the Variations between Submitted data and Retrieved data page.

3.1.0.4.3 Conditional update

Unlike this rest of this page, the conditional create, update, patch and delete are trial use until further experience is gained with their use. Their status will be reviewed in a future version of FHIR.

The conditional update interaction allows a client to update an existing resource based on some identification criteria, rather than by logical id. To accomplish this, the client issues a PUT as shown:

```
PUT [base]/[type]?[search parameters]
```

When the server processes this update, it performs a search using its standard search facilities for the resource type, with the goal of resolving a single logical id for this request. The action it takes depends on how many matches are found:

• No matches, no id provided: The server creates the resource.

• No matches, id provided: The server treats the interaction as an Update as Create interaction (or rejects it, if it does not support Update as Create)

• One Match, no resource id provided OR (resource id provided and it matches the found resource): The server performs the update against the matching resource

• One Match, resource id provided but does not match resource found: The server returns a 400 Bad Request error indicating the client id specification was a problem preferably with an OperationOutcome

• Multiple matches: The server returns a 412 Precondition Failed error indicating the client's criteria were not selective enough preferably with an OperationOutcome

This variant can be used to allow a stateless client (such as an interface engine) to submit updated results to a server, without having to remember the logical ids that the server has assigned. For example, a client updating the status of a lab result from "preliminary" to "final" might submit the finalized result using PUT path/Observation?identifier=http://my-lab-system|123

Note that transactions and conditional create/update/delete are complex interactions and it is not expected that every server will implement them. Servers that don't support the conditional update SHOULD return an HTTP 400 error and MAY include an OperationOutcome.

The resource MAY contain an id element, but does not need to (this is one of the few cases where a resource exists without an id element). If an id is provided, the server SHALL ignore it - see documentation for the update interaction.

3.1.0.5 Managing Resource Contention

Lost Updates , where two clients update the same resource, and the second overwrites the updates of the first, can be prevented using a combination of the ETag and If-Match header. This is also known as 'Optimistic Locking'.

To support this usage, servers SHOULD always return an ETag header with each resource:

```
HTTP 200 OK
Date: Sat, 09 Feb 2013 16:09:50 GMT
Last-Modified: Sat, 02 Feb 2013 12:02:47 GMT
ETag: W/"23"
Content-Type: application/fhir+json
```

If provided, the value of the ETag SHALL match the value of the version id for the resource. Servers are allowed to generate the version id in whatever fashion that they wish, so long as they are valid according to the id data type, and are unique within the address space of all versions of the same resource. When resources are returned as part of a bundle, there is no ETag, and the versionId of the resource is used directly.

If the client wishes to request a version aware update, it submits the request with an If-Match header that quotes the ETag from the server:

```
PUT /Patient/347 HTTP/1.1
If-Match: W/"23"
```

If the version id given in the If-Match header does not match, the server returns a 412 Precondition Failed status code instead of updating the resource.

Servers can require that clients provide an If-Match header by returning 400 Client Error status codes when no If-Match header is found. Note that a 409 Conflict can be returned when the server detects the update cannot

be done (e.g. due to server side pessimistic locking).

3.1.0.6 patch

As an alternative to updating an entire resource, clients can perform a patch operation. This can be useful when a client is seeking to minimize its bandwidth utilization, or in scenarios where a client has only partial access or support for a resource. The patch interaction is performed by an HTTP PATCH command as shown:

```
PATCH [base]/[type]/[id] {?_format=[mime-type]}
```

The body of a PATCH operation SHALL be either:

• a JSON Patch document with a content type of application/json-patch+json

• an XML Patch document with a content type of application/xml-patch+xml

• a FHIRPath Patch parameters resource with FHIR Content Type

In either case, the server SHALL process its own copy of the resource in the format indicated, applying the operations specified in the document, following the relevant PATCH specification. When the operations have all been processed, the server processes the resulting document as an Update operation; all the version and error handling etc. apply as specified, as does the Prefer Header.

Processing PATCH operations may be very version sensitive. For this reason, servers SHALL support conditional PATCH, which works exactly the same as specified for update in Concurrency Management. Clients SHOULD always consider using version specific PATCH operations so that inappropriate actions are not executed.

Unlike this rest of this page, the conditional create, update, patch and delete are trial use until further experience is gained with their use. Their status will be reviewed in a future version of FHIR.

In addition, servers that support PATCH, and that support Conditional Update SHOULD also support conditional PATCH. When the server processes a conditional PATCH, it performs a search using its standard search facilities for the resource type, with the goal of resolving a single logical id for this request. The action it takes depends on how many matches are found:

• No matches: The server returns a 404 Not Found

• One Match: The server performs the update against the matching resource

• Multiple matches: The server returns a 412 Precondition Failed error indicating the client's criteria were not selective enough

The server SHALL ensure that the narrative in a resource is not clinically unsafe after the PATCH operation is performed. Exactly how this is defined and can be achieved depends on the context, and how narrative is being maintained, but servers may wish to consider:

• If the existing narrative has a status != generated, the server could reject the PATCH operation

• The server could regenerate the narrative once the operation has been applied to the data

• In some limited circumstances, an XML PATCH operation could update the narrative

• The server could delete the narrative, on the basis that some later process will be able to populate it correctly

Processing XML Patch documents is tricky because of namespace handling. Servers SHALL handle namespaces correctly, but note that FHIR resources only contain two XML namespaces, for FHIR (http://hl7.org/fhir) and XHTML (http://www.w3.org/1999/xhtml).

For PATCH Examples, see the FHIR test cases.

Patch operations may be performed as part of Batch or Transaction Operations using the FHIRPath Patch format.

Patch is not defined for all resources - see not about PATCH on Binary.

3.1.0.7 delete

The delete interaction removes an existing resource. The interaction is performed by an HTTP DELETE command as shown:

```
DELETE [base]/[type]/[id]
```

The request body SHALL be empty.

A delete interaction means that subsequent non-version specific reads of a resource return a 410 HTTP status code and that the resource is no longer found through search interactions. Upon successful deletion, or if the resource does not exist at all, the server should return either a 200 OK if the response contains a payload, or a 204 No Content with no response payload, or a 202 Accepted if the server wishes to be non-commital about the outcome of the delete.

Whether to support delete at all, or for a particular resource type or a particular instance is at the discretion of the server based on the policy and business rules that apply in its context. If the server refuses to delete resources of that type as a blanket policy, then it should return the 405 Method not allowed status code. If the server refuses to delete a resource because of reasons specific to that resource, such as referential integrity, it should return the 409 Conflict status code. Note that the servers MAY choose to enforce business rules

regarding deletion of resources that are being referenced by other resources, but they also MAY NOT do so. Performing this interaction on a resource that is already deleted has no effect, and the server should return either a 200 OK if the response contains a payload, or a 204 No Content with no response payload. Resources that have been deleted may be "brought back to life" by a subsequent update interaction using an HTTP PUT.

Many resources have a status element that overlaps with the idea of deletion. Each resource type defines what the semantics of the deletion interactions are. If no documentation is provided, the deletion interaction should be understood as deleting the record of the resource, with nothing about the state of the real-world corresponding resource implied.

For servers that maintain a version history, the delete interaction does not remove a resource's version history. From a version history respect, deleting a resource is the equivalent of creating a special kind of history entry that has no content and is marked as deleted. Note that there is no support for deleting past versions - see notes on the history interaction. Since deleted resources may be brought back to life, servers MAY include an ETag on the delete response to allow version contention management when a resource is brought back to life.

Note that irrespective of this rule, servers are free to completely delete the resource and it's history if policy or business rules make this the appropriate action to take.

3.1.0.7.1 Conditional delete

Unlike this rest of this page, the conditional create, update, patch and delete are trial use until further experience is gained with their use. Their status will be reviewed in a future version of FHIR.

The conditional delete interaction allows a client to delete an existing resource based on some selection criteria, rather than by a specific logical id. To accomplish this, the client issues an HTTP DELETE as shown:

```
DELETE [base]/[type]/?[search parameters]
```

When the server processes this delete, it performs a search as specified using the standard search facilities for the resource type. The action it takes depends on how many matches are found:

• No matches or One Match: The server performs an ordinary delete on the matching resource

• Multiple matches: A server may choose to delete all the matching resources, or it may choose to return a 412 Precondition Failed error indicating the client's criteria were not selective enough. A server indicates whether it can delete multiple resources in its Capability Statement (.rest.resource.conditionalDelete)

This variant can be used to allow a stateless client (such as an interface engine) to delete a resource on a server, without having to remember the logical ids that the server has assigned. For example, a client deleting a lab atomic result might delete the resource using DELETE /Observation?identifier=http://my-lab-system|123.

Note that transactions and conditional create/update/delete are complex interactions and it is not expected that every server will implement them. Servers that don't support the conditional update SHOULD return an HTTP 400 error and MAY include an OperationOutcome.

3.1.0.8 create

The create interaction creates a new resource in a server-assigned location. If the client wishes to have control over the id of a newly submitted resource, it should use the update interaction instead. The create interaction is performed by an HTTP POST command as shown:

```
POST [base]/[type] {?_format=[mime-type]}
```

The request body SHALL be a FHIR Resource. The resource does not need to have an id element (this is one of the few cases where a resource exists without an id element). If an id is provided, the server SHALL ignore it. If the request body includes a meta, the server SHALL ignore the existing versionId and lastUpdated values. The server SHALL populate the id, meta.versionId and meta.lastUpdated with the new correct values. Servers are allowed to review and alter the other metadata values, but SHOULD refrain from doing so (see metadata description for further information).

A server SHOULD otherwise accept the resource as submitted when it accepts the create, and return the same content when it is subsequently read. However some systems might not be able to do this; see the note on transactional integrity for discussion.

The server returns a 201 Created HTTP status code, and SHALL also return a Location header which contains the new Logical Id and Version Id of the created resource version:

```
Location: [base]/[type]/[id]/_history/[vid]
```

where [id] and [vid] are the newly created id and version id for the resource version. The Location header should be as specific as possible - if the server understands versioning, the version is included. If a server does not track versions, the Location header will just contain [base]/[type]/[id]. The Location MAY be an absolute or relative URL.

Servers SHOULD return an ETag header with the versionId (if versioning is supported) and a Last-Modified header. The body of response is as described in Managing Return Content.

When the resource syntax or data is incorrect or invalid, and cannot be used to create a new resource, the server returns a 400 Bad Request HTTP status code. When the server rejects the content of the resource

because of business rules, the server returns a 422 Unprocessable Entity error HTTP status code. In either case, the server SHOULD include a response body containing an OperationOutcome with detailed error messages describing the reason for the error.

Note: Servers MAY choose to preserve XML comments, instructions, and formatting or JSON whitespace when accepting creates, but are not required to do so. The impact of this on digital signatures may need to be considered.

Common HTTP Status codes returned on FHIR-related errors (in addition to normal HTTP errors related to security, header and content type negotiation issues):

• 400 Bad Request - resource could not be parsed or failed basic FHIR validation rules

• 404 Not Found - resource type not supported, or not a FHIR end-point

• 422 Unprocessable Entity - the proposed resource violated applicable FHIR profiles or server business rules. This should be accompanied by an OperationOutcome resource providing additional detail

In general, if an instance fails the constraints documented in the CapabilityStatement then the response should be a 400, whereas if the instance fails other non-externally described business rules, the response would be a 422 error. However, there's no expectation that servers will tightly adhere to this differentiation (nor is it clear that it makes much difference whether they do or not). In practice, servers may also return 5xx errors in these cases without being deemed non-conformant.

For additional information on how systems may behave when processing updates, refer to the Variations between Submitted data and Retrieved data page.

3.1.0.8.1 Conditional create

Unlike this rest of this page, the conditional create, update, patche and delete are trial use until further experience is gained with their use. Their status will be reviewed in a future version of FHIR.

The conditional create interaction allows a client to create a new resource only if some equivalent resource does not already exist on the server. The client defines what equivalence means in this case by supplying a FHIR search query using an HL7 defined extension header "If-None-Exist" as shown:

```
If-None-Exist: [search parameters]
```

The parameter just contains the search parameters (what would be in the URL following the "?").

When the server processes this create, it performs a search as specified using its standard search facilities for the resource type. The action it takes depends on how many matches are found:

• No matches: The server processes the create as above

• One Match: The server ignores the post and returns 200 OK

• Multiple matches: The server returns a 412 Precondition Failed error indicating the client's criteria were not selective enough

This variant can be used to avoid the risk of two clients creating duplicate resources for the same record. For example, a client posting a new lab result might specify If-None-Exist: identifier=http://my-lab-system|123 to ensure it does not create a duplicate record.

Note that transactions and conditional create/update/delete are complex interactions and it is not expected that every server will implement them. Servers that don't support the conditional update SHOULD return an HTTP 400 error and MAY include an OperationOutcome.

3.1.0.9 search

This interaction searches a set of resources based on some filter criteria. The interaction can be performed by several different HTTP commands.

```
GET [base]/[type]{?[parameters]{&_format=[mime-type]}}
```

This searches all resources of a particular type using the criteria represented in the parameters.

Because of the way that some user agents and proxies treat GET and POST requests, in addition to the get based search method above, servers that support search SHALL also support a POST based search:

```
POST  [base]/[type]/_search{?[parameters]{&_format=[mime-type]}}
Content-Type: application/x-www-form-urlencoded

param1=value&param2=value
```

This has exactly the same semantics as the equivalent GET command. Note that in the POST variant, parameters may appear in both the URL and the body. Parameters have the same meaning in either place. Since parameters can repeat, putting them in both places is the same as repeating them (which is valid for some parameters and not for others).

Note: Supporting GET means that PHI (Personal health information) might appear in search parameters, and therefore in HTTP logs. For this reason logs should be regarded as being as sensitive as the resources themselves. This is a general requirement irrespective of the use of GET - see the security page for further commentary.

All these search interactions take a series of parameters that are a series of name=value pairs encoded in the URL (or as an application/x-www-form-urlencoded (Specification) submission for a POST). (See W3C HTML forms).

Note: application/x-www-form-urlencoded is supported for POST so that invoking a search by GET or POST can be done from HTML forms in a browser (though considerable active content might be required in the browser), although this is not the main usage.

A HEAD request can also be used - see below.

Searches are processed as specified for the Search handling mechanism.

If the search succeeds, the server SHALL return a 200 OK HTTP status code and the return content SHALL be a Bundle with type = searchset containing the results of the search as a collection of zero or more resources in a defined order. Note that resources returned in the search bundle MAY be located on the another server than the one that performs the search (i.e. the Bundle.entry.fullUrl may be different to the [base] from the search URL).

The result collection can be long, so servers may use paging. If they do, they SHALL use the method described below (adapted from RFC 5005 (Feed Paging and Archiving) for breaking the collection into pages if appropriate. The server MAY also return an OperationOutcome resource within the searchset Bundle entries that contains additional information about the search; if one is sent it SHALL NOT include any issues with a fatal or error severity, and it SHALL be marked with a Bundle.entry.search.mode of outcome.

If the search fails (cannot be executed, not that there are no matches), the return value return value SHALL be a status code 4xx or 5xx with an OperationOutcome.

Common HTTP Status codes returned on FHIR-related errors (in addition to normal HTTP errors related to security, header and content type negotiation issues):

• 400 Bad Request - search could not be processed or failed basic FHIR validation rules

• 401 Not Authorized - authorization is required for the interaction that was attempted

• 404 Not Found - resource type not supported, or not a FHIR end-point

3.1.0.9.1 Variant Searches

To search a compartment, for either all possible resources or for a particular resource type, respectively:

```
GET [base]/[Compartment]/[id]/*{?[parameters]{&_format=[mime-type]}}
GET [base]/[Compartment]/[id]/[type]{?[parameters]{&_format=[mime-type]}}
```

In the first URL the character "*" appears in the URL as a literal to mean 'all types'. So, for example, to retrieve all the observation resources for a particular LOINC code associated with a particular encounter:

```
GET [base]/Encounter/23423445/Observation?code=2951-2  {&_format=[mime-type]}
```

Note that there are specific operations defined to support fetching an entire patient record or all record for an encounter.

It is also possible to search across multiple resource types:

```
GET [base]?_type=Condition,Observation&[parameters]{&_format=[mime-type]}
```

This is a request to search on both Condition and Observation. In this case, the only parameters that can be used are those defined for both Condition and Observation (using SearchParameter.base - see Cross-resource Search Parameters), or the parameters defined for all resources. If a search lists types not listed in SearchParameter.base for any of the parameters, this is an error, and a server SHOULD return a 400 status. It is also possible to search on all types at once:

```
GET [base]?[parameters]{&_format=[mime-type]}
```

When searching all resources at once, the only search parameters that be can be used in global search like this are the base parameters that apply to all resources.

3.1.0.10 capabilities

The capabilities interaction retrieves the information about a server's capabilities - which portions of this specification it supports. The interaction is performed by an HTTP GET command as shown:

```
GET [base]/metadata{?mode=[mode]} {&_format=[mime-type]}
```

Applications SHALL return a resource that describes the functionality of the server end-point. The information returned depends on the value of the mode parameter:

full (or mode not present) | A Capability Statement that specifies which resource types and interactions are supported
normative | As above, but only the normative portions of the Capability Statement
terminology | A TerminologyCapabilities resource that provides further information about terminologies which are supported by the server

Servers MAY ignore the mode parameter and return a CapabilityStatement resource. Note: servers might be required to support this parameter in further versions of this specification.

If a 404 Unknown is returned from the GET, FHIR (or the specified version) is not supported on the nominated service url. An ETag header SHOULD be returned with the Response. The value of the ETag header SHALL change if the returned resource changes.

Servers SHOULD check for the fhirVersion MIME-type parameter when processing this request.

The resource returned typically has an arbitrary id, and no meta element, though it is not prohibited. Capability statements can become quite large; servers are encouraged to support the _summary and _elements parameters on the capabilities interaction, though this is not required. In addition, servers are encouraged to implement the $subset and $implements operations to make it easy for a client to check conformance.

In addition to this capabilities interaction, a server may also choose to provide the standard set of interactions (read, search, create, update) defined on this page for the CapabilityStatement Resource end-point. This is different from the capabilities interaction:

capabilities interaction | returns a capability statement describing the server's current operational functionality
CapabilityStatement end-point | manages a repository of capability statements (e.g. the HL7 capability statement registry)

All servers are required to support the capabilities interaction, but servers may choose whether they wish to support the CapabilityStatement end-point, just like any other end-point.

Implementation Note: In DSTU 2 and earlier, the resource that this interaction returned was named "Conformance". Clients often connect to a server, and use the capabilities interaction to check whether they are version and/or feature compatible with the server. Such clients should be able to process either a Conformance or a CapabilityStatement resource.

3.1.0.11 batch/transaction

The batch and transaction interactions submit a set of actions to perform on a server in a single HTTP request/response. The actions may be performed independently as a "batch", or as a single atomic "transaction" where the entire set of changes succeed or fail as a single entity. Multiple actions on multiple resources of the same or different types may be submitted, and they may be a mix of other interactions defined on this page (e.g. read, search, create, update, delete, etc.), or using the operations framework.

The transaction mode is especially useful where one would otherwise need multiple interactions, possibly with a risk of loss of referential integrity if a later interaction fails (e.g. when storing a Provenance resource and its corresponding target resource, or and IHE-MHD transaction "Provide Document Resources" with a DocumentManifest, and some number of DocumentReference, List, and Binary resources).

Note that transactions and conditional create/update/delete are complex interactions and it is not expected that every server will implement them. Servers that don't support the batches or transactions SHOULD return an HTTP 400 error and MAY include an OperationOutcome.

A batch or transaction interaction is performed by an HTTP POST command as shown:

```
POST [base] {?_format=[mime-type]}
```

The content of the post submission is a Bundle with Bundle.type = batch or transaction. Each entry SHALL carry request details (Bundle.entry.request) that provides the HTTP details of the action in order to inform the system processing the batch or transaction what to do for the entry. If the HTTP command is a PUT or POST, then the entry SHALL contain a resource for the body of the action. The resources in the bundle are each processed separately as if they were an individual interaction or operation as otherwise described on this page, or the Operations framework. The actions are subject to the normal processing for each, including the meta element, verification and version aware updates, and transactional integrity. In the case of a batch each entry is treated as if an individual interaction or operation, in the case of a transaction all interactions or operations either succeed or fail together (see below).

Examples:

• Transaction Example with Matching Response

• Batch request to fetch Meds & Allergies with Response

• Batch request to fetch simple Patient Summary with Response

3.1.0.11.1 Batch Processing Rules

For a batch, there SHALL be no interdependencies between the different entries in the Bundle that cause change on the server. The success or failure of one change SHOULD not alter the success or failure or resulting content of another change. Servers SHOULD validate that this is the case. Note that it is considered that servers execute the batch in the same order as that specified below for transactions, though the order of execution should not matter given the previous rule.

References within a Bundle.entry.resource to another Bundle.entry.resource that is being created within the batch are considered to be non-conformant.

When processing the batch, the HTTP response code is 200 Ok if the batch was processed correctly, regardless of the success of the operations within the Batch. To determine the status of the operations, look inside the returned Bundle. A response code on an entry of other than 2xx (200, 202 etc) indicates that processing the request in the entry failed.

3.1.0.11.2 Transaction Processing Rules

For a transaction, servers SHALL either accept all actions and return a 200 OK, along with a response bundle (see below), or reject all resources and return an HTTP 400 or 500 type response. It is not an error if the submitted bundle has no resources in it. The outcome of processing the transaction SHALL NOT depend on the order of the resources in the transaction. A resource can only appear in a transaction once (by identity).

Because of the rules that a transaction is atomic where all actions pass or fail together and the order of the entries doesn't matter, there is a particular order in which to process the actions:

• Process any DELETE interactions

• Process any POST interactions

• Process any PUT or PATCH interactions

• Process any GET or HEAD interactions

• Resolve any conditional references

If any resource identities (including resolved identities from conditional update/delete) overlap in steps 1-3, then the transaction SHALL fail.

A transaction may include references from one resource to another in the bundle, including circular references where resources refer to each other. When the server assigns a new id to any resource in the bundle which has a POST method as part of the processing rules above, it SHALL also update any references to that resource in the same bundle as they are processed (see about Ids in a bundle). References to resources that are not part of the bundle are left untouched. Version-specific references should remain as version-specific references after the references have been updated. Note that when building a transaction, a client can use arbitrarily chosen version references since they will all be re-assigned anyway. Servers SHALL replace all matching links in the bundle, whether they are found in the resource ids, resource references, elements of type uri, url, oid, uuid, and <a href="" & <img src="" in the narrative. Elements of type canonical are not replaced.

When processing a "POST" (create), the full URL is treated as the id of the resource on the source, and is ignored; the server generates an id for the resource. For updates, the server performs a mapping between the fullUrl specified and the local URL the server knows that instance as, if possible. If the server does not have a mapping for the fullUrl, the server ignores the base URL and attempts an update assuming the base is the same as the server base. This allows the same transaction bundle to be sent to multiple systems without changing the fullUrls for each target.

When processing a batch or transaction, a server MAY choose to honor existing logical ids (e.g. Observation/1234 remains as Observation/1234 on the server), but since this is only safe in controlled circumstances, servers may choose to assign new ids to all submitted resources, irrespective of any claimed logical id in the resource, or fullUrl on entries in the batch/transaction.

Conditional References

When constructing the bundle, the client might not know the logical id of a resource, but it may know identifying information - e.g. an identifier. This situation arises commonly when building transactions from v2 messages. The client could resolve that identifier to a logical id using a search, but that would mean that the resolution to a logical id does not occur within the same transaction as the commit (as well as significantly complicating the client). Because of this, in a transaction (and only in a transaction), references to resources may be replaced by a search URI that describes how to find the correct reference:

```
<Bundle xmlns="http://hl7.org/fhir">
  <id value="20160113160203" />
  <type value="transaction" />
  <entry>
    <fullUrl value="urn:uuid:c72aa430-2ddc-456e-7a09-dea8264671d8" />
    <resource>
      <Observation>
        <subject>
           <reference value="Patient?identifier=12345"/>
        </subject>
        <--! rest of resource omitted -->
      </Observation>
    </resource>
    <request>
      <method value="POST" />
    </request>
  </entry>
<Bundle>
```

The search URI is relative to the server's [base] path, and always starts with a resource type: [type]?parameters.... Only filtering parameters are allowed; none of the parameters that control the return of resources are relevant.

When processing transactions, servers SHALL:

• check all references for search URIs

• For search URIs, use the search to locate matching resources

• if there are no matches, or multiple matches, the transaction fails, and an error is returned to the user

• if there is a single match, the server replaces the search URI with a reference to the matching resource

3.1.0.11.3 Batch/Transaction Response

For a batch, or a successful transaction, the response the server SHALL return a Bundle with type set to batch-response or transaction-response that contains one entry for each entry in the request, in the same order, with the outcome of processing the entry. For a failed transaction, the server returns a single OperationOutcome instead of a Bundle.

A client may use the returned Bundle to track the outcomes of processing the entry, and the identities assigned to the resources by the server. Each entry element SHALL contain a response element which details the outcome of processing the entry - the HTTP status code, and the location and ETag header values, which are used for identifying and versioning the resources. In addition, a resource may be included in the entry, as specified by the Prefer header.

3.1.0.11.4 Accepting other Bundle types

A server may choose to accept bundle types other than batch or transaction when POSTed to the [base] URL.

Bundles of type history inherently have the same structure as a transaction, and can be treated as either a transaction or batch, so servers SHOULD accept a history Bundle - this makes it possible to replicate data from one server to another easily using a pub/sub model. Note, however, that the original transaction boundaries might not be represented in a history list, and a resource may occur more than once in a history list, so servers processing history bundles must have some strategy to manage this. When processing a history bundle via a transaction, any entries with the request method of POST must use the Bundle.entry.resource.id (which must match the Bundle.entry.response.location) for that resource so that references are preserved.

For other Bundle types, should the server choose to accept them, there will be no request element (note that every entry will have a resource). In this case, the server treats the entry as either a create or an update interaction, depending on whether it recognizes the identity of the resource - if the identity of the resource refers to a valid location on the server, it should treat it as an update to that location. Note: this option allows a client to delegate the matching process to the server.

3.1.0.12 history

The history interaction retrieves the history of either a particular resource, all resources of a given type, or all resources supported by the system. These three variations of the history interaction are performed by HTTP GET command as shown:

```
GET [base]/[type]/[id]/_history{?[parameters]&_format=[mime-type]}
GET [base]/[type]/_history{?[parameters]&_format=[mime-type]}
GET [base]/_history{?[parameters]&_format=[mime-type]}
```

The return content is a Bundle with type set to history containing the specified version history, sorted with oldest versions last, and including deleted resources. Each entry SHALL minimally contain at least one of: a resource which holds the resource as it is at the conclusion of the interaction, or a request with entry.request.method The request provides information about the result of the interaction that led to this new version, and allows, for instance, a subscriber system to differentiate between newly created resources and updates to existing resources. The principal reason a resource might be missing is that the resource was changed by some other channel rather than via the RESTful interface. If the entry.request.method is a PUT or a POST, the entry SHALL contain a resource.

The interactions create, update, and delete create history entries. Other interactions do not (note that these operations may produce side-effects such as new AuditEvent resources; these are represented as create interactions in their own right). New resources or updates to existing resources that are triggered by operations also appear in the history, as do updates to the resources that result from interactions outside the scope of the RESTful interface.

A HEAD request can also be used - see below.

A create interaction is represented in a history interaction in the following way:

```
<entry>
  <resource>
    <Patient>
      <!-- the id of the created resource -->
      <id value="23424"/>
      <!-- snip -->
    </Patient>
  </resource>
  <request>
    <!-- POST: this was a create -->
    <method value="POST"/>
    <url value="Patient"/>
  </request>
  <!-- response carries the instant the server processed the create -->
  <response>
    <lastModified value="2014-08-15T10:35:02.034Z"/>
  </response>
</entry>
```

A delete interaction is represented in a history interaction in the following way:

```
<entry>
  <!-- no resource included for a delete -->
  <request>
    <method value="DELETE"/>
    <url value="Patient/[id]"/>
  </request>
  <!-- response carries the instant the server processed the delete -->
  <response>
    <lastModified value="2014-08-20T11:05:34.174Z"/>
  </response>
</entry>
```

Notes:

• conditional creates, updates and deletes are converted to direct updates and deletes in a history list

• operations do not appear directly in the history log, but side effects (e.g. creation of audit logs. stored binaries, etc.) will appear where relevant

• The resource in the entry is the resource as processed by the server, not as submitted by the client (may be different)

• The server SHOULD populate at least response.lastModified so the time of processing is clear in the history record

• Servers may choose to only record successful interactions. Servers may choose to only use 200 OK instead of other more specific success codes

• There may be more than one version of a given resource in the history

In addition to the standard _format parameter, the parameters to this interaction may also include:

_count : integer | The maximum number of search results on a page, excluding related resources included by _include or _revinclude or OperationOutcomes. The server is not bound to return the number requested, but cannot return more

_since : instant | Only include resource versions that were created at or after the given instant in time

_at : date(Time) | Only include resource versions that were current at some point during the time period specified in the date time value (see Search notes on date searching)

_list : reference | Only include resource versions that are referenced in the specified list (current list references are allowed)

Each of these parameters SHALL NOT appear more than once.

The history list can be restricted to a limited period by specifying a _since parameter which contains a full date time with time zone. Clients should be aware that due to timing imprecision, they may receive notifications of a resource update on the boundary instant more than once. Servers are not required to support a precision finer than by second.

The updates list can be long, so servers may use paging. If they do, they SHALL use the method described below for breaking the list into pages if appropriate, and respect the specified _count across pages.

The history interaction can be used to set up a subscription from one system to another, so that resources are synchronized between them. Refer to the Subscription resource for an alternate means of system synchronization.

Additional Notes about maintaining a history of resources:

• The history is a record version history on a per-resource basis. It is not intended to support concurrent versions, or multi-branch version history

• Accordingly, there is no way to update or delete past versions of the record, except that the metadata can be modified (mainly for access control purposes)

• All past versions of a resource are considered to be superceded, and no longer active, but retained for audit/integrity purposes

• In the case that a past version of a resource needs to be explicitly documented as 'entered-in-error', use a Provenance resource pointing to the past version of the resource

• When tracing the history of a specific resource, applications should retrieve any provenance resources relating to the resource or its past versions

• If a request is made for a history that is not available (e.g. the system does not keep a history for the type, or the particular instance), the server should return a 404 Not Found along with an OperationOutcome explaining the problem

Thereis a caveat with the _list parameter, associated with changes to the list while making repeated periodic queries; if the list changes, the response will include changes to the resources in the list for the period specified, but will omit both later changes to items no longer in the list, or older changes associated with items in the list. This might not be a problem, but implementers should be aware of this issue.

3.1.0.13 Transactional Integrity

When processing create and update interactions, a FHIR server is not obliged to accept the entire resource as it is; when the resource is retrieved through a read interaction subsequently, the resource may be different. The difference may arise for several reasons:

• The server merged updated content with existing content

• The server applied business rules and altered the content

• The server does not fully support all the features or possible values of the resource

Note that there is no general-purpose method to make merging with existing content or altering the content by business rules safe or predictable - what is possible, safe and/or required is highly context dependent. These kinds of behaviors may be driven by security considerations. With regard to incomplete support, clients can consult the server's base CapabilityStatement profile references to determine which features or values the server does not support.

The PATCH operation offers some support for making changes to a part of a resource and should be used where a client wishes to change just part of a resource, though transactional integrity issues are still important.

To the degree that the server alters the resource for any of the 3 reasons above, the FHIR server will create implementation consequences for the eco-system that it is part of, which will need to be managed (i.e. it will cost more). For this reason, servers SHOULD change the resource as little as possible, given the constraints of the system exposing the FHIR resource. However due to the variability that exists within healthcare, this specification allows that servers MAY alter the resource on create/update.

Similarly, to the degree that an implementation context makes special rules about merging content or altering the content, that context will become more expensive to maintain.

Although these rules are stated with regard to servers, a similar concept applies to clients - to the degree that different client systems interacting with the server do not support the same feature set, the clients and/or the server will be forced to implement custom logic to prevent information from being lost or corrupted.

Some of these problems can be mitigated by following a pattern built on top of version-aware updates. In this pattern:

• The server provides a read interaction for any resource it accepts update interactions on

• Before updating, the client reads the latest version of the resource

• The client applies the changes it wants to the resource, leaving other information intact (note the extension related rules around this)

• The client writes the result back as an update interaction, and is able to handle a 409 or 412 response (usually by trying again)

If clients follow this pattern, then information from other systems that they do not understand will be maintained through the update.

Note that it is possible for a server to choose to maintain the information that would be lost, but there is no defined way for a server to determine whether the client omitted the information because it wasn't supported (perhaps in this case) or whether it wishes to delete the information.

3.1.0.13.1 Conformance

Both client and server systems SHOULD clearly document how transaction integrity is handled, in the documentation in the CapabilityStatement.

3.1.0.14 Paging

Servers SHOULD support paging for the results of a search or history interaction, and if they do, they SHALL conform to this method (adapted from RFC 5005 (Feed Paging and Archiving) for sending continuation links to the client when returning a Bundle (e.g. with history and search). If the server does not do this then there is no way to continue paging.

This example shows the third page of a search result:

```
<Bundle xmlns="http://hl7.org/fhir">
  <!-- snip metadata -->
  <!-- This Search url starts with base search, and adds the effective
    parameters, and additional parameters for search state. All searches
    SHALL return this value.

    In this case, the search continuation method is that the server
    maintains a state, with page references into the stateful list.
  -->
  <link>
    <relation value="self">
    <url value="http://example.org/Patient?name=peter&stateid=23&page=3"/>
  </link>
  <!-- 4 links for navigation in the search. All of these are optional, but recommended -->

  <link>
    <relation value="first"/>
    <url value="http://example.org/Patient?name=peter&stateid=23&page=1"/>
  </link>
  <link>
    <relation value="previous"/>
    <url value="http://example.org/Patient?name=peter&stateid=23&page=2"/>
  </link>
```

```
  <link>
    <relation value="next"/>
    <url value="http://example.org/Patient?name=peter&stateid=23&page=4"/>
  </link>
  <link>
    <relation value="last"/>
    <url value="http://example.org/Patient?name=peter&stateid=23&page=26"/>
  </link>

  <!-- then the search results... -->
</Bundle>
```

The links are opaque to the client, have no dictated structure, and only the server understands them. The client must use the server supplied links in order to traverse the pages. A server MAY add additional state tracking parameters to the links, as shown in the example above, though the server need not use a stateful paging method as shown in this example - it is at the discretion of the server how to best ensure that the continuation retains integrity in the context of ongoing changes to the resources. An alternative approach is to use version specific references to the records on the boundaries, but this is subject to continuity failures when records are updated.

A server MAY inform the client of the total number of resources returned by the interaction for which the results are paged using the Bundle.total.

Note that for search, where _include can be used to return additional related resources, the total number of resources in the feed may exceed the number indicated in totalResults.

In the case of a search, the initial request may be made via a POST, but the follow up page requests will be made via GET requests. However servers SHOULD allow for a client to convert the follow up requests to be made via a POST.

3.1.0.15 Support for HEAD

Anywhere that a GET request can be used, a HEAD request is also allowed. HEAD requests are treated as specified in HTTP: same response as a GET, but with no body.

Servers that do not support HEAD MUST respond in accordance with the HTTP specification, for example using a 405 ("method not allowed") or a 501 ("not implemented").

3.1.0.16 Custom Headers

This specification defines or recommends some custom headers that implementers can use to assist with deployment/debugging purposes:

X-Request-Id | A unique id to for the request/response assigned by either client or server. Request: assigned by the client. Response: assigned by the server
X-Correlation-Id | A client assigned request id echoed back in the response
X-Forwarded-For | Identifies the originating IP address of a client to an intermediary
X-Forwarded-Host | Identifies the original host requested by the client in the Host HTTP request header
X-Intermediary | Stamped by an active intermediary that changes the request or the response to alter it's content (see below)

The request id in X-Request-Id is purely to help connect between requests and logs/audit trails. The client can assign an id to the request, and send that in the X-Request-Id header. The server can either use that id or assign it's own, which it returns as the X-Request-Id header in the response. When the server assigned id is different to the client assigned id, the server SHOULD also return the X-Correlation-Id header with the client's original id in it.

The HTTP protocol may be routed through an HTTP proxy (e.g. as squid). Such proxies are transparent to the applications, though implementers should be alert to the effects of caching, particularly including the risk of receiving stale content. See the HTTP specification for further detail

Interface engines may also be placed between the consumer and the provider. These differ from proxies because they actively alter the content and/or destination of the HTTP exchange and are not bound by the rules that apply to HTTP proxies. Such agents are allowed, but SHALL mark the request with an X-Intermediary header to assist with debugging/troubleshooting. Any agent that modifies an HTTP request or response content other than under the rules for HTTP proxies SHALL add a stamp to the HTTP headers like this:

```
X-Intermediary : [identity - usually a FQDN]
```

End point systems SHALL NOT use this header for any purpose. Its aim is to assist with system troubleshooting.

3.1.0.17 Summary

These tables present a summary of the interactions described here. Note that all requests may include an optional Accept header to indicate the format used for the response (this is even true for DELETE since an OperationOutcome may be returned).

Interaction | Path | Request
| Verb | Content-Type | Body | Prefer | Conditional
read | /[type]/[id] | GET | N/A | N/A | N/A | O: ETag, If-Modified-Since, If-None-Match
vread | /[type]/[id]/_history/[vid] | GET | N/A | N/A | N/A | N/A

update | /[type]/[id] | PUT | R | Resource | O | O: If-Match
patch | /[type]/[id] | PATCH | R (may be a patch type) | Patch | O | O: If-Match
delete | /[type]/[id] | DELETE | N/A | N/A | N/A | N/A
create | /[type] | POST | R | Resource | O | O: If-None-Exist
search | /[type]? | GET | N/A | N/A | N/A | N/A
/[type]/_search? | POST | application/x-www-form-urlencoded | form data | N/A | N/A
search-all | ? | GET | N/A | N/A | N/A | N/A
capabilities | /metadata | GET | N/A | N/A | N/A | N/A
transaction | / | POST | R | Bundle | O | N/A
history | /[type]/[id]/_history | GET | N/A | N/A | N/A | N/A
history-type | /[type]/_history | GET | N/A | N/A | N/A | N/A
history-all | /_history | GET | N/A | N/A | N/A | N/A
(operation) | /$[name], /[type]/$[name] or /[type]/[id]/$[name] | POST | R | Parameters | N/A | N/A
GET | N/A | N/A | N/A | N/A
POST | application/x-www-form-urlencoded | form data | N/A | N/A

Notes:

• N/A = not present, R = Required, O = optional

• For operations defined on all resources, including direct access to the meta element, see Resource Operations

Interaction | Response
| Content-Type | Body | Location | Versioning | Status Codes
read | R | R: Resource | N/A | R: ETag, Last-Modified | 200, 404, 410
vread | R | R: Resource | N/A | R: ETag, Last-Modified | 200, 404
update | R if body | O: Resource (Prefer) | N/A | R: ETag, Last-Modified | 200, 201, 400, 404, 405, 409, 412, 422
patch | R if body | O: Resource (Prefer) | N/A | R: ETag, Last-Modified | 200, 201, 400, 404, 405, 409, 412, 422
delete | R if body | O: OperationOutcome | N/A | N/A | 200, 202, 204, 404, 405, 409, 412
create | R if body | O : Resource (Prefer) | R | R: ETag, Last-Modified | 201, 400, 404, 405, 422
search | R | R: Bundle | N/A | N/A | 200, 401?
search-all | R | R: Bundle | N/A | N/A | 200, 401?
capabilities | R | R: CapabilityStatement | N/A | N/A | 200, 404
transaction | R | R: Bundle | N/A | N/A | 200, 400, 404, 405, 409, 412, 422
history | R | R: Bundle | N/A | N/A | 200
history-type | R | R: Bundle | N/A | N/A | 200
history-all | R | R: Bundle | N/A | N/A | 200
(operation) | R | R: Parameters/Resource | N/A | N/A | 200

Note: this table lists the status codes described here, but other status codes are possible as described by the HTTP specification. Additional codes that are likely are server errors and various codes associated with authentication protocols. The security page notes several security related issues that may impact which codes to return.

3.1.1 Search

URL: https://hl7.org/fhir/R4/search.html

3.1.1 Search

FHIR Infrastructure Work Group | Maturity Level: Normative | Standards Status: Normative

| This page has been approved as part of an ANSI standard. See the Infrastructure Package for further details.

Searching for resources is fundamental to the mechanics of FHIR. Search operations traverse through an existing set of resources filtering by parameters supplied to the search operation. The text below describes the FHIR search framework, starting with simple cases moving to the more complex. Implementers need only implement the amount of complexity that they require for their implementations.

3.1.1.1 Summary Table

Search Parameter Types | Parameters for all resources | Search result parameters
Number Date/DateTime String Token Reference Composite Quantity URI Special | _id _lastUpdated _tag _profile _security _text _content _list _has _type | _sort _count _include _revinclude _summary _total _elements _contained _containedType

In addition, there is a special search parameters _query and _filter that allow for an alternative method of searching, and the parameters _format and _pretty defined for all interactions.

Also, there is a single page that lists all the search parameters. Note that search parameter names are case sensitive, though this specification never defines different parameters with names that differ only in case. Clients SHOULD use correct case, and servers SHALL not define additional parameters with different meanings with names that only differ in case.

There are several safety issues associated with the implementation of searching that implementers should always keep in mind.

3.1.1.2 Introduction

In the simplest case, a search is executed by performing a GET operation in the RESTful framework:

```
GET [base]/[type]?name=value&...{&_format=[mime-type]}}
```

For this RESTful search (see definition in RESTful API), the parameters are a series of name=[value] pairs encoded in the URL or as an application/x-www-form-urlencoded submission for a POST:

```
POST  [base]/[type]/_search{?[parameters]{&_format=[mime-type]}}
```

The server determines which of the set of resources it serves meet the specific criteria, and returns the results in the HTTP response as a bundle which includes the resources that are the results of the search. Note that the _format parameter works for search like for other interactions.

Search operations are executed in one of three defined contexts that control which set of resources are being searched:

• A specified resource type: GET [base]/[type]?parameter(s)

• A specified compartment, perhaps with a specified resource type in that compartment: GET [base]/Patient/[id]/[type]?parameter(s)

• All resource types: GET [base]?parameter(s) (parameters common to all types). If the _type parameter is included, all other search parameters SHALL be common to all provided types. If _type is not included, all parameters SHALL be common to all resource types.

Search operations can also be implemented in the messaging framework.

The server determines which of their resources meet the criteria contained in the search parameters as described below. However, the server has the prerogative to return additional search results if it believes them to be relevant. Note: There is a special search for the most relevant context in which the search set is indeterminate: Patient MPI Search.

Search using GET may include sensitive information in the search parameters. Therefore, secure communications and endpoint management are recommended, see Security Communications

The response to any search operation is always a list of resources in a Bundle. An alternative approach is to use GraphQL.

3.1.1.3 Handling Errors

If a server is unable to execute a search request, it may return an error. A HTTP status code of 403 signifies that the server refused to perform the search, while other 4xx and 5xx codes signify that some sort of error has occurred. When the search fails, a server SHOULD return an OperationOutcome detailing the cause of the failure. Note: An empty search result is not a failure.

In some cases, parameters may cause an error, or might not be able to match anything. For instance:

• A parameter may refer to a non-existent resource e.g. GET [base]/Observation?subject=101, where "101" does not exist

• A parameter may refer to a non-existent resource e.g. GET [base]/Observation?patient.identifier=http://example.com/fhir/identifier/mrn|123456, where there is no patient for MRN 123456

• A parameter may refer to an unknown code e.g. GET [base]/Observation?code=loinc|1234-1, where the LOINC code "1234-1" is not known to the server

• A parameter may refer to a time that is out of scope e.g. GET [base]/Condition?onset=le1995, where the system only has data going back to 2001

• A parameter may use an illegal or unacceptable modifier e.g. GET [base]/Condition?onset:text=1995, where the modifier cannot be processed by the server

• A data time parameter may have incorrect format e.g. GET [base]/Condition?onset=23%20May%202009

• A parameter may be unknown or unsupported (see below)

Note: Empty parameters are not an error - they are just ignored by the server.

Where the content of the parameter is syntactically incorrect, servers SHOULD return an error. However, where the issue is a logical condition (e.g. unknown subject or code), the server SHOULD process the search, including processing the parameter - with the result of returning an empty search set, since the parameter cannot be satisfied.

In such cases, the search process MAY include an OperationOutcome in the search set that contains additional hints and warnings about the search process. This is included in the search results as an entry with search mode = outcome. Clients can use this information to improve future searches. If, for example, a client performed the following search:

```
GET [base]/Observation?patient.identifier=http://example.com/fhir/identifier/mrn|123456
```

and there is no patient for MRN 123456, the server would return a bundle with a warning.

Unknown and unsupported parameters

Servers may receive parameters from the client that they do not recognize, or may receive parameters they recognize but do not support (either in general, or for a specific search). In general, servers SHOULD ignore unknown or unsupported parameters for the following reasons:

• Various HTTP stacks and proxies may add parameters that aren't under the control of the client

• The client can determine what parameters the server used by examining the self link in the return (see below)

Clients can specify how the server should behave, by using the prefer header

• Prefer: handling=strict: Client requests that the server return an error for any unknown or unsupported parameter

• Prefer: handling=lenient: Client requests that the server ignore any unknown or unsupported parameter

Servers SHOULD honor the client's request, but are not required to do so.

3.1.1.4 Standard Parameters

3.1.1.4.1 Parameters for all resources

The following parameters apply to all resources: _content, _id, _lastUpdated, _profile, _query, _security, _source, _tag. In addition, the search parameter _text and _filter, (documented below) also applies to all resources (as do the search result parameters).

The search parameter _id refers to the logical id of the resource, and can be used when the search context specifies a resource type:

```
GET [base]/Patient?_id=23
```

This search finds the patient resource with the given id (there can only be one resource for a given id). Functionally, this is equivalent to a simple read operation:

```
GET [base]/Patient/23
```

However, the search with parameter _id returns a bundle with the requested resource, instead of just the resource itself. Additional parameters can be added which may provide additional functionality on top of this base read equivalence (e.g. _include). Note that although the _id parameter has a type of token, because servers SHALL use exact match with it, there is no system for the _id parameter. Note that matches on _id are always case sensitive.

The search parameter _lastUpdated can be used to select resources based on the last time they were changed:

```
GET [base]/Observation?_lastUpdated=gt2010-10-01
```

This search finds any observations changed since 1-Oct 2010. When this search parameter is used, applications should consider synchronization approaches (RESTful history or the Subscription resource).

The search parameters _tag, _profile and _security parameters search on the equivalent elements in the meta element. For example,

```
GET [base]/Condition?_tag=http://acme.org/codes|needs-review
```

searches for all Condition resources with the tag:

```
{
  "system" : "http://acme.org/codes",
  "code" : "needs-review"
}
```

In the same manner:

```
GET [base]/DiagnosticReport?_profile=http://hl7.org/fhir/StructureDefinition/lipid
GET [base]/DiagnosticReport?_profile=Profile/lipid
```

restricts the search to only DiagnosticReport resources that are tagged as conforming to a particular profile. The second reference is relative and refers a local profile on the same server.

_tag, and _security parameters are token types (see below), and _profile is a reference search parameter.

3.1.1.4.2 Parameters for each resource

In addition to the _id parameter which exists for all resources, each FHIR resource type defines its own set of search parameters with their names, types, and meanings. These search parameters are on the same page as the resource definitions, and are also published as part of the standard Capability statement (XML or JSON).

In general, the defined search parameters correspond to a single element in the resource, but this is not required, and some search parameters refer to the same type of element in multiple places, or refer to derived values.

Some search parameters defined by resources are associated with more than one path in a resource. This means that the search parameter matches if any of the paths contain matching content. If a path matches, the whole resource is returned in the search results. The client may have to examine the resource to determine which path contains the match.

Servers are not required to implement any of the standard search parameters (except for the _id parameter described above). Servers may also define their own parameters.

3.1.1.4.3 Search Parameter Types

Each search parameter is defined by a type that specifies how the search parameter behaves. These are the defined parameter types:

number | Search parameter SHALL be a number (a whole number, or a decimal).

date | Search parameter is on a date/time. The date format is the standard XML format, though other formats may be supported.

string | Search parameter is a simple string, like a name part. Search is case-insensitive and accent-insensitive. May match just the start of a string. String parameters may contain spaces.

token | Search parameter on a coded element or identifier. May be used to search through the text, display, code and code/codesystem (for codes) and label, system and key (for identifier). Its value is either a string or a pair of namespace and value, separated by a "|", depending on the modifier used.

reference | A reference to another resource (Reference or canonical).

composite | A composite search parameter that combines a search on two values together.

quantity | A search parameter that searches on a quantity.

uri | A search parameter that searches on a URI (RFC 3986).

special | Special logic applies to this parameter per the description of the search parameter.

The search parameters can also append "modifiers" that control their behavior. The kinds of modifiers that available is dependent on the type of parameter being modified.

3.1.1.4.4 Modifiers

Parameters are defined per resource. Parameter names may specify a modifier as a suffix. The modifiers are separated from the parameter name by a colon. Modifiers are:

• For all parameters (except combination): :missing; e.g. gender:missing=true (or false). Searching for gender:missing=true will return all the resources that don't have a value for the gender parameter (which usually equates to not having the relevant element in the resource). Searching for gender:missing=false will return all the resources that have a value for the gender parameter. For simple data type elements, :missing=true will match on all elements where either the underlying element is omitted or where the element is present with extensions but no @value is specified

• For string: :exact returns results that match the entire supplied parameter, including casing and combining characters. Note that the handling of (extended) grapheme clusters is within the discretion of the server, i.e. the server decides if a string search parameter matches on canonically equivalent characters, or matches on the actual used Unicode code points, or :contains (case insensitive and combining character-insensitive, search text matched anywhere in the string), instead of the default behavior (case insensitive and accent-insensitive, partial matches at the start of the string).

• For token: :text (the match does a partial searches on the text portion of a CodeableConcept or the display portion of a Coding), instead of the default search which uses codes. Other defined modifiers are :in, :below, :above and :not-in which are described below.

• For reference: :[type] where [type] is the name of a type of resource, :identifier, and, for some parameters, :above and :below

• For uri: :below, :above indicate that instead of an exact match, either the search term left-matches the value, or vice-versa.

Server SHALL reject any search request that contains is suffixed by a modifier that the server does not support for that parameter. For example, if the server supports the name search parameter, but not the :exact modifier on the name, it should reject a search with the parameter name:exact=Bill, using an HTTP 400 error with an OperationOutcome with a clear error message.

3.1.1.4.5 Prefixes

For the ordered parameter types of number, date, and quantity, a prefix to the parameter value may be used to control the nature of the matching. To avoid URL escaping and visual confusion, the following prefixes are used:

eq | the value for the parameter in the resource is equal to the provided value | the range of the search value fully contains the range of the target value

ne | the value for the parameter in the resource is not equal to the provided value | the range of the search value does not fully contain the range of the target value

gt | the value for the parameter in the resource is greater than the provided value | the range above the search value intersects (i.e. overlaps) with the range of the target value

lt | the value for the parameter in the resource is less than the provided value | the range below the search value intersects (i.e. overlaps) with the range of the target value

ge | the value for the parameter in the resource is greater or equal to the provided value | the range above the search value intersects (i.e. overlaps) with the range of the target value, or the range of the search value fully contains the range of the target value

le | the value for the parameter in the resource is less or equal to the provided value | the range below the search value intersects (i.e. overlaps) with the range of the target value or the range of the search value fully contains the range of the target value

sa | the value for the parameter in the resource starts after the provided value | the range of the search value does not overlap with the range of the target value, and the range above the search value contains the range of the target value

eb | the value for the parameter in the resource ends before the provided value | the range of the search value does overlap not with the range of the target value, and the range below the search value contains the range of the target value

ap | the value for the parameter in the resource is approximately the same to the provided value. Note that the recommended value for the approximation is 10% of the stated value (or for a date, 10% of the gap between now and the date), but systems may choose other values where appropriate | the range of the search value overlaps with the range of the target value

If no prefix is present, the prefix eq is assumed. Note that the way search parameters operate is not the same as the way the operations on two numbers work in a mathematical sense. sa (starts-after) and eb (ends-before) are not used with integer values but are used for decimals.

For each prefix above, two interpretations are provided - the simple intent of the prefix and the interpretation of the parameter when applied to ranges. The range interpretation is provided for decimals and dates. Searches are always performed on values that are implicitly or explicitly a range. For instance, the number 2.0 has an implicit range of 1.95 to 2.05, and the date 2015-08-12 has an implicit range of all the time during that day. If the target value is a Range, a Period, or a Timing, then the target is explicitly a range. Three ranges are identified:

range of the value | The limits implied by the precision of the value | The number 2.0 has a range of 1.95 to 2.05 The date 2015-08-12 has a range from 00:00 to 00:00 exclusive

range below the value | Up to the specified value | The range below 2.0 includes any value less or equal to <2.00000000000000000000 The range before 2015-08-12T05:23:45 includes any time up to 2015-08-12T05:23:45.000000000000000

range above the value | The specified value and up | The range above 2.0 includes any value greater or equal to <2.00000000000000000000 The range after 2015-08-12T05:23:45 includes any time after 2015-08-12T05:23:45.000000000000000

The proper use of these ranges is discussed further below.

3.1.1.4.6 number

Searching on a simple numerical value in a resource. Examples:

[parameter]=100 | Values that equal 100, to 3 significant figures precision, so this is actually searching for values in the range [99.5 ... 100.5)

[parameter]=100.00 | Values that equal 100, to 5 significant figures precision, so this is actually searching for values in the range [99.995 ... 100.005)

[parameter]=1e2 | Values that equal 100, to 1 significant figures precision, so this is actually searching for values in the range [95 ... 105)

[parameter]=lt100 | Values that are less than exactly 100

[parameter]=le100 | Values that are less or equal to exactly 100

[parameter]=gt100 | Values that are greater than exactly 100

[parameter]=ge100 | Values that are greater or equal to exactly 100

[parameter]=ne100 | Values that are not equal to 100 (actually, in the range 99.5 to 100.5)

Notes about searching on Numbers:

• When a number search is used against a resource element that stores a simple integer (e.g. ImmunizationRecommendation.recommendation.doseNumber), and the search parameter is not expressed using the exponential forms, and does not include any non-zero digits after a decimal point, the significance issues cancel out and searching is based on exact matches. Note that if there are non-zero digits after a decimal point, there cannot be any matches

• When a comparison prefix in the set lgt, lt, ge, le, sa & eb is provided, the implicit precision of the number is ignored, and they are treated as if they have arbitrarily high precision

• The way search parameters operate in resources is not the same as whether two numbers are equal to each other in a mathematical sense

• Searching on decimals involves an implicit range. The number of significant digits of the implicit range is the number of digits specified in the search parameter value, excluding leading zeros. So 100 and 1.00e2 both have the same number of significant digits - three

Here are some example searches:

Search | Description

GET [base]/RiskAssessment?probability=gt0.8 | Search for all the Risk Assessments with probability great than 0.8 (could also be probability=gt8e-1 using exponential form)

GET [base]/ImmunizationRecommendation?dose-number=2 | Search for any immunization recommendation recommending a second dose

3.1.1.4.7 date

A date parameter searches on a date/time or period. As is usual for date/time related functionality, while the concepts are relatively straight-forward, there are a number of subtleties involved in ensuring consistent behavior.

The date parameter format is yyyy-mm-ddThh:mm:ss[Z|(+|-)hh:mm] (the standard XML format).

Technically, this is any of the date, dateTime, and instant data types; e.g. Any degree of precision can be provided, but it SHALL be populated from the left (e.g. can't specify a month without a year), except that the minutes SHALL be present if an hour is present, and you SHOULD provide a time zone if the time part is present. Note: Time can consist of hours and minutes with no seconds, unlike the XML Schema dateTime type. Some user agents may escape the : characters in the URL, and servers SHALL handle this correctly.

Date parameters may be used with the following data types:

date | The range of the value is the day, month, or year as specified

dateTime | The range of the value as defined above; e.g. For example, the date 2013-01-10 specifies all the time from 00:00 on 10-Jan 2013 to immediately before 00:00 on 11-Jan 2013

instant | An instant is considered a fixed point in time with an interval smaller than the precision of the system, i.e. an interval with an effective width of 0

Period | Explicit, though the upper or lower bound might not actually be specified in resources.

Timing | the specified scheduling details are ignored and only the outer limits matter. For instance, a schedule that specifies every second day between 31-Jan 2013 and 24-Mar 2013 includes 1-Feb 2013, even though that is on an odd day that is not specified by the period. This is to keep the server load processing queries reasonable.

Implicitly, a missing lower boundary is "less than" any actual date. A missing upper boundary is "greater than" any actual date. The use of the prefixes:

[parameter]=eq2013-01-14 | 2013-01-14T00:00 matches (obviously) 2013-01-14T10:00 matches 2013-01-15T00:00 does not match - it's not in the range

[parameter]=ne2013-01-14 | 2013-01-15T00:00 matches - it's not in the range 2013-01-14T00:00 does not match - it's in the range 2013-01-14T10:00 does not match - it's in the range

[parameter]=lt2013-01-14T10:00 | 2013-01-14 matches, because it includes the part of 14-Jan 2013 before 10am

[parameter]=gt2013-01-14T10:00 | 2013-01-14 matches, because it includes the part of 14-Jan 2013 after 10am

[parameter]=ge2013-03-14 | "from 21-Jan 2013 onwards" is included because that period may include times after 14-Mar 2013

[parameter]=le2013-03-14 | "from 21-Jan 2013 onwards" is included because that period may include times before 14-Mar 2013

[parameter]=sa2013-03-14 | "from 15-Mar 2013 onwards" is included because that period starts after 14-Mar 2013 "from 21-Jan 2013 onwards" is not included because that period starts before 14-Mar 2013 "before and including 21-Jan 2013" is not included because that period starts (and ends) before 14-Mar 2013

[parameter]=eb2013-03-14 | "from 15-Mar 2013 onwards" is not included because that period starts after 14-Mar 2013 "from 21-Jan 2013 onwards" is not included because that period starts before 14-Mar 2013, but does not end before it "before and including 21-Jan 2013" is included because that period ends before 14-Mar 2013

[parameter]=ap2013-03-14 | 14-Mar 2013 is included - as it exactly matches 21-Jan 2013 is not included because that is near 14-Mar 2013 15-Jun 2015 is not included - as it is not near 14-Mar 2013. Note that the exact value here is at the discretion of the system

Other notes:

• When the date parameter is not fully specified, matches against it are based on the behavior of intervals, where: Dates with only the year specified are equivalent to an interval that starts at the first instant of January 1st to the last instant of December 31st, e.g. 2000 is equivalent to an interval of [2000-01-01T00:00, 2000-12-31T23:59]. Dates with the year and month are equivalent to an interval that starts at the first instant of the first day of the month and ends on the last instant of the last day of the month, e.g. 2000-04 is equivalent to an interval of [2000-04-01T00:00, 2000-04-30T23:59].

• Where possible, the system should correct for time zones when performing queries. Dates do not have time zones, and time zones should not be considered. Where both search parameters and resource element date times do not have time zones, the servers local time zone should be assumed.

To search for all the procedures in a patient compartment that occurred over a 2-year period:

```
GET [base]/Patient/23/Procedure?date=ge2010-01-01&date=le2011-12-31
```

Managing time zones and offsets and their impact on search is a very difficult area. The FHIR implementation community is still investigating and debating the best way to handle time zones. Implementation guides may make additional rules in this regard.

Future versions of this specification may impose rules around the use of time zones with dates. Implementers and authors of implementation guides should be aware of ongoing work in this area.

Implementer feedback is welcome on the issue tracker or chat.fhir.org .

3.1.1.4.8 string

For a simple string search, a string parameter serves as the input for a search against sequences of characters. This search is insensitive to casing and included combining characters, like accents or other diacritical marks. Punctuation and non-significant whitespace (e.g. repeated space characters, tab vs space) should also be ignored. By default, a field matches a string query if the value of the field equals or starts with the supplied parameter value, after both have been normalized by case and combining characters. Therefore, the default string search only operates on the base characters of the string parameter. The :contains modifier returns results that include the supplied parameter value anywhere within the field being searched. The :exact modifier returns results that match the entire supplied parameter, including casing and accents.

Examples:

[base]/Patient?given=eve | Any patients with a name containing a given part with "eve" at the start of the name. This would include patients with the given name "Eve", "Evelyn".

[base]/Patient?given:contains=eve | Any patients with a name with a given part containing "eve" at any position. This would include patients with the given name "Eve", "Evelyn", and also "Severine".

[base]/Patient?given:exact=Eve | Any patients with a name with a given part that is exactly "Eve". Note: This would not include patients with the given name "eve" or "EVE".

An additional modifier :text can be used to specify a search with advanced text handling (see below) though only a few servers are expected to offer this facility.

When a string search parameter refers to the types HumanName and Address, the search covers the elements of type string, and does not cover elements such as use and period. For robust search, servers

should search the parts of a family name independently. E.g. searching either "Carreno" or "Quinones" should match a family name of "Carreno Quinones". HL7 affiliates may make more specific recommendations about how search should work in their specific culture.

It is at the discretion of the server whether to pre-process names, addresses, and contact details to remove separator characters prior to matching in order to ensure more consistent behavior. For example, a server might remove all spaces and - characters from phone numbers. What is most appropriate varies depending on culture and context. A server may also use a free-text style searches on this property to achieve the best results. When searching whole names and addresses (not parts), servers may also use flexible match or a free-text style searches on names to achieve the best results.

3.1.1.4.9 uri

The uri parameter refers to an element that contains a URI (RFC 3986). By default, matches are precise (e.g. case, accent, and escape) sensitive, and the entire URI must match. The modifier :above or :below can be used to indicate that partial matching is used. For example:

```
GET [base]/ValueSet?url=http://acme.org/fhir/ValueSet/123
GET [base]/ValueSet?url:below=http://acme.org/fhir/
GET [base]/ValueSet?url:above=http://acme.org/fhir/ValueSet/123/_history/5
GET [base]/ValueSet?url=urn:oid:1.2.3.4.5
```

• The first line is a request to find any value set with the exact url "http://acme.org/fhir/ValueSet/123"

• The second line performs a search that will return any value sets that have a URL that starts with "http://acme.org/fhir/"

• The third line shows the converse - search for any value set above a given specific URL. This will match on any value set with the specified URL, but also on http://acme.org/ValueSet/123. Note that there are not many use cases where :above is useful as compared to the :below search

• The fourth line shows an example of searching by an OID. Note that the :above and :below modifiers only apply to URLs, and not URNS such as OIDs

The search type uri is used with elements of type uri and url. The type reference is used for the types Reference and canonical. Note that for uri parameters that refer to the Canonical URLs of the conformance and knowledge resources (e.g. StructureDefinition, ValueSet, PlanDefinition etc), servers SHOULD support searching by canonical references, and SHOULD support automatically detecting a |[version] portion as part of the search parameter, and interpreting that portion as a search on the version.

3.1.1.4.10 token

A token type is a parameter that provides a close to exact match search on a string of characters, potentially scoped by a URI. It is mostly used against a code or identifier data type where the value may have a URI that scopes its meaning, where the search is performed against the pair from a Coding or an Identifier. Tokens are also used against other fields where exact matches are required - uris, booleans, and ContactPoints. In these cases, the URI portion is not used.

For tokens, matches are literal (e.g. not based on subsumption or other code system features). Match is case sensitive unless the underlying semantics for the context indicate that the token should be interpreted case-insensitively (see, e.g. CodeSystem.caseSensitive). Note that matches on _id are always case sensitive. If the underlying data type is string then the search is not case sensitive.

Note: There are many challenging issues around case sensitivity and token searches. Some code systems are case sensitive (e.g. UCUM) while others are known not to be. For many code systems, it's ambiguous. Other kinds of values are also ambiguous. When in doubt, servers SHOULD treat tokens in a case-insensitive manner, on the grounds that including undesired data has less safety implications than excluding desired behavior. Clients SHOULD always use the correct case when possible, and allow for the server to perform case-insensitive matching.

To use subsumption-based logic, use the modifiers below, or list all the codes in the hierarchy. The syntax for the value is one of the following:

• [parameter]=[code]: the value of [code] matches a Coding.code or Identifier.value irrespective of the value of the system property

• [parameter]=[system]|[code]: the value of [code] matches a Coding.code or Identifier.value, and the value of [system] matches the system property of the Identifier or Coding

• [parameter]=|[code]: the value of [code] matches a Coding.code or Identifier.value, and the Coding/Identifier has no system property

• [parameter]=[system]|: any element where the value of [system] matches the system property of the Identifier or Coding

Notes:

• The namespace URI and code both must be escaped correctly. If a system is not applicable (e.g. an element of type uri, then just the form [parameter]=[code] is used.

• For token parameters on elements of type ContactPoint, uri, or boolean, the presence of the pipe symbol SHALL NOT be used - only the [parameter]=[code] form is allowed

Token search parameters are used for the following data types:

Data Type | URI | Code | Comments
Coding | Coding.system | Coding.code
CodeableConcept | CodeableConcept.coding.system | CodeableConcept.coding.code | Matches against any coding in the CodeableConcept
Identifier | Identifier.system | Identifier.value | Clients can search by type not system using the :of-type modifier, see below. To search on a CDA II.root - which may appear in either Identifier.system or Identifier.value, use the syntax identifier=|[root],[root]
ContactPoint | | ContactPoint.value | At the discretion of the server, token searches on ContactPoint may use special handling, such as ignoring punctuation, performing partial searches etc.
code | (implicit) | code | the system is defined in the value set (though it's not usually needed)
boolean | | boolean | The implicit system for boolean values is http://hl7.org/fhir/special-values but this is never actually used
uri | | uri
string | n/a | string | Token is sometimes used for string to indicate that exact matching is the correct default search strategy

Note: The use of token search parameters for boolean fields: the boolean values "true" and "false" are also represented as formal codes in the Special Values code system, which is useful when boolean values need to be represented in a Coding data type. The namespace for these codes is http://hl7.org/fhir/special-values, though there is usually no reason to use this, as a simple true or false is sufficient.

Modifiers:

Modifier | Use
:text | The search parameter is processed as a string that searches text associated with the code/value - either CodeableConcept.text, Coding.display, or Identifier.type.text. In this case, the search functions as a normal string search
:not | Reverse the code matching described in the paragraph above: return all resources that do not have a matching item. Note that this includes resources that have no value for the parameter - e.g. ?gender:not=male includes all patients that do not have gender = male, including patients that do not have a gender at all
:above | The search parameter is a concept with the form [system]|[code], and the search parameter tests whether the coding in a resource subsumes the specified search code. For example, the search concept has an is-a relationship with the coding in the resource, and this includes the coding itself.
:below | the search parameter is a concept with the form [system]|[code], and the search parameter tests whether the coding in a resource is subsumed by the specified search code. For example, the coding in the resource has an is-a relationship with the search concept, and this includes the coding itself.
:in | The search parameter is a URI (relative or absolute) that identifies a value set, and the search parameter tests whether the coding is in the specified value set. The reference may be literal (to an address where the value set can be found) or logical (a reference to ValueSet.url). If the server can treat the reference as a literal URL, it does, else it tries to match known logical ValueSet.url values.
:not-in | The search parameter is a URI (relative or absolute) that identifies a value set, and the search parameter tests whether the coding is not in the specified value set.
:of-type | The search parameter has the format system|code|value, where the system and code refer to a Identifier.type.coding.system and .code, and match if any of the type codes match. All 3 parts must be present

Most servers will only process value sets that are already known/registered/supported internally. However, servers can elect to accept any valid reference to a value set. Servers may elect to consider concept mappings when testing for subsumption relationships.

Example searches:

Search | Description
GET [base]/Patient?identifier=http://acme.org/patient|2345 | Search for all the patients with an identifier with key = "2345" in the system "http://acme.org/patient"
GET [base]/Patient?gender=male | Search for any patient with a gender that has the code "male"
GET [base]/Patient?gender:not=male | Search for any patient with a gender that does not have the code "male". Note that for :not, the search does not return any resources that have a gen
GET [base]/Composition?section=48765-2 | Search for any Composition that contains an Allergies and adverse reaction section
GET [base]/Composition?section:not=48765-2 | Search for any Composition that does not contain an Allergies and adverse reaction section. Note that this search does not return "any document that has a section that is not an Allergies and adverse reaction section" (e.g. in the presence of multiple possible matches, the negation applies to the set, not each individual entry)
GET [base]/Patient?active=true | Search for any patients that are active
GET [base]/Condition?code=http://acme.org/conditions/codes|ha125 | Search for any condition with a code "ha125" in the code system "http://acme.org/conditions/codes"
GET [base]/Condition?code=ha125 | Search for any condition with a code "ha125". Note that there is not often any useful overlap in literal symbols between code systems, so the previous example is generally preferred
GET [base]/Condition?code:text=headache | Search for any Condition with a code that has a text "headache" associated with it (either in the text, or a display)
GET [base]/Condition?code:in=http%3A%2F%2Fsnomed.info%2Fsct%3Ffhir_vs%3Disa%2F126851005 | Search for any condition in the SNOMED CT value set "http://snomed.info/sct?fhir_vs=isa/126851005" that includes all descendants of "Neoplasm of liver"
GET [base]/Condition?code:below=126851005 | Search for any condition that is subsumed by the SNOMED CT Code "Neoplasm of liver". Note: This is the same outcome as the previous search
GET [base]/Condition?code:in=http://acme.org/fhir/ValueSet/cardiac-conditions | Search for any condition that is in the institutions list of cardiac conditions
GET [base]/Patient?identifier:otype=http://terminology.hl7.org/CodeSystem/v2-0203|MR|446053 | Search for the Medical Record Number 446053 - this is useful where the system id for the MRN is not known

3.1.1.4.10.1 Searching Mime Types

The :below modifier is also very useful with searching mime types, such as for DocumentReference.contenttype which refers to Attachment.contentType. A simple search such as:

```
GET [base]/DocumentReference?contenttype=text/xml
```

will miss documents with a mime type such as text/xml; charset=UTF-8. This search will find all text/xml documents:

```
GET [base]/DocumentReference?contenttype:below=text/xml
```

For ease of processing on the server, servers are only required to support :below on the base part of the mime type; servers are not required to sort between different parameters and do formal subsumption logic.

3.1.1.4.11 quantity

A quantity parameter searches on the Quantity data type. The syntax for the value follows the form:

• [parameter]=[prefix][number]|[system]|[code] matches a quantity with the given unit

The prefix is optional, and is as described above, both regarding how precision and comparator/range operators are interpreted. Like a number parameter, the number part of the search value can be a decimal in exponential format. The system and code follow the same pattern as token parameters are also optional. Example searches:

Search	Description
GET [base]/Observation?value-quantity=5.4|http://unitsofmeasure.org|mg	Search for all the observations with a value of 5.4(+/-0.05) mg where mg is understood as a UCUM unit (system/code)
GET [base]/Observation?value-quantity=5.40e-3|http://unitsofmeasure.org|g	Search for all the observations with a value of 0.0054(+/-0.000005) g where g is understood as a UCUM unit (system/code)
GET [base]/Observation?value-quantity=5.4||mg	Search for all the observations with a value of 5.4(+/-0.05) mg where the unit - either the code (code) or the stated human unit (unit) are "mg"
GET [base]/Observation?value-quantity=5.4	Search for all the observations with a value of 5.4(+/-0.05) irrespective of the unit
GET [base]/Observation?value-quantity=le5.4|http://unitsofmeasure.org|mg	Search for all the observations where the value of is less than 5.4 mg exactly where mg is understood as a UCUM unit
GET [base]/Observation?value-quantity=ap5.4|http://unitsofmeasure.org|mg	Search for all the observations where the value of is about 5.4 mg where mg is understood as a UCUM unit (typically, within 10% of the value - see above)

Specifying a system and a code for the search implies that the search is based on a particular code system - usually UCUM , and that a precise (and potentially canonical) match is desired. In this case, it is inappropriate to search on the human display for the unit, which can be is uncontrolled and may unpredictable.

The search processor may choose to perform a search based on canonical units (e.g. any value where the units can be converted to a value in mg in the case above). For example, an observation may have a value of 23 mm/hr. This is equal to 0.23 m/hr. The search processer can choose to normalise all the values to a canonical unit such as 6.4e-6 m/sec, and convert search terms to the same units (m/sec). Such conversions can be performed based on the semantics defined in UCUM

3.1.1.4.12 reference

A reference parameter refers to references between resources. For example, find all Conditions where the subject reference is a particular patient, where the patient is selected by name or identifier. The interpretation of a reference parameter is either:

• [parameter]=[id] the logical [id] of a resource using a local reference (i.e. a relative reference)

• [parameter]=[type]/[id] the logical [id] of a resource of a specified type using a local reference (i.e. a relative reference), for when the reference can point to different types of resources (e.g. Observation.subject)

• [parameter]=[url] where the [url] is an absolute URL - a reference to a resource by its absolute location, or by it's canonical URL

Note: A relative reference resolving to the same value as a specified absolute URL, or vice versa, qualifies as a match. For example, if the search parameter value is Patient/123, then this will find references like this:

```
<patient>
  <reference value="Patient/123"/>
</patient>
```

If the server base address is http://example.org/fhir, then the full URL for that reference is http://example.org/fhir/Patient/123, which means that the search term also matches patient references like this:

```
<patient>
  <reference value="http://example.org/fhir/Patient/123"/>
</patient>
```

In addition, searching for reference=http://example.org/fhir/Patient/123 will also match both references.

Some references may point to more than one type of resource; e.g. subject: Reference(Patient|Group|Device|..). In these cases, multiple resources may have the same logical identifier. Servers SHOULD reject a search where the logical id refers to more than one matching resource across different types. In order to allow the client to perform a search in these situations the type is specified explicitly:

```
GET [base]/Observation?subject=Patient/23
```

This searches for any observations where the subject refers to the patient resource with the logical identifier "23". A modifier is also defined to allow the client to be explicit about the intended type:

```
GET [base]/Observation?subject:Patient=23
```

This has the same effect as the previous search. The modifier becomes useful when used with chaining as explained in the next section. Note: The [type] modifier can't be used with a reference to a resource found on another server, since the server would not usually know what type that resource has. However, since these are absolute references, there can be no ambiguity about the type.

In some cases, search parameters are defined with an implicitly limited scope. For example, Observation has an element subject, which is a reference to one of a number of types. This has a matching search parameter subject, which refers to any of the possible types. In addition to this, there is another search parameter patient, which also refers to Observation.subject, but is limited to only include references of type Patient. When using the patient search parameter, there is no need to specify ":Patient" as a modifier, or "Patient/" in the search value, as this must always be true.

References are also allowed to have an identifier. The modifier :identifier allows for searching by the identifier rather than the literal reference:

```
GET [base]/Observation?subject:identifier=http://acme.org/fhir/identifier/mrn|123456
```

This is a search for all observations that reference a patient by a particular patient MRN. When the :identifier modifier is used, the search value works as a token search. The :identifier modifier is not supported on canonical elements since they do not have an identifier separate from the reference itself.

Chaining is not supported when using the :identifier modifier, nor are chaining, includes or reverse includes supported for reference elements that do not have a reference element.

The reference search parameter is mostly used for resource elements of type Reference or canonical. However, it is also be used to search resource elements of type Resource - i.e. where one resource is directly nested within another - see the Bundle search parameters 'message' and 'composition' as an example of this.

3.1.1.4.13 References and Versions

Elements of type Reference may contain a versioned reference:

```
  <evidence>
     <reference value="Observation/123/_history/234234"/>
  </evidence>
```

When searching on versioned references, the following rules apply:

• If a resource has a reference that is versioned, and chaining is performed, the criteria should ideally be evaluated against the version referenced, but most systems will not be capable of this because search is only defined to function against the current version of a resource

• Where a search does not act on the referenced version, search results SHOULD contain a OperationOutcome with a warning that indicates the discrepancy

• If a resource has a reference that is versioned and _include is performed, the specified version SHOULD be provided.

Elements of type canonical may contain a version specific reference, but this version is different in both meaning and format to version specific references that might be found in a Reference:

```
  <valueSet value="http://hl7.org/fhir/ValueSet/example|3.0"/>
```

This version is a reference to the business version of the resource.

For canonical references, servers SHOULD support searching by Canonical URLs, and SHOULD support automatically detecting a |[version] portion as part of the search parameter and interpreting that portion as a search on the business version of the target resource. The modifier :below is used with canonical references, to control whether the version is considered in the search. The search:

```
GET {base]/Observation?definition:below=http:http://acme.com/some-profile
```

matches all of these element values:

• http://acme.com/some-profile|1.0

• http://acme.com/some-profile|1.1

• http://acme.com/some-profile|2.0

The search:

```
GET {base]/Observation?definition:below=http:http://acme.com/some-profile|1
```

matches the first two element values.

3.1.1.4.14 Searching Hierarchies

Some references are circular - that is, the reference points to another resource of the same type. When the reference establishes a strict hierarchy, the modifiers :above and :below may be used to search transitively through the hierarchy:

```
GET [base]/Procedure?location:below=42
```

This search returns no only all procedures that occurred at location with id 42, but also any procedures that occurred in locations that are part of location with id 42.

```
GET [base]/MedicationAdministration?encounter:above=21
```

Returns all medication administrations that happened during encounter with id 21 or during any "parent" encounter of that encounter.

Servers indicate that :above/:below is supported on a search parameter by defining them as Modifiers on the Search Parameter definition.

3.1.1.4.15 Chained parameters

In order to save a client from performing a series of search operations, reference parameters may be "chained" by appending them with a period (.) followed by the name of a search parameter defined for the target resource. This can be done recursively, following a logical path through a graph of related resources, separated by .. For instance, given that the resource DiagnosticReport has a search parameter named subject, which is usually a reference to a Patient resource, and the Patient resource includes a parameter name which searches on patient name, then the search

```
GET [base]/DiagnosticReport?subject.name=peter
```

is a request to return all the lab reports that have a subject whose name includes "peter". Because the Diagnostic Report subject can be one of a set of different resources, it's necessary to limit the search to a particular type:

```
GET [base]/DiagnosticReport?subject:Patient.name=peter
```

This request returns all the lab reports that have a subject which is a patient, whose name includes "peter".

Note that chained parameters are applied independently to the target resource. For example,

```
GET Patient?general-practitioner.name=Joe&general-practitioner.address-state=MN
```

may return Patients cared for by Joe from CA and Jane from MN: no one practitioner need satisfy both conditions. E.g. the chains are evaluated separately. For use cases where the joins must be evaluated in groups, there are either Composite search parameters, or the _filter parameter.

Advanced Search Note: Where a chained parameter searches a resource reference that may have more than one type of resource as its target, the parameter chain may end up referring to search parameters with the same name on more than one kind of resource at once. Servers SHOULD reject a search where the logical id refers to more than one matching resource across different types. For example, the client has to specify the type explicitly using the syntax in the second example above.

3.1.1.4.16 Reverse Chaining

The _has parameter provides limited support for reverse chaining - that is, selecting resources based on the properties of resources that refer to them (instead of chaining, above, where resources can be selected based on the properties of resources that they refer to). Here is an example of the _has parameter:

```
GET [base]/Patient?_has:Observation:patient:code=1234-5
```

This requests the server to return Patient resources, where the patient resource is referred to by at least one Observation where the observation has a code of 1234, and where the Observation refers to the patient resource in the patient search parameter.

"Or" searches are allowed (e.g. GET [base]/Patient?_has:Observation:patient:code=123,456), and multiple _has parameters are allowed (e.g. GET [base]/Patient?_has:Observation:patient:code=123&_has:Observation:patient:code=456). Note that each _has parameter is processed independently of other _has parameters.

The _has parameter can be chained, like this:

```
GET [base]/Patient?_has:Observation:patient:_has:AuditEvent:entity:user=MyUserId
```

Fetch all the patients that have an Observation where the observation has an audit event from a specific user.

3.1.1.4.17 Composite Search Parameters

Composite search parameters support joining single values with a $. For example, the result of the search operation is the intersection of the resources that match the criteria specified by each individual search parameter. If a parameter repeats, such as /Patient?language=FR&language=NL, then this matches a patient who speaks both languages. This is known as an AND search parameter, since the server is expected to respond only with results which match both values.

If, instead, the search is to find patients that speak either language, then this is a single parameter with multiple values, separated by a ,. For example, /Patient?language=FR,NL. This is known as an OR search parameter, since the server is expected to respond with results which match either value. Every search parameter may be used with comma-separated values in this fashion; this includes the use of search parameters with modifiers, such as `?code:text=this,that.

AND parameters and OR parameters may also be combined, for example: /Patient?language=FR,NL&language=EN would refer to any patient who speaks English, as well as either French or Dutch.

This approach allows for simple combinations of and/or values, but doesn't allow a search based on a pair of values, such as all observations with a sodium value >150 mmol/L (particularly as the end criteria of a chained search), or searching on Group.characteristic where you need find a combination of key/value, not an

intersection of separate matches on key and value. Another example is spatial coordinates when doing geographical searches.

To allow these searches, a resource may also specify composite parameters that take sequences of single values that match other defined parameters as an argument. The matching parameter of each component in such a sequence is documented in the definition of the parameter. These sequences are formed by joining the single values with a $. Note: This sequence is a single value and itself can be composed into a set of values, so that, for example, multiple matching characteristic-value parameters can be specified as GET [base]/Group?characteristic-value=gender$mixed,owner$Eve.

Note: Modifiers are not used on composite parameters.

Examples of using composite parameters:

Search	Description
GET [base]/DiagnosticReport?result.code-value-quantity=http://loinc.org|2823-3$gt5.4|http://unitsofmeasure.org|mmol/L	Search for all diagnostic reports that contain on observation with a potassium value of >5.4 mmol/L (UCUM)
GET [base]/Observation?component-code-value-quantity=http://loinc.org|8480-6$lt60	Search for all the observations with a systolic blood pressure < 60. Note that in this case, the unit is assumed (everyone uses mmHg)
GET [base]/Group?characteristic-value=gender$mixed	Search for all groups that have a characteristic "gender" with a text value of "mixed"
GET [base]/Questionnaire?context-type-value=focus$http://snomed.info/sct|408934002	Search for all questionnaires that have a clinical focus = "Substance abuse prevention assessment (procedure)"

3.1.1.4.18 Handling Missing Data

Consider the case of searching for all AllergyIntolerance resources:

```
GET [base]/AllergyIntolerance?clinical-status=http://terminology.hl7.org/CodeSystem/allergyintolerance-clinical|active
```

This search will only return resources that have a value for clinicalStatus:

```
{
  "resourceType" : "AllergyIntolerance",
   "clinicalStatus": {
    "coding": [
      {
        "system": "http://terminology.hl7.org/CodeSystem/allergyintolerance-clinical",
        "code": "active"
      }
    ]
  }
}
```

Resources missing a clinicalStatus will not be returned. This is probably unsafe - it would not usually be appropriate to ignore AllergyIntolerance warnings with an unknown clinical status, and only return resources with an explicit clinicalStatus. Instead, it might be desired to return AllergyIntolerance resources with either an explicit value for clinicalStatus, or none:

```
GET [base]/AllergyIntolerance?clinical-status=http://terminology.hl7.org/CodeSystem/allergyintolerance-clinical|active
GET [base]/AllergyIntolerance?clinical-status:missing=true
```

Note that this is 2 separate queries. They can be combined in a batch, but not in a single operation. This query will always return an empty list, as no resource can satisfy both criteria at once:

```
GET [base]/AllergyIntolerance?clinical-status=http://terminology.hl7.org/CodeSystem/allergyintolerance-
    clinical|active&clinical-status:missing=true
```

There is no way to use the :missing modifier and mix with a value using the comma syntax documented above for composite search parameters.

An alternative approach is to use the _filter parameter, for servers that support this parameter.

3.1.1.4.19 Escaping Search Parameters

In the rules described above, special rules are defined for the characters $, ,, and |. As a consequence, if these characters appear in an actual parameter value, they must be differentiated from their use as separator characters. When any of these characters appear in an actual parameter value, they must be prepended by the character \, which also must be used to prepend itself. Therefore, param=xxx$xxx indicates that it is a composite parameter, while param=xx\$xx indicates that the parameter has the literal value xx$xx. The parameter value xx\xx is illegal, and the parameter value param=xx\\xx indicates a literal value of xx\xx. This means that:

```
GET [base]/Observation?code=a,b
```

is a request for any Observation that has a code of either a or b, whereas:

```
GET [base]/Observation?code=a\,b
```

is a request for any Observation that has a code of a,b.

This escaping is at a different level to the % encoding that applies to all URL parameters. Standard % escaping still applies, such that these URLs have the same meaning:

```
GET [base]/ValueSet?url=http://acme.org/fhir/ValueSet/123,http://acme.org/fhir/ValueSet/124,125
GET [base]/ValueSet?url=http://acme.org/fhir/ValueSet/123,http://acme.org/fhir/ValueSet/124%2CValueSet/125
```

As do these URLs:

```
GET [base]/ValueSet?url=http://acme.org/fhir/ValueSet/123,http://acme.org/fhir/ValueSet/124\,125
GET [base]/ValueSet?url=http%58%47%47acme%46org%47fhir%47ValueSet%47123%44http%58%47%47acme%46org%47fhir%47ValueSet%4712
    4%92%46125
```

This specification defines this additional form of escape because the escape syntax using \ applies to all parameter values after they have been 'unescaped' on the server while being read from the HTTP headers.

3.1.1.4.20 Text Search Parameters

The special text search parameters, _text and _content, search on the narrative of the resource, and the entire content of the resource respectively. Just like string parameters using the :text modifier, these parameters SHOULD support a sophisticated search functionality of the type offered by typical text indexing services. The value of the parameter is a text-based search, which may involve searching multiple words with thesaurus and proximity considerations, and logical operations such as AND, OR etc. For example:

```
GET [base]/Condition?_text=(bone OR liver) and metastases
```

This request returns all Condition resources with the word "metastases" and either "bone" or "liver" in the narrative. The server MAY choose to search for related words as well.

Implementers could consider using the rules specified by the OData specification for the $search parameter . Typical implementations would use Lucene, Solr, an SQL-based full text search, or some similar indexing service.

3.1.1.4.21 Special Parameters

A few parameters have the type 'special'. That indicates that the way this parameter works is unique to the parameter and described with the parameter. The general modifiers and comparators do not apply, except as stated in the description.

Implementers will generally need to do special implementations for these parameters. These parameters are special:

- _filter (all resources)
- near on Location

3.1.1.4.22 Searching by list

The _list parameter allows for the retrieval of resources that are referenced by a List resource.

```
GET [base]/Patient?_list=42
```

This request returns all Patient resources that are referenced from the list found at [base]/List/42) in List.entry.item. While it is possible to retrieve the list, and then iterate the entries in the list fetching each patient, using a list as a search criterion allows for additional search criteria to be specified. For instance:

```
GET [base]/Patient?_list=42&gender=female
```

This request will return all female patients in the list. The server can return the list referred to in the search parameter as an included resource, but is not required to do so. In addition, a system can support searching by lists by their logical function. For example:

```
GET [base]/AllergyIntolerance?patient=42&_list=$current-allergies
```

This request will return all allergies in patient 42's "Current Allergy List". The server returns all relevant AllergyIntolerance resources, and can also choose to return the list. For further information, refer to the definition of "$current-allergies", and the List Operation "Find". Note: Servers are not required to make these lists available to the clients as list resources, but may choose to do so.

3.1.1.4.23 Advanced filtering

The search mechanism described above is flexible, and easy to implement for simple cases, but is limited in its ability to express combination queries. To complement this mechanism, the "_filter" search expression parameter can be used.

For example, "Find all the observations for patient with a name including peter that have a LOINC code 1234-5":

```
GET [base]/Observation?code=http://loinc.org|1234-5&subject.name=peter
```

Using the _filter parameter, the search would be expressed like this:

```
GET [base]/Observation?_filter=code eq http://loinc.org|1234-5 and subject.name co "peter"
```

The _filter parameter is described in detail on the "_Filter Parameter" page.

3.1.1.4.24 Specifying the type of Resource

Normally, a search is initiated against a known type of resource, e.g.

```
GET [base]/Observation?params...
```

However, in some circumstances, a search is executed where there is no fixed type of resource:

- Using search across all resource types (GET [base]?params...)
- Using search with messaging
- Some search specifications inside other services e.g. decision support usage

In these circumstances, the search criteria may need to specify one or more resource types that the search applies to. This can be done by using the _type parameter:

```
GET [base]/?_type=Observation,Condition&other params...
```

If no type is specified, the only search parameters that be can be used in global search like this are the base parameters that apply to all resources. If multiple types are specified, any search parameters shared across the entire set of specified resources may be used (see search parameter registry).

Technically, the _type parameter is a token parameter on the Resource Types Value Set.

3.1.1.5 Managing Returned Resources

3.1.1.5.1 Sorting

The client can indicate which order to return the results by using the parameter _sort, which can contain a comma-separated list of sort rules in priority order:

```
GET [base]/Observation?_sort=status,-date,category
```

Each item in the comma separated list is a search parameter, optionally with a '-' prefix. The prefix indicates decreasing order; in its absence, the parameter is applied in increasing order.

Notes:

• When sorting, the actual sort value used is not returned explicitly by the server for each resource, just the resource contents.

• To sort by relevance, use _score.

• The server returns the sort it performs as part of the returned search parameters (see below).

• A search parameter can refer to an element that repeats, and therefore there can be multiple values for a given search parameter for a single resource. In this case, the sort is based on the item in the set of multiple parameters that comes earliest in the specified sort order when ordering the returned resources.

• When sorting on string search parameters, sorting SHOULD be performed on a case-insensitive basis. Accents may either be ignored or sorted as per realm convention.

• This specification does not specify exacts rules for consistency of sorting across servers. In general, this is deemed to be not as essential as consistency of filtering (though even that is a little variable). The purpose of sorting is to provide data in a "reasonable" order for end-users. "Reasonable" may vary by use case and realm, particularly for accented characters.

The _total parameter has the status trial-use pending real world experience of it's use.

3.1.1.5.2 Total number of matching resources

The return Bundle has an element total which is the number of resources that match the search parameters.

Note that Bundle.total represents the total number of matches, not how many resources are returned in a particular response (see paging, immediately below).

Providing a precise number of matching resources may be onerous for the server, depending on how the server is designed. To help reduce the server load, a client can provide the parameter _total to indicate it's preference with regard to the total, which can have one of the following values:

none	There is no need to populate the total count; the client will not use it
estimate	A rough estimate of the number of matching resources is sufficient
accurate	The client requests that the server provide an exact total of the number of matching resources

The Bundle.total element is still optional, and the servers can ignore the _total parameter: it is just an optimization hint, that might possibly save the server some work.

3.1.1.5.3 Page Count

In order to keep the load on clients, servers and the network minimized, the server may choose to return the results in a series of pages. The search result set contains the URLs that the client uses to request additional pages from the search set. For a simple RESTful search, the page links are contained in the returned bundle as links.

Typically, a server will provide its own parameters in the links that it uses to manage the state of the search as pages are retrieved. These parameters do not need to be understood or processed by the client.

The parameter _count is defined as an instruction to the server regarding how many resources should be returned in a single page. Servers SHALL NOT return more resources than requested, even if they don't support paging, but may return less than the client requested. The server should repeat the original _count parameter in its returned page links so that subsequent paging requests honor the original _count. Note: It is at the discretion of the search engine as to how to handle ongoing updates to the resources while the search is proceeding.

Note: The combination of _sort and _count can be used to return only the latest resource that meets a particular criteria - set the criteria, and then sort by date in descending order, with _count=1. This way, the last matching resource will be returned.

if _count has the value 0, this shall be treated the same as _summary=count: the server returns a bundle that reports the total number of resources that match in Bundle.total, but with no entries, and no prev/next/last links. Note that the Bundle.total only include the total number of matching resources. It does not count extra resources such as OperationOutcome or included resources that may also be returned. In the same way, the _count parameter only applies to resources with entry.search.mode = search, and does not include included resources or operation outcomes.

The _count parameter has no impact on the value of Bundle.total as the latter represents the total number of matches, not how many are returned in a single Bundle response.

3.1.1.5.4 Including other resources in result (_include and _revinclude)

Clients may request that the engine return resources related to the search results, in order to reduce the overall network delay of repeated retrievals of related resources. This is useful when the client is searching on a clinical resource, but for every such resource returned, the client will also need the subject (patient) resource that the clinical resource refers to. The client can use the _include parameter to indicate that the subject resources be included in the results. An alternative scenario is where the client wishes to fetch a particular resource, and any resources that refer to it. For example, the client may wish to fetch a MedicationRequest, and any provenance resources that refer to the prescription. This is known as a reverse include, and is specified by providing a _revinclude parameter.

Both _include and _revinclude are based on search parameters, rather than paths in the resource, since joins, such as chaining, are already done by search parameter.

Each _include parameter specifies a search parameter to join on:

```
GET [base]/MedicationRequest?_include=MedicationRequest:patient
GET [base]/MedicationRequest?_revinclude=Provenance:target
```

The first search requests all matching MedicationRequests, to include any patient that the medication prescriptions in the result set refer to. The second search requests all matching prescriptions, return all the provenance resources that refer to them.

Parameter values for both _include and _revinclude have three parts, separated by a : character:

• The name of the source resource from which the join comes

• The name of the search parameter which must be of type reference

• (Optional) A specific of type of target resource (for when the search parameter refers to multiple possible target types)

_include and _revinclude parameters do not include multiple values. Instead, the parameters are repeated for each different include criteria.

For each returned resource, the server identifies the resources that meet the criteria expressed in the join, and adds to the results, with the entry.search.mode set to "include" (in some searches, it is not obvious which resources are matches, and which are includes). If there is no reference, or no matching resource, the resource cannot be retrieved (e.g. on a different server), then the resource is omitted, and no error is returned.

The inclusion process can be iterative, if (and only if) the modifier :iterate is included. For example, this search returns all Medication Request resources and their prescribing Practitioner Resources for the matching Medication Dispense resources:

```
GET [base]/MedicationDispense?_include=MedicationDispense:prescription
    &_include:iterate=MedicationRequest:performer&criteria...
```

This technique applies to circular relationships as well. For example, the first of these two searches includes any related observations to the target relationships, but only those directly related. The second search asks for the _include based on related parameter to be executed iteratively, so it will retrieve observations that are directly related, and also any related observations to any other included observation.

```
GET [base]/Observation?_include=Observation:related-target&criteria...
GET [base]/Observation?_include:iterate=Observation:related-target&criteria...
```

Both _include and _revinclude use the wild card "*" for the search parameter name, indicating that any search parameter of type=reference be included. Though both clients and servers need to take care not to request or return too many resources when doing this. Most notably, using iterative wildcards inclusions might lead to the retrieval of the full patient's record, or even more than that: resources are organized into an interlinked network and broad _include paths may eventually traverse all possible paths on the server. For servers, these iterative and wildcard _includes are demanding and may slow the search response time significantly.

It is at the server's discretion how deep to iteratively evaluate the inclusions. Servers are expected to limit the number of iterations done to an appropriate level and are not obliged to honor requests to include additional resources in the search results. Because iterative search is generally resource intensive, it is not the default behavior.

When search results are paged, each page of search results should include the matching includes for the resources in each page, so that each page stands alone as a coherent package.

Note: when considering using _include and _revinclude, implementers should also consider whether using GraphQL and/or GraphDefinition are more appropriate approaches in their context.

3.1.1.5.5 Contained Resources

By default, search results only include resources that are not contained in other resources. A chained condition will be evaluated inside contained resources. To illustrate this, consider a MedicationRequest resource that has a contained Medication resource specifying a custom formulation that has ingredient with a itemCodeableConcept "abc" in "http://acme.com./medications". In this case, a search:

```
GET MedicationRequest?medication.ingredient-code=abc
```

will include the MedicationRequest resource in the results. However, this search:

```
GET Medication?ingredient-code=abc
```

will not include the contained Medication resource in the results, since either the wrong type of resource would be returned, or the contained resource would be returned without its container resource, which provides context to the contained resource.

Clients can modify this behavior using the _contained parameter, which can have one of the following values:

• false (default): Do not return contained resources

• true: return only contained resources

• both: return both contained and non-contained (normal) resources

When contained resources are being returned, the server should return either the container resource, or the contained resource alone. The client can specify which by using the _containedType parameter, which can have one of the following values:

• container (default): Return the container resources

• contained: return only the contained resource

When returning a container resource, the server simply puts this in the search results:

```
<Bundle>
  ...
  <entry>
    <resource>
      <MedicationRequest>
        <id value="23">
        ....
        <contained>
          <Medication>
            <id value="m1">
            ...
          </Medication>
        <contained>

      </MedicationRequest>
    </resource>
    <search>
      <mode value="match"/>
    </search>
  </entry>
</Bundle>
```

In the case of returning container resources, the server SHALL populate the entry.search.mode element, as shown, so that the client can pick apart matches and includes (since the usual approach of doing it by type might not work).

If the return type is the contained resource, this must be done slightly differently:

```
<Bundle>
  ...
  <entry>
    <fullUrl value="http://example.com/fhir/MedicationRequest/23#m1"/>
    <resource>
      <Medication>
        <id value="m1">
        ...
      </Medication>
    </resource>
    <search>
      <mode value="match"/>
    </search>
  </entry>
</Bundle>
```

In this case, the fullUrl informs the client that this is a contained resource, along with indicating the identity of the containing resource.

3.1.1.5.6 External References

If the _include path selects a reference that refers to a resource on another server, the server can elect to include that resource in the search results for the convenience of the client.

If the _include path selects a reference that refers to an entity that is not a Resource, such as an image attachment, the server may also elect to include this in the returned results as a Binary resource. For example, the include path may point to an attachment which is by reference, like this:

```
<content>
  <contentType>image/jpeg</contentType>
  <url>http://example.org/images/2343434/234234.jpg</url>
</content>
```

The server can retrieve the target of this reference, and add it to the results for the convenience of the client.

3.1.1.5.7 Paging

When returning paged results for a search with _include resources, all _include resources that are related to the primary resources returned for the page SHOULD also be returned as part of that same page, even if some of those resource instances have previously been returned on previous pages. This approach allows both sender and receiver to avoid caching results of other pages.

3.1.1.5.8 Summary

The client can request the server to return only a portion of the resources by using the parameter _summary:

```
GET [base]/ValueSet?_summary=true
```

The _summary parameter requests the server to return a subset of the resource. It can contain one of the following values:

true | Return a limited subset of elements from the resource. This subset SHOULD consist solely of all supported elements that are marked as "summary" in the base definition of the resource(s) (see ElementDefinition.isSummary)
text | Return only the "text" element, the 'id' element, the 'meta' element, and only top-level mandatory elements
data | Remove the text element
count | Search only: just return a count of the matching resources, without returning the actual matches
false | Return all parts of the resource(s)

The intent of the _summary parameter is to reduce the total processing load on server, client, and resources between them such as the network. It is most useful for resources that are large, particularly ones that include images or elements that may repeat many times. The purpose of the summary form is to allow a client to quickly retrieve a large set of resources, and let a user pick the appropriate one. The summary for an element is defined to allow a user to quickly sort and filter the resources, and typically omit important content on the basis that the entire resource will be retrieved when the user selects a resource.

Servers are not obliged to return just a summary as requested. There are only a limited number of summary forms defined for resources in order to allow servers to store the summarized form(s) in advance. Servers SHOULD mark the resources with the tag SUBSETTED to ensure that the incomplete resource is not accidentally used to overwrite a complete resource.

Note that the _include and _revinclude parameters cannot be mixed with _summary=text.

Implementation Note: There is some question about the inclusion of extensions in the summary. Additional rules may be made around this in the future.

3.1.1.5.9 Elements

If one of the summary views defined above is not appropriate, a client can request a specific set of elements be returned as part of a resource in the search results using the _elements parameter:

```
GET [base]/Patient?_elements=identifier,active,link
```

The _elements parameter consists of a comma-separated list of base element names such as, elements defined at the root level in the resource. Only elements that are listed are to be returned. Clients SHOULD list all mandatory and modifier elements in a resource as part of the list of elements. The list of elements does not apply to included resources.

Servers are not obliged to return just the requested elements. Servers SHOULD always return mandatory elements whether they are requested or not. Servers SHOULD mark the resources with the tag SUBSETTED to ensure that the incomplete resource is not actually used to overwrite a complete resource.

3.1.1.5.10 Relevance

Where a search specifies a non-deterministic sort, the search algorithm may generate a ranking score to indicate which resources meet the specified criteria better than others. The server can return this score in entry.score:

```
<entry>
  <score value=".45"/>
  <Patient>
    ... patient data ...
  </Patient>
</entry>
```

The score is a decimal number with a value between (and including) 0 and 1, where 1 is best match, and 0 is least match.

3.1.1.6 Server Conformance

In order to allow the client to be confident about what search parameters were used as criteria by the server, the server SHALL return the parameters that were actually used to process the search. Applications processing search results SHALL check these returned values where necessary. For example, if the server did not support some of the filters specified in the search, a client might manually apply those filters to the retrieved result set, display a warning message to the user or take some other action.

In the case of a RESTful search, these parameters are encoded in the self link in the bundle that is returned:

```
<link>
  <relation value="self"/>
  <url value="http://example.org/Patient?name=peter"/>
</link>
```

In other respects, servers have considerable discretion with regards to supporting search:

• Servers can choose which parameters to support (other than _id above).

• Servers can choose when and where to implement parameter chaining, and when and where they support the _include parameter.

• Servers can declare additional parameters in the profiles referenced from their Capability statements. Servers should define search parameters starting with a "-" character to ensure that the names they choose do not clash with future parameters defined by this specification.

• Servers are not required to enforce case sensitivity on parameter names, though the names are case sensitive (and URLs are generally case-sensitive).

• Servers may choose how many results to return, though the client can use _count as above

• Servers can choose how to sort the return results, though they SHOULD honor the _sort parameter.

3.1.1.7 Advanced Search

The search framework described above is a useful framework for providing a simple search based on indexed criteria, but more sophisticated query capability is needed to handle precise queries, complex decision support-based requests, and direct queries that have human resolution.

More advanced search operations are specified by the _query parameter:

```
GET [base]/Patient?_query=name&parameters...
```

The _query parameter names a custom search profile that describes a specific query operation. The named query may define additional named parameters that are used with that particular named query. Servers can define their own additional named queries to meet their own uses using an OperationDefinition.

There can only ever be one _query parameter in a set of search parameters. Servers processing search requests SHALL refuse to process a search request if they do not recognize the _query parameter value.

3.1.1.8 Search Result Currency

The results of a search operation are only guaranteed to be current at the instant the operation is executed. After the operation is executed, ongoing actions performed on the resources against which the search was executed will render the results increasingly stale. The significance of this depends on the nature of the search, and the kind of use that is being made of the results.

This is particularly relevant when the server is returning the results in a series of pages. It is at the discretion of the search engine of how to handle ongoing updates to the resources while the search is proceeding.

Performing a search operation does not change the set of resources on the server, with the exception of the creation of Audit Event resources auditing the search itself.

3.1.1.9 Summary Tables

Common Parameters defined for all resources:
Name | Type | Description | Paths
_id | token | Resource id (not a full URL) | Resource.id
_lastUpdated | date | Date last updated. Server has discretion on the boundary precision | Resource.meta.lastUpdated
_tag | token | Search by a resource tag | Resource.meta.tag
_profile | uri | Search for all resources tagged with a profile | Resource.meta.profile
_security | token | Search by a security label | Resource.meta.security
_text | string | Text search against the narrative
_content | string | Text search against the entire resource
_list | string | All resources in nominated list (by id, not a full URL)
_query | string | Custom named query
Search Control Parameters:
Name | Type | Description | Allowable Content
_sort | string | Order to sort results in (can repeat for inner sort orders) | Name of a valid search parameter
_count | number | Number of results per page | Whole number
_include | string | Other resources to include in the search results that search matches point to | SourceType:searchParam(:targetType)
_revinclude | string | Other resources to include in the search results when they refer to search matches | SourceType:searchParam(:targetType)
_summary | string | Just return the summary elements (for resources where this is defined) | true | false (false is default)

_contained | string | Whether to return resources contained in other resources in the search matches | true | false | both (false is default)
_containedType | string | If returning contained resources, whether to return the contained or container resources | container | contained

Cross-map between search parameter types and Data types:

Data Type | number | date | reference | quantity | uri | string | token
Primitive Types
base64Binary | Not used in search
boolean | N | N | N | N | N | Y . true|false (System = http://hl7.org/fhir/special-values but not usually needed)
canonical | N | N | Y | N | Y | N | N
code | N | N | N | N | N | Y . (System, if desired, is defined in the underlying value set for each code)
date | N | Y | N | N | N | N | N
dateTime | N | Y | N | N | N | N | N
decimal | Y | N | N | N | N | N | N
id | N | N | N | N | N | N | Y
instant | N | Y | N | N | N | N | N
integer | Y | N | N | N | N | N | N
markdown | Not used in search
oid | Not used in search (but see uri)
positiveInt | Not used in search (but see integer)
string | N | N | N | N | N | Y | Y
time | Not used in search
unsignedInt | Not used in search (but see integer)
uri | N | N | Y | N | Y | N | N
url | Not used in search (but see uri)
uuid | Not used in search (but see uri)
Data Types
Address | N | N | N | N | N | Y search on any string element in the address | N
Age | N | N | N | Y | N | N | N
Annotation | Not used in search
Attachment | Not used in search
CodeableConcept | N | N | N | N | N | N | Y
Coding | N | N | N | N | N | N | Y
Count | Not used in search
ContactPoint | N | N | N | N | N | N | Y
Distance | Not used in search
Duration | N | N | N | Y | N | N | N
HumanName | N | N | N | N | N | Y Search on any string element in the name | N
Identifier | N | N | N | N | N | N | Y
Money | N | N | N | Y | N | N | N
Period | N | Y | N | N | N | N | N
Quantity | N | N | N | Y | N | N | N
Range | N | N | N | Y | N | N | N
Ratio | Not used in search
Reference | N | N | Y | N | N | N | N
SampledData | Not used in search
Signature | Not used in search
Timing | N | Y | N | N | N | N | N

3.1.2 Using GraphQL with FHIR

URL: https://hl7.org/fhir/R4/graphql.html

3.1.2 Using GraphQL with FHIR

FHIR Infrastructure Work Group | Maturity Level: 0 | Standards Status: Trial Use

GraphQL is a query language for APIs and a runtime for fulfilling those queries with your existing data. This page describes how to use GraphQL with FHIR. The GraphQL interface may be implemented by any server, and may be provided as a facade service in front of a conformant RESTful API. For GraphQL documentation, see http://graphql.org .

Implementation Note: This page is a draft. For ongoing discussion, see chat.fhir.org .

3.1.2.1 Status of GraphQL

Implementers should note that GraphQL is not a formal standard, and no standards path has been dscribed for it. At present, GraphQL is copyright by Facebook , and licensed under a standard BSD-3 license. The GraphQL language is still under development.

3.1.2.2 Invoking GraphQL

The standard end points for graphQL are as defined on the $graphql operation:

```
[base]/$graphql
```

A System level query. The query must start by selecting a resource type and some search criteria, as defined below.

```
[base]/[Type]/[id]/$graphql
```

A Resource Instance level the query assumes a single resource is in scope, and just queries on the data available in the given resource.

GraphQL can be invoked by get with a query parameter, or a POST with the graphQL as the body, or a JSON body. (see the GraphQL documentation).

The mime type of the response is application/json. Other formats are not described by the specification. The mimetype is *not* application/fhir+json - the response is not a FHIR resource, though it may look like one closely.

3.1.2.3 Conformance

Servers that conform to FHIR/GraphQL specification:

• SHALL use types, queries and mutations that conform to those provided in the GraphQL definitions in the specification

• SHOULD support the @skip and @include directives. The data re-organization directives defined at the end of this page SHOULD be considered too

• SHALL indicate in their conformance statement that they support graphQL queries (todo: how)

• SHOULD provide the same functionality as that defined in their conformance statement for read/search/create/update/delete/patch operations

• SHALL provide introspection services that correctly describe the services they offer

• MAY provide additional queries and mutations beyond those defined listed here

Note that most servers only provide a subset of the full functionality described by the FHIR interface. Where servers support FHIR features, and implement GraphQL, they must make them available as defined by this page, but these rules do not require that servers implement all (or even any) of the functionality defined here. Also, these rules do not prevent servers providing additional end-points, queries or mutations that provide additional functionality.

3.1.2.3.1 Error Handling

With regard to handling errors, FHIR GraphQL servers:

• SHALL return a 200 ok with a "data" field, or a 3xx, 4xx or 5xx response with a graphQL error as described below

• SHALL return an error when fields that are not valid or not supported are used, or when queries, mutations, arguments, types, and directives that are not recognized or supported are encountered

• MAY include an operation outcome in a mutation response in the "extensions" property along with a "data" element, if the mutation succeeded. The OperationOutcome SHALL only include information and warning messages

A graphQL error response with an OperationOutcome looks like this:

```
{
  "data": null,
  "errors": [
    {
      "extensions": {
        "resource" : {
          "resourceType": "OperationOutcome",
          "issue": [
            {
              "severity": "error",
              "code": "exception",
              "diagnostics": "500: Internal server error"
            }
          ]
        },
      },
      "message": "500: Internal server error"
    }
  ]
}
```

The error response SHALL include at least one error. Each error SHALL include a message, and MAY include the graphQL locations and path properties. Each error SHOULD include an OperationOutcome in the resource property in the graphQL extensions element. Note that in this usage, there MAY be multiple OperationOutcome, one for each errors item, with one issue matching the message.

3.1.2.4 Field Selection

Any FHIR defined field can be used directly e.g. this graphql against the Patient resource (r3)

```
 {
  name { text given family }
 }
```

Example: http://test.fhir.org/r3/Patient/example/$graphql?query={name{text,given,family}} (note: examples are only informative).

Polymorphic fields are represented by their JSON property name E.g. for Observation.value[x]:

```
{
 valueQuantity { value unit }
}
```

Example: http://test.fhir.org/r3/Observation/example/$graphql?query={valueQuantity{value,unit}} (note: examples are only informative).

Note: This is because the leaf names have to correspond to scalar types, so there is no use selecting all the variants at once

Extensions on primitives: the JSON convention for primitives is observed. e.g. use _[name] for accessing extensions on primitives. So

```
{
 birthDate _birthDate { extension {valueDateTime} }
}
```

results in

```
{
 "birthDate":"2016-05-18",
 "_birthDate":{
   "extension":[{
     "valueDateTime":"2016-05-18T10:28:45Z"
    }]
 }
}
```

Example: http://test.fhir.org/r3/Patient/example/$graphql?query={birthDate,_birthDate{extension{valueDateTime}}} (note: examples are only informative).

3.1.2.4.1 Field Arguments: List Navigation

Primitive fields SHALL NOT have any arguments at all. Complex fields may have one of more of the following parameters, all of which help select a subset of a repeating element:

• fhirpath - a FHIRPath statement selecting which of the subnodes is to be included

• [field] - the name of a sub-property with a specified value that must be matched for the field to be included

• _offset - specify the offset to start at for a repeating element (see below)

• _count - specify how many to elements to return from a repeating list

FHIRPath:

Use a FHIRPath statement selecting which of the set of notes in a list is to be included:

```
{
 name(fhirpath: "family.exists()") { text given family }
}
```

Example: http://test.fhir.org/r3/Patient/example/$graphql?query={name(fhirpath:%22family.exists()%22){text,given,family}} (as compared to http://test.fhir.org/r3/Patient/example/$graphql?query={name{text,given,family}}) (note: examples are only informative).

Field Filter:

Not all systems support FHIRPath, so a simpler syntax is provided, where the client specifies a single sub-element by name, and a specified value that must be matched for the field to be included:

```
{
 name(use: official) { text given family }
}
```

Example: http://test.fhir.org/r3/Patient/example/$graphql?query={name(use:official){text,given,family}} (note: examples are only informative).

One way this can be used is to select particular extensions:

```
{
 myext: extension(url: "http://myextension.url/details") { value : valueString }
}
```

will result in

```
{
 "myext" : {
   "value" : "some value"
  }
}
```

List offsets: _offset and _count

For some large resources, it may be desired to select only a subset of a long list of elements, and to make repeated requests reading further into the list as long as required. This can be done with the _count and offset filters:

```
{
 entry(_count: 5, _offset: 5) { deleted item { reference } }
}
```

Example: http://test.fhir.org/r4/List/long/$graphql?query={entry(_count:5,_offset:5){deleted item{reference}}} (note: examples are only informative).

This will select the second set of 5 entries in the nominated List resource. Clients can use this approach to iterate through a list. If a filter is applied to this (as documented above), the offset and count apply to the filtered list. Clients would iterate the list until there are no more entries returned.

3.1.2.4.2 Additional Selectors

3.1.2.4.2.1 Resource references

An object of type Reference can have an additional selection resource. This is an instruction to the server to resolve the reference, and then include the contents of the resource as specified by sub-selections in the property name "resource" (can be aliased). e.g. On Observation:

```
{
  subject { reference, resource {active} }
}
```

The resource selector has two arguments:

• optional : true | false. (default is false). If the server cannot resolve the reference (e.g. resource does not exist, or user security rights or choices do not permit resource to be seen), and optional is not true, the server returns an error instead of the graph output

```
{
 id
 subject {
  reference
  resource {
     ...on Patient {
         birthDate
   }
   ...on Group {
       name
   }
  }
 }
 code {coding {system code} }
}
```

Example: http://test.fhir.org/r4/Observation/example/$graphql?query={id,subject{reference,resource{...on%20Patient{birthDate}...on%20Practioner{practitionerRole{speciality}}}}code{coding{system,code}}} (note: examples are only informative).

but slightly denser:

```
{
 id
 subject {
  reference
   resource(type : Patient) { birthDate }
   resource(type : Practioner) { practitionerRole {  speciality } }
 }
 code {coding {system code} }
}
```

Example: http://test.fhir.org/r4/Observation/example/$graphql?query={id,subject{reference,resource(type:Patient){birthDate}resource(type:Practioner){practitionerRole{speciality}}}code{coding{system,code}}} (note: examples are only informative).

Clients can use either approach - type selection as a parameter, of using a fragment type condition - the first is shorter while the second aligns with the GraphQL schema more explicitly.

3.1.2.5 Searching resources

When a GraphQL statement is run at the system level, rather than against a particular resource, the first thing the query must do is select the resource(s) of interest. The logic of this follows the FHIR search API, but reimplements in a GraphQL centric way.

There are 3 ways to query, for a single resource, for a simple list of resources, and a full API. For a single resource, the client names the type of the resource, and provides an id:

```
{
  Patient(id: example) { id, active }
}
```

This returns a single Patient with the name id. The output from this is a single resource:

```
{
  "data" : {
    "Patient" {
      "id" : "example",
      "active" : "true",
    }
```

```
  }
}
```

Example: http://test.fhir.org/r4/$graphql?query={Patient(id:example){id,name{given,family}}} (note: examples are only informative).

Alternatively, the client can ask for a list of resources. Here, the client simply asks for a list of resources:

```
{
  ConditionList(clinical_status: relapse, patient: example) { id, clinicalStatus }
}
```

This is a request to list all the Condition resources that have a status if 'relapsed' for the patient with id = example. The arguments are most of the search parameters defined on or for the specified resource. Notes:

• the parameter's names are changed by replacing '-' in the name with '_' (for graphql syntax requirements) (FHIR defines parameters starting with _ but will never define search parameters starting with '-')

• The search parameters _include, _revinclude, _contained, _containedType are not supported

• Unlike the RESTful search API, the composite syntax (multiple arguments separated by comma) is not used. Instead, supply an array of parameters (_id: [1,2,example]). Values in an array or ORed

• For And functionality, or to use chain searches or modifiers, use the _filter parameter

• There is one additional possible argument, "fhirpath" which the server evaluates on all the possible matches

The output of this is a list:

```
{
  "data" : {
    "ConditionList" : [{
      "id" : "100",
      "clinicalStatus" : "relapse"
    },{
      "id" : "100",
      "clinicalStatus" : "relapse"
    }]
  }
}
```

Example: http://test.fhir.org/r4/$graphql?query={PatientList(name:%22pet%22){name{family,given}}} (note: examples are only informative).

Servers may reject the request if there are too many matches to return in a single request. If they do so, they SHALL return an error indicating that the query could not be fulfilled. (rather than silently filtering the list).

The simple list approach does not allow for the management of long lists. To do this, clients are able to request a Connection based approach (based on http://graphql.org/learn/pagination/ , but adapted to the existing FHIR Search API).

```
{
  ConditionConnection (clinical_status: active, patient: example) {
    count offset pagesize
    edges {
      mode, score, resource { id, active }
    }
    first previous next last
  }
}
```

The arguments are the same as for the simple List case, with the addition of the special argument 'cursor' (see below). The server returns a connection object that contains information about the search, along with a list of 'edges', one for each match. Each edge has 3 properties: mode, score (match the same properties on Bundle.entry.search) and "resource" which is the actual matches.

```
{
 "data" : {
   "ConditionConnection" : {
     "count": 50,
     "offset" : 0,
     "pageSize" : 25,
     "next" : "45f9ada8-db37-4498-ba7d-75a044668387:3"
     "edges" : [{
        "resource" : {
          "id" : "100",
          "clinicalStatus" : "relapse",
        }
    },{
      "resource" : {
        "id" : "100",
        "clinicalStatus" : "relapse",
       }
    }]
  }
 }
}
```

Example 1: http://test.fhir.org/r4/$graphql?query={PatientConnection(name:%22pet%22){count,offset,pagesize,first,previous,next,last,edges{mode,score,resource{name{family,given}}}}} (note: examples are only informative).

Example 2 (cursor): http://test.fhir.org/r4/$graphql?query={PatientConnection(){count,offset,pagesize,first,previous,next,last,edges{mode,score,resource{resourceType,id,name{family,given}}}}}}} (note: examples are only informative).

Notes:

• When using the fhirpath argument with Connection based searches, the server may choose to apply the fhirpath filter after the search paging is performed, so that individual pages may be shorter (or empty)

The client can follow up on the first/previous/next/last links using the argument 'cursor':

```
{
  ConditionConnection (_cursor : "45f9ada8-db37-4498-ba7d-75a044668387") {
    count offset pagesize
    edges {
      id, clinicalStatus
    }
    first previous next last
  }
}
```

Example: http://test.fhir.org/r4/$graphql?query={PatientConnection(cursor:%225b0719b5-0f01-442e-9576-5b0514c19a:50%22){count,offset,pagesize,first,previous,next,last,edges{mode,score,resource{resourceType,id,name{family,given}}}}}}} (note: examples are only informative). (note, though, that to make this one work, you have to replace the cursor token with one that you got from the link above).

Notes:

• The client cannot change the parameters for the search (and should not specify them) when providing a cursor. The value of the token is opaque to the client, and only understood by the server

3.1.2.5.0.1 List vs Connection Searches

The specification defines the 2 forms of searches to meet differing requirements. For the List search:

• the returned graph is simpler to iterate

• in particular, the returned graph doesn't need to be considered with logic asking 'is there more pages to get' (the server returns everything, or an error)

• servers can fulfill the simpler search with much less resources, because they do not need to remember state about the searches in order to correctly construct follow up pages

In general, clients should use the simple search, for efficiency purposes, where appropriate (e.g. where the user is not going see a 'next' button). This particularly applies when doing reverse reference resolution.

3.1.2.5.0.2 Reverse References

It's also possible to use search is a special mode, doing reverse lookups - e.g. list all the resources that refer to this resource. An example of this use is to look up a patient, and also retrieve all the Condition resources for the patient.

This is a special case of search, above, but with an additional mandatory parameter _reference. For example:

```
{
  name { [some fields] }
  ConditionList(_reference: patient) {
   [some fields from Condition]
  }
}
```

There must be at least the argument "_reference" which identifies which of the search parameters for the target resource is used to match the resource that has focus. In addition, there may be other arguments as defined above in search (except that the "id" argument is prohibited here as nonsensical)

The response for the query above would be

```
{
  "name: [ [some fields] ],
  "ConditionList" : [
    { [some fields from matching Condition resource] }
  ]
}
```

Example: http://test.fhir.org/r4/Patient/example/$graphql?query={name{family,given}ConditionList(_reference:patient){id,clinicalStatus}} (note: examples are only informative).

The "connection" based search option is also supported, as described above, with the addition of _reference. If the client wishes to pursue any of the cursor based links in the graphQL results it asks for back, then it initiates this a new separate query as defined above, rather than repeating the nested query. I.e. the 'cursor' argument is prohibited, except at the root of query.

Example: http://test.fhir.org/r4/Patient/example/$graphql?query={name{family,given}ConditionConnection(_reference:patient){count,offset,pagesize,first,previous,next,last,edges{mode,score,resource{...onCondition{resourceType,id,clinicalStatus}}}}} (note: examples are only informative).

3.1.2.6 Operations

It's also possible to apply graphql to the results of other operations. This is done by adding the parameter "_graphql" when invoking the operation. e.g.

```
GET [base]/ValueSet/doc-typecodes/$validate-
  code?system=http://loinc.org&code=1963-8&display=test&_graphql={result:parameter(name:"result"){value:valueBoolean}}
```

Example: http://test.fhir.org/r4/ValueSet/doc-typecodes/$validate-code?system=http://loinc.org&code=1963-8&display=test&_graphql={result:parameter(name:%22result%22){value:valueBoolean}} (note: examples are only informative).

The graphQL is executed on the output of the operation. If the operation fails, and returns an operation outcome, then the graphQL is not executed.

3.1.2.7 Data Flattening Directives

GraphQL is a very effective language for navigating a graph and selecting subset of information from it. However for some uses, the physical structure of the result set is important. This is most relevant when extracting data for statistical analysis in languages such as R . In order to facilitate these kind of uses, FHIR servers should consider supporting the following directives that allow implementers to flatten the return graph for easier analysis

3.1.2.7.1 Flattening a node

`@flatten`

This directive indicates that the field to which it is attached is not actually produced in the output graph. Instead, its children will be processed and added to the output graph as specified in its place.

Notes:

• If @flatten is used on an element with repeating cardinality, then by default, all the children will become lists

• When using @flatten, all the collated children must have the same FHIR type. The server SHALL return an error if they don't

For an example, take this graphQL, and apply it to the patient example:

```
{
  identifier { system value }
  active
  name { text given family }
}
```

This will give the output:

```
{
  "identifier": [{
      "system": "urn:oid:1.2.36.146.595.217.0.1",
      "value": "12345"
  }],
  "active": true,
  "name": [{
    "given": ["Peter","James"],
    "family": "Chalmers"
  },{
    "given": ["Jim"]
  },{
    "given": ["Peter","James"],
    "family": "Windsor"
  }]
}
```

Adding the @flatten directive changes the output:

```
{
  identifier @flatten { system value }
  active
  name @flatten { text given family }
}
```

This has the output:

```
{
  "system":["urn:oid:1.2.36.146.595.217.0.1"],
  "value":["12345"],
  "active":true,
  "given":["Peter","James","Jim","Peter","James"],
  "family":["Chalmers","Windsor"]
}
```

3.1.2.7.2 Short cut for selecting only the first element

`@first`

This is a shortcut for a FHIR path filter [$index = 0] and indicates to only take the first match of the elements. Note that the selection of the first element only applies to the immediate context of the field in the source graph, not to the output graph

Example:

```
{
```

```
  identifier @flatten { system value }
  active
  name @flatten { text given @first family }
}
```

Gives the output:

```
{
  "system":["urn:oid:1.2.36.146.595.217.0.1"],
  "value":["12345"],
  "active":true,
  "given":["Peter","Jim","Peter"],
  "family":["Chalmers","Windsor"]
}
```

3.1.2.7.3 Managing output cardinality

`@singleton`

This directive indicates that an field collates to a single node, not a list. It is only used in association with fields on which a parent has @flatten, and overrides the impact of flattening the parent in making it a list. The server SHALL return an error if there is more than one value when flattening

Extending the previous example, adding @singleton:

```
{
  identifier @flatten { system @singleton value @singleton }
  active
  name @flatten @first { text given family @singleton }
}
```

Gives the output:

```
{
  "system":"urn:oid:1.2.36.146.595.217.0.1",
  "value":"12345",
  "active":true,
  "given":["Peter","James"],
  "family":"Chalmers"
}
```

3.1.2.7.4 Converting Lists to singletons

`@slice(fhirpath)`

This indicates that in the output graph, each element in the source will have "." and the result of the FHIRPath as a string appended to the specified name. This slices a list up into multiple single values. For example

```
{ name @slice(path: "$index") @flatten {given @first @singleton family}}
```

For a resource that has 2 names will result in the output

```
{
"Given.0" : "first name, first given",
"Family.0" : ["first name family name"],
"Given.1" : "second name, first given",
"Family.1" : ["second name family name"]
}
```

Other uses might be e.g. Telecom @slice(use) to generate telecom.home for instance.

Notes:

• In general, the intent of @slice is to break a list into multiple singletons. However servers SHALL NOT treat the outputs as singletons unless this is explicitly specified using @singleton

• The suffixes added by this method are cumulative when nesting e.g. .suffix1.suffix2

• The same general outcome can be achieved by a set of fields, each with an alias and a filter, if the possible values are known in advance

Examples:

```
{
  identifier @flatten { system value }
  active
  name @flatten @slice(path: "use") { given family @singleton }
}
```

produces

```
{
  "system":["urn:oid:1.2.36.146.595.217.0.1"],
  "value":["12345"],
  "active":true,
  "given.official":["Peter","James"],
  "family.official":"Chalmers",
  "given.usual":["Jim"],
  "given.maiden":["Peter","James"],
  "family.maiden":"Windsor"
}
```

and

```
{
  identifier @flatten { system value }
  active
```

```
  name @flatten @slice(path: "$index") { given family @singleton }
}
```

produces

```
{
  "system":["urn:oid:1.2.36.146.595.217.0.1"],
  "value":["12345"],
  "active":true,
  "given.0":["Peter","James"],
  "family.0":"Chalmers",
  "given.1":["Jim"],
  "given.2":["Peter","James"],
  "family.2":"Windsor"
}
```

3.1.2.8 Mutations

Mutations are defined for the operations Create, Update, and Delete.

Create:

```
PatientCreate(res : Patient) {
  returns a Patient
}
```

Like the API, the Patient resource might or might not have an ID; if it is present, the value is overwritten by the server.

Update:

```
PatientUpdate(id : ID, res : Patient) {
  returns a Patient
}
```

Delete:

```
PatientDelete(id : ID) {
  [no return]
}
```

The parameters are input types that are resources that exactly match the output types. If these operations fail, an http error is returned.

Todo: mutations for patch, batch, transaction?

3.1.3 _filter Parameter

URL: https://hl7.org/fhir/R4/search_filter.html

3.1.3 _filter Parameter

FHIR Infrastructure Work Group | Maturity Level: 2 | Standards Status: Trial Use

3.1.3.1 Introduction

_filter is a parameter that can be used with the Search Operation. It uses the same set of standard parameters defined for the resources, and provides a syntax for expressing a set of query expressions on the underlying resources.

Examples of filters:

• Patient: name co "pet" - all patients with the characters "pet" in a given or family name

• Patient: given eq "peter" and birthdate ge 2014-10-10 - all patients with a given name of peter, born on or after 10-Oct 2014

• Observation: code eq http://loinc.org|1234-5 - all observations with the LOINC code "1234-5"

• Observation: subject.name co "pet" - all observations on a patient with the characters "pet" in a given or family name

• Observation: related[type eq "has-component"].target pr true - all observations that have component observations (note: this uses one of the search parameters defined for this mechanism, see below)

• Observation: related[type eq has-component].target re Observation/4 - all observations that have Observation/v as a component

The _filter syntax has the following features:

• A filter can be a logical one (x or x, or x and x, or not x).

• A filter can contain other filters in a set of parentheses: "()".

• A filter can be a test - path operation value, where operation is taken from the table below, and value is either a "true", "false", a JSON string, or a token (any sequence of non-whitespace characters, excluding ")" and "]". Values are never case sensitive.

• A 'path' is a name, with chained searches done by name.name etc.as per existing source. There can also be a filter: name[filter].name.

• The name is one of the defined search parameters that are used with the other search mechanism, with some special exemptions defined below.

Note: The only difference between a "string" value and a "token" value is that a string can contain spaces and ')' and ']'. There is otherwise no significant difference between them.

Formal grammar for the syntax:

```
filter        = paramExp / logExp / ("not") "(" filter ")"
logExp        = filter ("and" / "or" filter)+
paramExp      = paramPath SP compareOp SP compValue
compareOp     = (see table below)
compValue     = string / numberOrDate / token
string        = json string
token         = any sequence of non-whitespace characters (by Unicode rules) except "]" and ")"
paramPath     = paramName (("[" filter "]") "." paramPath)
paramName     = nameCharStart (nameChar)*
nameCharStart = "_" / ALPHA
nameChar      = "_" / "-" / DIGIT / ALPHA
numberOrDate  = DIGIT (DateChar)*
dateChar      = DIGIT / "T" / "-" / "." / "+"
```

Notes about using the syntax:

• Logical expressions are evaluated left to right, with no precedence between "and" and "or". If there is ambiguity, use parentheses to be explicit.

• Rhe compareOp is always evaluated against the set of values produced by evaluating the param path.

• Rhe parameter names are those defined by the specification for search parameters, except for those defined below.

• Rhe date format is a standard XML (i.e. XSD) dateTime (including time zone).

3.1.3.2 Operators

This table summarizes the comparison operations available:

Operation | Definition
eq | an item in the set has an equal value
ne | An item in the set has an unequal value
co | An item in the set contains this value
sw | An item in the set starts with this value
ew | An item in the set ends with this value
gt / lt / ge / le | A value in the set is (greater than, less than, greater or equal, less or equal) the given value
ap | A value in the set is approximately the same as this value. Note that the recommended value for the approximation is 10% of the stated value (or for a date, 10% of the gap between now and the date), but systems may choose other values where appropriate
sa | The value starts after the specified value
eb | The value ends before the specified value
pr | The set is empty or not (value is false or true)
po | True if a (implied) date period in the set overlaps with the implied period in the value
ss | True if the value subsumes a concept in the set
sb | True if the value is subsumed by a concept in the set
in | True if one of the concepts is in the nominated value set by URI, either a relative, literal or logical vs
ni | True if none of the concepts are in the nominated value set by URI, either a relative, literal or logical vs
re | True if one of the references in set points to the given URL

For detailed rules about the operators eq, ne, le, ge, lt, gt, sa, and eb see Search Prefixes.

The interpretation of the operation depends on the type of the search parameter it is being evaluated against. This table contains those details:

Operation | String | Number | Date | Token | Reference | Quantity
eq | Character sequence is the same (case insensitive) | Number is the same incl same precision | Date is the same including same precision and time zone if provided | Token is the same, including namespace if specified (case insensitive) | n/a | Unit and value are the same
ne | (same)
Co | Character sequence matches somewhere (case insensitive) | An item in the set's implicit imprecision includes the stated value | An item in the set's implicit period includes the stated value | n/a | n/a | n/a?
sw | Character sequence matches from first digit (left most, when L->R) (case insensitive) | n/a | n/a | n/a | n/a | n/a
ew | Character sequence matches up to last digit (right most, when L->R) (case insensitive) | n/a | n/a | n/a | n/a | n/a
gt / lt / ge / le | Based on Integer comparison of Unicode code points of starting character (trimmed) (case insensitive) | Based on numerical comparison | Based on date period comparison per 2.2.2.3 | n/a | n/a | Based on numerical comparison if units are the same (or are canonicalized)
pr
po | n/a | n/a | Based on date period comparison per 2.2.2.3 | | n/a | n/a
ss | n/a | n/a | n/a | Based on logical subsumption; potentially catering for mapping between tx | n/a | n/a
sb | n/a | n/a | n/a | Based on logical subsumption; potentially catering for mapping between tx | n/a | n/a
in | n/a | n/a | n/a | Based on logical subsumption; potentially catering for mapping between tx | n/a | n/a
re | n/a | n/a | n/a | n/a | Relative or absolute url | n/a

Note:

• For token, the format is the same as the existing search parameter.

• For convenience, the codes "loinc", "snomed", "rxnorm" and "ucum" are predefined and can be used in place of their associated full namespace.

3.1.3.3 Additional Parameters

Some additional parameters are defined for the filter parameter (to do: move these into the standard parameters):

Resource Type | Parameter Name | Children
Observation | related | target = related-target Type = related-type
Group | characteristic | value = value code = characteristic
DocumentReference | relatesTo | code = relation target = relatesTo
ServiceRequest | event | status = event-status date = event-date
ServiceRequest | item | status = item-status code = item-code site = bodysite event = item-event
ServiceRequest | item-event | status = item-past-status date = item-date actor = actor

Note:

• Any time these names are used in a parameter, they must have a filter and a chained name under them.

• The first column is the resource type against which this name can be used.

• The second column is the parameter name that is used.

• The third column defines the names that can be used in the chained parameter, and in the filter, and shows which existing search parameters they equate to.

• For example, you could search on Observation for '_filter=related[type eq has-component].target re url'. "type" refers to the search parameter "related-type", and "target" to the search parameter "related-target".

3.1.4 Defined Search Parameters

URL: https://hl7.org/fhir/R4/searchparameter-registry.html

3.1.4 Defined Search Parameters

FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative

This page lists all search parameters defined as part of this specification. For further details regarding using search parameters, see searching on the RESTful API.

All resources have the canonical URL http://hl7.org/fhir/SearchParameter/[id] where [id] is taken from the table below. You can download XML and JSON source for this page.

Parameter | Type | Id | Description | Paths (Expression)
Resource N
_content TU | string | Resource-content | Search on the entire content of the resource
_id TU | token | Resource-id | Logical id of this artifact | Resource.id
_lastUpdated TU | date | Resource-lastUpdated | When the resource version last changed | Resource.meta.lastUpdated
_profile TU | uri | Resource-profile | Profiles this resource claims to conform to | Resource.meta.profile
_query TU | token | Resource-query | A custom search profile that describes a specific defined query operation
_security TU | token | Resource-security | Security Labels applied to this resource | Resource.meta.security
_source TU | uri | Resource-source | Identifies where the resource comes from | Resource.meta.source
_tag TU | token | Resource-tag | Tags applied to this resource | Resource.meta.tag
DomainResource N
_text TU | string | DomainResource-text | Search on the narrative of the resource
Common Search Parameters
address TU | string | individual-address | Multiple Resources: Patient: A server defined search that may match any of the string fields in the Address, including line, city, district, state, country, postalCode, and/or text Person: A server defined search that may match any of the string fields in the Address, including line, city, district, state, country, postalCode, and/or text Practitioner: A server defined search that may match any of the string fields in the Address, including line, city, district, state, country, postalCode, and/or text RelatedPerson: A server defined search that may match any of the string fields in the Address, including line, city, district, state, country, postalCode, and/or text | Patient.address | Person.address | Practitioner.address | RelatedPerson.address
address-city TU | string | individual-address-city | Multiple Resources: Patient: A city specified in an address Person: A city specified in an address Practitioner: A city specified in an address RelatedPerson: A city specified in an address | Patient.address.city | Person.address.city | Practitioner.address.city | RelatedPerson.address.city
address-country TU | string | individual-address-country | Multiple Resources: Patient: A country specified in an address Person: A country specified in an address Practitioner: A country specified in an address RelatedPerson: A country specified in an address | Patient.address.country | Person.address.country | Practitioner.address.country | RelatedPerson.address.country
address-postalcode TU | string | individual-address-postalcode | Multiple Resources: Patient: A postalCode specified in an address Person: A postal code specified in an address Practitioner: A postalCode specified in an address RelatedPerson: A postal code specified in an address | Patient.address.postalCode | Person.address.postalCode | Practitioner.address.postalCode | RelatedPerson.address.postalCode
address-state TU | string | individual-address-state | Multiple Resources: Patient: A state specified in an address Person: A state specified in an address Practitioner: A state specified in an address RelatedPerson: A state specified in an address | Patient.address.state | Person.address.state | Practitioner.address.state | RelatedPerson.address.state
address-use TU | token | individual-address-use | Multiple Resources: Patient: A use code specified in an address Person: A use code specified in an address Practitioner: A use code specified in an address RelatedPerson: A use code specified in an address |

Patient.address.use | Person.address.use | Practitioner.address.use | RelatedPerson.address.use

birthdate TU | date | individual-birthdate | Multiple Resources: Patient: The patient's date of birth Person: The person's date of birth RelatedPerson: The Related Person's date of birth | Patient.birthDate | Person.birthDate | RelatedPerson.birthDate

code TU | token | clinical-code | Multiple Resources: AllergyIntolerance: Code that identifies the allergy or intolerance Condition: Code for the condition DeviceRequest: Code for what is being requested/ordered DiagnosticReport: The code for the report, as opposed to codes for the atomic results, which are the names on the observation resource referred to from the result FamilyMemberHistory: A search by a condition code List: What the purpose of this list is Medication: Returns medications for a specific code MedicationAdministration: Return administrations of this medication code MedicationDispense: Returns dispenses of this medicine code MedicationRequest: Return prescriptions of this medication code MedicationStatement: Return statements of this medication code Observation: The code of the observation type Procedure: A code to identify a procedure ServiceRequest: What is being requested/ordered | AllergyIntolerance.code | AllergyIntolerance.reaction.substance | Condition.code | (DeviceRequest.code as CodeableConcept) | DiagnosticReport.code | FamilyMemberHistory.condition.code | List.code | Medication.code | (MedicationAdministration.medication as CodeableConcept) | (MedicationDispense.medication as CodeableConcept) | (MedicationRequest.medication as CodeableConcept) | (MedicationStatement.medication as CodeableConcept) | Observation.code | Procedure.code | ServiceRequest.code

context TU | token | conformance-context | Multiple Resources: CapabilityStatement: A use context assigned to the capability statement CodeSystem: A use context assigned to the code system CompartmentDefinition: A use context assigned to the compartment definition ConceptMap: A use context assigned to the concept map GraphDefinition: A use context assigned to the graph definition ImplementationGuide: A use context assigned to the implementation guide MessageDefinition: A use context assigned to the message definition NamingSystem: A use context assigned to the naming system OperationDefinition: A use context assigned to the operation definition SearchParameter: A use context assigned to the search parameter StructureDefinition: A use context assigned to the structure definition StructureMap: A use context assigned to the structure map TerminologyCapabilities: A use context assigned to the terminology capabilities ValueSet: A use context assigned to the value set | (CapabilityStatement.useContext.value as CodeableConcept) | (CodeSystem.useContext.value as CodeableConcept) | (CompartmentDefinition.useContext.value as CodeableConcept) | (ConceptMap.useContext.value as CodeableConcept) | (GraphDefinition.useContext.value as CodeableConcept) | (ImplementationGuide.useContext.value as CodeableConcept) | (MessageDefinition.useContext.value as CodeableConcept) | (NamingSystem.useContext.value as CodeableConcept) | (OperationDefinition.useContext.value as CodeableConcept) | (SearchParameter.useContext.value as CodeableConcept) | (StructureDefinition.useContext.value as CodeableConcept) | (StructureMap.useContext.value as CodeableConcept) | (TerminologyCapabilities.useContext.value as CodeableConcept) | (ValueSet.useContext.value as CodeableConcept)

context-quantity TU | quantity | conformance-context-quantity | Multiple Resources: CapabilityStatement: A quantity- or range-valued use context assigned to the capability statement CodeSystem: A quantity- or range-valued use context assigned to the code system CompartmentDefinition: A quantity- or range-valued use context assigned to the compartment definition ConceptMap: A quantity- or range-valued use context assigned to the concept map GraphDefinition: A quantity- or range-valued use context assigned to the graph definition ImplementationGuide: A quantity- or range-valued use context assigned to the implementation guide MessageDefinition: A quantity- or range-valued use context assigned to the message definition NamingSystem: A quantity- or range-valued use context assigned to the naming system OperationDefinition: A quantity- or range-valued use context assigned to the operation definition SearchParameter: A quantity- or range-valued use context assigned to the search parameter StructureDefinition: A quantity- or range-valued use context assigned to the structure definition StructureMap: A quantity- or range-valued use context assigned to the structure map TerminologyCapabilities: A quantity- or range-valued use context assigned to the terminology capabilities ValueSet: A quantity- or range-valued use context assigned to the value set | (CapabilityStatement.useContext.value as Quantity) | (CapabilityStatement.useContext.value as Range) | (CodeSystem.useContext.value as Quantity) | (CodeSystem.useContext.value as Range) | (CompartmentDefinition.useContext.value as Quantity) | (CompartmentDefinition.useContext.value as Range) | (ConceptMap.useContext.value as Quantity) | (ConceptMap.useContext.value as Range) | (GraphDefinition.useContext.value as Quantity) | (GraphDefinition.useContext.value as Range) | (ImplementationGuide.useContext.value as Quantity) | (ImplementationGuide.useContext.value as Range) | (MessageDefinition.useContext.value as Quantity) | (MessageDefinition.useContext.value as Range) | (NamingSystem.useContext.value as Quantity) | (NamingSystem.useContext.value as Range) | (OperationDefinition.useContext.value as Quantity) | (OperationDefinition.useContext.value as Range) | (SearchParameter.useContext.value as Quantity) | (SearchParameter.useContext.value as Range) | (StructureDefinition.useContext.value as Quantity) | (StructureDefinition.useContext.value as Range) | (StructureMap.useContext.value as Quantity) | (StructureMap.useContext.value as Range) | (TerminologyCapabilities.useContext.value as Quantity) | (TerminologyCapabilities.useContext.value as Range) | (ValueSet.useContext.value as Quantity) | (ValueSet.useContext.value as Range)

context-type TU | token | conformance-context-type | Multiple Resources: CapabilityStatement: A type of use context assigned to the capability statement CodeSystem: A type of use context assigned to the code system CompartmentDefinition: A type of use context assigned to the compartment definition ConceptMap: A type of use context assigned to the concept map GraphDefinition: A type of use context assigned to the graph definition ImplementationGuide: A type of use context assigned to the implementation guide MessageDefinition: A type of use context assigned to the message definition NamingSystem: A type of use context assigned to the naming system OperationDefinition: A type of use context assigned to the operation definition SearchParameter: A type of use context assigned to the search parameter StructureDefinition: A type of use context assigned to the structure definition StructureMap: A type of use context assigned to the structure map TerminologyCapabilities: A type of use context assigned to the terminology capabilities ValueSet: A type of use context assigned to the value set | CapabilityStatement.useContext.code | CodeSystem.useContext.code | CompartmentDefinition.useContext.code | ConceptMap.useContext.code | GraphDefinition.useContext.code | ImplementationGuide.useContext.code | MessageDefinition.useContext.code | NamingSystem.useContext.code | OperationDefinition.useContext.code | SearchParameter.useContext.code | StructureDefinition.useContext.code | StructureMap.useContext.code | TerminologyCapabilities.useContext.code | ValueSet.useContext.code

context-type-quantity TU | composite | conformance-context-type-quantity | Multiple Resources: CapabilityStatement: A use context type and quantity- or range-based value assigned to the capability statement CodeSystem: A use context type and quantity- or range-based value assigned to the code system CompartmentDefinition: A use context type and quantity- or range-based value assigned to the compartment definition ConceptMap: A use context type and quantity- or range-based value assigned to the concept map GraphDefinition: A use context type and quantity- or range-based value assigned to the graph definition ImplementationGuide: A use context type and quantity- or range-based value assigned to the implementation guide MessageDefinition: A use context type and quantity- or range-based value assigned to the message definition NamingSystem: A use context type and quantity- or range-based value assigned to the naming system OperationDefinition: A use context type and quantity- or range-based value assigned to the operation definition SearchParameter: A use context type and quantity- or range-based value assigned to the search parameter StructureDefinition: A use context type and quantity- or range-based value assigned to the structure definition StructureMap: A use context type and quantity- or range-based value assigned to the structure

map TerminologyCapabilities: A use context type and quantity- or range-based value assigned to the terminology capabilities ValueSet: A use context type and quantity- or range-based value assigned to the value set | CapabilityStatement.useContext | CodeSystem.useContext | CompartmentDefinition.useContext | ConceptMap.useContext | GraphDefinition.useContext | ImplementationGuide.useContext | MessageDefinition.useContext | NamingSystem.useContext | OperationDefinition.useContext | SearchParameter.useContext | StructureDefinition.useContext | StructureMap.useContext | TerminologyCapabilities.useContext | ValueSet.useContext

context-type-value TU | composite | conformance-context-type-value | Multiple Resources: CapabilityStatement: A use context type and value assigned to the capability statement CodeSystem: A use context type and value assigned to the code system CompartmentDefinition: A use context type and value assigned to the compartment definition ConceptMap: A use context type and value assigned to the concept map GraphDefinition: A use context type and value assigned to the graph definition ImplementationGuide: A use context type and value assigned to the implementation guide MessageDefinition: A use context type and value assigned to the message definition NamingSystem: A use context type and value assigned to the naming system OperationDefinition: A use context type and value assigned to the operation definition SearchParameter: A use context type and value assigned to the search parameter StructureDefinition: A use context type and value assigned to the structure definition StructureMap: A use context type and value assigned to the structure map TerminologyCapabilities: A use context type and value assigned to the terminology capabilities ValueSet: A use context type and value assigned to the value set | CapabilityStatement.useContext | CodeSystem.useContext | CompartmentDefinition.useContext | ConceptMap.useContext | GraphDefinition.useContext | ImplementationGuide.useContext | MessageDefinition.useContext | NamingSystem.useContext | OperationDefinition.useContext | SearchParameter.useContext | StructureDefinition.useContext | StructureMap.useContext | TerminologyCapabilities.useContext | ValueSet.useContext

date TU | date | conformance-date | Multiple Resources: CapabilityStatement: The capability statement publication date CodeSystem: The code system publication date CompartmentDefinition: The compartment definition publication date ConceptMap: The concept map publication date GraphDefinition: The graph definition publication date ImplementationGuide: The implementation guide publication date MessageDefinition: The message definition publication date NamingSystem: The naming system publication date OperationDefinition: The operation definition publication date SearchParameter: The search parameter publication date StructureDefinition: The structure definition publication date StructureMap: The structure map publication date TerminologyCapabilities: The terminology capabilities publication date ValueSet: The value set publication date | CapabilityStatement.date | CodeSystem.date | CompartmentDefinition.date | ConceptMap.date | GraphDefinition.date | ImplementationGuide.date | MessageDefinition.date | NamingSystem.date | OperationDefinition.date | SearchParameter.date | StructureDefinition.date | StructureMap.date | TerminologyCapabilities.date | ValueSet.date

date TU | date | clinical-date | Multiple Resources: AllergyIntolerance: Date first version of the resource instance was recorded CarePlan: Time period plan covers CareTeam: Time period team covers ClinicalImpression: When the assessment was documented Composition: Composition editing time Consent: When this Consent was created or indexed DiagnosticReport: The clinically relevant time of the report Encounter: A date within the period the Encounter lasted EpisodeOfCare: The provided date search value falls within the episode of care's period FamilyMemberHistory: When history was recorded or last updated Flag: Time period when flag is active Immunization: Vaccination (non)-Administration Date List: When the list was prepared Observation: Obtained date/time. If the obtained element is a period, a date that falls in the period Procedure: When the procedure was performed RiskAssessment: When was assessment made? SupplyRequest: When the request was made | AllergyIntolerance.recordedDate | CarePlan.period | CareTeam.period | ClinicalImpression.date | Composition.date | Consent.dateTime | DiagnosticReport.effective | Encounter.period | EpisodeOfCare.period | FamilyMemberHistory.date | Flag.period | Immunization.occurrence | List.date | Observation.effective | Procedure.performed | (RiskAssessment.occurrence as dateTime) | SupplyRequest.authoredOn

description TU | string | conformance-description | Multiple Resources: CapabilityStatement: The description of the capability statement CodeSystem: The description of the code system CompartmentDefinition: The description of the compartment definition ConceptMap: The description of the concept map GraphDefinition: The description of the graph definition ImplementationGuide: The description of the implementation guide MessageDefinition: The description of the message definition NamingSystem: The description of the naming system OperationDefinition: The description of the operation definition SearchParameter: The description of the search parameter StructureDefinition: The description of the structure definition StructureMap: The description of the structure map TerminologyCapabilities: The description of the terminology capabilities ValueSet: The description of the value set | CapabilityStatement.description | CodeSystem.description | CompartmentDefinition.description | ConceptMap.description | GraphDefinition.description | ImplementationGuide.description | MessageDefinition.description | NamingSystem.description | OperationDefinition.description | SearchParameter.description | StructureDefinition.description | StructureMap.description | TerminologyCapabilities.description | ValueSet.description

email TU | token | individual-email | Multiple Resources: Patient: A value in an email contact Person: A value in an email contact Practitioner: A value in an email contact PractitionerRole: A value in an email contact RelatedPerson: A value in an email contact | Patient.telecom.where(system='email') | Person.telecom.where(system='email') | Practitioner.telecom.where(system='email') | PractitionerRole.telecom.where(system='email') | RelatedPerson.telecom.where(system='email')

encounter TU | reference | clinical-encounter | Multiple Resources: Composition: Context of the Composition DeviceRequest: Encounter during which request was created DiagnosticReport: The Encounter when the order was made DocumentReference: Context of the document content Flag: Alert relevant during encounter List: Context in which list created NutritionOrder: Return nutrition orders with this encounter identifier Observation: Encounter related to the observation Procedure: Encounter created as part of RiskAssessment: Where was assessment performed? ServiceRequest: An encounter in which this request is made VisionPrescription: Return prescriptions with this encounter identifier | Composition.encounter | DeviceRequest.encounter | DiagnosticReport.encounter | DocumentReference.context.encounter | Flag.encounter | List.encounter | NutritionOrder.encounter | Observation.encounter | Procedure.encounter | RiskAssessment.encounter | ServiceRequest.encounter | VisionPrescription.encounter

family TU | string | individual-family | Multiple Resources: Patient: A portion of the family name of the patient Practitioner: A portion of the family name | Patient.name.family | Practitioner.name.family

gender TU | token | individual-gender | Multiple Resources: Patient: Gender of the patient Person: The gender of the person Practitioner: Gender of the practitioner RelatedPerson: Gender of the related person | Patient.gender | Person.gender | Practitioner.gender | RelatedPerson.gender

given TU | string | individual-given | Multiple Resources: Patient: A portion of the given name of the patient Practitioner: A portion of the given name | Patient.name.given | Practitioner.name.given

identifier TU | token | clinical-identifier | Multiple Resources: AllergyIntolerance: External ids for this item CarePlan: External Ids for this plan CareTeam: External Ids for this team Composition: Version-independent identifier for the Composition Condition: A unique identifier of the condition record Consent: Identifier for this record (external references) DetectedIssue: Unique id for the detected issue DeviceRequest: Business identifier for request/order DiagnosticReport: An identifier for the report DocumentManifest: Unique Identifier for the set of documents DocumentReference: Master Version Specific Identifier Encounter: Identifier(s) by which this encounter is known EpisodeOfCare: Business Identifier(s) relevant for this EpisodeOfCare FamilyMemberHistory: A search by a record identifier Goal: External Ids for this goal ImagingStudy: Identifiers for the Study, such

as DICOM Study Instance UID and Accession number Immunization: Business identifier List: Business identifier MedicationAdministration: Return administrations with this external identifier MedicationDispense: Returns dispenses with this external identifier MedicationRequest: Return prescriptions with this external identifier MedicationStatement: Return statements with this external identifier NutritionOrder: Return nutrition orders with this external identifier Observation: The unique id for a particular observation Procedure: A unique identifier for a procedure RiskAssessment: Unique identifier for the assessment ServiceRequest: Identifiers assigned to this order SupplyDelivery: External identifier SupplyRequest: Business Identifier for SupplyRequest VisionPrescription: Return prescriptions with this external identifier | AllergyIntolerance.identifier | CarePlan.identifier | CareTeam.identifier | Composition.identifier | Condition.identifier | Consent.identifier | DetectedIssue.identifier | DeviceRequest.identifier | DiagnosticReport.identifier | DocumentManifest.masterIdentifier | DocumentManifest.identifier | DocumentReference.masterIdentifier | DocumentReference.identifier | Encounter.identifier | EpisodeOfCare.identifier | FamilyMemberHistory.identifier | Goal.identifier | ImagingStudy.identifier | Immunization.identifier | List.identifier | MedicationAdministration.identifier | MedicationDispense.identifier | MedicationRequest.identifier | MedicationStatement.identifier | NutritionOrder.identifier | Observation.identifier | Procedure.identifier | RiskAssessment.identifier | ServiceRequest.identifier | SupplyDelivery.identifier | SupplyRequest.identifier | VisionPrescription.identifier

identifier TU | token | conformance-identifier | Multiple Resources: CodeSystem: External identifier for the code system ConceptMap: External identifier for the concept map MessageDefinition: External identifier for the message definition StructureDefinition: External identifier for the structure definition StructureMap: External identifier for the structure map ValueSet: External identifier for the value set | CodeSystem.identifier | ConceptMap.identifier | MessageDefinition.identifier | StructureDefinition.identifier | StructureMap.identifier | ValueSet.identifier

jurisdiction TU | token | conformance-jurisdiction | Multiple Resources: CapabilityStatement: Intended jurisdiction for the capability statement CodeSystem: Intended jurisdiction for the code system ConceptMap: Intended jurisdiction for the concept map GraphDefinition: Intended jurisdiction for the graph definition ImplementationGuide: Intended jurisdiction for the implementation guide MessageDefinition: Intended jurisdiction for the message definition NamingSystem: Intended jurisdiction for the naming system OperationDefinition: Intended jurisdiction for the operation definition SearchParameter: Intended jurisdiction for the search parameter StructureDefinition: Intended jurisdiction for the structure definition StructureMap: Intended jurisdiction for the structure map TerminologyCapabilities: Intended jurisdiction for the terminology capabilities ValueSet: Intended jurisdiction for the value set | CapabilityStatement.jurisdiction | CodeSystem.jurisdiction | ConceptMap.jurisdiction | GraphDefinition.jurisdiction | ImplementationGuide.jurisdiction | MessageDefinition.jurisdiction | NamingSystem.jurisdiction | OperationDefinition.jurisdiction | SearchParameter.jurisdiction | StructureDefinition.jurisdiction | StructureMap.jurisdiction | TerminologyCapabilities.jurisdiction | ValueSet.jurisdiction

medication TU | reference | medications-medication | Multiple Resources: MedicationAdministration: Return administrations of this medication resource MedicationDispense: Returns dispenses of this medicine resource MedicationRequest: Return prescriptions for this medication reference MedicationStatement: Return statements of this medication reference | (MedicationAdministration.medication as Reference) | (MedicationDispense.medication as Reference) | (MedicationRequest.medication as Reference) | (MedicationStatement.medication as Reference)

name TU | string | conformance-name | Multiple Resources: CapabilityStatement: Computationally friendly name of the capability statement CodeSystem: Computationally friendly name of the code system CompartmentDefinition: Computationally friendly name of the compartment definition ConceptMap: Computationally friendly name of the concept map GraphDefinition: Computationally friendly name of the graph definition ImplementationGuide: Computationally friendly name of the implementation guide MessageDefinition: Computationally friendly name of the message definition NamingSystem: Computationally friendly name of the naming system OperationDefinition: Computationally friendly name of the operation definition SearchParameter: Computationally friendly name of the search parameter StructureDefinition: Computationally friendly name of the structure definition StructureMap: Computationally friendly name of the structure map TerminologyCapabilities: Computationally friendly name of the terminology capabilities ValueSet: Computationally friendly name of the value set | CapabilityStatement.name | CodeSystem.name | CompartmentDefinition.name | ConceptMap.name | GraphDefinition.name | ImplementationGuide.name | MessageDefinition.name | NamingSystem.name | OperationDefinition.name | SearchParameter.name | StructureDefinition.name | StructureMap.name | TerminologyCapabilities.name | ValueSet.name

patient TU | reference | clinical-patient | Multiple Resources: AllergyIntolerance: Who the sensitivity is for CarePlan: Who the care plan is for CareTeam: Who care team is for ClinicalImpression: Patient or group assessed Composition: Who and/or what the composition is about Condition: Who has the condition? Consent: Who the consent applies to DetectedIssue: Associated patient DeviceRequest: Individual the service is ordered for DeviceUseStatement: Search by subject - a patient DiagnosticReport: The subject of the report if a patient DocumentManifest: The subject of the set of documents DocumentReference: Who/what is the subject of the document Encounter: The patient or group present at the encounter EpisodeOfCare: The patient who is the focus of this episode of care FamilyMemberHistory: The identity of a subject to list family member history items for Flag: The identity of a subject to list flags for Goal: Who this goal is intended for ImagingStudy: Who the study is about Immunization: The patient for the vaccination record List: If all resources have the same subject MedicationAdministration: The identity of a patient to list administrations for MedicationDispense: The identity of a patient to list dispenses for MedicationRequest: Returns prescriptions for a specific patient MedicationStatement: Returns statements for a specific patient. NutritionOrder: The identity of the person who requires the diet, formula or nutritional supplement Observation: The subject that the observation is about (if patient) Procedure: Search by subject - a patient RiskAssessment: Who/what does assessment apply to? ServiceRequest: Search by subject - a patient SupplyDelivery: Patient for whom the item is supplied VisionPrescription: The identity of a patient to list dispenses for | AllergyIntolerance.patient | CarePlan.subject.where(resolve() is Patient) | CareTeam.subject.where(resolve() is Patient) | ClinicalImpression.subject.where(resolve() is Patient) | Composition.subject.where(resolve() is Patient) | Condition.subject.where(resolve() is Patient) | Consent.patient | DetectedIssue.patient | DeviceRequest.subject.where(resolve() is Patient) | DeviceUseStatement.subject | DiagnosticReport.subject.where(resolve() is Patient) | DocumentManifest.subject.where(resolve() is Patient) | DocumentReference.subject.where(resolve() is Patient) | Encounter.subject.where(resolve() is Patient) | EpisodeOfCare.patient | FamilyMemberHistory.patient | Flag.subject.where(resolve() is Patient) | Goal.subject.where(resolve() is Patient) | ImagingStudy.subject.where(resolve() is Patient) | Immunization.patient | List.subject.where(resolve() is Patient) | MedicationAdministration.subject.where(resolve() is Patient) | MedicationDispense.subject.where(resolve() is Patient) | MedicationRequest.subject.where(resolve() is Patient) | MedicationStatement.subject.where(resolve() is Patient) | NutritionOrder.patient | Observation.subject.where(resolve() is Patient) | Procedure.subject.where(resolve() is Patient) | RiskAssessment.subject.where(resolve() is Patient) | ServiceRequest.subject.where(resolve() is Patient) | SupplyDelivery.patient | VisionPrescription.patient

phone TU | token | individual-phone | Multiple Resources: Patient: A value in a phone contact Person: A value in a phone contact Practitioner: A value in a phone contact PractitionerRole: A value in a phone contact RelatedPerson: A value in a phone contact | Patient.telecom.where(system='phone') | Person.telecom.where(system='phone') | Practitioner.telecom.where(system='phone') | PractitionerRole.telecom.where(system='phone') | RelatedPerson.telecom.where(system='phone')

phonetic TU | string | individual-phonetic | Multiple Resources: Patient: A portion of either family or given name using some kind of phonetic matching algorithm Person: A portion of name using some kind of phonetic matching algorithm Practitioner: A portion of

either family or given name using some kind of phonetic matching algorithm RelatedPerson: A portion of name using some kind of phonetic matching algorithm | Patient.name | Person.name | Practitioner.name | RelatedPerson.name

publisher TU | string | conformance-publisher | Multiple Resources: CapabilityStatement: Name of the publisher of the capability statement CodeSystem: Name of the publisher of the code system CompartmentDefinition: Name of the publisher of the compartment definition ConceptMap: Name of the publisher of the concept map GraphDefinition: Name of the publisher of the graph definition ImplementationGuide: Name of the publisher of the implementation guide MessageDefinition: Name of the publisher of the message definition NamingSystem: Name of the publisher of the naming system OperationDefinition: Name of the publisher of the operation definition SearchParameter: Name of the publisher of the search parameter StructureDefinition: Name of the publisher of the structure definition StructureMap: Name of the publisher of the structure map TerminologyCapabilities: Name of the publisher of the terminology capabilities ValueSet: Name of the publisher of the value set | CapabilityStatement.publisher | CodeSystem.publisher | CompartmentDefinition.publisher | ConceptMap.publisher | GraphDefinition.publisher | ImplementationGuide.publisher | MessageDefinition.publisher | NamingSystem.publisher | OperationDefinition.publisher | SearchParameter.publisher | StructureDefinition.publisher | StructureMap.publisher | TerminologyCapabilities.publisher | ValueSet.publisher

status TU | token | conformance-status | Multiple Resources: CapabilityStatement: The current status of the capability statement CodeSystem: The current status of the code system CompartmentDefinition: The current status of the compartment definition ConceptMap: The current status of the concept map GraphDefinition: The current status of the graph definition ImplementationGuide: The current status of the implementation guide MessageDefinition: The current status of the message definition NamingSystem: The current status of the naming system OperationDefinition: The current status of the operation definition SearchParameter: The current status of the search parameter StructureDefinition: The current status of the structure definition StructureMap: The current status of the structure map TerminologyCapabilities: The current status of the terminology capabilities ValueSet: The current status of the value set | CapabilityStatement.status | CodeSystem.status | CompartmentDefinition.status | ConceptMap.status | GraphDefinition.status | ImplementationGuide.status | MessageDefinition.status | NamingSystem.status | OperationDefinition.status | SearchParameter.status | StructureDefinition.status | StructureMap.status | TerminologyCapabilities.status | ValueSet.status

status TU | token | medications-status | Multiple Resources: MedicationAdministration: MedicationAdministration event status (for example one of active/paused/completed/nullified) MedicationDispense: Returns dispenses with a specified dispense status MedicationRequest: Status of the prescription MedicationStatement: Return statements that match the given status | MedicationAdministration.status | MedicationDispense.status | MedicationRequest.status | MedicationStatement.status

telecom TU | token | individual-telecom | Multiple Resources: Patient: The value in any kind of telecom details of the patient Person: The value in any kind of contact Practitioner: The value in any kind of contact PractitionerRole: The value in any kind of contact RelatedPerson: The value in any kind of contact | Patient.telecom | Person.telecom | Practitioner.telecom | PractitionerRole.telecom | RelatedPerson.telecom

title TU | string | conformance-title | Multiple Resources: CapabilityStatement: The human-friendly name of the capability statement CodeSystem: The human-friendly name of the code system ConceptMap: The human-friendly name of the concept map ImplementationGuide: The human-friendly name of the implementation guide MessageDefinition: The human-friendly name of the message definition OperationDefinition: The human-friendly name of the operation definition StructureDefinition: The human-friendly name of the structure definition StructureMap: The human-friendly name of the structure map TerminologyCapabilities: The human-friendly name of the terminology capabilities ValueSet: The human-friendly name of the value set | CapabilityStatement.title | CodeSystem.title | ConceptMap.title | ImplementationGuide.title | MessageDefinition.title | OperationDefinition.title | StructureDefinition.title | StructureMap.title | TerminologyCapabilities.title | ValueSet.title

type TU | token | clinical-type | Multiple Resources: AllergyIntolerance: allergy | intolerance - Underlying mechanism (if known) Composition: Kind of composition (LOINC if possible) DocumentManifest: Kind of document set DocumentReference: Kind of document (LOINC if possible) Encounter: Specific type of encounter EpisodeOfCare: Type/class - e.g. specialist referral, disease management | AllergyIntolerance.type | Composition.type | DocumentManifest.type | DocumentReference.type | Encounter.type | EpisodeOfCare.type

url TU | uri | conformance-url | Multiple Resources: CapabilityStatement: The uri that identifies the capability statement CodeSystem: The uri that identifies the code system CompartmentDefinition: The uri that identifies the compartment definition ConceptMap: The uri that identifies the concept map GraphDefinition: The uri that identifies the graph definition ImplementationGuide: The uri that identifies the implementation guide MessageDefinition: The uri that identifies the message definition OperationDefinition: The uri that identifies the operation definition SearchParameter: The uri that identifies the search parameter StructureDefinition: The uri that identifies the structure definition StructureMap: The uri that identifies the structure map TerminologyCapabilities: The uri that identifies the terminology capabilities ValueSet: The uri that identifies the value set | CapabilityStatement.url | CodeSystem.url | CompartmentDefinition.url | ConceptMap.url | GraphDefinition.url | ImplementationGuide.url | MessageDefinition.url | OperationDefinition.url | SearchParameter.url | StructureDefinition.url | StructureMap.url | TerminologyCapabilities.url | ValueSet.url

version TU | token | conformance-version | Multiple Resources: CapabilityStatement: The business version of the capability statement CodeSystem: The business version of the code system CompartmentDefinition: The business version of the compartment definition ConceptMap: The business version of the concept map GraphDefinition: The business version of the graph definition ImplementationGuide: The business version of the implementation guide MessageDefinition: The business version of the message definition OperationDefinition: The business version of the operation definition SearchParameter: The business version of the search parameter StructureDefinition: The business version of the structure definition StructureMap: The business version of the structure map TerminologyCapabilities: The business version of the terminology capabilities ValueSet: The business version of the value set | CapabilityStatement.version | CodeSystem.version | CompartmentDefinition.version | ConceptMap.version | GraphDefinition.version | ImplementationGuide.version | MessageDefinition.version | OperationDefinition.version | SearchParameter.version | StructureDefinition.version | StructureMap.version | TerminologyCapabilities.version | ValueSet.version

Account TU

identifier | token | Account-identifier | Account number | Account.identifier

name | string | Account-name | Human-readable label | Account.name

owner | reference | Account-owner | Entity managing the Account | Account.owner

patient | reference | Account-patient | The entity that caused the expenses | Account.subject.where(resolve() is Patient)

period | date | Account-period | Transaction window | Account.servicePeriod

status | token | Account-status | active | inactive | entered-in-error | on-hold | unknown | Account.status

subject | reference | Account-subject | The entity that caused the expenses | Account.subject

type | token | Account-type | E.g. patient, expense, depreciation | Account.type

ActivityDefinition TU

composed-of | reference | ActivityDefinition-composed-of | What resource is being referenced | ActivityDefinition.relatedArtifact.where(type='composed-of').resource

context | token | ActivityDefinition-context | A use context assigned to the activity definition | (ActivityDefinition.useContext.value as CodeableConcept)
context-quantity | quantity | ActivityDefinition-context-quantity | A quantity- or range-valued use context assigned to the activity definition | (ActivityDefinition.useContext.value as Quantity) | (ActivityDefinition.useContext.value as Range)
context-type | token | ActivityDefinition-context-type | A type of use context assigned to the activity definition | ActivityDefinition.useContext.code
context-type-quantity | composite | ActivityDefinition-context-type-quantity | A use context type and quantity- or range-based value assigned to the activity definition | ActivityDefinition.useContext
context-type-value | composite | ActivityDefinition-context-type-value | A use context type and value assigned to the activity definition | ActivityDefinition.useContext
date | date | ActivityDefinition-date | The activity definition publication date | ActivityDefinition.date
depends-on | reference | ActivityDefinition-depends-on | What resource is being referenced | ActivityDefinition.relatedArtifact.where(type='depends-on').resource | ActivityDefinition.library
derived-from | reference | ActivityDefinition-derived-from | What resource is being referenced | ActivityDefinition.relatedArtifact.where(type='derived-from').resource
description | string | ActivityDefinition-description | The description of the activity definition | ActivityDefinition.description
effective | date | ActivityDefinition-effective | The time during which the activity definition is intended to be in use | ActivityDefinition.effectivePeriod
identifier | token | ActivityDefinition-identifier | External identifier for the activity definition | ActivityDefinition.identifier
jurisdiction | token | ActivityDefinition-jurisdiction | Intended jurisdiction for the activity definition | ActivityDefinition.jurisdiction
name | string | ActivityDefinition-name | Computationally friendly name of the activity definition | ActivityDefinition.name
predecessor | reference | ActivityDefinition-predecessor | What resource is being referenced | ActivityDefinition.relatedArtifact.where(type='predecessor').resource
publisher | string | ActivityDefinition-publisher | Name of the publisher of the activity definition | ActivityDefinition.publisher
status | token | ActivityDefinition-status | The current status of the activity definition | ActivityDefinition.status
successor | reference | ActivityDefinition-successor | What resource is being referenced | ActivityDefinition.relatedArtifact.where(type='successor').resource
title | string | ActivityDefinition-title | The human-friendly name of the activity definition | ActivityDefinition.title
topic | token | ActivityDefinition-topic | Topics associated with the module | ActivityDefinition.topic
url | uri | ActivityDefinition-url | The uri that identifies the activity definition | ActivityDefinition.url
version | token | ActivityDefinition-version | The business version of the activity definition | ActivityDefinition.version
AdverseEvent TU
actuality | token | AdverseEvent-actuality | actual | potential | AdverseEvent.actuality
category | token | AdverseEvent-category | product-problem | product-quality | product-use-error | wrong-dose | incorrect-prescribing-information | wrong-technique | wrong-route-of-administration | wrong-rate | wrong-duration | wrong-time | expired-drug | medical-device-use-error | problem-different-manufacturer | unsafe-physical-environment | AdverseEvent.category
date | date | AdverseEvent-date | When the event occurred | AdverseEvent.date
event | token | AdverseEvent-event | Type of the event itself in relation to the subject | AdverseEvent.event
location | reference | AdverseEvent-location | Location where adverse event occurred | AdverseEvent.location
recorder | reference | AdverseEvent-recorder | Who recorded the adverse event | AdverseEvent.recorder
resultingcondition | reference | AdverseEvent-resultingcondition | Effect on the subject due to this event | AdverseEvent.resultingCondition
seriousness | token | AdverseEvent-seriousness | Seriousness of the event | AdverseEvent.seriousness
severity | token | AdverseEvent-severity | mild | moderate | severe | AdverseEvent.severity
study | reference | AdverseEvent-study | AdverseEvent.study | AdverseEvent.study
subject | reference | AdverseEvent-subject | Subject impacted by event | AdverseEvent.subject
substance | reference | AdverseEvent-substance | Refers to the specific entity that caused the adverse event | AdverseEvent.suspectEntity.instance
AllergyIntolerance TU
asserter | reference | AllergyIntolerance-asserter | Source of the information about the allergy | AllergyIntolerance.asserter
category | token | AllergyIntolerance-category | food | medication | environment | biologic | AllergyIntolerance.category
clinical-status | token | AllergyIntolerance-clinical-status | active | inactive | resolved | AllergyIntolerance.clinicalStatus
code | token | clinical-code | Code that identifies the allergy or intolerance | AllergyIntolerance.code | AllergyIntolerance.reaction.substance
criticality | token | AllergyIntolerance-criticality | low | high | unable-to-assess | AllergyIntolerance.criticality
date | date | clinical-date | Date first version of the resource instance was recorded | AllergyIntolerance.recordedDate
identifier | token | clinical-identifier | External ids for this item | AllergyIntolerance.identifier
last-date | date | AllergyIntolerance-last-date | Date(/time) of last known occurrence of a reaction | AllergyIntolerance.lastOccurrence
manifestation | token | AllergyIntolerance-manifestation | Clinical symptoms/signs associated with the Event | AllergyIntolerance.reaction.manifestation
onset | date | AllergyIntolerance-onset | Date(/time) when manifestations showed | AllergyIntolerance.reaction.onset
patient | reference | clinical-patient | Who the sensitivity is for | AllergyIntolerance.patient
recorder | reference | AllergyIntolerance-recorder | Who recorded the sensitivity | AllergyIntolerance.recorder
route | token | AllergyIntolerance-route | How the subject was exposed to the substance | AllergyIntolerance.reaction.exposureRoute
severity | token | AllergyIntolerance-severity | mild | moderate | severe (of event as a whole) | AllergyIntolerance.reaction.severity
type | token | clinical-type | allergy | intolerance - Underlying mechanism (if known) | AllergyIntolerance.type
verification-status | token | AllergyIntolerance-verification-status | unconfirmed | confirmed | refuted | entered-in-error | AllergyIntolerance.verificationStatus
Appointment TU
actor | reference | Appointment-actor | Any one of the individuals participating in the appointment | Appointment.participant.actor
appointment-type | token | Appointment-appointment-type | The style of appointment or patient that has been booked in the slot (not service type) | Appointment.appointmentType

based-on | reference | Appointment-based-on | The service request this appointment is allocated to assess | Appointment.basedOn
date | date | Appointment-date | Appointment date/time. | Appointment.start
identifier | token | Appointment-identifier | An Identifier of the Appointment | Appointment.identifier
location | reference | Appointment-location | This location is listed in the participants of the appointment | Appointment.participant.actor.where(resolve() is Location)
part-status | token | Appointment-part-status | The Participation status of the subject, or other participant on the appointment. Can be used to locate participants that have not responded to meeting requests. | Appointment.participant.status
patient | reference | Appointment-patient | One of the individuals of the appointment is this patient | Appointment.participant.actor.where(resolve() is Patient)
practitioner | reference | Appointment-practitioner | One of the individuals of the appointment is this practitioner | Appointment.participant.actor.where(resolve() is Practitioner)
reason-code | token | Appointment-reason-code | Coded reason this appointment is scheduled | Appointment.reasonCode
reason-reference | reference | Appointment-reason-reference | Reason the appointment is to take place (resource) | Appointment.reasonReference
service-category | token | Appointment-service-category | A broad categorization of the service that is to be performed during this appointment | Appointment.serviceCategory
service-type | token | Appointment-service-type | The specific service that is to be performed during this appointment | Appointment.serviceType
slot | reference | Appointment-slot | The slots that this appointment is filling | Appointment.slot
specialty | token | Appointment-specialty | The specialty of a practitioner that would be required to perform the service requested in this appointment | Appointment.specialty
status | token | Appointment-status | The overall status of the appointment | Appointment.status
supporting-info | reference | Appointment-supporting-info | Additional information to support the appointment | Appointment.supportingInformation
AppointmentResponse TU
actor | reference | AppointmentResponse-actor | The Person, Location/HealthcareService or Device that this appointment response replies for | AppointmentResponse.actor
appointment | reference | AppointmentResponse-appointment | The appointment that the response is attached to | AppointmentResponse.appointment
identifier | token | AppointmentResponse-identifier | An Identifier in this appointment response | AppointmentResponse.identifier
location | reference | AppointmentResponse-location | This Response is for this Location | AppointmentResponse.actor.where(resolve() is Location)
part-status | token | AppointmentResponse-part-status | The participants acceptance status for this appointment | AppointmentResponse.participantStatus
patient | reference | AppointmentResponse-patient | This Response is for this Patient | AppointmentResponse.actor.where(resolve() is Patient)
practitioner | reference | AppointmentResponse-practitioner | This Response is for this Practitioner | AppointmentResponse.actor.where(resolve() is Practitioner)
AuditEvent TU
action | token | AuditEvent-action | Type of action performed during the event | AuditEvent.action
address | string | AuditEvent-address | Identifier for the network access point of the user device | AuditEvent.agent.network.address
agent | reference | AuditEvent-agent | Identifier of who | AuditEvent.agent.who
agent-name | string | AuditEvent-agent-name | Human friendly name for the agent | AuditEvent.agent.name
agent-role | token | AuditEvent-agent-role | Agent role in the event | AuditEvent.agent.role
altid | token | AuditEvent-altid | Alternative User identity | AuditEvent.agent.altId
date | date | AuditEvent-date | Time when the event was recorded | AuditEvent.recorded
entity | reference | AuditEvent-entity | Specific instance of resource | AuditEvent.entity.what
entity-name | string | AuditEvent-entity-name | Descriptor for entity | AuditEvent.entity.name
entity-role | token | AuditEvent-entity-role | What role the entity played | AuditEvent.entity.role
entity-type | token | AuditEvent-entity-type | Type of entity involved | AuditEvent.entity.type
outcome | token | AuditEvent-outcome | Whether the event succeeded or failed | AuditEvent.outcome
patient | reference | AuditEvent-patient | Identifier of who | AuditEvent.agent.who.where(resolve() is Patient) | AuditEvent.entity.what.where(resolve() is Patient)
policy | uri | AuditEvent-policy | Policy that authorized event | AuditEvent.agent.policy
site | token | AuditEvent-site | Logical source location within the enterprise | AuditEvent.source.site
source | reference | AuditEvent-source | The identity of source detecting the event | AuditEvent.source.observer
subtype | token | AuditEvent-subtype | More specific type/id for the event | AuditEvent.subtype
type | token | AuditEvent-type | Type/identifier of event | AuditEvent.type
Basic TU
author | reference | Basic-author | Who created | Basic.author
code | token | Basic-code | Kind of Resource | Basic.code
created | date | Basic-created | When created | Basic.created
identifier | token | Basic-identifier | Business identifier | Basic.identifier
patient | reference | Basic-patient | Identifies the focus of this resource | Basic.subject.where(resolve() is Patient)
subject | reference | Basic-subject | Identifies the focus of this resource | Basic.subject
BodyStructure TU
identifier | token | BodyStructure-identifier | Bodystructure identifier | BodyStructure.identifier
location | token | BodyStructure-location | Body site | BodyStructure.location
morphology | token | BodyStructure-morphology | Kind of Structure | BodyStructure.morphology
patient | reference | BodyStructure-patient | Who this is about | BodyStructure.patient
Bundle N
composition TU | reference | Bundle-composition | The first resource in the bundle, if the bundle type is "document" - this is a composition, and this parameter provides access to search its contents | Bundle.entry[0].resource

identifier TU | token | Bundle-identifier | Persistent identifier for the bundle | Bundle.identifier
message TU | reference | Bundle-message | The first resource in the bundle, if the bundle type is "message" - this is a message header, and this parameter provides access to search its contents | Bundle.entry[0].resource
timestamp TU | date | Bundle-timestamp | When the bundle was assembled | Bundle.timestamp
type TU | token | Bundle-type | document | message | transaction | transaction-response | batch | batch-response | history | searchset | collection | Bundle.type
CapabilityStatement N
context TU | token | conformance-context | A use context assigned to the capability statement | (CapabilityStatement.useContext.value as CodeableConcept)
context-quantity TU | quantity | conformance-context-quantity | A quantity- or range-valued use context assigned to the capability statement | (CapabilityStatement.useContext.value as Quantity) | (CapabilityStatement.useContext.value as Range)
context-type TU | token | conformance-context-type | A type of use context assigned to the capability statement | CapabilityStatement.useContext.code
context-type-quantity TU | composite | conformance-context-type-quantity | A use context type and quantity- or range-based value assigned to the capability statement | CapabilityStatement.useContext
context-type-value TU | composite | conformance-context-type-value | A use context type and value assigned to the capability statement | CapabilityStatement.useContext
date TU | date | conformance-date | The capability statement publication date | CapabilityStatement.date
description TU | string | conformance-description | The description of the capability statement | CapabilityStatement.description
fhirversion TU | token | CapabilityStatement-fhirversion | The version of FHIR | CapabilityStatement.version
format TU | token | CapabilityStatement-format | formats supported (xml | json | ttl | mime type) | CapabilityStatement.format
guide TU | reference | CapabilityStatement-guide | Implementation guides supported | CapabilityStatement.implementationGuide
jurisdiction TU | token | conformance-jurisdiction | Intended jurisdiction for the capability statement | CapabilityStatement.jurisdiction
mode TU | token | CapabilityStatement-mode | Mode - restful (server/client) or messaging (sender/receiver) | CapabilityStatement.rest.mode
name TU | string | conformance-name | Computationally friendly name of the capability statement | CapabilityStatement.name
publisher TU | string | conformance-publisher | Name of the publisher of the capability statement | CapabilityStatement.publisher
resource TU | token | CapabilityStatement-resource | Name of a resource mentioned in a capability statement | CapabilityStatement.rest.resource.type
resource-profile TU | reference | CapabilityStatement-resource-profile | A profile id invoked in a capability statement | CapabilityStatement.rest.resource.profile
security-service TU | token | CapabilityStatement-security-service | OAuth | SMART-on-FHIR | NTLM | Basic | Kerberos | Certificates | CapabilityStatement.rest.security.service
software TU | string | CapabilityStatement-software | Part of the name of a software application | CapabilityStatement.software.name
status TU | token | conformance-status | The current status of the capability statement | CapabilityStatement.status
supported-profile TU | reference | CapabilityStatement-supported-profile | Profiles for use cases supported | CapabilityStatement.rest.resource.supportedProfile
title TU | string | conformance-title | The human-friendly name of the capability statement | CapabilityStatement.title
url TU | uri | conformance-url | The uri that identifies the capability statement | CapabilityStatement.url
version TU | token | conformance-version | The business version of the capability statement | CapabilityStatement.version
CarePlan TU
activity-code | token | CarePlan-activity-code | Detail type of activity | CarePlan.activity.detail.code
activity-date | date | CarePlan-activity-date | Specified date occurs within period specified by CarePlan.activity.detail.scheduled[x] | CarePlan.activity.detail.scheduled
activity-reference | reference | CarePlan-activity-reference | Activity details defined in specific resource | CarePlan.activity.reference
based-on | reference | CarePlan-based-on | Fulfills CarePlan | CarePlan.basedOn
care-team | reference | CarePlan-care-team | Who's involved in plan? | CarePlan.careTeam
category | token | CarePlan-category | Type of plan | CarePlan.category
condition | reference | CarePlan-condition | Health issues this plan addresses | CarePlan.addresses
date | date | clinical-date | Time period plan covers | CarePlan.period
encounter | reference | CarePlan-encounter | Encounter created as part of | CarePlan.encounter
goal | reference | CarePlan-goal | Desired outcome of plan | CarePlan.goal
identifier | token | clinical-identifier | External Ids for this plan | CarePlan.identifier
instantiates-canonical | reference | CarePlan-instantiates-canonical | Instantiates FHIR protocol or definition | CarePlan.instantiatesCanonical
instantiates-uri | uri | CarePlan-instantiates-uri | Instantiates external protocol or definition | CarePlan.instantiatesUri
intent | token | CarePlan-intent | proposal | plan | order | option | CarePlan.intent
part-of | reference | CarePlan-part-of | Part of referenced CarePlan | CarePlan.partOf
patient | reference | clinical-patient | Who the care plan is for | CarePlan.subject.where(resolve() is Patient)
performer | reference | CarePlan-performer | Matches if the practitioner is listed as a performer in any of the "simple" activities. (For performers of the detailed activities, chain through the activitydetail search parameter.) | CarePlan.activity.detail.performer
replaces | reference | CarePlan-replaces | CarePlan replaced by this CarePlan | CarePlan.replaces
status | token | CarePlan-status | draft | active | on-hold | revoked | completed | entered-in-error | unknown | CarePlan.status
subject | reference | CarePlan-subject | Who the care plan is for | CarePlan.subject
CareTeam TU
category | token | CareTeam-category | Type of team | CareTeam.category
date | date | clinical-date | Time period team covers | CareTeam.period
encounter | reference | CareTeam-encounter | Encounter created as part of | CareTeam.encounter
identifier | token | clinical-identifier | External Ids for this team | CareTeam.identifier
participant | reference | CareTeam-participant | Who is involved | CareTeam.participant.member
patient | reference | clinical-patient | Who care team is for | CareTeam.subject.where(resolve() is Patient)

status | token | CareTeam-status | proposed | active | suspended | inactive | entered-in-error | CareTeam.status
subject | reference | CareTeam-subject | Who care team is for | CareTeam.subject
ChargeItem TU
account | reference | ChargeItem-account | Account to place this charge | ChargeItem.account
code | token | ChargeItem-code | A code that identifies the charge, like a billing code | ChargeItem.code
context | reference | ChargeItem-context | Encounter / Episode associated with event | ChargeItem.context
entered-date | date | ChargeItem-entered-date | Date the charge item was entered | ChargeItem.enteredDate
enterer | reference | ChargeItem-enterer | Individual who was entering | ChargeItem.enterer
factor-override | number | ChargeItem-factor-override | Factor overriding the associated rules | ChargeItem.factorOverride
identifier | token | ChargeItem-identifier | Business Identifier for item | ChargeItem.identifier
occurrence | date | ChargeItem-occurrence | When the charged service was applied | ChargeItem.occurrence
patient | reference | ChargeItem-patient | Individual service was done for/to | ChargeItem.subject.where(resolve() is Patient)
performer-actor | reference | ChargeItem-performer-actor | Individual who was performing | ChargeItem.performer.actor
performer-function | token | ChargeItem-performer-function | What type of performance was done | ChargeItem.performer.function
performing-organization | reference | ChargeItem-performing-organization | Organization providing the charged service | ChargeItem.performingOrganization
price-override | quantity | ChargeItem-price-override | Price overriding the associated rules | ChargeItem.priceOverride
quantity | quantity | ChargeItem-quantity | Quantity of which the charge item has been serviced | ChargeItem.quantity
requesting-organization | reference | ChargeItem-requesting-organization | Organization requesting the charged service | ChargeItem.requestingOrganization
service | reference | ChargeItem-service | Which rendered service is being charged? | ChargeItem.service
subject | reference | ChargeItem-subject | Individual service was done for/to | ChargeItem.subject
ChargeItemDefinition TU
context | token | ChargeItemDefinition-context | A use context assigned to the charge item definition | (ChargeItemDefinition.useContext.value as CodeableConcept)
context-quantity | quantity | ChargeItemDefinition-context-quantity | A quantity- or range-valued use context assigned to the charge item definition | (ChargeItemDefinition.useContext.value as Quantity) | (ChargeItemDefinition.useContext.value as Range)
context-type | token | ChargeItemDefinition-context-type | A type of use context assigned to the charge item definition | ChargeItemDefinition.useContext.code
context-type-quantity | composite | ChargeItemDefinition-context-type-quantity | A use context type and quantity- or range-based value assigned to the charge item definition | ChargeItemDefinition.useContext
context-type-value | composite | ChargeItemDefinition-context-type-value | A use context type and value assigned to the charge item definition | ChargeItemDefinition.useContext
date | date | ChargeItemDefinition-date | The charge item definition publication date | ChargeItemDefinition.date
description | string | ChargeItemDefinition-description | The description of the charge item definition | ChargeItemDefinition.description
effective | date | ChargeItemDefinition-effective | The time during which the charge item definition is intended to be in use | ChargeItemDefinition.effectivePeriod
identifier | token | ChargeItemDefinition-identifier | External identifier for the charge item definition | ChargeItemDefinition.identifier
jurisdiction | token | ChargeItemDefinition-jurisdiction | Intended jurisdiction for the charge item definition | ChargeItemDefinition.jurisdiction
publisher | string | ChargeItemDefinition-publisher | Name of the publisher of the charge item definition | ChargeItemDefinition.publisher
status | token | ChargeItemDefinition-status | The current status of the charge item definition | ChargeItemDefinition.status
title | string | ChargeItemDefinition-title | The human-friendly name of the charge item definition | ChargeItemDefinition.title
url | uri | ChargeItemDefinition-url | The uri that identifies the charge item definition | ChargeItemDefinition.url
version | token | ChargeItemDefinition-version | The business version of the charge item definition | ChargeItemDefinition.version
Claim TU
care-team | reference | Claim-care-team | Member of the CareTeam | Claim.careTeam.provider
created | date | Claim-created | The creation date for the Claim | Claim.created
detail-udi | reference | Claim-detail-udi | UDI associated with a line item, detail product or service | Claim.item.detail.udi
encounter | reference | Claim-encounter | Encounters associated with a billed line item | Claim.item.encounter
enterer | reference | Claim-enterer | The party responsible for the entry of the Claim | Claim.enterer
facility | reference | Claim-facility | Facility where the products or services have been or will be provided | Claim.facility
identifier | token | Claim-identifier | The primary identifier of the financial resource | Claim.identifier
insurer | reference | Claim-insurer | The target payor/insurer for the Claim | Claim.insurer
item-udi | reference | Claim-item-udi | UDI associated with a line item product or service | Claim.item.udi
patient | reference | Claim-patient | Patient receiving the products or services | Claim.patient
payee | reference | Claim-payee | The party receiving any payment for the Claim | Claim.payee.party
priority | token | Claim-priority | Processing priority requested | Claim.priority
procedure-udi | reference | Claim-procedure-udi | UDI associated with a procedure | Claim.procedure.udi
provider | reference | Claim-provider | Provider responsible for the Claim | Claim.provider
status | token | Claim-status | The status of the Claim instance. | Claim.status
subdetail-udi | reference | Claim-subdetail-udi | UDI associated with a line item, detail, subdetail product or service | Claim.item.detail.subDetail.udi
use | token | Claim-use | The kind of financial resource | Claim.use
ClaimResponse TU
created | date | ClaimResponse-created | The creation date | ClaimResponse.created
disposition | string | ClaimResponse-disposition | The contents of the disposition message | ClaimResponse.disposition
identifier | token | ClaimResponse-identifier | The identity of the ClaimResponse | ClaimResponse.identifier
insurer | reference | ClaimResponse-insurer | The organization which generated this resource | ClaimResponse.insurer

outcome | token | ClaimResponse-outcome | The processing outcome | ClaimResponse.outcome
patient | reference | ClaimResponse-patient | The subject of care | ClaimResponse.patient
payment-date | date | ClaimResponse-payment-date | The expected payment date | ClaimResponse.payment.date
request | reference | ClaimResponse-request | The claim reference | ClaimResponse.request
requestor | reference | ClaimResponse-requestor | The Provider of the claim | ClaimResponse.requestor
status | token | ClaimResponse-status | The status of the ClaimResponse | ClaimResponse.status
use | token | ClaimResponse-use | The type of claim | ClaimResponse.use
ClinicalImpression TU
assessor | reference | ClinicalImpression-assessor | The clinician performing the assessment | ClinicalImpression.assessor
date | date | clinical-date | When the assessment was documented | ClinicalImpression.date
encounter | reference | ClinicalImpression-encounter | Encounter created as part of | ClinicalImpression.encounter
finding-code | token | ClinicalImpression-finding-code | What was found | ClinicalImpression.finding.itemCodeableConcept
finding-ref | reference | ClinicalImpression-finding-ref | What was found | ClinicalImpression.finding.itemReference
identifier | token | ClinicalImpression-identifier | Business identifier | ClinicalImpression.identifier
investigation | reference | ClinicalImpression-investigation | Record of a specific investigation | ClinicalImpression.investigation.item
patient | reference | clinical-patient | Patient or group assessed | ClinicalImpression.subject.where(resolve() is Patient)
previous | reference | ClinicalImpression-previous | Reference to last assessment | ClinicalImpression.previous
problem | reference | ClinicalImpression-problem | Relevant impressions of patient state | ClinicalImpression.problem
status | token | ClinicalImpression-status | in-progress | completed | entered-in-error | ClinicalImpression.status
subject | reference | ClinicalImpression-subject | Patient or group assessed | ClinicalImpression.subject
supporting-info | reference | ClinicalImpression-supporting-info | Information supporting the clinical impression | ClinicalImpression.supportingInfo
CodeSystem N
code TU | token | CodeSystem-code | A code defined in the code system | CodeSystem.concept.code
content-mode TU | token | CodeSystem-content-mode | not-present | example | fragment | complete | supplement | CodeSystem.content
context TU | token | conformance-context | A use context assigned to the code system | (CodeSystem.useContext.value as CodeableConcept)
context-quantity TU | quantity | conformance-context-quantity | A quantity- or range-valued use context assigned to the code system | (CodeSystem.useContext.value as Quantity) | (CodeSystem.useContext.value as Range)
context-type TU | token | conformance-context-type | A type of use context assigned to the code system | CodeSystem.useContext.code
context-type-quantity TU | composite | conformance-context-type-quantity | A use context type and quantity- or range-based value assigned to the code system | CodeSystem.useContext
context-type-value TU | composite | conformance-context-type-value | A use context type and value assigned to the code system | CodeSystem.useContext
date TU | date | conformance-date | The code system publication date | CodeSystem.date
description TU | string | conformance-description | The description of the code system | CodeSystem.description
identifier TU | token | conformance-identifier | External identifier for the code system | CodeSystem.identifier
jurisdiction TU | token | conformance-jurisdiction | Intended jurisdiction for the code system | CodeSystem.jurisdiction
language TU | token | CodeSystem-language | A language in which a designation is provided | CodeSystem.concept.designation.language
name TU | string | conformance-name | Computationally friendly name of the code system | CodeSystem.name
publisher TU | string | conformance-publisher | Name of the publisher of the code system | CodeSystem.publisher
status TU | token | conformance-status | The current status of the code system | CodeSystem.status
supplements TU | reference | CodeSystem-supplements | Find code system supplements for the referenced code system | CodeSystem.supplements
system TU | uri | CodeSystem-system | The system for any codes defined by this code system (same as 'url') | CodeSystem.url
title TU | string | conformance-title | The human-friendly name of the code system | CodeSystem.title
url TU | uri | conformance-url | The uri that identifies the code system | CodeSystem.url
version TU | token | conformance-version | The business version of the code system | CodeSystem.version
Communication TU
based-on | reference | Communication-based-on | Request fulfilled by this communication | Communication.basedOn
category | token | Communication-category | Message category | Communication.category
encounter | reference | Communication-encounter | Encounter created as part of | Communication.encounter
identifier | token | Communication-identifier | Unique identifier | Communication.identifier
instantiates-canonical | reference | Communication-instantiates-canonical | Instantiates FHIR protocol or definition | Communication.instantiatesCanonical
instantiates-uri | uri | Communication-instantiates-uri | Instantiates external protocol or definition | Communication.instantiatesUri
medium | token | Communication-medium | A channel of communication | Communication.medium
part-of | reference | Communication-part-of | Part of this action | Communication.partOf
patient | reference | Communication-patient | Focus of message | Communication.subject.where(resolve() is Patient)
received | date | Communication-received | When received | Communication.received
recipient | reference | Communication-recipient | Message recipient | Communication.recipient
sender | reference | Communication-sender | Message sender | Communication.sender
sent | date | Communication-sent | When sent | Communication.sent
status | token | Communication-status | preparation | in-progress | not-done | on-hold | stopped | completed | entered-in-error | unknown | Communication.status
subject | reference | Communication-subject | Focus of message | Communication.subject
CommunicationRequest TU
authored | date | CommunicationRequest-authored | When request transitioned to being actionable | CommunicationRequest.authoredOn

based-on | reference | CommunicationRequest-based-on | Fulfills plan or proposal | CommunicationRequest.basedOn
category | token | CommunicationRequest-category | Message category | CommunicationRequest.category
encounter | reference | CommunicationRequest-encounter | Encounter created as part of | CommunicationRequest.encounter
group-identifier | token | CommunicationRequest-group-identifier | Composite request this is part of | CommunicationRequest.groupIdentifier
identifier | token | CommunicationRequest-identifier | Unique identifier | CommunicationRequest.identifier
medium | token | CommunicationRequest-medium | A channel of communication | CommunicationRequest.medium
occurrence | date | CommunicationRequest-occurrence | When scheduled | (CommunicationRequest.occurrence as dateTime)
patient | reference | CommunicationRequest-patient | Focus of message | CommunicationRequest.subject.where(resolve() is Patient)
priority | token | CommunicationRequest-priority | routine | urgent | asap | stat | CommunicationRequest.priority
recipient | reference | CommunicationRequest-recipient | Message recipient | CommunicationRequest.recipient
replaces | reference | CommunicationRequest-replaces | Request(s) replaced by this request | CommunicationRequest.replaces
requester | reference | CommunicationRequest-requester | Who/what is requesting service | CommunicationRequest.requester
sender | reference | CommunicationRequest-sender | Message sender | CommunicationRequest.sender
status | token | CommunicationRequest-status | draft | active | on-hold | revoked | completed | entered-in-error | unknown | CommunicationRequest.status
subject | reference | CommunicationRequest-subject | Focus of message | CommunicationRequest.subject
CompartmentDefinition TU
code | token | CompartmentDefinition-code | Patient | Encounter | RelatedPerson | Practitioner | Device | CompartmentDefinition.code
context | token | conformance-context | A use context assigned to the compartment definition | (CompartmentDefinition.useContext.value as CodeableConcept)
context-quantity | quantity | conformance-context-quantity | A quantity- or range-valued use context assigned to the compartment definition | (CompartmentDefinition.useContext.value as Quantity) | (CompartmentDefinition.useContext.value as Range)
context-type | token | conformance-context-type | A type of use context assigned to the compartment definition | CompartmentDefinition.useContext.code
context-type-quantity | composite | conformance-context-type-quantity | A use context type and quantity- or range-based value assigned to the compartment definition | CompartmentDefinition.useContext
context-type-value | composite | conformance-context-type-value | A use context type and value assigned to the compartment definition | CompartmentDefinition.useContext
date | date | conformance-date | The compartment definition publication date | CompartmentDefinition.date
description | string | conformance-description | The description of the compartment definition | CompartmentDefinition.description
name | string | conformance-name | Computationally friendly name of the compartment definition | CompartmentDefinition.name
publisher | string | conformance-publisher | Name of the publisher of the compartment definition | CompartmentDefinition.publisher
resource | token | CompartmentDefinition-resource | Name of resource type | CompartmentDefinition.resource.code
status | token | conformance-status | The current status of the compartment definition | CompartmentDefinition.status
url | uri | conformance-url | The uri that identifies the compartment definition | CompartmentDefinition.url
version | token | conformance-version | The business version of the compartment definition | CompartmentDefinition.version
Composition TU
attester | reference | Composition-attester | Who attested the composition | Composition.attester.party
author | reference | Composition-author | Who and/or what authored the composition | Composition.author
category | token | Composition-category | Categorization of Composition | Composition.category
confidentiality | token | Composition-confidentiality | As defined by affinity domain | Composition.confidentiality
context | token | Composition-context | Code(s) that apply to the event being documented | Composition.event.code
date | date | clinical-date | Composition editing time | Composition.date
encounter | reference | clinical-encounter | Context of the Composition | Composition.encounter
entry | reference | Composition-entry | A reference to data that supports this section | Composition.section.entry
identifier | token | clinical-identifier | Version-independent identifier for the Composition | Composition.identifier
patient | reference | clinical-patient | Who and/or what the composition is about | Composition.subject.where(resolve() is Patient)
period | date | Composition-period | The period covered by the documentation | Composition.event.period
related-id | token | Composition-related-id | Target of the relationship | (Composition.relatesTo.target as Identifier)
related-ref | reference | Composition-related-ref | Target of the relationship | (Composition.relatesTo.target as Reference)
section | token | Composition-section | Classification of section (recommended) | Composition.section.code
status | token | Composition-status | preliminary | final | amended | entered-in-error | Composition.status
subject | reference | Composition-subject | Who and/or what the composition is about | Composition.subject
title | string | Composition-title | Human Readable name/title | Composition.title
type | token | clinical-type | Kind of composition (LOINC if possible) | Composition.type
ConceptMap TU
context | token | conformance-context | A use context assigned to the concept map | (ConceptMap.useContext.value as CodeableConcept)
context-quantity | quantity | conformance-context-quantity | A quantity- or range-valued use context assigned to the concept map | (ConceptMap.useContext.value as Quantity) | (ConceptMap.useContext.value as Range)
context-type | token | conformance-context-type | A type of use context assigned to the concept map | ConceptMap.useContext.code
context-type-quantity | composite | conformance-context-type-quantity | A use context type and quantity- or range-based value assigned to the concept map | ConceptMap.useContext
context-type-value | composite | conformance-context-type-value | A use context type and value assigned to the concept map | ConceptMap.useContext
date | date | conformance-date | The concept map publication date | ConceptMap.date
dependson | uri | ConceptMap-dependson | Reference to property mapping depends on | ConceptMap.group.element.target.dependsOn.property
description | string | conformance-description | The description of the concept map | ConceptMap.description

identifier | token | conformance-identifier | External identifier for the concept map | ConceptMap.identifier
jurisdiction | token | conformance-jurisdiction | Intended jurisdiction for the concept map | ConceptMap.jurisdiction
name | string | conformance-name | Computationally friendly name of the concept map | ConceptMap.name
other | reference | ConceptMap-other | canonical reference to an additional ConceptMap to use for mapping if the source concept is unmapped | ConceptMap.group.unmapped.url
product | uri | ConceptMap-product | Reference to property mapping depends on | ConceptMap.group.element.target.product.property
publisher | string | conformance-publisher | Name of the publisher of the concept map | ConceptMap.publisher
source | reference | ConceptMap-source | The source value set that contains the concepts that are being mapped | (ConceptMap.source as canonical)
source-code | token | ConceptMap-source-code | Identifies element being mapped | ConceptMap.group.element.code
source-system | uri | ConceptMap-source-system | Source system where concepts to be mapped are defined | ConceptMap.group.source
source-uri | reference | ConceptMap-source-uri | The source value set that contains the concepts that are being mapped | (ConceptMap.source as uri)
status | token | conformance-status | The current status of the concept map | ConceptMap.status
target | reference | ConceptMap-target | The target value set which provides context for the mappings | (ConceptMap.target as canonical)
target-code | token | ConceptMap-target-code | Code that identifies the target element | ConceptMap.group.element.target.code
target-system | uri | ConceptMap-target-system | Target system that the concepts are to be mapped to | ConceptMap.group.target
target-uri | reference | ConceptMap-target-uri | The target value set which provides context for the mappings | (ConceptMap.target as uri)
title | string | conformance-title | The human-friendly name of the concept map | ConceptMap.title
url | uri | conformance-url | The uri that identifies the concept map | ConceptMap.url
version | token | conformance-version | The business version of the concept map | ConceptMap.version
Condition TU
abatement-age | quantity | Condition-abatement-age | Abatement as age or age range | Condition.abatement.as(Age) | Condition.abatement.as(Range)
abatement-date | date | Condition-abatement-date | Date-related abatements (dateTime and period) | Condition.abatement.as(dateTime) | Condition.abatement.as(Period)
abatement-string | string | Condition-abatement-string | Abatement as a string | Condition.abatement.as(string)
asserter | reference | Condition-asserter | Person who asserts this condition | Condition.asserter
body-site | token | Condition-body-site | Anatomical location, if relevant | Condition.bodySite
category | token | Condition-category | The category of the condition | Condition.category
clinical-status | token | Condition-clinical-status | The clinical status of the condition | Condition.clinicalStatus
code | token | clinical-code | Code for the condition | Condition.code
encounter | reference | Condition-encounter | Encounter created as part of | Condition.encounter
evidence | token | Condition-evidence | Manifestation/symptom | Condition.evidence.code
evidence-detail | reference | Condition-evidence-detail | Supporting information found elsewhere | Condition.evidence.detail
identifier | token | clinical-identifier | A unique identifier of the condition record | Condition.identifier
onset-age | quantity | Condition-onset-age | Onsets as age or age range | Condition.onset.as(Age) | Condition.onset.as(Range)
onset-date | date | Condition-onset-date | Date related onsets (dateTime and Period) | Condition.onset.as(dateTime) | Condition.onset.as(Period)
onset-info | string | Condition-onset-info | Onsets as a string | Condition.onset.as(string)
patient | reference | clinical-patient | Who has the condition? | Condition.subject.where(resolve() is Patient)
recorded-date | date | Condition-recorded-date | Date record was first recorded | Condition.recordedDate
severity | token | Condition-severity | The severity of the condition | Condition.severity
stage | token | Condition-stage | Simple summary (disease specific) | Condition.stage.summary
subject | reference | Condition-subject | Who has the condition? | Condition.subject
verification-status | token | Condition-verification-status | unconfirmed | provisional | differential | confirmed | refuted | entered-in-error | Condition.verificationStatus
Consent TU
action | token | Consent-action | Actions controlled by this rule | Consent.provision.action
actor | reference | Consent-actor | Resource for the actor (or group, by role) | Consent.provision.actor.reference
category | token | Consent-category | Classification of the consent statement - for indexing/retrieval | Consent.category
consentor | reference | Consent-consentor | Who is agreeing to the policy and rules | Consent.performer
data | reference | Consent-data | The actual data reference | Consent.provision.data.reference
date | date | clinical-date | When this Consent was created or indexed | Consent.dateTime
identifier | token | clinical-identifier | Identifier for this record (external references) | Consent.identifier
organization | reference | Consent-organization | Custodian of the consent | Consent.organization
patient | reference | clinical-patient | Who the consent applies to | Consent.patient
period | date | Consent-period | Timeframe for this rule | Consent.provision.period
purpose | token | Consent-purpose | Context of activities covered by this rule | Consent.provision.purpose
scope | token | Consent-scope | Which of the four areas this resource covers (extensible) | Consent.scope
security-label | token | Consent-security-label | Security Labels that define affected resources | Consent.provision.securityLabel
source-reference | reference | Consent-source-reference | Search by reference to a Consent, DocumentReference, Contract or QuestionnaireResponse | Consent.source
status | token | Consent-status | draft | proposed | active | rejected | inactive | entered-in-error | Consent.status
Contract TU
authority | reference | Contract-authority | The authority of the contract | Contract.authority
domain | reference | Contract-domain | The domain of the contract | Contract.domain
identifier | token | Contract-identifier | The identity of the contract | Contract.identifier
instantiates | uri | Contract-instantiates | A source definition of the contract | Contract.instantiatesUri

issued | date | Contract-issued | The date/time the contract was issued | Contract.issued
patient | reference | Contract-patient | The identity of the subject of the contract (if a patient) | Contract.subject.where(resolve() is Patient)
signer | reference | Contract-signer | Contract Signatory Party | Contract.signer.party
status | token | Contract-status | The status of the contract | Contract.status
subject | reference | Contract-subject | The identity of the subject of the contract | Contract.subject
url | uri | Contract-url | The basal contract definition | Contract.url
Coverage TU
beneficiary | reference | Coverage-beneficiary | Covered party | Coverage.beneficiary
class-type | token | Coverage-class-type | Coverage class (eg. plan, group) | Coverage.class.type
class-value | string | Coverage-class-value | Value of the class (eg. Plan number, group number) | Coverage.class.value
dependent | string | Coverage-dependent | Dependent number | Coverage.dependent
identifier | token | Coverage-identifier | The primary identifier of the insured and the coverage | Coverage.identifier
patient | reference | Coverage-patient | Retrieve coverages for a patient | Coverage.beneficiary
payor | reference | Coverage-payor | The identity of the insurer or party paying for services | Coverage.payor
policy-holder | reference | Coverage-policy-holder | Reference to the policyholder | Coverage.policyHolder
status | token | Coverage-status | The status of the Coverage | Coverage.status
subscriber | reference | Coverage-subscriber | Reference to the subscriber | Coverage.subscriber
type | token | Coverage-type | The kind of coverage (health plan, auto, Workers Compensation) | Coverage.type
CoverageEligibilityRequest TU
created | date | CoverageEligibilityRequest-created | The creation date for the EOB | CoverageEligibilityRequest.created
enterer | reference | CoverageEligibilityRequest-enterer | The party who is responsible for the request | CoverageEligibilityRequest.enterer
facility | reference | CoverageEligibilityRequest-facility | Facility responsible for the goods and services | CoverageEligibilityRequest.facility
identifier | token | CoverageEligibilityRequest-identifier | The business identifier of the Eligibility | CoverageEligibilityRequest.identifier
patient | reference | CoverageEligibilityRequest-patient | The reference to the patient | CoverageEligibilityRequest.patient
provider | reference | CoverageEligibilityRequest-provider | The reference to the provider | CoverageEligibilityRequest.provider
status | token | CoverageEligibilityRequest-status | The status of the EligibilityRequest | CoverageEligibilityRequest.status
CoverageEligibilityResponse TU
created | date | CoverageEligibilityResponse-created | The creation date | CoverageEligibilityResponse.created
disposition | string | CoverageEligibilityResponse-disposition | The contents of the disposition message | CoverageEligibilityResponse.disposition
identifier | token | CoverageEligibilityResponse-identifier | The business identifier | CoverageEligibilityResponse.identifier
insurer | reference | CoverageEligibilityResponse-insurer | The organization which generated this resource | CoverageEligibilityResponse.insurer
outcome | token | CoverageEligibilityResponse-outcome | The processing outcome | CoverageEligibilityResponse.outcome
patient | reference | CoverageEligibilityResponse-patient | The reference to the patient | CoverageEligibilityResponse.patient
request | reference | CoverageEligibilityResponse-request | The EligibilityRequest reference | CoverageEligibilityResponse.request
requestor | reference | CoverageEligibilityResponse-requestor | The EligibilityRequest provider | CoverageEligibilityResponse.requestor
status | token | CoverageEligibilityResponse-status | The EligibilityRequest status | CoverageEligibilityResponse.status
DetectedIssue TU
author | reference | DetectedIssue-author | The provider or device that identified the issue | DetectedIssue.author
code | token | DetectedIssue-code | Issue Category, e.g. drug-drug, duplicate therapy, etc. | DetectedIssue.code
identified | date | DetectedIssue-identified | When identified | DetectedIssue.identified
identifier | token | clinical-identifier | Unique id for the detected issue | DetectedIssue.identifier
implicated | reference | DetectedIssue-implicated | Problem resource | DetectedIssue.implicated
patient | reference | clinical-patient | Associated patient | DetectedIssue.patient
Device TU
device-name | string | Device-device-name | A server defined search that may match any of the string fields in Device.deviceName or Device.type. | Device.deviceName.name | Device.type.coding.display | Device.type.text
identifier | token | Device-identifier | Instance id from manufacturer, owner, and others | Device.identifier
location | reference | Device-location | A location, where the resource is found | Device.location
manufacturer | string | Device-manufacturer | The manufacturer of the device | Device.manufacturer
model | string | Device-model | The model of the device | Device.modelNumber
organization | reference | Device-organization | The organization responsible for the device | Device.owner
patient | reference | Device-patient | Patient information, if the resource is affixed to a person | Device.patient
status | token | Device-status | active | inactive | entered-in-error | unknown | Device.status
type | token | Device-type | The type of the device | Device.type
udi-carrier | string | Device-udi-carrier | UDI Barcode (RFID or other technology) string in HRF format. | Device.udiCarrier.carrierHRF
udi-di | string | Device-udi-di | The udi Device Identifier (DI) | Device.udiCarrier.deviceIdentifier
url | uri | Device-url | Network address to contact device | Device.url
DeviceDefinition TU
identifier | token | DeviceDefinition-identifier | The identifier of the component | DeviceDefinition.identifier
parent | reference | DeviceDefinition-parent | The parent DeviceDefinition resource | DeviceDefinition.parentDevice
type | token | DeviceDefinition-type | The device component type | DeviceDefinition.type
DeviceMetric TU
category | token | DeviceMetric-category | The category of the metric | DeviceMetric.category
identifier | token | DeviceMetric-identifier | The identifier of the metric | DeviceMetric.identifier
parent | reference | DeviceMetric-parent | The parent DeviceMetric resource | DeviceMetric.parent
source | reference | DeviceMetric-source | The device resource | DeviceMetric.source

type | token | DeviceMetric-type | The component type | DeviceMetric.type
DeviceRequest TU
authored-on | date | DeviceRequest-authored-on | When the request transitioned to being actionable | DeviceRequest.authoredOn
based-on | reference | DeviceRequest-based-on | Plan/proposal/order fulfilled by this request | DeviceRequest.basedOn
code | token | clinical-code | Code for what is being requested/ordered | (DeviceRequest.code as CodeableConcept)
device | reference | DeviceRequest-device | Reference to resource that is being requested/ordered | (DeviceRequest.code as Reference)
encounter | reference | clinical-encounter | Encounter during which request was created | DeviceRequest.encounter
event-date | date | DeviceRequest-event-date | When service should occur | (DeviceRequest.occurrence as dateTime) | (DeviceRequest.occurrence as Period)
group-identifier | token | DeviceRequest-group-identifier | Composite request this is part of | DeviceRequest.groupIdentifier
identifier | token | clinical-identifier | Business identifier for request/order | DeviceRequest.identifier
instantiates-canonical | reference | DeviceRequest-instantiates-canonical | Instantiates FHIR protocol or definition | DeviceRequest.instantiatesCanonical
instantiates-uri | uri | DeviceRequest-instantiates-uri | Instantiates external protocol or definition | DeviceRequest.instantiatesUri
insurance | reference | DeviceRequest-insurance | Associated insurance coverage | DeviceRequest.insurance
intent | token | DeviceRequest-intent | proposal | plan | original-order |reflex-order | DeviceRequest.intent
patient | reference | clinical-patient | Individual the service is ordered for | DeviceRequest.subject.where(resolve() is Patient)
performer | reference | DeviceRequest-performer | Desired performer for service | DeviceRequest.performer
prior-request | reference | DeviceRequest-prior-request | Request takes the place of referenced completed or terminated requests | DeviceRequest.priorRequest
requester | reference | DeviceRequest-requester | Who/what is requesting service | DeviceRequest.requester
status | token | DeviceRequest-status | entered-in-error | draft | active |suspended | completed | DeviceRequest.status
subject | reference | DeviceRequest-subject | Individual the service is ordered for | DeviceRequest.subject
DeviceUseStatement TU
device | reference | DeviceUseStatement-device | Search by device | DeviceUseStatement.device
identifier | token | DeviceUseStatement-identifier | Search by identifier | DeviceUseStatement.identifier
patient | reference | clinical-patient | Search by subject - a patient | DeviceUseStatement.subject
subject | reference | DeviceUseStatement-subject | Search by subject | DeviceUseStatement.subject
DiagnosticReport TU
based-on | reference | DiagnosticReport-based-on | Reference to the service request. | DiagnosticReport.basedOn
category | token | DiagnosticReport-category | Which diagnostic discipline/department created the report | DiagnosticReport.category
code | token | clinical-code | The code for the report, as opposed to codes for the atomic results, which are the names on the observation resource referred to from the result | DiagnosticReport.code
conclusion | token | DiagnosticReport-conclusion | A coded conclusion (interpretation/impression) on the report | DiagnosticReport.conclusionCode
date | date | clinical-date | The clinically relevant time of the report | DiagnosticReport.effective
encounter | reference | clinical-encounter | The Encounter when the order was made | DiagnosticReport.encounter
identifier | token | clinical-identifier | An identifier for the report | DiagnosticReport.identifier
issued | date | DiagnosticReport-issued | When the report was issued | DiagnosticReport.issued
media | reference | DiagnosticReport-media | A reference to the image source. | DiagnosticReport.media.link
patient | reference | clinical-patient | The subject of the report if a patient | DiagnosticReport.subject.where(resolve() is Patient)
performer | reference | DiagnosticReport-performer | Who is responsible for the report | DiagnosticReport.performer
result | reference | DiagnosticReport-result | Link to an atomic result (observation resource) | DiagnosticReport.result
results-interpreter | reference | DiagnosticReport-results-interpreter | Who was the source of the report | DiagnosticReport.resultsInterpreter
specimen | reference | DiagnosticReport-specimen | The specimen details | DiagnosticReport.specimen
status | token | DiagnosticReport-status | The status of the report | DiagnosticReport.status
subject | reference | DiagnosticReport-subject | The subject of the report | DiagnosticReport.subject
DocumentManifest TU
author | reference | DocumentManifest-author | Who and/or what authored the DocumentManifest | DocumentManifest.author
created | date | DocumentManifest-created | When this document manifest created | DocumentManifest.created
description | string | DocumentManifest-description | Human-readable description (title) | DocumentManifest.description
identifier | token | clinical-identifier | Unique Identifier for the set of documents | DocumentManifest.masterIdentifier | DocumentManifest.identifier
item | reference | DocumentManifest-item | Items in manifest | DocumentManifest.content
patient | reference | clinical-patient | The subject of the set of documents | DocumentManifest.subject.where(resolve() is Patient)
recipient | reference | DocumentManifest-recipient | Intended to get notified about this set of documents | DocumentManifest.recipient
related-id | token | DocumentManifest-related-id | Identifiers of things that are related | DocumentManifest.related.identifier
related-ref | reference | DocumentManifest-related-ref | Related Resource | DocumentManifest.related.ref
source | uri | DocumentManifest-source | The source system/application/software | DocumentManifest.source
status | token | DocumentManifest-status | current | superseded | entered-in-error | DocumentManifest.status
subject | reference | DocumentManifest-subject | The subject of the set of documents | DocumentManifest.subject
type | token | clinical-type | Kind of document set | DocumentManifest.type
DocumentReference TU
authenticator | reference | DocumentReference-authenticator | Who/what authenticated the document | DocumentReference.authenticator
author | reference | DocumentReference-author | Who and/or what authored the document | DocumentReference.author
category | token | DocumentReference-category | Categorization of document | DocumentReference.category
contenttype | token | DocumentReference-contenttype | Mime type of the content, with charset etc. | DocumentReference.content.attachment.contentType

custodian | reference | DocumentReference-custodian | Organization which maintains the document | DocumentReference.custodian
date | date | DocumentReference-date | When this document reference was created | DocumentReference.date
description | string | DocumentReference-description | Human-readable description | DocumentReference.description
encounter | reference | clinical-encounter | Context of the document content | DocumentReference.context.encounter
event | token | DocumentReference-event | Main clinical acts documented | DocumentReference.context.event
facility | token | DocumentReference-facility | Kind of facility where patient was seen | DocumentReference.context.facilityType
format | token | DocumentReference-format | Format/content rules for the document | DocumentReference.content.format
identifier | token | clinical-identifier | Master Version Specific Identifier | DocumentReference.masterIdentifier | DocumentReference.identifier
language | token | DocumentReference-language | Human language of the content (BCP-47) | DocumentReference.content.attachment.language
location | uri | DocumentReference-location | Uri where the data can be found | DocumentReference.content.attachment.url
patient | reference | clinical-patient | Who/what is the subject of the document | DocumentReference.subject.where(resolve() is Patient)
period | date | DocumentReference-period | Time of service that is being documented | DocumentReference.context.period
related | reference | DocumentReference-related | Related identifiers or resources | DocumentReference.context.related
relatesto | reference | DocumentReference-relatesto | Target of the relationship | DocumentReference.relatesTo.target
relation | token | DocumentReference-relation | replaces | transforms | signs | appends | DocumentReference.relatesTo.code
relationship | composite | DocumentReference-relationship | Combination of relation and relatesTo | DocumentReference.relatesTo
security-label | token | DocumentReference-security-label | Document security-tags | DocumentReference.securityLabel
setting | token | DocumentReference-setting | Additional details about where the content was created (e.g. clinical specialty) | DocumentReference.context.practiceSetting
status | token | DocumentReference-status | current | superseded | entered-in-error | DocumentReference.status
subject | reference | DocumentReference-subject | Who/what is the subject of the document | DocumentReference.subject
type | token | clinical-type | Kind of document (LOINC if possible) | DocumentReference.type

EffectEvidenceSynthesis TU

context | token | EffectEvidenceSynthesis-context | A use context assigned to the effect evidence synthesis | (EffectEvidenceSynthesis.useContext.value as CodeableConcept)
context-quantity | quantity | EffectEvidenceSynthesis-context-quantity | A quantity- or range-valued use context assigned to the effect evidence synthesis | (EffectEvidenceSynthesis.useContext.value as Quantity) | (EffectEvidenceSynthesis.useContext.value as Range)
context-type | token | EffectEvidenceSynthesis-context-type | A type of use context assigned to the effect evidence synthesis | EffectEvidenceSynthesis.useContext.code
context-type-quantity | composite | EffectEvidenceSynthesis-context-type-quantity | A use context type and quantity- or range-based value assigned to the effect evidence synthesis | EffectEvidenceSynthesis.useContext
context-type-value | composite | EffectEvidenceSynthesis-context-type-value | A use context type and value assigned to the effect evidence synthesis | EffectEvidenceSynthesis.useContext
date | date | EffectEvidenceSynthesis-date | The effect evidence synthesis publication date | EffectEvidenceSynthesis.date
description | string | EffectEvidenceSynthesis-description | The description of the effect evidence synthesis | EffectEvidenceSynthesis.description
effective | date | EffectEvidenceSynthesis-effective | The time during which the effect evidence synthesis is intended to be in use | EffectEvidenceSynthesis.effectivePeriod
identifier | token | EffectEvidenceSynthesis-identifier | External identifier for the effect evidence synthesis | EffectEvidenceSynthesis.identifier
jurisdiction | token | EffectEvidenceSynthesis-jurisdiction | Intended jurisdiction for the effect evidence synthesis | EffectEvidenceSynthesis.jurisdiction
name | string | EffectEvidenceSynthesis-name | Computationally friendly name of the effect evidence synthesis | EffectEvidenceSynthesis.name
publisher | string | EffectEvidenceSynthesis-publisher | Name of the publisher of the effect evidence synthesis | EffectEvidenceSynthesis.publisher
status | token | EffectEvidenceSynthesis-status | The current status of the effect evidence synthesis | EffectEvidenceSynthesis.status
title | string | EffectEvidenceSynthesis-title | The human-friendly name of the effect evidence synthesis | EffectEvidenceSynthesis.title
url | uri | EffectEvidenceSynthesis-url | The uri that identifies the effect evidence synthesis | EffectEvidenceSynthesis.url
version | token | EffectEvidenceSynthesis-version | The business version of the effect evidence synthesis | EffectEvidenceSynthesis.version

Encounter TU

account | reference | Encounter-account | The set of accounts that may be used for billing for this Encounter | Encounter.account
appointment | reference | Encounter-appointment | The appointment that scheduled this encounter | Encounter.appointment
based-on | reference | Encounter-based-on | The ServiceRequest that initiated this encounter | Encounter.basedOn
class | token | Encounter-class | Classification of patient encounter | Encounter.class
date | date | clinical-date | A date within the period the Encounter lasted | Encounter.period
diagnosis | reference | Encounter-diagnosis | The diagnosis or procedure relevant to the encounter | Encounter.diagnosis.condition
episode-of-care | reference | Encounter-episode-of-care | Episode(s) of care that this encounter should be recorded against | Encounter.episodeOfCare
identifier | token | clinical-identifier | Identifier(s) by which this encounter is known | Encounter.identifier
length | quantity | Encounter-length | Length of encounter in days | Encounter.length
location | reference | Encounter-location | Location the encounter takes place | Encounter.location.location
location-period | date | Encounter-location-period | Time period during which the patient was present at the location | Encounter.location.period
part-of | reference | Encounter-part-of | Another Encounter this encounter is part of | Encounter.partOf

participant | reference | Encounter-participant | Persons involved in the encounter other than the patient | Encounter.participant.individual
participant-type | token | Encounter-participant-type | Role of participant in encounter | Encounter.participant.type
patient | reference | clinical-patient | The patient or group present at the encounter | Encounter.subject.where(resolve() is Patient)
practitioner | reference | Encounter-practitioner | Persons involved in the encounter other than the patient | Encounter.participant.individual.where(resolve() is Practitioner)
reason-code | token | Encounter-reason-code | Coded reason the encounter takes place | Encounter.reasonCode
reason-reference | reference | Encounter-reason-reference | Reason the encounter takes place (reference) | Encounter.reasonReference
service-provider | reference | Encounter-service-provider | The organization (facility) responsible for this encounter | Encounter.serviceProvider
special-arrangement | token | Encounter-special-arrangement | Wheelchair, translator, stretcher, etc. | Encounter.hospitalization.specialArrangement
status | token | Encounter-status | planned | arrived | triaged | in-progress | onleave | finished | cancelled + | Encounter.status
subject | reference | Encounter-subject | The patient or group present at the encounter | Encounter.subject
type | token | clinical-type | Specific type of encounter | Encounter.type
Endpoint TU
connection-type | token | Endpoint-connection-type | Protocol/Profile/Standard to be used with this endpoint connection | Endpoint.connectionType
identifier | token | Endpoint-identifier | Identifies this endpoint across multiple systems | Endpoint.identifier
name | string | Endpoint-name | A name that this endpoint can be identified by | Endpoint.name
organization | reference | Endpoint-organization | The organization that is managing the endpoint | Endpoint.managingOrganization
payload-type | token | Endpoint-payload-type | The type of content that may be used at this endpoint (e.g. XDS Discharge summaries) | Endpoint.payloadType
status | token | Endpoint-status | The current status of the Endpoint (usually expected to be active) | Endpoint.status
EnrollmentRequest TU
identifier | token | EnrollmentRequest-identifier | The business identifier of the Enrollment | EnrollmentRequest.identifier
patient | reference | EnrollmentRequest-patient | The party to be enrolled | EnrollmentRequest.candidate
status | token | EnrollmentRequest-status | The status of the enrollment | EnrollmentRequest.status
subject | reference | EnrollmentRequest-subject | The party to be enrolled | EnrollmentRequest.candidate
EnrollmentResponse TU
identifier | token | EnrollmentResponse-identifier | The business identifier of the EnrollmentResponse | EnrollmentResponse.identifier
request | reference | EnrollmentResponse-request | The reference to the claim | EnrollmentResponse.request
status | token | EnrollmentResponse-status | The status of the enrollment response | EnrollmentResponse.status
EpisodeOfCare TU
care-manager | reference | EpisodeOfCare-care-manager | Care manager/care coordinator for the patient | EpisodeOfCare.careManager.where(resolve() is Practitioner)
condition | reference | EpisodeOfCare-condition | Conditions/problems/diagnoses this episode of care is for | EpisodeOfCare.diagnosis.condition
date | date | clinical-date | The provided date search value falls within the episode of care's period | EpisodeOfCare.period
identifier | token | clinical-identifier | Business Identifier(s) relevant for this EpisodeOfCare | EpisodeOfCare.identifier
incoming-referral | reference | EpisodeOfCare-incoming-referral | Incoming Referral Request | EpisodeOfCare.referralRequest
organization | reference | EpisodeOfCare-organization | The organization that has assumed the specific responsibilities of this EpisodeOfCare | EpisodeOfCare.managingOrganization
patient | reference | clinical-patient | The patient who is the focus of this episode of care | EpisodeOfCare.patient
status | token | EpisodeOfCare-status | The current status of the Episode of Care as provided (does not check the status history collection) | EpisodeOfCare.status
type | token | clinical-type | Type/class - e.g. specialist referral, disease management | EpisodeOfCare.type
EventDefinition TU
composed-of | reference | EventDefinition-composed-of | What resource is being referenced | EventDefinition.relatedArtifact.where(type='composed-of').resource
context | token | EventDefinition-context | A use context assigned to the event definition | (EventDefinition.useContext.value as CodeableConcept)
context-quantity | quantity | EventDefinition-context-quantity | A quantity- or range-valued use context assigned to the event definition | (EventDefinition.useContext.value as Quantity) | (EventDefinition.useContext.value as Range)
context-type | token | EventDefinition-context-type | A type of use context assigned to the event definition | EventDefinition.useContext.code
context-type-quantity | composite | EventDefinition-context-type-quantity | A use context type and quantity- or range-based value assigned to the event definition | EventDefinition.useContext
context-type-value | composite | EventDefinition-context-type-value | A use context type and value assigned to the event definition | EventDefinition.useContext
date | date | EventDefinition-date | The event definition publication date | EventDefinition.date
depends-on | reference | EventDefinition-depends-on | What resource is being referenced | EventDefinition.relatedArtifact.where(type='depends-on').resource
derived-from | reference | EventDefinition-derived-from | What resource is being referenced | EventDefinition.relatedArtifact.where(type='derived-from').resource
description | string | EventDefinition-description | The description of the event definition | EventDefinition.description
effective | date | EventDefinition-effective | The time during which the event definition is intended to be in use | EventDefinition.effectivePeriod
identifier | token | EventDefinition-identifier | External identifier for the event definition | EventDefinition.identifier
jurisdiction | token | EventDefinition-jurisdiction | Intended jurisdiction for the event definition | EventDefinition.jurisdiction
name | string | EventDefinition-name | Computationally friendly name of the event definition | EventDefinition.name

predecessor | reference | EventDefinition-predecessor | What resource is being referenced | EventDefinition.relatedArtifact.where(type='predecessor').resource
publisher | string | EventDefinition-publisher | Name of the publisher of the event definition | EventDefinition.publisher
status | token | EventDefinition-status | The current status of the event definition | EventDefinition.status
successor | reference | EventDefinition-successor | What resource is being referenced | EventDefinition.relatedArtifact.where(type='successor').resource
title | string | EventDefinition-title | The human-friendly name of the event definition | EventDefinition.title
topic | token | EventDefinition-topic | Topics associated with the module | EventDefinition.topic
url | uri | EventDefinition-url | The uri that identifies the event definition | EventDefinition.url
version | token | EventDefinition-version | The business version of the event definition | EventDefinition.version
Evidence TU
composed-of | reference | Evidence-composed-of | What resource is being referenced | Evidence.relatedArtifact.where(type='composed-of').resource
context | token | Evidence-context | A use context assigned to the evidence | (Evidence.useContext.value as CodeableConcept)
context-quantity | quantity | Evidence-context-quantity | A quantity- or range-valued use context assigned to the evidence | (Evidence.useContext.value as Quantity) | (Evidence.useContext.value as Range)
context-type | token | Evidence-context-type | A type of use context assigned to the evidence | Evidence.useContext.code
context-type-quantity | composite | Evidence-context-type-quantity | A use context type and quantity- or range-based value assigned to the evidence | Evidence.useContext
context-type-value | composite | Evidence-context-type-value | A use context type and value assigned to the evidence | Evidence.useContext
date | date | Evidence-date | The evidence publication date | Evidence.date
depends-on | reference | Evidence-depends-on | What resource is being referenced | Evidence.relatedArtifact.where(type='depends-on').resource
derived-from | reference | Evidence-derived-from | What resource is being referenced | Evidence.relatedArtifact.where(type='derived-from').resource
description | string | Evidence-description | The description of the evidence | Evidence.description
effective | date | Evidence-effective | The time during which the evidence is intended to be in use | Evidence.effectivePeriod
identifier | token | Evidence-identifier | External identifier for the evidence | Evidence.identifier
jurisdiction | token | Evidence-jurisdiction | Intended jurisdiction for the evidence | Evidence.jurisdiction
name | string | Evidence-name | Computationally friendly name of the evidence | Evidence.name
predecessor | reference | Evidence-predecessor | What resource is being referenced | Evidence.relatedArtifact.where(type='predecessor').resource
publisher | string | Evidence-publisher | Name of the publisher of the evidence | Evidence.publisher
status | token | Evidence-status | The current status of the evidence | Evidence.status
successor | reference | Evidence-successor | What resource is being referenced | Evidence.relatedArtifact.where(type='successor').resource
title | string | Evidence-title | The human-friendly name of the evidence | Evidence.title
topic | token | Evidence-topic | Topics associated with the Evidence | Evidence.topic
url | uri | Evidence-url | The uri that identifies the evidence | Evidence.url
version | token | Evidence-version | The business version of the evidence | Evidence.version
EvidenceVariable TU
composed-of | reference | EvidenceVariable-composed-of | What resource is being referenced | EvidenceVariable.relatedArtifact.where(type='composed-of').resource
context | token | EvidenceVariable-context | A use context assigned to the evidence variable | (EvidenceVariable.useContext.value as CodeableConcept)
context-quantity | quantity | EvidenceVariable-context-quantity | A quantity- or range-valued use context assigned to the evidence variable | (EvidenceVariable.useContext.value as Quantity) | (EvidenceVariable.useContext.value as Range)
context-type | token | EvidenceVariable-context-type | A type of use context assigned to the evidence variable | EvidenceVariable.useContext.code
context-type-quantity | composite | EvidenceVariable-context-type-quantity | A use context type and quantity- or range-based value assigned to the evidence variable | EvidenceVariable.useContext
context-type-value | composite | EvidenceVariable-context-type-value | A use context type and value assigned to the evidence variable | EvidenceVariable.useContext
date | date | EvidenceVariable-date | The evidence variable publication date | EvidenceVariable.date
depends-on | reference | EvidenceVariable-depends-on | What resource is being referenced | EvidenceVariable.relatedArtifact.where(type='depends-on').resource
derived-from | reference | EvidenceVariable-derived-from | What resource is being referenced | EvidenceVariable.relatedArtifact.where(type='derived-from').resource
description | string | EvidenceVariable-description | The description of the evidence variable | EvidenceVariable.description
effective | date | EvidenceVariable-effective | The time during which the evidence variable is intended to be in use | EvidenceVariable.effectivePeriod
identifier | token | EvidenceVariable-identifier | External identifier for the evidence variable | EvidenceVariable.identifier
jurisdiction | token | EvidenceVariable-jurisdiction | Intended jurisdiction for the evidence variable | EvidenceVariable.jurisdiction
name | string | EvidenceVariable-name | Computationally friendly name of the evidence variable | EvidenceVariable.name
predecessor | reference | EvidenceVariable-predecessor | What resource is being referenced | EvidenceVariable.relatedArtifact.where(type='predecessor').resource
publisher | string | EvidenceVariable-publisher | Name of the publisher of the evidence variable | EvidenceVariable.publisher
status | token | EvidenceVariable-status | The current status of the evidence variable | EvidenceVariable.status
successor | reference | EvidenceVariable-successor | What resource is being referenced | EvidenceVariable.relatedArtifact.where(type='successor').resource
title | string | EvidenceVariable-title | The human-friendly name of the evidence variable | EvidenceVariable.title
topic | token | EvidenceVariable-topic | Topics associated with the EvidenceVariable | EvidenceVariable.topic
url | uri | EvidenceVariable-url | The uri that identifies the evidence variable | EvidenceVariable.url

version | token | EvidenceVariable-version | The business version of the evidence variable | EvidenceVariable.version
ExampleScenario TU
context | token | ExampleScenario-context | A use context assigned to the example scenario | (ExampleScenario.useContext.value as CodeableConcept)
context-quantity | quantity | ExampleScenario-context-quantity | A quantity- or range-valued use context assigned to the example scenario | (ExampleScenario.useContext.value as Quantity) | (ExampleScenario.useContext.value as Range)
context-type | token | ExampleScenario-context-type | A type of use context assigned to the example scenario | ExampleScenario.useContext.code
context-type-quantity | composite | ExampleScenario-context-type-quantity | A use context type and quantity- or range-based value assigned to the example scenario | ExampleScenario.useContext
context-type-value | composite | ExampleScenario-context-type-value | A use context type and value assigned to the example scenario | ExampleScenario.useContext
date | date | ExampleScenario-date | The example scenario publication date | ExampleScenario.date
identifier | token | ExampleScenario-identifier | External identifier for the example scenario | ExampleScenario.identifier
jurisdiction | token | ExampleScenario-jurisdiction | Intended jurisdiction for the example scenario | ExampleScenario.jurisdiction
name | string | ExampleScenario-name | Computationally friendly name of the example scenario | ExampleScenario.name
publisher | string | ExampleScenario-publisher | Name of the publisher of the example scenario | ExampleScenario.publisher
status | token | ExampleScenario-status | The current status of the example scenario | ExampleScenario.status
url | uri | ExampleScenario-url | The uri that identifies the example scenario | ExampleScenario.url
version | token | ExampleScenario-version | The business version of the example scenario | ExampleScenario.version
ExplanationOfBenefit TU
care-team | reference | ExplanationOfBenefit-care-team | Member of the CareTeam | ExplanationOfBenefit.careTeam.provider
claim | reference | ExplanationOfBenefit-claim | The reference to the claim | ExplanationOfBenefit.claim
coverage | reference | ExplanationOfBenefit-coverage | The plan under which the claim was adjudicated | ExplanationOfBenefit.insurance.coverage
created | date | ExplanationOfBenefit-created | The creation date for the EOB | ExplanationOfBenefit.created
detail-udi | reference | ExplanationOfBenefit-detail-udi | UDI associated with a line item detail product or service | ExplanationOfBenefit.item.detail.udi
disposition | string | ExplanationOfBenefit-disposition | The contents of the disposition message | ExplanationOfBenefit.disposition
encounter | reference | ExplanationOfBenefit-encounter | Encounters associated with a billed line item | ExplanationOfBenefit.item.encounter
enterer | reference | ExplanationOfBenefit-enterer | The party responsible for the entry of the Claim | ExplanationOfBenefit.enterer
facility | reference | ExplanationOfBenefit-facility | Facility responsible for the goods and services | ExplanationOfBenefit.facility
identifier | token | ExplanationOfBenefit-identifier | The business identifier of the Explanation of Benefit | ExplanationOfBenefit.identifier
item-udi | reference | ExplanationOfBenefit-item-udi | UDI associated with a line item product or service | ExplanationOfBenefit.item.udi
patient | reference | ExplanationOfBenefit-patient | The reference to the patient | ExplanationOfBenefit.patient
payee | reference | ExplanationOfBenefit-payee | The party receiving any payment for the Claim | ExplanationOfBenefit.payee.party
procedure-udi | reference | ExplanationOfBenefit-procedure-udi | UDI associated with a procedure | ExplanationOfBenefit.procedure.udi
provider | reference | ExplanationOfBenefit-provider | The reference to the provider | ExplanationOfBenefit.provider
status | token | ExplanationOfBenefit-status | Status of the instance | ExplanationOfBenefit.status
subdetail-udi | reference | ExplanationOfBenefit-subdetail-udi | UDI associated with a line item detail subdetail product or service | ExplanationOfBenefit.item.detail.subDetail.udi
FamilyMemberHistory TU
code | token | clinical-code | A search by a condition code | FamilyMemberHistory.condition.code
date | date | clinical-date | When history was recorded or last updated | FamilyMemberHistory.date
identifier | token | clinical-identifier | A search by a record identifier | FamilyMemberHistory.identifier
instantiates-canonical | reference | FamilyMemberHistory-instantiates-canonical | Instantiates FHIR protocol or definition | FamilyMemberHistory.instantiatesCanonical
instantiates-uri | uri | FamilyMemberHistory-instantiates-uri | Instantiates external protocol or definition | FamilyMemberHistory.instantiatesUri
patient | reference | clinical-patient | The identity of a subject to list family member history items for | FamilyMemberHistory.patient
relationship | token | FamilyMemberHistory-relationship | A search by a relationship type | FamilyMemberHistory.relationship
sex | token | FamilyMemberHistory-sex | A search by a sex code of a family member | FamilyMemberHistory.sex
status | token | FamilyMemberHistory-status | partial | completed | entered-in-error | health-unknown | FamilyMemberHistory.status
Flag TU
author | reference | Flag-author | Flag creator | Flag.author
date | date | clinical-date | Time period when flag is active | Flag.period
encounter | reference | clinical-encounter | Alert relevant during encounter | Flag.encounter
identifier | token | Flag-identifier | Business identifier | Flag.identifier
patient | reference | clinical-patient | The identity of a subject to list flags for | Flag.subject.where(resolve() is Patient)
subject | reference | Flag-subject | The identity of a subject to list flags for | Flag.subject
Goal TU
achievement-status | token | Goal-achievement-status | in-progress | improving | worsening | no-change | achieved | sustaining | not-achieved | no-progress | not-attainable | Goal.achievementStatus
category | token | Goal-category | E.g. Treatment, dietary, behavioral, etc. | Goal.category
identifier | token | clinical-identifier | External Ids for this goal | Goal.identifier
lifecycle-status | token | Goal-lifecycle-status | proposed | planned | accepted | active | on-hold | completed | cancelled | entered-in-error | rejected | Goal.lifecycleStatus

patient | reference | clinical-patient | Who this goal is intended for | Goal.subject.where(resolve() is Patient)
start-date | date | Goal-start-date | When goal pursuit begins | (Goal.start as date)
subject | reference | Goal-subject | Who this goal is intended for | Goal.subject
target-date | date | Goal-target-date | Reach goal on or before | (Goal.target.due as date)
GraphDefinition TU
context | token | conformance-context | A use context assigned to the graph definition | (GraphDefinition.useContext.value as CodeableConcept)
context-quantity | quantity | conformance-context-quantity | A quantity- or range-valued use context assigned to the graph definition | (GraphDefinition.useContext.value as Quantity) | (GraphDefinition.useContext.value as Range)
context-type | token | conformance-context-type | A type of use context assigned to the graph definition | GraphDefinition.useContext.code
context-type-quantity | composite | conformance-context-type-quantity | A use context type and quantity- or range-based value assigned to the graph definition | GraphDefinition.useContext
context-type-value | composite | conformance-context-type-value | A use context type and value assigned to the graph definition | GraphDefinition.useContext
date | date | conformance-date | The graph definition publication date | GraphDefinition.date
description | string | conformance-description | The description of the graph definition | GraphDefinition.description
jurisdiction | token | conformance-jurisdiction | Intended jurisdiction for the graph definition | GraphDefinition.jurisdiction
name | string | conformance-name | Computationally friendly name of the graph definition | GraphDefinition.name
publisher | string | conformance-publisher | Name of the publisher of the graph definition | GraphDefinition.publisher
start | token | GraphDefinition-start | Type of resource at which the graph starts | GraphDefinition.start
status | token | conformance-status | The current status of the graph definition | GraphDefinition.status
url | uri | conformance-url | The uri that identifies the graph definition | GraphDefinition.url
version | token | conformance-version | The business version of the graph definition | GraphDefinition.version
Group TU
actual | token | Group-actual | Descriptive or actual | Group.actual
characteristic | token | Group-characteristic | Kind of characteristic | Group.characteristic.code
characteristic-value | composite | Group-characteristic-value | A composite of both characteristic and value | Group.characteristic
code | token | Group-code | The kind of resources contained | Group.code
exclude | token | Group-exclude | Group includes or excludes | Group.characteristic.exclude
identifier | token | Group-identifier | Unique id | Group.identifier
managing-entity | reference | Group-managing-entity | Entity that is the custodian of the Group's definition | Group.managingEntity
member | reference | Group-member | Reference to the group member | Group.member.entity
type | token | Group-type | The type of resources the group contains | Group.type
value | token | Group-value | Value held by characteristic | (Group.characteristic.value as CodeableConcept) | (Group.characteristic.value as boolean)
GuidanceResponse TU
identifier | token | GuidanceResponse-identifier | The identifier of the guidance response | GuidanceResponse.identifier
patient | reference | GuidanceResponse-patient | The identity of a patient to search for guidance response results | GuidanceResponse.subject.where(resolve() is Patient)
request | token | GuidanceResponse-request | The identifier of the request associated with the response | GuidanceResponse.requestIdentifier
subject | reference | GuidanceResponse-subject | The subject that the guidance response is about | GuidanceResponse.subject
HealthcareService TU
active | token | HealthcareService-active | The Healthcare Service is currently marked as active | HealthcareService.active
characteristic | token | HealthcareService-characteristic | One of the HealthcareService's characteristics | HealthcareService.characteristic
coverage-area | reference | HealthcareService-coverage-area | Location(s) service is intended for/available to | HealthcareService.coverageArea
endpoint | reference | HealthcareService-endpoint | Technical endpoints providing access to electronic services operated for the healthcare service | HealthcareService.endpoint
identifier | token | HealthcareService-identifier | External identifiers for this item | HealthcareService.identifier
location | reference | HealthcareService-location | The location of the Healthcare Service | HealthcareService.location
name | string | HealthcareService-name | A portion of the Healthcare service name | HealthcareService.name
organization | reference | HealthcareService-organization | The organization that provides this Healthcare Service | HealthcareService.providedBy
program | token | HealthcareService-program | One of the Programs supported by this HealthcareService | HealthcareService.program
service-category | token | HealthcareService-service-category | Service Category of the Healthcare Service | HealthcareService.category
service-type | token | HealthcareService-service-type | The type of service provided by this healthcare service | HealthcareService.type
specialty | token | HealthcareService-specialty | The specialty of the service provided by this healthcare service | HealthcareService.specialty
ImagingStudy TU
basedon | reference | ImagingStudy-basedon | The order for the image | ImagingStudy.basedOn
bodysite | token | ImagingStudy-bodysite | The body site studied | ImagingStudy.series.bodySite
dicom-class | token | ImagingStudy-dicom-class | The type of the instance | ImagingStudy.series.instance.sopClass
encounter | reference | ImagingStudy-encounter | The context of the study | ImagingStudy.encounter
endpoint | reference | ImagingStudy-endpoint | The endpoint for the study or series | ImagingStudy.endpoint | ImagingStudy.series.endpoint
identifier | token | clinical-identifier | Identifiers for the Study, such as DICOM Study Instance UID and Accession number | ImagingStudy.identifier

instance | token | ImagingStudy-instance | SOP Instance UID for an instance | ImagingStudy.series.instance.uid
interpreter | reference | ImagingStudy-interpreter | Who interpreted the images | ImagingStudy.interpreter
modality | token | ImagingStudy-modality | The modality of the series | ImagingStudy.series.modality
patient | reference | clinical-patient | Who the study is about | ImagingStudy.subject.where(resolve() is Patient)
performer | reference | ImagingStudy-performer | The person who performed the study | ImagingStudy.series.performer.actor
reason | token | ImagingStudy-reason | The reason for the study | ImagingStudy.reasonCode
referrer | reference | ImagingStudy-referrer | The referring physician | ImagingStudy.referrer
series | token | ImagingStudy-series | DICOM Series Instance UID for a series | ImagingStudy.series.uid
started | date | ImagingStudy-started | When the study was started | ImagingStudy.started
status | token | ImagingStudy-status | The status of the study | ImagingStudy.status
subject | reference | ImagingStudy-subject | Who the study is about | ImagingStudy.subject
Immunization TU
date | date | clinical-date | Vaccination (non)-Administration Date | Immunization.occurrence
identifier | token | clinical-identifier | Business identifier | Immunization.identifier
location | reference | Immunization-location | The service delivery location or facility in which the vaccine was / was to be administered | Immunization.location
lot-number | string | Immunization-lot-number | Vaccine Lot Number | Immunization.lotNumber
manufacturer | reference | Immunization-manufacturer | Vaccine Manufacturer | Immunization.manufacturer
patient | reference | clinical-patient | The patient for the vaccination record | Immunization.patient
performer | reference | Immunization-performer | The practitioner or organization who played a role in the vaccination | Immunization.performer.actor
reaction | reference | Immunization-reaction | Additional information on reaction | Immunization.reaction.detail
reaction-date | date | Immunization-reaction-date | When reaction started | Immunization.reaction.date
reason-code | token | Immunization-reason-code | Reason why the vaccine was administered | Immunization.reasonCode
reason-reference | reference | Immunization-reason-reference | Why immunization occurred | Immunization.reasonReference
series | string | Immunization-series | The series being followed by the provider | Immunization.protocolApplied.series
status | token | Immunization-status | Immunization event status | Immunization.status
status-reason | token | Immunization-status-reason | Reason why the vaccine was not administered | Immunization.statusReason
target-disease | token | Immunization-target-disease | The target disease the dose is being administered against | Immunization.protocolApplied.targetDisease
vaccine-code | token | Immunization-vaccine-code | Vaccine Product Administered | Immunization.vaccineCode
ImmunizationEvaluation TU
date | date | ImmunizationEvaluation-date | Date the evaluation was generated | ImmunizationEvaluation.date
dose-status | token | ImmunizationEvaluation-dose-status | The status of the dose relative to published recommendations | ImmunizationEvaluation.doseStatus
identifier | token | ImmunizationEvaluation-identifier | ID of the evaluation | ImmunizationEvaluation.identifier
immunization-event | reference | ImmunizationEvaluation-immunization-event | The vaccine administration event being evaluated | ImmunizationEvaluation.immunizationEvent
patient | reference | ImmunizationEvaluation-patient | The patient being evaluated | ImmunizationEvaluation.patient
status | token | ImmunizationEvaluation-status | Immunization evaluation status | ImmunizationEvaluation.status
target-disease | token | ImmunizationEvaluation-target-disease | The vaccine preventable disease being evaluated against | ImmunizationEvaluation.targetDisease
ImmunizationRecommendation TU
date | date | ImmunizationRecommendation-date | Date recommendation(s) created | ImmunizationRecommendation.date
identifier | token | ImmunizationRecommendation-identifier | Business identifier | ImmunizationRecommendation.identifier
information | reference | ImmunizationRecommendation-information | Patient observations supporting recommendation | ImmunizationRecommendation.recommendation.supportingPatientInformation
patient | reference | ImmunizationRecommendation-patient | Who this profile is for | ImmunizationRecommendation.patient
status | token | ImmunizationRecommendation-status | Vaccine recommendation status | ImmunizationRecommendation.recommendation.forecastStatus
support | reference | ImmunizationRecommendation-support | Past immunizations supporting recommendation | ImmunizationRecommendation.recommendation.supportingImmunization
target-disease | token | ImmunizationRecommendation-target-disease | Disease to be immunized against | ImmunizationRecommendation.recommendation.targetDisease
vaccine-type | token | ImmunizationRecommendation-vaccine-type | Vaccine or vaccine group recommendation applies to | ImmunizationRecommendation.recommendation.vaccineCode
ImplementationGuide TU
context | token | conformance-context | A use context assigned to the implementation guide | (ImplementationGuide.useContext.value as CodeableConcept)
context-quantity | quantity | conformance-context-quantity | A quantity- or range-valued use context assigned to the implementation guide | (ImplementationGuide.useContext.value as Quantity) | (ImplementationGuide.useContext.value as Range)
context-type | token | conformance-context-type | A type of use context assigned to the implementation guide | ImplementationGuide.useContext.code
context-type-quantity | composite | conformance-context-type-quantity | A use context type and quantity- or range-based value assigned to the implementation guide | ImplementationGuide.useContext
context-type-value | composite | conformance-context-type-value | A use context type and value assigned to the implementation guide | ImplementationGuide.useContext
date | date | conformance-date | The implementation guide publication date | ImplementationGuide.date
depends-on | reference | ImplementationGuide-depends-on | Identity of the IG that this depends on | ImplementationGuide.dependsOn.uri
description | string | conformance-description | The description of the implementation guide | ImplementationGuide.description
experimental | token | ImplementationGuide-experimental | For testing purposes, not real usage | ImplementationGuide.experimental
global | reference | ImplementationGuide-global | Profile that all resources must conform to | ImplementationGuide.global.profile

jurisdiction | token | conformance-jurisdiction | Intended jurisdiction for the implementation guide | ImplementationGuide.jurisdiction
name | string | conformance-name | Computationally friendly name of the implementation guide | ImplementationGuide.name
publisher | string | conformance-publisher | Name of the publisher of the implementation guide | ImplementationGuide.publisher
resource | reference | ImplementationGuide-resource | Location of the resource | ImplementationGuide.definition.resource.reference
status | token | conformance-status | The current status of the implementation guide | ImplementationGuide.status
title | string | conformance-title | The human-friendly name of the implementation guide | ImplementationGuide.title
url | uri | conformance-url | The uri that identifies the implementation guide | ImplementationGuide.url
version | token | conformance-version | The business version of the implementation guide | ImplementationGuide.version
InsurancePlan TU
address | string | InsurancePlan-address | A server defined search that may match any of the string fields in the Address, including line, city, district, state, country, postalCode, and/or text | InsurancePlan.contact.address
address-city | string | InsurancePlan-address-city | A city specified in an address | InsurancePlan.contact.address.city
address-country | string | InsurancePlan-address-country | A country specified in an address | InsurancePlan.contact.address.country
address-postalcode | string | InsurancePlan-address-postalcode | A postal code specified in an address | InsurancePlan.contact.address.postalCode
address-state | string | InsurancePlan-address-state | A state specified in an address | InsurancePlan.contact.address.state
address-use | token | InsurancePlan-address-use | A use code specified in an address | InsurancePlan.contact.address.use
administered-by | reference | InsurancePlan-administered-by | Product administrator | InsurancePlan.administeredBy
endpoint | reference | InsurancePlan-endpoint | Technical endpoint | InsurancePlan.endpoint
identifier | token | InsurancePlan-identifier | Any identifier for the organization (not the accreditation issuer's identifier) | InsurancePlan.identifier
name | string | InsurancePlan-name | A portion of the organization's name or alias | name | alias
owned-by | reference | InsurancePlan-owned-by | An organization of which this organization forms a part | InsurancePlan.ownedBy
phonetic | string | InsurancePlan-phonetic | A portion of the organization's name using some kind of phonetic matching algorithm | InsurancePlan.name
status | token | InsurancePlan-status | Is the Organization record active | InsurancePlan.status
type | token | InsurancePlan-type | A code for the type of organization | InsurancePlan.type
Invoice TU
account | reference | Invoice-account | Account that is being balanced | Invoice.account
date | date | Invoice-date | Invoice date / posting date | Invoice.date
identifier | token | Invoice-identifier | Business Identifier for item | Invoice.identifier
issuer | reference | Invoice-issuer | Issuing Organization of Invoice | Invoice.issuer
participant | reference | Invoice-participant | Individual who was involved | Invoice.participant.actor
participant-role | token | Invoice-participant-role | Type of involvement in creation of this Invoice | Invoice.participant.role
patient | reference | Invoice-patient | Recipient(s) of goods and services | Invoice.subject.where(resolve() is Patient)
recipient | reference | Invoice-recipient | Recipient of this invoice | Invoice.recipient
status | token | Invoice-status | draft | issued | balanced | cancelled | entered-in-error | Invoice.status
subject | reference | Invoice-subject | Recipient(s) of goods and services | Invoice.subject
totalgross | quantity | Invoice-totalgross | Gross total of this Invoice | Invoice.totalGross
totalnet | quantity | Invoice-totalnet | Net total of this Invoice | Invoice.totalNet
type | token | Invoice-type | Type of Invoice | Invoice.type
Library TU
composed-of | reference | Library-composed-of | What resource is being referenced | Library.relatedArtifact.where(type='composed-of').resource
content-type | token | Library-content-type | The type of content in the library (e.g. text/cql) | Library.content.contentType
context | token | Library-context | A use context assigned to the library | (Library.useContext.value as CodeableConcept)
context-quantity | quantity | Library-context-quantity | A quantity- or range-valued use context assigned to the library | (Library.useContext.value as Quantity) | (Library.useContext.value as Range)
context-type | token | Library-context-type | A type of use context assigned to the library | Library.useContext.code
context-type-quantity | composite | Library-context-type-quantity | A use context type and quantity- or range-based value assigned to the library | Library.useContext
context-type-value | composite | Library-context-type-value | A use context type and value assigned to the library | Library.useContext
date | date | Library-date | The library publication date | Library.date
depends-on | reference | Library-depends-on | What resource is being referenced | Library.relatedArtifact.where(type='depends-on').resource
derived-from | reference | Library-derived-from | What resource is being referenced | Library.relatedArtifact.where(type='derived-from').resource
description | string | Library-description | The description of the library | Library.description
effective | date | Library-effective | The time during which the library is intended to be in use | Library.effectivePeriod
identifier | token | Library-identifier | External identifier for the library | Library.identifier
jurisdiction | token | Library-jurisdiction | Intended jurisdiction for the library | Library.jurisdiction
name | string | Library-name | Computationally friendly name of the library | Library.name
predecessor | reference | Library-predecessor | What resource is being referenced | Library.relatedArtifact.where(type='predecessor').resource
publisher | string | Library-publisher | Name of the publisher of the library | Library.publisher
status | token | Library-status | The current status of the library | Library.status
successor | reference | Library-successor | What resource is being referenced | Library.relatedArtifact.where(type='successor').resource
title | string | Library-title | The human-friendly name of the library | Library.title
topic | token | Library-topic | Topics associated with the module | Library.topic

type | token | Library-type | The type of the library (e.g. logic-library, model-definition, asset-collection, module-definition) | Library.type
url | uri | Library-url | The uri that identifies the library | Library.url
version | token | Library-version | The business version of the library | Library.version
Linkage TU
author | reference | Linkage-author | Author of the Linkage | Linkage.author
item | reference | Linkage-item | Matches on any item in the Linkage | Linkage.item.resource
source | reference | Linkage-source | Matches on any item in the Linkage with a type of 'source' | Linkage.item.resource
List TU
code | token | clinical-code | What the purpose of this list is | List.code
date | date | clinical-date | When the list was prepared | List.date
empty-reason | token | List-empty-reason | Why list is empty | List.emptyReason
encounter | reference | clinical-encounter | Context in which list created | List.encounter
identifier | token | clinical-identifier | Business identifier | List.identifier
item | reference | List-item | Actual entry | List.entry.item
notes | string | List-notes | The annotation - text content (as markdown) | List.note.text
patient | reference | clinical-patient | If all resources have the same subject | List.subject.where(resolve() is Patient)
source | reference | List-source | Who and/or what defined the list contents (aka Author) | List.source
status | token | List-status | current | retired | entered-in-error | List.status
subject | reference | List-subject | If all resources have the same subject | List.subject
title | string | List-title | Descriptive name for the list | List.title
Location TU
address | string | Location-address | A (part of the) address of the location | Location.address
address-city | string | Location-address-city | A city specified in an address | Location.address.city
address-country | string | Location-address-country | A country specified in an address | Location.address.country
address-postalcode | string | Location-address-postalcode | A postal code specified in an address | Location.address.postalCode
address-state | string | Location-address-state | A state specified in an address | Location.address.state
address-use | token | Location-address-use | A use code specified in an address | Location.address.use
endpoint | reference | Location-endpoint | Technical endpoints providing access to services operated for the location | Location.endpoint
identifier | token | Location-identifier | An identifier for the location | Location.identifier
name | string | Location-name | A portion of the location's name or alias | Location.name | Location.alias
near | special | Location-near | Search for locations where the location.position is near to, or within a specified distance of, the provided coordinates expressed as [latitude]|[longitude]|[distance]|[units] (using the WGS84 datum, see notes). If the units are omitted, then kms should be assumed. If the distance is omitted, then the server can use its own discretion as to what distances should be considered near (and units are irrelevant) Servers may search using various techniques that might have differing accuracies, depending on implementation efficiency. Requires the near-distance parameter to be provided also | Location.position
operational-status | token | Location-operational-status | Searches for locations (typically bed/room) that have an operational status (e.g. contaminated, housekeeping) | Location.operationalStatus
organization | reference | Location-organization | Searches for locations that are managed by the provided organization | Location.managingOrganization
partof | reference | Location-partof | A location of which this location is a part | Location.partOf
status | token | Location-status | Searches for locations with a specific kind of status | Location.status
type | token | Location-type | A code for the type of location | Location.type
Measure TU
composed-of | reference | Measure-composed-of | What resource is being referenced | Measure.relatedArtifact.where(type='composed-of').resource
context | token | Measure-context | A use context assigned to the measure | (Measure.useContext.value as CodeableConcept)
context-quantity | quantity | Measure-context-quantity | A quantity- or range-valued use context assigned to the measure | (Measure.useContext.value as Quantity) | (Measure.useContext.value as Range)
context-type | token | Measure-context-type | A type of use context assigned to the measure | Measure.useContext.code
context-type-quantity | composite | Measure-context-type-quantity | A use context type and quantity- or range-based value assigned to the measure | Measure.useContext
context-type-value | composite | Measure-context-type-value | A use context type and value assigned to the measure | Measure.useContext
date | date | Measure-date | The measure publication date | Measure.date
depends-on | reference | Measure-depends-on | What resource is being referenced | Measure.relatedArtifact.where(type='depends-on').resource | Measure.library
derived-from | reference | Measure-derived-from | What resource is being referenced | Measure.relatedArtifact.where(type='derived-from').resource
description | string | Measure-description | The description of the measure | Measure.description
effective | date | Measure-effective | The time during which the measure is intended to be in use | Measure.effectivePeriod
identifier | token | Measure-identifier | External identifier for the measure | Measure.identifier
jurisdiction | token | Measure-jurisdiction | Intended jurisdiction for the measure | Measure.jurisdiction
name | string | Measure-name | Computationally friendly name of the measure | Measure.name
predecessor | reference | Measure-predecessor | What resource is being referenced | Measure.relatedArtifact.where(type='predecessor').resource
publisher | string | Measure-publisher | Name of the publisher of the measure | Measure.publisher
status | token | Measure-status | The current status of the measure | Measure.status
successor | reference | Measure-successor | What resource is being referenced | Measure.relatedArtifact.where(type='successor').resource
title | string | Measure-title | The human-friendly name of the measure | Measure.title
topic | token | Measure-topic | Topics associated with the measure | Measure.topic
url | uri | Measure-url | The uri that identifies the measure | Measure.url

version | token | Measure-version | The business version of the measure | Measure.version
MeasureReport TU
date | date | MeasureReport-date | The date of the measure report | MeasureReport.date
evaluated-resource | reference | MeasureReport-evaluated-resource | An evaluated resource referenced by the measure report | MeasureReport.evaluatedResource
identifier | token | MeasureReport-identifier | External identifier of the measure report to be returned | MeasureReport.identifier
measure | reference | MeasureReport-measure | The measure to return measure report results for | MeasureReport.measure
patient | reference | MeasureReport-patient | The identity of a patient to search for individual measure report results for | MeasureReport.subject.where(resolve() is Patient)
period | date | MeasureReport-period | The period of the measure report | MeasureReport.period
reporter | reference | MeasureReport-reporter | The reporter to return measure report results for | MeasureReport.reporter
status | token | MeasureReport-status | The status of the measure report | MeasureReport.status
subject | reference | MeasureReport-subject | The identity of a subject to search for individual measure report results for | MeasureReport.subject
Media TU
based-on | reference | Media-based-on | Procedure that caused this media to be created | Media.basedOn
created | date | Media-created | When Media was collected | Media.created
device | reference | Media-device | Observing Device | Media.device
encounter | reference | Media-encounter | Encounter associated with media | Media.encounter
identifier | token | Media-identifier | Identifier(s) for the image | Media.identifier
modality | token | Media-modality | The type of acquisition equipment/process | Media.modality
operator | reference | Media-operator | The person who generated the image | Media.operator
patient | reference | Media-patient | Who/What this Media is a record of | Media.subject.where(resolve() is Patient)
site | token | Media-site | Observed body part | Media.bodySite
status | token | Media-status | preparation | in-progress | not-done | on-hold | stopped | completed | entered-in-error | unknown | Media.status
subject | reference | Media-subject | Who/What this Media is a record of | Media.subject
type | token | Media-type | Classification of media as image, video, or audio | Media.type
view | token | Media-view | Imaging view, e.g. Lateral or Antero-posterior | Media.view
Medication TU
code | token | clinical-code | Returns medications for a specific code | Medication.code
expiration-date | date | Medication-expiration-date | Returns medications in a batch with this expiration date | Medication.batch.expirationDate
form | token | Medication-form | Returns medications for a specific dose form | Medication.form
identifier | token | Medication-identifier | Returns medications with this external identifier | Medication.identifier
ingredient | reference | Medication-ingredient | Returns medications for this ingredient reference | (Medication.ingredient.item as Reference)
ingredient-code | token | Medication-ingredient-code | Returns medications for this ingredient code | (Medication.ingredient.item as CodeableConcept)
lot-number | token | Medication-lot-number | Returns medications in a batch with this lot number | Medication.batch.lotNumber
manufacturer | reference | Medication-manufacturer | Returns medications made or sold for this manufacturer | Medication.manufacturer
status | token | Medication-status | Returns medications for this status | Medication.status
MedicationAdministration TU
code | token | clinical-code | Return administrations of this medication code | (MedicationAdministration.medication as CodeableConcept)
context | reference | MedicationAdministration-context | Return administrations that share this encounter or episode of care | MedicationAdministration.context
device | reference | MedicationAdministration-device | Return administrations with this administration device identity | MedicationAdministration.device
effective-time | date | MedicationAdministration-effective-time | Date administration happened (or did not happen) | MedicationAdministration.effective
identifier | token | clinical-identifier | Return administrations with this external identifier | MedicationAdministration.identifier
medication | reference | medications-medication | Return administrations of this medication resource | (MedicationAdministration.medication as Reference)
patient | reference | clinical-patient | The identity of a patient to list administrations for | MedicationAdministration.subject.where(resolve() is Patient)
performer | reference | MedicationAdministration-performer | The identity of the individual who administered the medication | MedicationAdministration.performer.actor
reason-given | token | MedicationAdministration-reason-given | Reasons for administering the medication | MedicationAdministration.reasonCode
reason-not-given | token | MedicationAdministration-reason-not-given | Reasons for not administering the medication | MedicationAdministration.statusReason
request | reference | MedicationAdministration-request | The identity of a request to list administrations from | MedicationAdministration.request
status | token | medications-status | MedicationAdministration event status (for example one of active/paused/completed/nullified) | MedicationAdministration.status
subject | reference | MedicationAdministration-subject | The identity of the individual or group to list administrations for | MedicationAdministration.subject
MedicationDispense TU
code | token | clinical-code | Returns dispenses of this medicine code | (MedicationDispense.medication as CodeableConcept)
context | reference | MedicationDispense-context | Returns dispenses with a specific context (episode or episode of care) | MedicationDispense.context

destination | reference | MedicationDispense-destination | Returns dispenses that should be sent to a specific destination | MedicationDispense.destination
identifier | token | clinical-identifier | Returns dispenses with this external identifier | MedicationDispense.identifier
medication | reference | medications-medication | Returns dispenses of this medicine resource | (MedicationDispense.medication as Reference)
patient | reference | clinical-patient | The identity of a patient to list dispenses for | MedicationDispense.subject.where(resolve() is Patient)
performer | reference | MedicationDispense-performer | Returns dispenses performed by a specific individual | MedicationDispense.performer.actor
prescription | reference | medications-prescription | Multiple Resources: MedicationDispense: The identity of a prescription to list dispenses from | MedicationDispense.authorizingPrescription
receiver | reference | MedicationDispense-receiver | The identity of a receiver to list dispenses for | MedicationDispense.receiver
responsibleparty | reference | MedicationDispense-responsibleparty | Returns dispenses with the specified responsible party | MedicationDispense.substitution.responsibleParty
status | token | medications-status | Returns dispenses with a specified dispense status | MedicationDispense.status
subject | reference | MedicationDispense-subject | The identity of a patient for whom to list dispenses | MedicationDispense.subject
type | token | MedicationDispense-type | Returns dispenses of a specific type | MedicationDispense.type
whenhandedover | date | MedicationDispense-whenhandedover | Returns dispenses handed over on this date | MedicationDispense.whenHandedOver
whenprepared | date | MedicationDispense-whenprepared | Returns dispenses prepared on this date | MedicationDispense.whenPrepared
MedicationKnowledge TU
classification | token | MedicationKnowledge-classification | Specific category assigned to the medication | MedicationKnowledge.medicineClassification.classification
classification-type | token | MedicationKnowledge-classification-type | The type of category for the medication (for example, therapeutic classification, therapeutic sub-classification) | MedicationKnowledge.medicineClassification.type
code | token | MedicationKnowledge-code | Code that identifies this medication | MedicationKnowledge.code
doseform | token | MedicationKnowledge-doseform | powder | tablets | capsule + | MedicationKnowledge.doseForm
ingredient | reference | MedicationKnowledge-ingredient | Medication(s) or substance(s) contained in the medication | (MedicationKnowledge.ingredient.item as Reference)
ingredient-code | token | MedicationKnowledge-ingredient-code | Medication(s) or substance(s) contained in the medication | (MedicationKnowledge.ingredient.item as CodeableConcept)
manufacturer | reference | MedicationKnowledge-manufacturer | Manufacturer of the item | MedicationKnowledge.manufacturer
monitoring-program-name | token | MedicationKnowledge-monitoring-program-name | Name of the reviewing program | MedicationKnowledge.monitoringProgram.name
monitoring-program-type | token | MedicationKnowledge-monitoring-program-type | Type of program under which the medication is monitored | MedicationKnowledge.monitoringProgram.type
monograph | reference | MedicationKnowledge-monograph | Associated documentation about the medication | MedicationKnowledge.monograph.source
monograph-type | token | MedicationKnowledge-monograph-type | The category of medication document | MedicationKnowledge.monograph.type
source-cost | token | MedicationKnowledge-source-cost | The source or owner for the price information | MedicationKnowledge.cost.source
status | token | MedicationKnowledge-status | active | inactive | entered-in-error | MedicationKnowledge.status
MedicationRequest TU
authoredon | date | MedicationRequest-authoredon | Return prescriptions written on this date | MedicationRequest.authoredOn
category | token | MedicationRequest-category | Returns prescriptions with different categories | MedicationRequest.category
code | token | clinical-code | Return prescriptions of this medication code | (MedicationRequest.medication as CodeableConcept)
date | date | medications-date | Multiple Resources: MedicationRequest: Returns medication request to be administered on a specific date | MedicationRequest.dosageInstruction.timing.event
encounter | reference | medications-encounter | Multiple Resources: MedicationRequest: Return prescriptions with this encounter identifier | MedicationRequest.encounter
identifier | token | clinical-identifier | Return prescriptions with this external identifier | MedicationRequest.identifier
intended-dispenser | reference | MedicationRequest-intended-dispenser | Returns prescriptions intended to be dispensed by this Organization | MedicationRequest.dispenseRequest.performer
intended-performer | reference | MedicationRequest-intended-performer | Returns the intended performer of the administration of the medication request | MedicationRequest.performer
intended-performertype | token | MedicationRequest-intended-performertype | Returns requests for a specific type of performer | MedicationRequest.performerType
intent | token | MedicationRequest-intent | Returns prescriptions with different intents | MedicationRequest.intent
medication | reference | medications-medication | Return prescriptions for this medication reference | (MedicationRequest.medication as Reference)
patient | reference | clinical-patient | Returns prescriptions for a specific patient | MedicationRequest.subject.where(resolve() is Patient)
priority | token | MedicationRequest-priority | Returns prescriptions with different priorities | MedicationRequest.priority
requester | reference | MedicationRequest-requester | Returns prescriptions prescribed by this prescriber | MedicationRequest.requester
status | token | medications-status | Status of the prescription | MedicationRequest.status
subject | reference | MedicationRequest-subject | The identity of a patient to list orders for | MedicationRequest.subject
MedicationStatement TU
category | token | MedicationStatement-category | Returns statements of this category of medicationstatement | MedicationStatement.category
code | token | clinical-code | Return statements of this medication code | (MedicationStatement.medication as CodeableConcept)

context | reference | MedicationStatement-context | Returns statements for a specific context (episode or episode of Care). | MedicationStatement.context
effective | date | MedicationStatement-effective | Date when patient was taking (or not taking) the medication | MedicationStatement.effective
identifier | token | clinical-identifier | Return statements with this external identifier | MedicationStatement.identifier
medication | reference | medications-medication | Return statements of this medication reference | (MedicationStatement.medication as Reference)
part-of | reference | MedicationStatement-part-of | Returns statements that are part of another event. | MedicationStatement.partOf
patient | reference | clinical-patient | Returns statements for a specific patient. | MedicationStatement.subject.where(resolve() is Patient)
source | reference | MedicationStatement-source | Who or where the information in the statement came from | MedicationStatement.informationSource
status | token | medications-status | Return statements that match the given status | MedicationStatement.status
subject | reference | MedicationStatement-subject | The identity of a patient, animal or group to list statements for | MedicationStatement.subject
MedicinalProduct TU
identifier | token | MedicinalProduct-identifier | Business identifier for this product. Could be an MPID | MedicinalProduct.identifier
name | string | MedicinalProduct-name | The full product name | MedicinalProduct.name.productName
name-language | token | MedicinalProduct-name-language | Language code for this name | MedicinalProduct.name.countryLanguage.language
MedicinalProductAuthorization TU
country | token | MedicinalProductAuthorization-country | The country in which the marketing authorization has been granted | MedicinalProductAuthorization.country
holder | reference | MedicinalProductAuthorization-holder | Marketing Authorization Holder | MedicinalProductAuthorization.holder
identifier | token | MedicinalProductAuthorization-identifier | Business identifier for the marketing authorization, as assigned by a regulator | MedicinalProductAuthorization.identifier
status | token | MedicinalProductAuthorization-status | The status of the marketing authorization | MedicinalProductAuthorization.status
subject | reference | MedicinalProductAuthorization-subject | The medicinal product that is being authorized | MedicinalProductAuthorization.subject
MedicinalProductContraindication TU
subject | reference | MedicinalProductContraindication-subject | The medication for which this is an contraindication | MedicinalProductContraindication.subject
MedicinalProductIndication TU
subject | reference | MedicinalProductIndication-subject | The medication for which this is an indication | MedicinalProductIndication.subject
MedicinalProductInteraction TU
subject | reference | MedicinalProductInteraction-subject | The medication for which this is an interaction | MedicinalProductInteraction.subject
MedicinalProductPackaged TU
identifier | token | MedicinalProductPackaged-identifier | Unique identifier | MedicinalProductPackaged.identifier
subject | reference | MedicinalProductPackaged-subject | The product with this is a pack for | MedicinalProductPackaged.subject
MedicinalProductPharmaceutical TU
identifier | token | MedicinalProductPharmaceutical-identifier | An identifier for the pharmaceutical medicinal product | MedicinalProductPharmaceutical.identifier
route | token | MedicinalProductPharmaceutical-route | Coded expression for the route | MedicinalProductPharmaceutical.routeOfAdministration.code
target-species | token | MedicinalProductPharmaceutical-target-species | Coded expression for the species | MedicinalProductPharmaceutical.routeOfAdministration.targetSpecies.code
MedicinalProductUndesirableEffect TU
subject | reference | MedicinalProductUndesirableEffect-subject | The medication for which this is an undesirable effect | MedicinalProductUndesirableEffect.subject
MessageDefinition TU
category | token | MessageDefinition-category | The behavior associated with the message | MessageDefinition.category
context | token | conformance-context | A use context assigned to the message definition | (MessageDefinition.useContext.value as CodeableConcept)
context-quantity | quantity | conformance-context-quantity | A quantity- or range-valued use context assigned to the message definition | (MessageDefinition.useContext.value as Quantity) | (MessageDefinition.useContext.value as Range)
context-type | token | conformance-context-type | A type of use context assigned to the message definition | MessageDefinition.useContext.code
context-type-quantity | composite | conformance-context-type-quantity | A use context type and quantity- or range-based value assigned to the message definition | MessageDefinition.useContext
context-type-value | composite | conformance-context-type-value | A use context type and value assigned to the message definition | MessageDefinition.useContext
date | date | conformance-date | The message definition publication date | MessageDefinition.date
description | string | conformance-description | The description of the message definition | MessageDefinition.description
event | token | MessageDefinition-event | The event that triggers the message or link to the event definition. | MessageDefinition.event
focus | token | MessageDefinition-focus | A resource that is a permitted focus of the message | MessageDefinition.focus.code
identifier | token | conformance-identifier | External identifier for the message definition | MessageDefinition.identifier
jurisdiction | token | conformance-jurisdiction | Intended jurisdiction for the message definition | MessageDefinition.jurisdiction
name | string | conformance-name | Computationally friendly name of the message definition | MessageDefinition.name
parent | reference | MessageDefinition-parent | A resource that is the parent of the definition | MessageDefinition.parent
publisher | string | conformance-publisher | Name of the publisher of the message definition | MessageDefinition.publisher

status | token | conformance-status | The current status of the message definition | MessageDefinition.status
title | string | conformance-title | The human-friendly name of the message definition | MessageDefinition.title
url | uri | conformance-url | The uri that identifies the message definition | MessageDefinition.url
version | token | conformance-version | The business version of the message definition | MessageDefinition.version
MessageHeader TU
author | reference | MessageHeader-author | The source of the decision | MessageHeader.author
code | token | MessageHeader-code | ok | transient-error | fatal-error | MessageHeader.response.code
destination | string | MessageHeader-destination | Name of system | MessageHeader.destination.name
destination-uri | uri | MessageHeader-destination-uri | Actual destination address or id | MessageHeader.destination.endpoint
enterer | reference | MessageHeader-enterer | The source of the data entry | MessageHeader.enterer
event | token | MessageHeader-event | Code for the event this message represents or link to event definition | MessageHeader.event
focus | reference | MessageHeader-focus | The actual content of the message | MessageHeader.focus
receiver | reference | MessageHeader-receiver | Intended "real-world" recipient for the data | MessageHeader.destination.receiver
response-id | token | MessageHeader-response-id | Id of original message | MessageHeader.response.identifier
responsible | reference | MessageHeader-responsible | Final responsibility for event | MessageHeader.responsible
sender | reference | MessageHeader-sender | Real world sender of the message | MessageHeader.sender
source | string | MessageHeader-source | Name of system | MessageHeader.source.name
source-uri | uri | MessageHeader-source-uri | Actual message source address or id | MessageHeader.source.endpoint
target | reference | MessageHeader-target | Particular delivery destination within the destination | MessageHeader.destination.target
MolecularSequence TU
chromosome | token | MolecularSequence-chromosome | Chromosome number of the reference sequence | MolecularSequence.referenceSeq.chromosome
chromosome-variant-coordinate | composite | MolecularSequence-chromosome-variant-coordinate | Search parameter by chromosome and variant coordinate. This will refer to part of a locus or part of a gene where search region will be represented in 1-based system. Since the coordinateSystem can either be 0-based or 1-based, this search query will include the result of both coordinateSystem that contains the equivalent segment of the gene or whole genome sequence. For example, a search for sequence can be represented as chromosome-variant-coordinate=1$lt345$gt123, this means it will search for the MolecularSequence resource with variants on chromosome 1 and with position >123 and <345, where in 1-based system resource, all strings within region 1:124-344 will be revealed, while in 0-based system resource, all strings within region 1:123-344 will be revealed. You may want to check detail about 0-based v.s. 1-based above. | MolecularSequence.variant
chromosome-window-coordinate | composite | MolecularSequence-chromosome-window-coordinate | Search parameter by chromosome and window. This will refer to part of a locus or part of a gene where search region will be represented in 1-based system. Since the coordinateSystem can either be 0-based or 1-based, this search query will include the result of both coordinateSystem that contains the equivalent segment of the gene or whole genome sequence. For example, a search for sequence can be represented as chromosome-window-coordinate=1$lt345$gt123, this means it will search for the MolecularSequence resource with a window on chromosome 1 and with position >123 and <345, where in 1-based system resource, all strings within region 1:124-344 will be revealed, while in 0-based system resource, all strings within region 1:123-344 will be revealed. You may want to check detail about 0-based v.s. 1-based above. | MolecularSequence.referenceSeq
identifier | token | MolecularSequence-identifier | The unique identity for a particular sequence | MolecularSequence.identifier
patient | reference | MolecularSequence-patient | The subject that the observation is about | MolecularSequence.patient
referenceseqid | token | MolecularSequence-referenceseqid | Reference Sequence of the sequence | MolecularSequence.referenceSeq.referenceSeqId
referenceseqid-variant-coordinate | composite | MolecularSequence-referenceseqid-variant-coordinate | Search parameter by reference sequence and variant coordinate. This will refer to part of a locus or part of a gene where search region will be represented in 1-based system. Since the coordinateSystem can either be 0-based or 1-based, this search query will include the result of both coordinateSystem that contains the equivalent segment of the gene or whole genome sequence. For example, a search for sequence can be represented as referenceSeqId-variant-coordinate=NC_000001.11$lt345$gt123, this means it will search for the MolecularSequence resource with variants on NC_000001.11 and with position >123 and <345, where in 1-based system resource, all strings within region NC_000001.11:124-344 will be revealed, while in 0-based system resource, all strings within region NC_000001.11:123-344 will be revealed. You may want to check detail about 0-based v.s. 1-based above. | MolecularSequence.variant
referenceseqid-window-coordinate | composite | MolecularSequence-referenceseqid-window-coordinate | Search parameter by reference sequence and window. This will refer to part of a locus or part of a gene where search region will be represented in 1-based system. Since the coordinateSystem can either be 0-based or 1-based, this search query will include the result of both coordinateSystem that contains the equivalent segment of the gene or whole genome sequence. For example, a search for sequence can be represented as referenceSeqId-window-coordinate=NC_000001.11$lt345$gt123, this means it will search for the MolecularSequence resource with a window on NC_000001.11 and with position >123 and <345, where in 1-based system resource, all strings within region NC_000001.11:124-344 will be revealed, while in 0-based system resource, all strings within region NC_000001.11:123-344 will be revealed. You may want to check detail about 0-based v.s. 1-based above. | MolecularSequence.referenceSeq
type | token | MolecularSequence-type | Amino Acid Sequence/ DNA Sequence / RNA Sequence | MolecularSequence.type
variant-end | number | MolecularSequence-variant-end | End position (0-based exclusive, which menas the acid at this position will not be included, 1-based inclusive, which means the acid at this position will be included) of the variant. | MolecularSequence.variant.end
variant-start | number | MolecularSequence-variant-start | Start position (0-based inclusive, 1-based inclusive, that means the nucleic acid or amino acid at this position will be included) of the variant. | MolecularSequence.variant.start
window-end | number | MolecularSequence-window-end | End position (0-based exclusive, which menas the acid at this position will not be included, 1-based inclusive, which means the acid at this position will be included) of the reference sequence. | MolecularSequence.referenceSeq.windowEnd
window-start | number | MolecularSequence-window-start | Start position (0-based inclusive, 1-based inclusive, that means the nucleic acid or amino acid at this position will be included) of the reference sequence. | MolecularSequence.referenceSeq.windowStart
NamingSystem TU
contact | string | NamingSystem-contact | Name of an individual to contact | NamingSystem.contact.name

context | token | conformance-context | A use context assigned to the naming system | (NamingSystem.useContext.value as CodeableConcept)
context-quantity | quantity | conformance-context-quantity | A quantity- or range-valued use context assigned to the naming system | (NamingSystem.useContext.value as Quantity) | (NamingSystem.useContext.value as Range)
context-type | token | conformance-context-type | A type of use context assigned to the naming system | NamingSystem.useContext.code
context-type-quantity | composite | conformance-context-type-quantity | A use context type and quantity- or range-based value assigned to the naming system | NamingSystem.useContext
context-type-value | composite | conformance-context-type-value | A use context type and value assigned to the naming system | NamingSystem.useContext
date | date | conformance-date | The naming system publication date | NamingSystem.date
description | string | conformance-description | The description of the naming system | NamingSystem.description
id-type | token | NamingSystem-id-type | oid | uuid | uri | other | NamingSystem.uniqueId.type
jurisdiction | token | conformance-jurisdiction | Intended jurisdiction for the naming system | NamingSystem.jurisdiction
kind | token | NamingSystem-kind | codesystem | identifier | root | NamingSystem.kind
name | string | conformance-name | Computationally friendly name of the naming system | NamingSystem.name
period | date | NamingSystem-period | When is identifier valid? | NamingSystem.uniqueId.period
publisher | string | conformance-publisher | Name of the publisher of the naming system | NamingSystem.publisher
responsible | string | NamingSystem-responsible | Who maintains system namespace? | NamingSystem.responsible
status | token | conformance-status | The current status of the naming system | NamingSystem.status
telecom | token | NamingSystem-telecom | Contact details for individual or organization | NamingSystem.contact.telecom
type | token | NamingSystem-type | e.g. driver, provider, patient, bank etc. | NamingSystem.type
value | string | NamingSystem-value | The unique identifier | NamingSystem.uniqueId.value
NutritionOrder TU
additive | token | NutritionOrder-additive | Type of module component to add to the feeding | NutritionOrder.enteralFormula.additiveType
datetime | date | NutritionOrder-datetime | Return nutrition orders requested on this date | NutritionOrder.dateTime
encounter | reference | clinical-encounter | Return nutrition orders with this encounter identifier | NutritionOrder.encounter
formula | token | NutritionOrder-formula | Type of enteral or infant formula | NutritionOrder.enteralFormula.baseFormulaType
identifier | token | clinical-identifier | Return nutrition orders with this external identifier | NutritionOrder.identifier
instantiates-canonical | reference | NutritionOrder-instantiates-canonical | Instantiates FHIR protocol or definition | NutritionOrder.instantiatesCanonical
instantiates-uri | uri | NutritionOrder-instantiates-uri | Instantiates external protocol or definition | NutritionOrder.instantiatesUri
oraldiet | token | NutritionOrder-oraldiet | Type of diet that can be consumed orally (i.e., take via the mouth). | NutritionOrder.oralDiet.type
patient | reference | clinical-patient | The identity of the person who requires the diet, formula or nutritional supplement | NutritionOrder.patient
provider | reference | NutritionOrder-provider | The identity of the provider who placed the nutrition order | NutritionOrder.orderer
status | token | NutritionOrder-status | Status of the nutrition order. | NutritionOrder.status
supplement | token | NutritionOrder-supplement | Type of supplement product requested | NutritionOrder.supplement.type
Observation N
based-on TU | reference | Observation-based-on | Reference to the service request. | Observation.basedOn
category TU | token | Observation-category | The classification of the type of observation | Observation.category
code TU | token | clinical-code | The code of the observation type | Observation.code
code-value-concept TU | composite | Observation-code-value-concept | Code and coded value parameter pair | Observation
code-value-date TU | composite | Observation-code-value-date | Code and date/time value parameter pair | Observation
code-value-quantity TU | composite | Observation-code-value-quantity | Code and quantity value parameter pair | Observation
code-value-string TU | composite | Observation-code-value-string | Code and string value parameter pair | Observation
combo-code TU | token | Observation-combo-code | The code of the observation type or component type | Observation.code | Observation.component.code
combo-code-value-concept TU | composite | Observation-combo-code-value-concept | Code and coded value parameter pair, including in components | Observation | Observation.component
combo-code-value-quantity TU | composite | Observation-combo-code-value-quantity | Code and quantity value parameter pair, including in components | Observation | Observation.component
combo-data-absent-reason TU | token | Observation-combo-data-absent-reason | The reason why the expected value in the element Observation.value[x] or Observation.component.value[x] is missing. | Observation.dataAbsentReason | Observation.component.dataAbsentReason
combo-value-concept TU | token | Observation-combo-value-concept | The value or component value of the observation, if the value is a CodeableConcept | (Observation.value as CodeableConcept) | (Observation.component.value as CodeableConcept)
combo-value-quantity TU | quantity | Observation-combo-value-quantity | The value or component value of the observation, if the value is a Quantity, or a SampledData (just search on the bounds of the values in sampled data) | (Observation.value as Quantity) | (Observation.value as SampledData) | (Observation.component.value as Quantity) | (Observation.component.value as SampledData)
component-code TU | token | Observation-component-code | The component code of the observation type | Observation.component.code
component-code-value-concept TU | composite | Observation-component-code-value-concept | Component code and component coded value parameter pair | Observation.component
component-code-value-quantity TU | composite | Observation-component-code-value-quantity | Component code and component quantity value parameter pair | Observation.component
component-data-absent-reason TU | token | Observation-component-data-absent-reason | The reason why the expected value in the element Observation.component.value[x] is missing. | Observation.component.dataAbsentReason
component-value-concept TU | token | Observation-component-value-concept | The value of the component observation, if the value is a CodeableConcept | (Observation.component.value as CodeableConcept)

component-value-quantity TU | quantity | Observation-component-value-quantity | The value of the component observation, if the value is a Quantity, or a SampledData (just search on the bounds of the values in sampled data) | (Observation.component.value as Quantity) | (Observation.component.value as SampledData)
data-absent-reason TU | token | Observation-data-absent-reason | The reason why the expected value in the element Observation.value[x] is missing. | Observation.dataAbsentReason
date TU | date | clinical-date | Obtained date/time. If the obtained element is a period, a date that falls in the period | Observation.effective
derived-from TU | reference | Observation-derived-from | Related measurements the observation is made from | Observation.derivedFrom
device TU | reference | Observation-device | The Device that generated the observation data. | Observation.device
encounter TU | reference | clinical-encounter | Encounter related to the observation | Observation.encounter
focus TU | reference | Observation-focus | The focus of an observation when the focus is not the patient of record. | Observation.focus
has-member TU | reference | Observation-has-member | Related resource that belongs to the Observation group | Observation.hasMember
identifier TU | token | clinical-identifier | The unique id for a particular observation | Observation.identifier
method TU | token | Observation-method | The method used for the observation | Observation.method
part-of TU | reference | Observation-part-of | Part of referenced event | Observation.partOf
patient TU | reference | clinical-patient | The subject that the observation is about (if patient) | Observation.subject.where(resolve() is Patient)
performer TU | reference | Observation-performer | Who performed the observation | Observation.performer
specimen TU | reference | Observation-specimen | Specimen used for this observation | Observation.specimen
status TU | token | Observation-status | The status of the observation | Observation.status
subject TU | reference | Observation-subject | The subject that the observation is about | Observation.subject
value-concept TU | token | Observation-value-concept | The value of the observation, if the value is a CodeableConcept | (Observation.value as CodeableConcept)
value-date TU | date | Observation-value-date | The value of the observation, if the value is a date or period of time | (Observation.value as dateTime) | (Observation.value as Period)
value-quantity TU | quantity | Observation-value-quantity | The value of the observation, if the value is a Quantity, or a SampledData (just search on the bounds of the values in sampled data) | (Observation.value as Quantity) | (Observation.value as SampledData)
value-string TU | string | Observation-value-string | The value of the observation, if the value is a string, and also searches in CodeableConcept.text | (Observation.value as string) | (Observation.value as CodeableConcept).text
OperationDefinition N
base TU | reference | OperationDefinition-base | Marks this as a profile of the base | OperationDefinition.base
code TU | token | OperationDefinition-code | Name used to invoke the operation | OperationDefinition.code
context TU | token | conformance-context | A use context assigned to the operation definition | (OperationDefinition.useContext.value as CodeableConcept)
context-quantity TU | quantity | conformance-context-quantity | A quantity- or range-valued use context assigned to the operation definition | (OperationDefinition.useContext.value as Quantity) | (OperationDefinition.useContext.value as Range)
context-type TU | token | conformance-context-type | A type of use context assigned to the operation definition | OperationDefinition.useContext.code
context-type-quantity TU | composite | conformance-context-type-quantity | A use context type and quantity- or range-based value assigned to the operation definition | OperationDefinition.useContext
context-type-value TU | composite | conformance-context-type-value | A use context type and value assigned to the operation definition | OperationDefinition.useContext
date TU | date | conformance-date | The operation definition publication date | OperationDefinition.date
description TU | string | conformance-description | The description of the operation definition | OperationDefinition.description
input-profile TU | reference | OperationDefinition-input-profile | Validation information for in parameters | OperationDefinition.inputProfile
instance TU | token | OperationDefinition-instance | Invoke on an instance? | OperationDefinition.instance
jurisdiction TU | token | conformance-jurisdiction | Intended jurisdiction for the operation definition | OperationDefinition.jurisdiction
kind TU | token | OperationDefinition-kind | operation | query | OperationDefinition.kind
name TU | string | conformance-name | Computationally friendly name of the operation definition | OperationDefinition.name
output-profile TU | reference | OperationDefinition-output-profile | Validation information for out parameters | OperationDefinition.outputProfile
publisher TU | string | conformance-publisher | Name of the publisher of the operation definition | OperationDefinition.publisher
status TU | token | conformance-status | The current status of the operation definition | OperationDefinition.status
system TU | token | OperationDefinition-system | Invoke at the system level? | OperationDefinition.system
title TU | string | conformance-title | The human-friendly name of the operation definition | OperationDefinition.title
type TU | token | OperationDefinition-type | Invoke at the type level? | OperationDefinition.type
url TU | uri | conformance-url | The uri that identifies the operation definition | OperationDefinition.url
version TU | token | conformance-version | The business version of the operation definition | OperationDefinition.version
Organization TU
active | token | Organization-active | Is the Organization record active | Organization.active
address | string | Organization-address | A server defined search that may match any of the string fields in the Address, including line, city, district, state, country, postalCode, and/or text | Organization.address
address-city | string | Organization-address-city | A city specified in an address | Organization.address.city
address-country | string | Organization-address-country | A country specified in an address | Organization.address.country
address-postalcode | string | Organization-address-postalcode | A postal code specified in an address | Organization.address.postalCode
address-state | string | Organization-address-state | A state specified in an address | Organization.address.state
address-use | token | Organization-address-use | A use code specified in an address | Organization.address.use

endpoint | reference | Organization-endpoint | Technical endpoints providing access to services operated for the organization | Organization.endpoint
identifier | token | Organization-identifier | Any identifier for the organization (not the accreditation issuer's identifier) | Organization.identifier
name | string | Organization-name | A portion of the organization's name or alias | Organization.name | Organization.alias
partof | reference | Organization-partof | An organization of which this organization forms a part | Organization.partOf
phonetic | string | Organization-phonetic | A portion of the organization's name using some kind of phonetic matching algorithm | Organization.name
type | token | Organization-type | A code for the type of organization | Organization.type
OrganizationAffiliation TU
active | token | OrganizationAffiliation-active | Whether this organization affiliation record is in active use | OrganizationAffiliation.active
date | date | OrganizationAffiliation-date | The period during which the participatingOrganization is affiliated with the primary organization | OrganizationAffiliation.period
email | token | OrganizationAffiliation-email | A value in an email contact | OrganizationAffiliation.telecom.where(system='email')
endpoint | reference | OrganizationAffiliation-endpoint | Technical endpoints providing access to services operated for this role | OrganizationAffiliation.endpoint
identifier | token | OrganizationAffiliation-identifier | An organization affiliation's Identifier | OrganizationAffiliation.identifier
location | reference | OrganizationAffiliation-location | The location(s) at which the role occurs | OrganizationAffiliation.location
network | reference | OrganizationAffiliation-network | Health insurance provider network in which the participatingOrganization provides the role's services (if defined) at the indicated locations (if defined) | OrganizationAffiliation.network
participating-organization | reference | OrganizationAffiliation-participating-organization | The organization that provides services to the primary organization | OrganizationAffiliation.participatingOrganization
phone | token | OrganizationAffiliation-phone | A value in a phone contact | OrganizationAffiliation.telecom.where(system='phone')
primary-organization | reference | OrganizationAffiliation-primary-organization | The organization that receives the services from the participating organization | OrganizationAffiliation.organization
role | token | OrganizationAffiliation-role | Definition of the role the participatingOrganization plays | OrganizationAffiliation.code
service | reference | OrganizationAffiliation-service | Healthcare services provided through the role | OrganizationAffiliation.healthcareService
specialty | token | OrganizationAffiliation-specialty | Specific specialty of the participatingOrganization in the context of the role | OrganizationAffiliation.specialty
telecom | token | OrganizationAffiliation-telecom | The value in any kind of contact | OrganizationAffiliation.telecom
Patient N
active TU | token | Patient-active | Whether the patient record is active | Patient.active
address TU | string | individual-address | A server defined search that may match any of the string fields in the Address, including line, city, district, state, country, postalCode, and/or text | Patient.address
address-city TU | string | individual-address-city | A city specified in an address | Patient.address.city
address-country TU | string | individual-address-country | A country specified in an address | Patient.address.country
address-postalcode TU | string | individual-address-postalcode | A postalCode specified in an address | Patient.address.postalCode
address-state TU | string | individual-address-state | A state specified in an address | Patient.address.state
address-use TU | token | individual-address-use | A use code specified in an address | Patient.address.use
birthdate TU | date | individual-birthdate | The patient's date of birth | Patient.birthDate
death-date TU | date | Patient-death-date | The date of death has been provided and satisfies this search value | (Patient.deceased as dateTime)
deceased TU | token | Patient-deceased | This patient has been marked as deceased, or as a death date entered | Patient.deceased.exists() and Patient.deceased != false
email TU | token | individual-email | A value in an email contact | Patient.telecom.where(system='email')
family TU | string | individual-family | A portion of the family name of the patient | Patient.name.family
gender TU | token | individual-gender | Gender of the patient | Patient.gender
general-practitioner TU | reference | Patient-general-practitioner | Patient's nominated general practitioner, not the organization that manages the record | Patient.generalPractitioner
given TU | string | individual-given | A portion of the given name of the patient | Patient.name.given
identifier TU | token | Patient-identifier | A patient identifier | Patient.identifier
language TU | token | Patient-language | Language code (irrespective of use value) | Patient.communication.language
link TU | reference | Patient-link | All patients linked to the given patient | Patient.link.other
name TU | string | Patient-name | A server defined search that may match any of the string fields in the HumanName, including family, give, prefix, suffix, suffix, and/or text | Patient.name
organization TU | reference | Patient-organization | The organization that is the custodian of the patient record | Patient.managingOrganization
phone TU | token | individual-phone | A value in a phone contact | Patient.telecom.where(system='phone')
phonetic TU | string | individual-phonetic | A portion of either family or given name using some kind of phonetic matching algorithm | Patient.name
telecom TU | token | individual-telecom | The value in any kind of telecom details of the patient | Patient.telecom
PaymentNotice TU
created | date | PaymentNotice-created | Creation date fro the notice | PaymentNotice.created
identifier | token | PaymentNotice-identifier | The business identifier of the notice | PaymentNotice.identifier
payment-status | token | PaymentNotice-payment-status | The type of payment notice | PaymentNotice.paymentStatus
provider | reference | PaymentNotice-provider | The reference to the provider | PaymentNotice.provider
request | reference | PaymentNotice-request | The Claim | PaymentNotice.request
response | reference | PaymentNotice-response | The ClaimResponse | PaymentNotice.response
status | token | PaymentNotice-status | The status of the payment notice | PaymentNotice.status
PaymentReconciliation TU
created | date | PaymentReconciliation-created | The creation date | PaymentReconciliation.created

disposition | string | PaymentReconciliation-disposition | The contents of the disposition message | PaymentReconciliation.disposition
identifier | token | PaymentReconciliation-identifier | The business identifier of the ExplanationOfBenefit | PaymentReconciliation.identifier
outcome | token | PaymentReconciliation-outcome | The processing outcome | PaymentReconciliation.outcome
payment-issuer | reference | PaymentReconciliation-payment-issuer | The organization which generated this resource | PaymentReconciliation.paymentIssuer
request | reference | PaymentReconciliation-request | The reference to the claim | PaymentReconciliation.request
requestor | reference | PaymentReconciliation-requestor | The reference to the provider who submitted the claim | PaymentReconciliation.requestor
status | token | PaymentReconciliation-status | The status of the payment reconciliation | PaymentReconciliation.status
Person TU
address | string | individual-address | A server defined search that may match any of the string fields in the Address, including line, city, district, state, country, postalCode, and/or text | Person.address
address-city | string | individual-address-city | A city specified in an address | Person.address.city
address-country | string | individual-address-country | A country specified in an address | Person.address.country
address-postalcode | string | individual-address-postalcode | A postal code specified in an address | Person.address.postalCode
address-state | string | individual-address-state | A state specified in an address | Person.address.state
address-use | token | individual-address-use | A use code specified in an address | Person.address.use
birthdate | date | individual-birthdate | The person's date of birth | Person.birthDate
email | token | individual-email | A value in an email contact | Person.telecom.where(system='email')
gender | token | individual-gender | The gender of the person | Person.gender
identifier | token | Person-identifier | A person Identifier | Person.identifier
link | reference | Person-link | Any link has this Patient, Person, RelatedPerson or Practitioner reference | Person.link.target
name | string | Person-name | A server defined search that may match any of the string fields in the HumanName, including family, give, prefix, suffix, suffix, and/or text | Person.name
organization | reference | Person-organization | The organization at which this person record is being managed | Person.managingOrganization
patient | reference | Person-patient | The Person links to this Patient | Person.link.target.where(resolve() is Patient)
phone | token | individual-phone | A value in a phone contact | Person.telecom.where(system='phone')
phonetic | string | individual-phonetic | A portion of name using some kind of phonetic matching algorithm | Person.name
practitioner | reference | Person-practitioner | The Person links to this Practitioner | Person.link.target.where(resolve() is Practitioner)
relatedperson | reference | Person-relatedperson | The Person links to this RelatedPerson | Person.link.target.where(resolve() is RelatedPerson)
telecom | token | individual-telecom | The value in any kind of contact | Person.telecom
PlanDefinition TU
composed-of | reference | PlanDefinition-composed-of | What resource is being referenced | PlanDefinition.relatedArtifact.where(type='composed-of').resource
context | token | PlanDefinition-context | A use context assigned to the plan definition | (PlanDefinition.useContext.value as CodeableConcept)
context-quantity | quantity | PlanDefinition-context-quantity | A quantity- or range-valued use context assigned to the plan definition | (PlanDefinition.useContext.value as Quantity) | (PlanDefinition.useContext.value as Range)
context-type | token | PlanDefinition-context-type | A type of use context assigned to the plan definition | PlanDefinition.useContext.code
context-type-quantity | composite | PlanDefinition-context-type-quantity | A use context type and quantity- or range-based value assigned to the plan definition | PlanDefinition.useContext
context-type-value | composite | PlanDefinition-context-type-value | A use context type and value assigned to the plan definition | PlanDefinition.useContext
date | date | PlanDefinition-date | The plan definition publication date | PlanDefinition.date
definition | reference | PlanDefinition-definition | Activity or plan definitions used by plan definition | PlanDefinition.action.definition
depends-on | reference | PlanDefinition-depends-on | What resource is being referenced | PlanDefinition.relatedArtifact.where(type='depends-on').resource | PlanDefinition.library
derived-from | reference | PlanDefinition-derived-from | What resource is being referenced | PlanDefinition.relatedArtifact.where(type='derived-from').resource
description | string | PlanDefinition-description | The description of the plan definition | PlanDefinition.description
effective | date | PlanDefinition-effective | The time during which the plan definition is intended to be in use | PlanDefinition.effectivePeriod
identifier | token | PlanDefinition-identifier | External identifier for the plan definition | PlanDefinition.identifier
jurisdiction | token | PlanDefinition-jurisdiction | Intended jurisdiction for the plan definition | PlanDefinition.jurisdiction
name | string | PlanDefinition-name | Computationally friendly name of the plan definition | PlanDefinition.name
predecessor | reference | PlanDefinition-predecessor | What resource is being referenced | PlanDefinition.relatedArtifact.where(type='predecessor').resource
publisher | string | PlanDefinition-publisher | Name of the publisher of the plan definition | PlanDefinition.publisher
status | token | PlanDefinition-status | The current status of the plan definition | PlanDefinition.status
successor | reference | PlanDefinition-successor | What resource is being referenced | PlanDefinition.relatedArtifact.where(type='successor').resource
title | string | PlanDefinition-title | The human-friendly name of the plan definition | PlanDefinition.title
topic | token | PlanDefinition-topic | Topics associated with the module | PlanDefinition.topic
type | token | PlanDefinition-type | The type of artifact the plan (e.g. order-set, eca-rule, protocol) | PlanDefinition.type
url | uri | PlanDefinition-url | The uri that identifies the plan definition | PlanDefinition.url
version | token | PlanDefinition-version | The business version of the plan definition | PlanDefinition.version
Practitioner TU
active | token | Practitioner-active | Whether the practitioner record is active | Practitioner.active

address | string | individual-address | A server defined search that may match any of the string fields in the Address, including line, city, district, state, country, postalCode, and/or text | Practitioner.address
address-city | string | individual-address-city | A city specified in an address | Practitioner.address.city
address-country | string | individual-address-country | A country specified in an address | Practitioner.address.country
address-postalcode | string | individual-address-postalcode | A postalCode specified in an address | Practitioner.address.postalCode
address-state | string | individual-address-state | A state specified in an address | Practitioner.address.state
address-use | token | individual-address-use | A use code specified in an address | Practitioner.address.use
communication | token | Practitioner-communication | One of the languages that the practitioner can communicate with | Practitioner.communication
email | token | individual-email | A value in an email contact | Practitioner.telecom.where(system='email')
family | string | individual-family | A portion of the family name | Practitioner.name.family
gender | token | individual-gender | Gender of the practitioner | Practitioner.gender
given | string | individual-given | A portion of the given name | Practitioner.name.given
identifier | token | Practitioner-identifier | A practitioner's Identifier | Practitioner.identifier
name | string | Practitioner-name | A server defined search that may match any of the string fields in the HumanName, including family, give, prefix, suffix, suffix, and/or text | Practitioner.name
phone | token | individual-phone | A value in a phone contact | Practitioner.telecom.where(system='phone')
phonetic | string | individual-phonetic | A portion of either family or given name using some kind of phonetic matching algorithm | Practitioner.name
telecom | token | individual-telecom | The value in any kind of contact | Practitioner.telecom
PractitionerRole TU
active | token | PractitionerRole-active | Whether this practitioner role record is in active use | PractitionerRole.active
date | date | PractitionerRole-date | The period during which the practitioner is authorized to perform in these role(s) | PractitionerRole.period
email | token | individual-email | A value in an email contact | PractitionerRole.telecom.where(system='email')
endpoint | reference | PractitionerRole-endpoint | Technical endpoints providing access to services operated for the practitioner with this role | PractitionerRole.endpoint
identifier | token | PractitionerRole-identifier | A practitioner's Identifier | PractitionerRole.identifier
location | reference | PractitionerRole-location | One of the locations at which this practitioner provides care | PractitionerRole.location
organization | reference | PractitionerRole-organization | The identity of the organization the practitioner represents / acts on behalf of | PractitionerRole.organization
phone | token | individual-phone | A value in a phone contact | PractitionerRole.telecom.where(system='phone')
practitioner | reference | PractitionerRole-practitioner | Practitioner that is able to provide the defined services for the organization | PractitionerRole.practitioner
role | token | PractitionerRole-role | The practitioner can perform this role at for the organization | PractitionerRole.code
service | reference | PractitionerRole-service | The list of healthcare services that this worker provides for this role's Organization/Location(s) | PractitionerRole.healthcareService
specialty | token | PractitionerRole-specialty | The practitioner has this specialty at an organization | PractitionerRole.specialty
telecom | token | individual-telecom | The value in any kind of contact | PractitionerRole.telecom
Procedure TU
based-on | reference | Procedure-based-on | A request for this procedure | Procedure.basedOn
category | token | Procedure-category | Classification of the procedure | Procedure.category
code | token | clinical-code | A code to identify a procedure | Procedure.code
date | date | clinical-date | When the procedure was performed | Procedure.performed
encounter | reference | clinical-encounter | Encounter created as part of | Procedure.encounter
identifier | token | clinical-identifier | A unique identifier for a procedure | Procedure.identifier
instantiates-canonical | reference | Procedure-instantiates-canonical | Instantiates FHIR protocol or definition | Procedure.instantiatesCanonical
instantiates-uri | uri | Procedure-instantiates-uri | Instantiates external protocol or definition | Procedure.instantiatesUri
location | reference | Procedure-location | Where the procedure happened | Procedure.location
part-of | reference | Procedure-part-of | Part of referenced event | Procedure.partOf
patient | reference | clinical-patient | Search by subject - a patient | Procedure.subject.where(resolve() is Patient)
performer | reference | Procedure-performer | The reference to the practitioner | Procedure.performer.actor
reason-code | token | Procedure-reason-code | Coded reason procedure performed | Procedure.reasonCode
reason-reference | reference | Procedure-reason-reference | The justification that the procedure was performed | Procedure.reasonReference
status | token | Procedure-status | preparation | in-progress | not-done | on-hold | stopped | completed | entered-in-error | unknown | Procedure.status
subject | reference | Procedure-subject | Search by subject | Procedure.subject
Provenance TU
agent | reference | Provenance-agent | Who participated | Provenance.agent.who
agent-role | token | Provenance-agent-role | What the agents role was | Provenance.agent.role
agent-type | token | Provenance-agent-type | How the agent participated | Provenance.agent.type
entity | reference | Provenance-entity | Identity of entity | Provenance.entity.what
location | reference | Provenance-location | Where the activity occurred, if relevant | Provenance.location
patient | reference | Provenance-patient | Target Reference(s) (usually version specific) | Provenance.target.where(resolve() is Patient)
recorded | date | Provenance-recorded | When the activity was recorded / updated | Provenance.recorded
signature-type | token | Provenance-signature-type | Indication of the reason the entity signed the object(s) | Provenance.signature.type
target | reference | Provenance-target | Target Reference(s) (usually version specific) | Provenance.target
when | date | Provenance-when | When the activity occurred | (Provenance.occurred as dateTime)

Questionnaire TU
code | token | Questionnaire-code | A code that corresponds to one of its items in the questionnaire | Questionnaire.item.code
context | token | Questionnaire-context | A use context assigned to the questionnaire | (Questionnaire.useContext.value as CodeableConcept)
context-quantity | quantity | Questionnaire-context-quantity | A quantity- or range-valued use context assigned to the questionnaire | (Questionnaire.useContext.value as Quantity) | (Questionnaire.useContext.value as Range)
context-type | token | Questionnaire-context-type | A type of use context assigned to the questionnaire | Questionnaire.useContext.code
context-type-quantity | composite | Questionnaire-context-type-quantity | A use context type and quantity- or range-based value assigned to the questionnaire | Questionnaire.useContext
context-type-value | composite | Questionnaire-context-type-value | A use context type and value assigned to the questionnaire | Questionnaire.useContext
date | date | Questionnaire-date | The questionnaire publication date | Questionnaire.date
definition | uri | Questionnaire-definition | ElementDefinition - details for the item | Questionnaire.item.definition
description | string | Questionnaire-description | The description of the questionnaire | Questionnaire.description
effective | date | Questionnaire-effective | The time during which the questionnaire is intended to be in use | Questionnaire.effectivePeriod
identifier | token | Questionnaire-identifier | External identifier for the questionnaire | Questionnaire.identifier
jurisdiction | token | Questionnaire-jurisdiction | Intended jurisdiction for the questionnaire | Questionnaire.jurisdiction
name | string | Questionnaire-name | Computationally friendly name of the questionnaire | Questionnaire.name
publisher | string | Questionnaire-publisher | Name of the publisher of the questionnaire | Questionnaire.publisher
status | token | Questionnaire-status | The current status of the questionnaire | Questionnaire.status
subject-type | token | Questionnaire-subject-type | Resource that can be subject of QuestionnaireResponse | Questionnaire.subjectType
title | string | Questionnaire-title | The human-friendly name of the questionnaire | Questionnaire.title
url | uri | Questionnaire-url | The uri that identifies the questionnaire | Questionnaire.url
version | token | Questionnaire-version | The business version of the questionnaire | Questionnaire.version
QuestionnaireResponse TU
author | reference | QuestionnaireResponse-author | The author of the questionnaire response | QuestionnaireResponse.author
authored | date | QuestionnaireResponse-authored | When the questionnaire response was last changed | QuestionnaireResponse.authored
based-on | reference | QuestionnaireResponse-based-on | Plan/proposal/order fulfilled by this questionnaire response | QuestionnaireResponse.basedOn
encounter | reference | QuestionnaireResponse-encounter | Encounter associated with the questionnaire response | QuestionnaireResponse.encounter
identifier | token | QuestionnaireResponse-identifier | The unique identifier for the questionnaire response | QuestionnaireResponse.identifier
part-of | reference | QuestionnaireResponse-part-of | Procedure or observation this questionnaire response was performed as a part of | QuestionnaireResponse.partOf
patient | reference | QuestionnaireResponse-patient | The patient that is the subject of the questionnaire response | QuestionnaireResponse.subject.where(resolve() is Patient)
questionnaire | reference | QuestionnaireResponse-questionnaire | The questionnaire the answers are provided for | QuestionnaireResponse.questionnaire
source | reference | QuestionnaireResponse-source | The individual providing the information reflected in the questionnaire respose | QuestionnaireResponse.source
status | token | QuestionnaireResponse-status | The status of the questionnaire response | QuestionnaireResponse.status
subject | reference | QuestionnaireResponse-subject | The subject of the questionnaire response | QuestionnaireResponse.subject
RelatedPerson TU
active | token | RelatedPerson-active | Indicates if the related person record is active | RelatedPerson.active
address | string | individual-address | A server defined search that may match any of the string fields in the Address, including line, city, district, state, country, postalCode, and/or text | RelatedPerson.address
address-city | string | individual-address-city | A city specified in an address | RelatedPerson.address.city
address-country | string | individual-address-country | A country specified in an address | RelatedPerson.address.country
address-postalcode | string | individual-address-postalcode | A postal code specified in an address | RelatedPerson.address.postalCode
address-state | string | individual-address-state | A state specified in an address | RelatedPerson.address.state
address-use | token | individual-address-use | A use code specified in an address | RelatedPerson.address.use
birthdate | date | individual-birthdate | The Related Person's date of birth | RelatedPerson.birthDate
email | token | individual-email | A value in an email contact | RelatedPerson.telecom.where(system='email')
gender | token | individual-gender | Gender of the related person | RelatedPerson.gender
identifier | token | RelatedPerson-identifier | An Identifier of the RelatedPerson | RelatedPerson.identifier
name | string | RelatedPerson-name | A server defined search that may match any of the string fields in the HumanName, including family, give, prefix, suffix, suffix, and/or text | RelatedPerson.name
patient | reference | RelatedPerson-patient | The patient this related person is related to | RelatedPerson.patient
phone | token | individual-phone | A value in a phone contact | RelatedPerson.telecom.where(system='phone')
phonetic | string | individual-phonetic | A portion of name using some kind of phonetic matching algorithm | RelatedPerson.name
relationship | token | RelatedPerson-relationship | The relationship between the patient and the relatedperson | RelatedPerson.relationship
telecom | token | individual-telecom | The value in any kind of contact | RelatedPerson.telecom
RequestGroup TU
author | reference | RequestGroup-author | The author of the request group | RequestGroup.author
authored | date | RequestGroup-authored | The date the request group was authored | RequestGroup.authoredOn
code | token | RequestGroup-code | The code of the request group | RequestGroup.code
encounter | reference | RequestGroup-encounter | The encounter the request group applies to | RequestGroup.encounter

group-identifier | token | RequestGroup-group-identifier | The group identifier for the request group | RequestGroup.groupIdentifier
identifier | token | RequestGroup-identifier | External identifiers for the request group | RequestGroup.identifier
instantiates-canonical | reference | RequestGroup-instantiates-canonical | The FHIR-based definition from which the request group is realized | RequestGroup.instantiatesCanonical
instantiates-uri | uri | RequestGroup-instantiates-uri | The external definition from which the request group is realized | RequestGroup.instantiatesUri
intent | token | RequestGroup-intent | The intent of the request group | RequestGroup.intent
participant | reference | RequestGroup-participant | The participant in the requests in the group | RequestGroup.action.participant
patient | reference | RequestGroup-patient | The identity of a patient to search for request groups | RequestGroup.subject.where(resolve() is Patient)
priority | token | RequestGroup-priority | The priority of the request group | RequestGroup.priority
status | token | RequestGroup-status | The status of the request group | RequestGroup.status
subject | reference | RequestGroup-subject | The subject that the request group is about | RequestGroup.subject
ResearchDefinition TU
composed-of | reference | ResearchDefinition-composed-of | What resource is being referenced | ResearchDefinition.relatedArtifact.where(type='composed-of').resource
context | token | ResearchDefinition-context | A use context assigned to the research definition | (ResearchDefinition.useContext.value as CodeableConcept)
context-quantity | quantity | ResearchDefinition-context-quantity | A quantity- or range-valued use context assigned to the research definition | (ResearchDefinition.useContext.value as Quantity) | (ResearchDefinition.useContext.value as Range)
context-type | token | ResearchDefinition-context-type | A type of use context assigned to the research definition | ResearchDefinition.useContext.code
context-type-quantity | composite | ResearchDefinition-context-type-quantity | A use context type and quantity- or range-based value assigned to the research definition | ResearchDefinition.useContext
context-type-value | composite | ResearchDefinition-context-type-value | A use context type and value assigned to the research definition | ResearchDefinition.useContext
date | date | ResearchDefinition-date | The research definition publication date | ResearchDefinition.date
depends-on | reference | ResearchDefinition-depends-on | What resource is being referenced | ResearchDefinition.relatedArtifact.where(type='depends-on').resource | ResearchDefinition.library
derived-from | reference | ResearchDefinition-derived-from | What resource is being referenced | ResearchDefinition.relatedArtifact.where(type='derived-from').resource
description | string | ResearchDefinition-description | The description of the research definition | ResearchDefinition.description
effective | date | ResearchDefinition-effective | The time during which the research definition is intended to be in use | ResearchDefinition.effectivePeriod
identifier | token | ResearchDefinition-identifier | External identifier for the research definition | ResearchDefinition.identifier
jurisdiction | token | ResearchDefinition-jurisdiction | Intended jurisdiction for the research definition | ResearchDefinition.jurisdiction
name | string | ResearchDefinition-name | Computationally friendly name of the research definition | ResearchDefinition.name
predecessor | reference | ResearchDefinition-predecessor | What resource is being referenced | ResearchDefinition.relatedArtifact.where(type='predecessor').resource
publisher | string | ResearchDefinition-publisher | Name of the publisher of the research definition | ResearchDefinition.publisher
status | token | ResearchDefinition-status | The current status of the research definition | ResearchDefinition.status
successor | reference | ResearchDefinition-successor | What resource is being referenced | ResearchDefinition.relatedArtifact.where(type='successor').resource
title | string | ResearchDefinition-title | The human-friendly name of the research definition | ResearchDefinition.title
topic | token | ResearchDefinition-topic | Topics associated with the ResearchDefinition | ResearchDefinition.topic
url | uri | ResearchDefinition-url | The uri that identifies the research definition | ResearchDefinition.url
version | token | ResearchDefinition-version | The business version of the research definition | ResearchDefinition.version
ResearchElementDefinition TU
composed-of | reference | ResearchElementDefinition-composed-of | What resource is being referenced | ResearchElementDefinition.relatedArtifact.where(type='composed-of').resource
context | token | ResearchElementDefinition-context | A use context assigned to the research element definition | (ResearchElementDefinition.useContext.value as CodeableConcept)
context-quantity | quantity | ResearchElementDefinition-context-quantity | A quantity- or range-valued use context assigned to the research element definition | (ResearchElementDefinition.useContext.value as Quantity) | (ResearchElementDefinition.useContext.value as Range)
context-type | token | ResearchElementDefinition-context-type | A type of use context assigned to the research element definition | ResearchElementDefinition.useContext.code
context-type-quantity | composite | ResearchElementDefinition-context-type-quantity | A use context type and quantity- or range-based value assigned to the research element definition | ResearchElementDefinition.useContext
context-type-value | composite | ResearchElementDefinition-context-type-value | A use context type and value assigned to the research element definition | ResearchElementDefinition.useContext
date | date | ResearchElementDefinition-date | The research element definition publication date | ResearchElementDefinition.date
depends-on | reference | ResearchElementDefinition-depends-on | What resource is being referenced | ResearchElementDefinition.relatedArtifact.where(type='depends-on').resource | ResearchElementDefinition.library
derived-from | reference | ResearchElementDefinition-derived-from | What resource is being referenced | ResearchElementDefinition.relatedArtifact.where(type='derived-from').resource
description | string | ResearchElementDefinition-description | The description of the research element definition | ResearchElementDefinition.description
effective | date | ResearchElementDefinition-effective | The time during which the research element definition is intended to be in use | ResearchElementDefinition.effectivePeriod
identifier | token | ResearchElementDefinition-identifier | External identifier for the research element definition | ResearchElementDefinition.identifier
jurisdiction | token | ResearchElementDefinition-jurisdiction | Intended jurisdiction for the research element definition | ResearchElementDefinition.jurisdiction

name | string | ResearchElementDefinition-name | Computationally friendly name of the research element definition | ResearchElementDefinition.name
predecessor | reference | ResearchElementDefinition-predecessor | What resource is being referenced | ResearchElementDefinition.relatedArtifact.where(type='predecessor').resource
publisher | string | ResearchElementDefinition-publisher | Name of the publisher of the research element definition | ResearchElementDefinition.publisher
status | token | ResearchElementDefinition-status | The current status of the research element definition | ResearchElementDefinition.status
successor | reference | ResearchElementDefinition-successor | What resource is being referenced | ResearchElementDefinition.relatedArtifact.where(type='successor').resource
title | string | ResearchElementDefinition-title | The human-friendly name of the research element definition | ResearchElementDefinition.title
topic | token | ResearchElementDefinition-topic | Topics associated with the ResearchElementDefinition | ResearchElementDefinition.topic
url | uri | ResearchElementDefinition-url | The uri that identifies the research element definition | ResearchElementDefinition.url
version | token | ResearchElementDefinition-version | The business version of the research element definition | ResearchElementDefinition.version
ResearchStudy TU
category | token | ResearchStudy-category | Classifications for the study | ResearchStudy.category
date | date | ResearchStudy-date | When the study began and ended | ResearchStudy.period
focus | token | ResearchStudy-focus | Drugs, devices, etc. under study | ResearchStudy.focus
identifier | token | ResearchStudy-identifier | Business Identifier for study | ResearchStudy.identifier
keyword | token | ResearchStudy-keyword | Used to search for the study | ResearchStudy.keyword
location | token | ResearchStudy-location | Geographic region(s) for study | ResearchStudy.location
partof | reference | ResearchStudy-partof | Part of larger study | ResearchStudy.partOf
principalinvestigator | reference | ResearchStudy-principalinvestigator | Researcher who oversees multiple aspects of the study | ResearchStudy.principalInvestigator
protocol | reference | ResearchStudy-protocol | Steps followed in executing study | ResearchStudy.protocol
site | reference | ResearchStudy-site | Facility where study activities are conducted | ResearchStudy.site
sponsor | reference | ResearchStudy-sponsor | Organization that initiates and is legally responsible for the study | ResearchStudy.sponsor
status | token | ResearchStudy-status | active | administratively-completed | approved | closed-to-accrual | closed-to-accrual-and-intervention | completed | disapproved | in-review | temporarily-closed-to-accrual | temporarily-closed-to-accrual-and-intervention | withdrawn | ResearchStudy.status
title | string | ResearchStudy-title | Name for this study | ResearchStudy.title
ResearchSubject TU
date | date | ResearchSubject-date | Start and end of participation | ResearchSubject.period
identifier | token | ResearchSubject-identifier | Business Identifier for research subject in a study | ResearchSubject.identifier
individual | reference | ResearchSubject-individual | Who is part of study | ResearchSubject.individual
patient | reference | ResearchSubject-patient | Who is part of study | ResearchSubject.individual
status | token | ResearchSubject-status | candidate | eligible | follow-up | ineligible | not-registered | off-study | on-study | on-study-intervention | on-study-observation | pending-on-study | potential-candidate | screening | withdrawn | ResearchSubject.status
study | reference | ResearchSubject-study | Study subject is part of | ResearchSubject.study
RiskAssessment TU
condition | reference | RiskAssessment-condition | Condition assessed | RiskAssessment.condition
date | date | clinical-date | When was assessment made? | (RiskAssessment.occurrence as dateTime)
encounter | reference | clinical-encounter | Where was assessment performed? | RiskAssessment.encounter
identifier | token | clinical-identifier | Unique identifier for the assessment | RiskAssessment.identifier
method | token | RiskAssessment-method | Evaluation mechanism | RiskAssessment.method
patient | reference | clinical-patient | Who/what does assessment apply to? | RiskAssessment.subject.where(resolve() is Patient)
performer | reference | RiskAssessment-performer | Who did assessment? | RiskAssessment.performer
probability | number | RiskAssessment-probability | Likelihood of specified outcome | RiskAssessment.prediction.probability
risk | token | RiskAssessment-risk | Likelihood of specified outcome as a qualitative value | RiskAssessment.prediction.qualitativeRisk
subject | reference | RiskAssessment-subject | Who/what does assessment apply to? | RiskAssessment.subject
RiskEvidenceSynthesis TU
context | token | RiskEvidenceSynthesis-context | A use context assigned to the risk evidence synthesis | (RiskEvidenceSynthesis.useContext.value as CodeableConcept)
context-quantity | quantity | RiskEvidenceSynthesis-context-quantity | A quantity- or range-valued use context assigned to the risk evidence synthesis | (RiskEvidenceSynthesis.useContext.value as Quantity) | (RiskEvidenceSynthesis.useContext.value as Range)
context-type | token | RiskEvidenceSynthesis-context-type | A type of use context assigned to the risk evidence synthesis | RiskEvidenceSynthesis.useContext.code
context-type-quantity | composite | RiskEvidenceSynthesis-context-type-quantity | A use context type and quantity- or range-based value assigned to the risk evidence synthesis | RiskEvidenceSynthesis.useContext
context-type-value | composite | RiskEvidenceSynthesis-context-type-value | A use context type and value assigned to the risk evidence synthesis | RiskEvidenceSynthesis.useContext
date | date | RiskEvidenceSynthesis-date | The risk evidence synthesis publication date | RiskEvidenceSynthesis.date
description | string | RiskEvidenceSynthesis-description | The description of the risk evidence synthesis | RiskEvidenceSynthesis.description
effective | date | RiskEvidenceSynthesis-effective | The time during which the risk evidence synthesis is intended to be in use | RiskEvidenceSynthesis.effectivePeriod
identifier | token | RiskEvidenceSynthesis-identifier | External identifier for the risk evidence synthesis | RiskEvidenceSynthesis.identifier

jurisdiction | token | RiskEvidenceSynthesis-jurisdiction | Intended jurisdiction for the risk evidence synthesis | RiskEvidenceSynthesis.jurisdiction
name | string | RiskEvidenceSynthesis-name | Computationally friendly name of the risk evidence synthesis | RiskEvidenceSynthesis.name
publisher | string | RiskEvidenceSynthesis-publisher | Name of the publisher of the risk evidence synthesis | RiskEvidenceSynthesis.publisher
status | token | RiskEvidenceSynthesis-status | The current status of the risk evidence synthesis | RiskEvidenceSynthesis.status
title | string | RiskEvidenceSynthesis-title | The human-friendly name of the risk evidence synthesis | RiskEvidenceSynthesis.title
url | uri | RiskEvidenceSynthesis-url | The uri that identifies the risk evidence synthesis | RiskEvidenceSynthesis.url
version | token | RiskEvidenceSynthesis-version | The business version of the risk evidence synthesis | RiskEvidenceSynthesis.version
Schedule TU
active | token | Schedule-active | Is the schedule in active use | Schedule.active
actor | reference | Schedule-actor | The individual(HealthcareService, Practitioner, Location, ...) to find a Schedule for | Schedule.actor
date | date | Schedule-date | Search for Schedule resources that have a period that contains this date specified | Schedule.planningHorizon
identifier | token | Schedule-identifier | A Schedule Identifier | Schedule.identifier
service-category | token | Schedule-service-category | High-level category | Schedule.serviceCategory
service-type | token | Schedule-service-type | The type of appointments that can be booked into associated slot(s) | Schedule.serviceType
specialty | token | Schedule-specialty | Type of specialty needed | Schedule.specialty
SearchParameter TU
base | token | SearchParameter-base | The resource type(s) this search parameter applies to | SearchParameter.base
code | token | SearchParameter-code | Code used in URL | SearchParameter.code
component | reference | SearchParameter-component | Defines how the part works | SearchParameter.component.definition
context | token | conformance-context | A use context assigned to the search parameter | (SearchParameter.useContext.value as CodeableConcept)
context-quantity | quantity | conformance-context-quantity | A quantity- or range-valued use context assigned to the search parameter | (SearchParameter.useContext.value as Quantity) | (SearchParameter.useContext.value as Range)
context-type | token | conformance-context-type | A type of use context assigned to the search parameter | SearchParameter.useContext.code
context-type-quantity | composite | conformance-context-type-quantity | A use context type and quantity- or range-based value assigned to the search parameter | SearchParameter.useContext
context-type-value | composite | conformance-context-type-value | A use context type and value assigned to the search parameter | SearchParameter.useContext
date | date | conformance-date | The search parameter publication date | SearchParameter.date
derived-from | reference | SearchParameter-derived-from | Original definition for the search parameter | SearchParameter.derivedFrom
description | string | conformance-description | The description of the search parameter | SearchParameter.description
jurisdiction | token | conformance-jurisdiction | Intended jurisdiction for the search parameter | SearchParameter.jurisdiction
name | string | conformance-name | Computationally friendly name of the search parameter | SearchParameter.name
publisher | string | conformance-publisher | Name of the publisher of the search parameter | SearchParameter.publisher
status | token | conformance-status | The current status of the search parameter | SearchParameter.status
target | token | SearchParameter-target | Types of resource (if a resource reference) | SearchParameter.target
type | token | SearchParameter-type | number | date | string | token | reference | composite | quantity | uri | special | SearchParameter.type
url | uri | conformance-url | The uri that identifies the search parameter | SearchParameter.url
version | token | conformance-version | The business version of the search parameter | SearchParameter.version
ServiceRequest TU
authored | date | ServiceRequest-authored | Date request signed | ServiceRequest.authoredOn
based-on | reference | ServiceRequest-based-on | What request fulfills | ServiceRequest.basedOn
body-site | token | ServiceRequest-body-site | Where procedure is going to be done | ServiceRequest.bodySite
category | token | ServiceRequest-category | Classification of service | ServiceRequest.category
code | token | clinical-code | What is being requested/ordered | ServiceRequest.code
encounter | reference | clinical-encounter | An encounter in which this request is made | ServiceRequest.encounter
identifier | token | clinical-identifier | Identifiers assigned to this order | ServiceRequest.identifier
instantiates-canonical | reference | ServiceRequest-instantiates-canonical | Instantiates FHIR protocol or definition | ServiceRequest.instantiatesCanonical
instantiates-uri | uri | ServiceRequest-instantiates-uri | Instantiates external protocol or definition | ServiceRequest.instantiatesUri
intent | token | ServiceRequest-intent | proposal | plan | directive | order | original-order | reflex-order | filler-order | instance-order | option | ServiceRequest.intent
occurrence | date | ServiceRequest-occurrence | When service should occur | ServiceRequest.occurrence
patient | reference | clinical-patient | Search by subject - a patient | ServiceRequest.subject.where(resolve() is Patient)
performer | reference | ServiceRequest-performer | Requested performer | ServiceRequest.performer
performer-type | token | ServiceRequest-performer-type | Performer role | ServiceRequest.performerType
priority | token | ServiceRequest-priority | routine | urgent | asap | stat | ServiceRequest.priority
replaces | reference | ServiceRequest-replaces | What request replaces | ServiceRequest.replaces
requester | reference | ServiceRequest-requester | Who/what is requesting service | ServiceRequest.requester
requisition | token | ServiceRequest-requisition | Composite Request ID | ServiceRequest.requisition
specimen | reference | ServiceRequest-specimen | Specimen to be tested | ServiceRequest.specimen
status | token | ServiceRequest-status | draft | active | on-hold | revoked | completed | entered-in-error | unknown | ServiceRequest.status
subject | reference | ServiceRequest-subject | Search by subject | ServiceRequest.subject

Slot TU
appointment-type | token | Slot-appointment-type | The style of appointment or patient that may be booked in the slot (not service type) | Slot.appointmentType
identifier | token | Slot-identifier | A Slot Identifier | Slot.identifier
schedule | reference | Slot-schedule | The Schedule Resource that we are seeking a slot within | Slot.schedule
service-category | token | Slot-service-category | A broad categorization of the service that is to be performed during this appointment | Slot.serviceCategory
service-type | token | Slot-service-type | The type of appointments that can be booked into the slot | Slot.serviceType
specialty | token | Slot-specialty | The specialty of a practitioner that would be required to perform the service requested in this appointment | Slot.specialty
start | date | Slot-start | Appointment date/time. | Slot.start
status | token | Slot-status | The free/busy status of the appointment | Slot.status
Specimen TU
accession | token | Specimen-accession | The accession number associated with the specimen | Specimen.accessionIdentifier
bodysite | token | Specimen-bodysite | The code for the body site from where the specimen originated | Specimen.collection.bodySite
collected | date | Specimen-collected | The date the specimen was collected | Specimen.collection.collected
collector | reference | Specimen-collector | Who collected the specimen | Specimen.collection.collector
container | token | Specimen-container | The kind of specimen container | Specimen.container.type
container-id | token | Specimen-container-id | The unique identifier associated with the specimen container | Specimen.container.identifier
identifier | token | Specimen-identifier | The unique identifier associated with the specimen | Specimen.identifier
parent | reference | Specimen-parent | The parent of the specimen | Specimen.parent
patient | reference | Specimen-patient | The patient the specimen comes from | Specimen.subject.where(resolve() is Patient)
status | token | Specimen-status | available | unavailable | unsatisfactory | entered-in-error | Specimen.status
subject | reference | Specimen-subject | The subject of the specimen | Specimen.subject
type | token | Specimen-type | The specimen type | Specimen.type
SpecimenDefinition TU
container | token | SpecimenDefinition-container | The type of specimen conditioned in container expected by the lab | SpecimenDefinition.typeTested.container.type
identifier | token | SpecimenDefinition-identifier | The unique identifier associated with the specimen | SpecimenDefinition.identifier
type | token | SpecimenDefinition-type | The type of collected specimen | SpecimenDefinition.typeCollected
StructureDefinition N
abstract TU | token | StructureDefinition-abstract | Whether the structure is abstract | StructureDefinition.abstract
base TU | reference | StructureDefinition-base | Definition that this type is constrained/specialized from | StructureDefinition.baseDefinition
base-path TU | token | StructureDefinition-base-path | Path that identifies the base element | StructureDefinition.snapshot.element.base.path | StructureDefinition.differential.element.base.path
context TU | token | conformance-context | A use context assigned to the structure definition | (StructureDefinition.useContext.value as CodeableConcept)
context-quantity TU | quantity | conformance-context-quantity | A quantity- or range-valued use context assigned to the structure definition | (StructureDefinition.useContext.value as Quantity) | (StructureDefinition.useContext.value as Range)
context-type TU | token | conformance-context-type | A type of use context assigned to the structure definition | StructureDefinition.useContext.code
context-type-quantity TU | composite | conformance-context-type-quantity | A use context type and quantity- or range-based value assigned to the structure definition | StructureDefinition.useContext
context-type-value TU | composite | conformance-context-type-value | A use context type and value assigned to the structure definition | StructureDefinition.useContext
date TU | date | conformance-date | The structure definition publication date | StructureDefinition.date
derivation TU | token | StructureDefinition-derivation | specialization | constraint - How relates to base definition | StructureDefinition.derivation
description TU | string | conformance-description | The description of the structure definition | StructureDefinition.description
experimental TU | token | StructureDefinition-experimental | For testing purposes, not real usage | StructureDefinition.experimental
ext-context TU | token | StructureDefinition-ext-context | The system is the URL for the context-type: e.g. http://hl7.org/fhir/extension-context-type#element|CodeableConcept.text | StructureDefinition.context.type
identifier TU | token | conformance-identifier | External identifier for the structure definition | StructureDefinition.identifier
jurisdiction TU | token | conformance-jurisdiction | Intended jurisdiction for the structure definition | StructureDefinition.jurisdiction
keyword TU | token | StructureDefinition-keyword | A code for the StructureDefinition | StructureDefinition.keyword
kind TU | token | StructureDefinition-kind | primitive-type | complex-type | resource | logical | StructureDefinition.kind
name TU | string | conformance-name | Computationally friendly name of the structure definition | StructureDefinition.name
path TU | token | StructureDefinition-path | A path that is constrained in the StructureDefinition | StructureDefinition.snapshot.element.path | StructureDefinition.differential.element.path
publisher TU | string | conformance-publisher | Name of the publisher of the structure definition | StructureDefinition.publisher
status TU | token | conformance-status | The current status of the structure definition | StructureDefinition.status
title TU | string | conformance-title | The human-friendly name of the structure definition | StructureDefinition.title
type TU | uri | StructureDefinition-type | Type defined or constrained by this structure | StructureDefinition.type
url TU | uri | conformance-url | The uri that identifies the structure definition | StructureDefinition.url
valueset TU | reference | StructureDefinition-valueset | A vocabulary binding reference | StructureDefinition.snapshot.element.binding.valueSet
version TU | token | conformance-version | The business version of the structure definition | StructureDefinition.version
StructureMap TU

context | token | conformance-context | A use context assigned to the structure map | (StructureMap.useContext.value as CodeableConcept)
context-quantity | quantity | conformance-context-quantity | A quantity- or range-valued use context assigned to the structure map | (StructureMap.useContext.value as Quantity) | (StructureMap.useContext.value as Range)
context-type | token | conformance-context-type | A type of use context assigned to the structure map | StructureMap.useContext.code
context-type-quantity | composite | conformance-context-type-quantity | A use context type and quantity- or range-based value assigned to the structure map | StructureMap.useContext
context-type-value | composite | conformance-context-type-value | A use context type and value assigned to the structure map | StructureMap.useContext
date | date | conformance-date | The structure map publication date | StructureMap.date
description | string | conformance-description | The description of the structure map | StructureMap.description
identifier | token | conformance-identifier | External identifier for the structure map | StructureMap.identifier
jurisdiction | token | conformance-jurisdiction | Intended jurisdiction for the structure map | StructureMap.jurisdiction
name | string | conformance-name | Computationally friendly name of the structure map | StructureMap.name
publisher | string | conformance-publisher | Name of the publisher of the structure map | StructureMap.publisher
status | token | conformance-status | The current status of the structure map | StructureMap.status
title | string | conformance-title | The human-friendly name of the structure map | StructureMap.title
url | uri | conformance-url | The uri that identifies the structure map | StructureMap.url
version | token | conformance-version | The business version of the structure map | StructureMap.version
Subscription TU
contact | token | Subscription-contact | Contact details for the subscription | Subscription.contact
criteria | string | Subscription-criteria | The search rules used to determine when to send a notification | Subscription.criteria
payload | token | Subscription-payload | The mime-type of the notification payload | Subscription.channel.payload
status | token | Subscription-status | The current state of the subscription | Subscription.status
type | token | Subscription-type | The type of channel for the sent notifications | Subscription.channel.type
url | uri | Subscription-url | The uri that will receive the notifications | Subscription.channel.endpoint
Substance TU
category | token | Substance-category | The category of the substance | Substance.category
code | token | Substance-code | The code of the substance or ingredient | Substance.code | (Substance.ingredient.substance as CodeableConcept)
container-identifier | token | Substance-container-identifier | Identifier of the package/container | Substance.instance.identifier
expiry | date | Substance-expiry | Expiry date of package or container of substance | Substance.instance.expiry
identifier | token | Substance-identifier | Unique identifier for the substance | Substance.identifier
quantity | quantity | Substance-quantity | Amount of substance in the package | Substance.instance.quantity
status | token | Substance-status | active | inactive | entered-in-error | Substance.status
substance-reference | reference | Substance-substance-reference | A component of the substance | (Substance.ingredient.substance as Reference)
SubstanceSpecification TU
code | token | SubstanceSpecification-code | The specific code | SubstanceSpecification.code.code
SupplyDelivery TU
identifier | token | clinical-identifier | External identifier | SupplyDelivery.identifier
patient | reference | clinical-patient | Patient for whom the item is supplied | SupplyDelivery.patient
receiver | reference | SupplyDelivery-receiver | Who collected the Supply | SupplyDelivery.receiver
status | token | SupplyDelivery-status | in-progress | completed | abandoned | entered-in-error | SupplyDelivery.status
supplier | reference | SupplyDelivery-supplier | Dispenser | SupplyDelivery.supplier
SupplyRequest TU
category | token | SupplyRequest-category | The kind of supply (central, non-stock, etc.) | SupplyRequest.category
date | date | clinical-date | When the request was made | SupplyRequest.authoredOn
identifier | token | clinical-identifier | Business Identifier for SupplyRequest | SupplyRequest.identifier
requester | reference | SupplyRequest-requester | Individual making the request | SupplyRequest.requester
status | token | SupplyRequest-status | draft | active | suspended + | SupplyRequest.status
subject | reference | SupplyRequest-subject | The destination of the supply | SupplyRequest.deliverTo
supplier | reference | SupplyRequest-supplier | Who is intended to fulfill the request | SupplyRequest.supplier
Task TU
authored-on | date | Task-authored-on | Search by creation date | Task.authoredOn
based-on | reference | Task-based-on | Search by requests this task is based on | Task.basedOn
business-status | token | Task-business-status | Search by business status | Task.businessStatus
code | token | Task-code | Search by task code | Task.code
encounter | reference | Task-encounter | Search by encounter | Task.encounter
focus | reference | Task-focus | Search by task focus | Task.focus
group-identifier | token | Task-group-identifier | Search by group identifier | Task.groupIdentifier
identifier | token | Task-identifier | Search for a task instance by its business identifier | Task.identifier
intent | token | Task-intent | Search by task intent | Task.intent
modified | date | Task-modified | Search by last modification date | Task.lastModified
owner | reference | Task-owner | Search by task owner | Task.owner
part-of | reference | Task-part-of | Search by task this task is part of | Task.partOf
patient | reference | Task-patient | Search by patient | Task.for.where(resolve() is Patient)
performer | token | Task-performer | Search by recommended type of performer (e.g., Requester, Performer, Scheduler). | Task.performerType
period | date | Task-period | Search by period Task is/was underway | Task.executionPeriod
priority | token | Task-priority | Search by task priority | Task.priority
requester | reference | Task-requester | Search by task requester | Task.requester

status | token | Task-status | Search by task status | Task.status
subject | reference | Task-subject | Search by subject | Task.for
TerminologyCapabilities TU
context | token | conformance-context | A use context assigned to the terminology capabilities | (TerminologyCapabilities.useContext.value as CodeableConcept)
context-quantity | quantity | conformance-context-quantity | A quantity- or range-valued use context assigned to the terminology capabilities | (TerminologyCapabilities.useContext.value as Quantity) | (TerminologyCapabilities.useContext.value as Range)
context-type | token | conformance-context-type | A type of use context assigned to the terminology capabilities | TerminologyCapabilities.useContext.code
context-type-quantity | composite | conformance-context-type-quantity | A use context type and quantity- or range-based value assigned to the terminology capabilities | TerminologyCapabilities.useContext
context-type-value | composite | conformance-context-type-value | A use context type and value assigned to the terminology capabilities | TerminologyCapabilities.useContext
date | date | conformance-date | The terminology capabilities publication date | TerminologyCapabilities.date
description | string | conformance-description | The description of the terminology capabilities | TerminologyCapabilities.description
jurisdiction | token | conformance-jurisdiction | Intended jurisdiction for the terminology capabilities | TerminologyCapabilities.jurisdiction
name | string | conformance-name | Computationally friendly name of the terminology capabilities | TerminologyCapabilities.name
publisher | string | conformance-publisher | Name of the publisher of the terminology capabilities | TerminologyCapabilities.publisher
status | token | conformance-status | The current status of the terminology capabilities | TerminologyCapabilities.status
title | string | conformance-title | The human-friendly name of the terminology capabilities | TerminologyCapabilities.title
url | uri | conformance-url | The uri that identifies the terminology capabilities | TerminologyCapabilities.url
version | token | conformance-version | The business version of the terminology capabilities | TerminologyCapabilities.version
TestReport TU
identifier | token | TestReport-identifier | An external identifier for the test report | TestReport.identifier
issued | date | TestReport-issued | The test report generation date | TestReport.issued
participant | uri | TestReport-participant | The reference to a participant in the test execution | TestReport.participant.uri
result | token | TestReport-result | The result disposition of the test execution | TestReport.result
tester | string | TestReport-tester | The name of the testing organization | TestReport.tester
testscript | reference | TestReport-testscript | The test script executed to produce this report | TestReport.testScript
TestScript TU
context | token | TestScript-context | A use context assigned to the test script | (TestScript.useContext.value as CodeableConcept)
context-quantity | quantity | TestScript-context-quantity | A quantity- or range-valued use context assigned to the test script | (TestScript.useContext.value as Quantity) | (TestScript.useContext.value as Range)
context-type | token | TestScript-context-type | A type of use context assigned to the test script | TestScript.useContext.code
context-type-quantity | composite | TestScript-context-type-quantity | A use context type and quantity- or range-based value assigned to the test script | TestScript.useContext
context-type-value | composite | TestScript-context-type-value | A use context type and value assigned to the test script | TestScript.useContext
date | date | TestScript-date | The test script publication date | TestScript.date
description | string | TestScript-description | The description of the test script | TestScript.description
identifier | token | TestScript-identifier | External identifier for the test script | TestScript.identifier
jurisdiction | token | TestScript-jurisdiction | Intended jurisdiction for the test script | TestScript.jurisdiction
name | string | TestScript-name | Computationally friendly name of the test script | TestScript.name
publisher | string | TestScript-publisher | Name of the publisher of the test script | TestScript.publisher
status | token | TestScript-status | The current status of the test script | TestScript.status
testscript-capability | string | TestScript-testscript-capability | TestScript required and validated capability | TestScript.metadata.capability.description
title | string | TestScript-title | The human-friendly name of the test script | TestScript.title
url | uri | TestScript-url | The uri that identifies the test script | TestScript.url
version | token | TestScript-version | The business version of the test script | TestScript.version
ValueSet N
code TU | token | ValueSet-code | This special parameter searches for codes in the value set. See additional notes on the ValueSet resource | ValueSet.expansion.contains.code | ValueSet.compose.include.concept.code
context TU | token | conformance-context | A use context assigned to the value set | (ValueSet.useContext.value as CodeableConcept)
context-quantity TU | quantity | conformance-context-quantity | A quantity- or range-valued use context assigned to the value set | (ValueSet.useContext.value as Quantity) | (ValueSet.useContext.value as Range)
context-type TU | token | conformance-context-type | A type of use context assigned to the value set | ValueSet.useContext.code
context-type-quantity TU | composite | conformance-context-type-quantity | A use context type and quantity- or range-based value assigned to the value set | ValueSet.useContext
context-type-value TU | composite | conformance-context-type-value | A use context type and value assigned to the value set | ValueSet.useContext
date TU | date | conformance-date | The value set publication date | ValueSet.date
description TU | string | conformance-description | The description of the value set | ValueSet.description
expansion TU | uri | ValueSet-expansion | Identifies the value set expansion (business identifier) | ValueSet.expansion.identifier
identifier TU | token | conformance-identifier | External identifier for the value set | ValueSet.identifier
jurisdiction TU | token | conformance-jurisdiction | Intended jurisdiction for the value set | ValueSet.jurisdiction
name TU | string | conformance-name | Computationally friendly name of the value set | ValueSet.name
publisher TU | string | conformance-publisher | Name of the publisher of the value set | ValueSet.publisher
reference TU | uri | ValueSet-reference | A code system included or excluded in the value set or an imported value set | ValueSet.compose.include.system

status TU | token | conformance-status | The current status of the value set | ValueSet.status
title TU | string | conformance-title | The human-friendly name of the value set | ValueSet.title
url TU | uri | conformance-url | The uri that identifies the value set | ValueSet.url
version TU | token | conformance-version | The business version of the value set | ValueSet.version
VerificationResult TU
target | reference | VerificationResult-target | A resource that was validated | VerificationResult.target
VisionPrescription TU
datewritten | date | VisionPrescription-datewritten | Return prescriptions written on this date | VisionPrescription.dateWritten
encounter | reference | clinical-encounter | Return prescriptions with this encounter identifier | VisionPrescription.encounter
identifier | token | clinical-identifier | Return prescriptions with this external identifier | VisionPrescription.identifier
patient | reference | clinical-patient | The identity of a patient to list dispenses for | VisionPrescription.patient
prescriber | reference | VisionPrescription-prescriber | Who authorized the vision prescription | VisionPrescription.prescriber
status | token | VisionPrescription-status | The status of the vision prescription | VisionPrescription.status

3.1.5 FHIRPath Patch

URL: https://hl7.org/fhir/R4/fhirpatch.html

3.1.5 FHIRPath Patch

FHIR Infrastructure Work Group | Maturity Level: 2 | Standards Status: Trial Use

This page documents how the Parameters resource is used to define a set of FHIRPath based patch operations.

3.1.5.1 Conceptual FHIRPath Operations

This table documents the FHIRPath operations that may be used to specify changes to a resource:

Type | Path | Name | Value | Index | source | destination | Details
add | Path at which to add the content | Name of the property to add | Data type to add at nominated place | | | | The content will be appended to the element identified in the path, using the name specified. Add can used for non-repeating elements as long as they do not already exist
insert | Path of the collection in which to insert the content | | value (Data type) to add at nominated place | index at which to insert | | | The content will be inserted into the nominated list at the index specified (0 based). The index is mandatory and must be equal or less than the number of elements in the list. Note: add is easier than insert at the end of the list
delete | Path of the element to delete (if found) | | | | | | Only a single element can be deleted
replace | Path of the element to replace | | value (Data type) to replace it with
move | Path of the collection in which to move the content | | | | list index to move from | list index to move to | Move an element within a single list

There are a few base rules that apply for all operations:

• The FHIRPath statement must return a single element

• The FHIRPath statement SHALL NOT cross resources using the resolve() function (e.g. like Observation.subject.resolve().identifier). resolve() SHALL only be used to refer to contained resource within the resource being patched. Servers SHALL NOT allow patch operations to alter other resources than the nominated target, and SHOULD return an error if the patch operation tries

• The type of the value must be correct for the place at which it will be added/inserted. Servers SHALL return an error if the type is wrong

• Servers SHALL return an error if the outcome of the patch operation is a not a valid resource

• Except for the delete operation, it is an error if no element matches the specified path

• Operations are applied in order, with each operation applied to the result of the previous patch operation

3.1.5.2 Parameters Format

The FHIRPath patch operations are encoded in a Parameters resource as follows:

• Each operation is a Parameter named "operation"

• Each operation has a series of parts, using the following parameter names from the table above with these types:

Parameter | Type
type | code
path | string
name | string
value | *
index | integer
source | integer
destination | integer

Here is an example of adding an element:

```
<Parameters xmlns="http://hl7.org/fhir">
  <parameter>
    <name value="operation"/>
    <part>
      <name value="type"/>
```

```
      <valueCode value="add"/>
    </part>
    <part>
      <name value="path"/>
      <valueString value="Patient"/>
    </part>
    <part>
      <name value="name"/>
      <valueString value="birthDate"/>
    </part>
    <part>
      <name value="value"/>
      <valueDate value="1930-01-01"/>
    </part>
  </parameter>
</Parameters>
```

3.1.5.2.1 Anonymous Types

Only some named data types (see the list) are allowed to be used directly in parameters. In order to add or insert other kinds of types - including anonymous elements (e.g. Observation.component, Timing.repeat), the content is defined by defining the name as described above, and instead of providing a value, a set of parts that are values are provided. Here is an example:

```
<Parameters xmlns="http://hl7.org/fhir">
  <parameter>
    <name value="operation"/>
    <part>
      <name value="type"/>
      <valueCode value="add"/>
    </part>
    <part>
      <name value="path"/>
      <valueString value="Patient"/>
    </part>
    <part>
      <name value="name"/>
      <valueString value="contact"/>
    </part>
    <part>
      <name value="value"/>
      <part>
        <name value="name"/>
        <valueHumanName>
          <text value="a name"/>
        </valueHumanName>
      </part>
    </part>
  </parameter>
</Parameters>
```

This pattern repeats as deep as necessary.

There is a set of test cases for implementers as part of the Test Case Downloads.

3.1.6 Asynchronous Request Pattern

URL: https://hl7.org/fhir/R4/async.html

3.1.6 Asynchronous Request Pattern

FHIR Infrastructure Work Group | Maturity Level: 2 | Standards Status: Draft

Implementation Note: the FHIR Asynchronous API and the $export Operation are under active development:

- Visit the Draft FHIR Bulk Data Repository for the most recent draft documentation and open issues
- Participate in design discussions at chat.fhir.org

3.1.6.1 Use Case

All of the interactions defined in the RESTful API are described as synchronous operations - that is, the client makes a query and waits for the server to respond with the outcome in the HTTP response. This pattern is not suitable once significant server side processing becomes necessary or when substantial amounts of data must be returned.

A good example of this is the $export operation , where simple searches may result in large amounts of data.

The asynchronous request pattern, based on rfc 7240 , caters to this use case and is applicable for all the Defined Interactions and for Operations, although for many of these uses it brings no benefit. Servers may choose which interactions the pattern should be supported on (if at all), and servers may choose to only support some of the operations using the asynchronous pattern.

3.1.6.2 Kick Off Request

The request will have whatever URL and other parameters would normally apply, but must include the headers described below.

GET [FHIR API Request]

3.1.6.2.0.1 Headers

Prefer (required)

Specifies whether the response is immediate or asynchronous. Setting this to respond-async triggers the async pattern.

Accept (required)

Specifies the format of the optional OperationOutcome response to the kick-off request. Any of the Serialization Format Representations are supported.

3.1.6.2.0.2 Query String Parameters

_outputFormat (string, optional, defaults to application/fhir+ndjson)

The format for the generated bulk data files. Currently, ndjson must be supported, though servers may choose to also support other output formats. Servers should support the full content type of application/fhir+ndjson as well as abbreviated representations including application/ndjson and ndjson.

3.1.6.2.0.3 Response - Success

• HTTP Status Code of 202 Accepted

• Content-Location header with a url for subsequent status requests

• Optionally a FHIR OperationOutcome in the body

3.1.6.2.0.4 Response - Error (e.g. unsupported search parameter)

• HTTP Status Code of 4XX or 5XX

• Optionally a FHIR OperationOutcome in the body

3.1.6.3 Bulk Data Delete Request:

After a bulk data request has been kicked-off, clients can send a delete request to the url provided in the Content-Location header to cancel the request.

3.1.6.3.0.1 Endpoint

DELETE [polling content location]

3.1.6.3.0.2 Response - Success

• HTTP Status Code of 202 Accepted

• Optionally a FHIR OperationOutcome in the body

3.1.6.3.0.3 Response - Error Status

• HTTP status code of 4XX or 5XX

• Optionally a FHIR OperationOutcome in the body

3.1.6.4 Bulk Data Status Request:

After a bulk data request has been kicked-off, clients can poll the url provided in the Content-Location header to obtain the status of the request.

Note: Clients should follow an exponential backoff approach when polling for status. Servers may supply a Retry-After header with a http date or a delay time in seconds. When provided, clients should use this information to inform the timing of future polling requests.

Note: The Accept header for this request should be application/json. In the case that errors prevent the export from completing, the response will contain a JSON-encoded FHIR OperationOutcome resource.

3.1.6.4.0.1 Endpoint

GET [polling content location]

3.1.6.4.0.2 Response - In-Progress Status

• HTTP Status Code of 202 Accepted

• Optionally an X-Progress header with a text description of the status of the request that's less than 100 characters. The format of this description is at the server's discretion and may be a percentage complete value or a more general status such as "in progress". Clients can try to parse this value, display it to the user, or log it.

3.1.6.4.0.3 Response - Error Status

• HTTP status code of 5XX

• Optionally a JSON FHIR OperationOutcome in the body

• Even if some resources cannot successfully be exported, the overall export operation may still succeed. In this case, the Response.error array of the completion Response must be populated (see below) with one or

more files in ndjson format containing OperationOutcome resources to indicate what went wrong.

3.1.6.4.0.4 Response - Complete Status

HTTP status of 200 OK

Content-Type header of application/json

Optionally an Expires header indicating when the files listed will no longer be available.

A body containing a json object providing metadata and links to the generated bulk data files.

Required Fields:

transactionTime - a FHIR instant type that indicates the server's time when the query is run. No resources that have a modified data after this instant should be in the response.

request - the full url of the original bulk data kick-off request

requiresAccessToken - boolean value indicating whether downloading the generated files will require an authentication token. Note: This may be false in the case of signed S3 urls or an internal file server within an organization's firewall.

output - array of bulk data file items with one entry for each generated file. Note: If no data is returned from the kick-off request, the server should return an empty array.

error - array of error file items following the same structure as the output array. Note: If no errors occurred, the server should return an empty array. Note: Only the OperationOutcome resource type is currently supported, so a server will generate ndjson files where each row is an OperationOutcome resource.

Each file item should contain the following fields:

type - the FHIR resource type that is contained in the file. Note: Each file may only contain resources of one type, but a server may create more than one file for each resources type returned. The number of resources contained in a file is may vary between servers. If no data is found for a resource, the server should not return an output item for it in the response.

url - the path to the file. The format of the file should reflect that requested in the _outputFormat parameter of the initial kick-off request.

Each file item may optionally contain the following field:

count - the number of resources in the file, represented as a JSON number.

Example response body:

```
{
  "transactionTime": "[instant]",
  "request" : "[base]/Patient/$export?_type=Patient,Observation",
  "requiresAccessToken" : true,
  "output" : [{
    "type" : "Patient",
    "url" : "http://serverpath2/patient_file_1.ndjson"
  },{
    "type" : "Patient",
    "url" : "http://serverpath2/patient_file_2.ndjson"
  },{
    "type" : "Observation",
    "url" : "http://serverpath2/observation_file_1.ndjson"
  }],
  "error" : [{
    "type" : "OperationOutcome",
    "url" : "http://serverpath2/err_file_1.ndjson"
  }]
}
```

3.1.6.5 File Requests:

Using the urls supplied in the completed status request body, clients can download the generated bulk data files (one or more per resource type). Note: These files may be served by a file server rather than a FHIR specific server. Also, if the requiresAccessToken field in the status body is set to true the request must include a valid access token in the Authorization header in these requests (i.e., Authorization: Bearer {{token}}).

3.1.6.5.0.1 Endpoint

GET [url from status request output field]

3.1.6.5.0.2 Headers

• Accept (optional, defaults to application/fhir+ndjson)

Specifies the format of the file being returned. Optional, but currently only application/fhir+ndjson is supported.

3.1.6.5.0.3 Response - Success

• HTTP status of 200 OK

• Content-Type header of application/fhir+ndjson

• Body of FHIR resources in newline delimited json - ndjson format

3.1.6.5.0.4 Response - Error

• HTTP Status Code of 4XX or 5XX

3.2 Extended Operations on the RESTful API

URL: https://hl7.org/fhir/R4/operations.html

3.2.0 Extended Operations on the RESTful API

FHIR Infrastructure Work Group | Maturity Level: Normative | Standards Status: Normative

| This page has been approved as part of an ANSI standard. See the Infrastructure Package for further details.

The RESTful API defines a set of common interactions (read, update, search, etc.) performed on a repository of typed resources. These interactions follow the RESTful paradigm of managing state by Create/Read/Update/Delete actions on a set of identified resources. While this approach solves many use cases, there is some functionality that can be met more efficiently using an RPC-like paradigm, where named operations are performed with inputs and outputs (Execute).

Operations are appropriately used where:

• the server needs to play an active role in formulating the content of the response, not merely return existing information

• the intended purpose is to cause side effects such as the modification of existing resources or creation of new resources, or other changes out of scope of the RESTful interface (e.g. merging patient records)

• The task involves business rules to be enforced across multiple different resources and/or

• the task involves updating several resources in a coordinated manner (note: this can also be done by a Transaction, but an operation can be more focused on the task

This specification describes a lightweight operation framework that seamlessly extends the RESTful API. The framework covers both how to execute such an operation (this page) and how to define an operation.

Operations have the following general properties:

• Each operation has a name

• Each operation has a list of 'in' and 'out' parameters

• Parameters are either resources, data types, or search parameters

• Operations are subject to the same security constraints and requirements as the RESTful API

• The URIs for the operation end-points are based on the existing RESTful API address scheme

• Operations may make use of all types of resources existing in the repository

• Operations may be performed on a specific resource, a resource type, or a whole system

3.2.0.1 Executing an Operation

Operations are executed using a URL derived from the FHIR endpoint, where the name of the operation is prefixed by a "dollar sign" ('$') character. For example:

```
POST http://fhir.someserver.org/fhir/Patient/1/$everything
```

When an operation has affectsState = false, and the parameters are all primitive data types with no extensions (as is the case with the example above), it may be invoked using GET as well. (Note: A HEAD request can also be used - see Support for HEAD).

Operations can be invoked on four types of FHIR endpoints:

• The "base" FHIR service endpoint (e.g. http://fhir.someserver.org/fhir): These are operations that operate on the full scale of the server. For example, "return me all extensions known by this server"

• A Resource type (e.g. http://fhir.someserver.org/fhir/Patient): These are operations that operate across all instances of a given resource type

• A Resource instance (e.g. http://fhir.someserver.org/fhir/Patient/1): These are operations that involve only a single instance of a Resource, like the $everything operation above does

The body of the invocation contains a special infrastructure resource called Parameters, which represents a collection of named parameters as <key,value> pairs, where the value may be any primitive or complex datatype or even a full Resource. It may also include strings formatted as search parameter types.

Upon completion, the operation returns another Parameters resource, containing one or more output parameters. This means that a FHIR operation can take a set of zero or more parameters in and return a set of zero or more result parameters out. Both the body of the POST and the returned result are always a Resource.

Operations may be invoked using a GET, with parameters as HTTP URL parameters, if:

• there are only simple input parameters - i.e. no complex datatypes like 'Identifier' or 'Reference', and

• and the operation does not affect the state of the server

If there is a single output parameter named 'return' then the response MAY be the resource that is the return value, with no Parameters resource. These kinds of usage are discussed further below.

If the response is a Bundle, the correct Bundle.type is 'collection', unless it has search semantics, such as matching resource counts, and page links (next etc).

3.2.0.1.1 Operations with no parameters

Executing operations without any parameters is a special case. For an operation that doesn't cause any state change, the operation is invoked in a straight forward fashion:

```
GET [base]/Composition/example/$document
```

For operations that call state changes, they must be invoked by a POST. There is no parameters resource in this case because a parameters resource cannot be empty. So the operation is invoked with a POST with an empty body:

```
POST [base]/Claim/example/$submit
Content-Length: 0
```

3.2.0.2 FHIR defined Operations

See the list of defined operations.

3.2.0.3 Implementation Defined Operations

Implementations are able to define their own operations in addition to those defined here. Name clashes between operations defined by different implementers can be resolved by the use of the server's Capability Statement.

Also, the definition of these or additional run time operations does not prevent the use of other kinds of operations that are not dependent on and/or not integrated with the RESTful API, provided that their addressing scheme does not clash with the scheme defined here.

3.2.0.4 Defining an Operation

Each Operation is defined by:

• A context for the Operation - system, resource type, or resource instance

• A name for the Operation

• A list of parameters along with their definitions

For each parameter, the following information is needed:

• Name - the name of the parameter. For implementer convenience, the name should be a valid token in common implementation languages

• Use - In | Out | Both

• Type - a data type or a Resource type

• Search Type - for string search parameters, what kind of search parameter they are (and what kind of modifiers they have)

• Profile - a StructureDefinition that describes additional restrictions about the parameter - only used if the parameter type is a resource or data type

• Documentation - a description of the parameter's use

• (Optional) Search Type - if the type is a string, and the parameter is being used like a search parameter, which kind of search type applies

Parameters may be nested into multi-part parameters. Each part has the same information as a parameter, except for use, which is taken from the parameter it is part of.

The resource Operation Definition is used to provide a computable definition of the Operation.

3.2.0.5 Extending an Operation

Implementations are able to extend an operation by defining new named parameters. Implementations can publish their own extended definitions using the Operation Definition resource, and this variant definition can use OperationDefinition.base to refer to the underlying definition.

Note that the FHIR specification will never define any parameter names starting with "x-".

3.2.0.6 Executing an Operation Synchronously

Operations are typically executed synchronously: a client sends a request to a server that includes the operation's in parameters and the server replies with the operation's out parameters.

The URL for an operation end-point depends on its context:

• system: the URL is [base]/$[name]

• resource type: the URL is [base]/[type]/$[name]

• resource instance: the URL is [base]/[type]/[id]/$[name]

3.2.0.6.1 Operation Request

An operation is generally invoked by performing an HTTP POST to the operation's end-point. The submitted content is the special Parameters format (the "in" parameters) - a list of named parameters. For an example, see the value set expansion request example. Note that when parameters have a search type, the search modifiers are available and are used on the parameter name in the Parameters resource (e.g. "code:in").

Note that the same arrangement as for the RESTful interface applies with respect to content types.

If all the parameters for the operation are primitive types and the operation has affectsState = false, the operation may be invoked by performing an HTTP GET operation where all of the values of the parameters are appended to the URL in the search portion of the URL (e.g. after the '?' character). Servers SHALL support this method of invocation. E.g.

```
GET [base]/ValueSet/$expand?url=http://hl7.org/fhir/ValueSet/body-site&filter=abdo
```

When using the HTTP GET operation, if there is a repeating parameter for the extended operation the values for that parameter are repeated by repeating the named parameter. E.g. Observation $stats statistic parameter

```
GET [base]/Observation/$stats?subject=Patient/123&code=55284-
    4&system=http://loinc.org&duration=1&statistic=average&statistic=min&statistic=max&statistic=count
```

If, when invoking the operation, there is exactly one input parameter of type Resource (irrespective of whether other possible parameters are defined), that the operation can also be executed by a POST with that resource as the body of the request (and no parameters on the url).

Servers MAY choose to support submission of the parameters represented in multi-part/form-data format as well, which can be useful when testing an operation using HTML forms.

3.2.0.6.2 Operation Response

If an operation succeeds, an HTTP Status success code is returned. This will usually be a 2xx code, though it may also be a 303 See Other. Other kinds of 3xx codes should be understood to indicate that the operation did not proceed, and the client will need to re-issue the operation if it can perform the redirection (e.g. may get redirected to an authentication step). User agents should note that servers may issue redirects, etc. to authenticate the client in response to an operation request. An HTTP status code of 4xx or 5xx indicates an error, and an OperationOutcome SHOULD be returned with details.

In general, an operation response uses the same Parameters format whether there is only one or there are multiple named out parameters.

If there is only one out parameter, which is a Resource with the parameter name "return" then the parameter format is not used, and the response is simply the resource itself.

The result of an operation is subject to content negotiation like any other interaction. Specifically, if the returned resource is a Binary, the response SHALL behave in the same manner as if a 'read' operation had been performed on the resource. I.e. The content will be returned as either a FHIR resourse with base64-encoded content or as a raw binary, depending on the Accept header specified when invoking the operation (see Serving Binary Resources using the RESTful API).

The resources that are returned by the operation may be retained and made available in the resource repository on the operation server. In that case, the server will provide the identity of the resource in the returned resources. When resources that are not persisted are returned in the response, they will have no id property.

3.2.0.7 Executing an Operation Asynchronously

Use the standard RESTful API Asynchronous pattern to execute operations asynchronously.

3.2.1 Operation Request Example

URL: https://hl7.org/fhir/R4/op-example-request.html

3.2.1 Operation Request Example

FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative

```
POST [base]/ValueSet/$expand HTTP/1.1
Content-Type: application/fhir+xml

<!--
   This is an example of an operation request for a value set
   expansion where the value set is submitted on the fly
-->
<Parameters xmlns="http://hl7.org/fhir">
  <parameter>
    <name value="filter"/>
    <valueString name="abdo"/>
  </parameter>
  <parameter>
    <name value="valueset"/>
    <resource>
      <ValueSet>
        <text>
```

```
          <status value="generated"/>
          <div xmlns="http://www.w3.org/1999/xhtml"><!-- Snipped for brevity --></div>
        </text>
        <identifier value="http://hl7.org/fhir/ValueSet/body-site"/>
        <name value="SNOMED CT Body Structures"/>
        <publisher value="FHIR Project team"/>
        <telecom>
          <system value="url"/>
          <value value="http://hl7.org/fhir"/>
        </telecom>
        <description value="This value set includes all the &quot;Clinical finding&quot; SNOMED CT codes (i.e. codes
         with an is-a relationship with 91723000: Anatomical structure)"/>
        <status value="draft"/>
        <compose>
          <include><!--   all the descendants of clinical finding, not include itself   -->
            <system value="http://snomed.info/sct"/>
            <filter><!--   todo: work this over. what this means is any concepts where they have is-a with 91723000.
               how should this be done?   -->
              <property value="concept"/>
              <op value="is-a"/>
              <value value="91723000"/>
            </filter>
          </include>
        </compose>
      </ValueSet>
    </resource>
  </parameter>
</Parameters>
```

3.3 FHIR Documents

URL: https://hl7.org/fhir/R4/documents.html
FHIR Infrastructure Work Group | Maturity Level: 3 | Standards Status: Trial Use

FHIR resources can be used to build documents that represent a composition: a coherent set of information that is a statement of healthcare information, including clinical observations and services. A document is an immutable set of resources with a fixed presentation that is authored and/or attested by humans, organizations and devices.

Documents built in this fashion may be exchanged between systems and persisted in document storage and management systems, including systems such as IHE XDS.

Applications claiming conformance to this framework claim to be conformant to "FHIR documents" (see Conformance).

FHIR documents may be 'clinical' (focused on patient healthcare information) but may also serve non-clinical purposes (e.g. FHIR Implementation guides, practice guidelines, patient handouts, etc.) HL7 will develop profiles in the future giving additional guidance on appropriate representation of clinical documents in general as well as specific types of clinical documents (e.g. Consolidated CDA).

Note that FHIR defines both this document format and a document reference resource. FHIR documents are for documents that are authored and assembled in FHIR, while the document reference resource is for general references to pre-existing documents.

• Example discharge summary: XML or JSON

3.3.1 Document Content

All documents have the same structure: a Bundle of resources of type "document" that has a Composition resource as the first resource in the bundle, followed by a series of other resources, referenced from the Composition resource, that provide supporting evidence for the document. The bundle gathers all the content of the document into a single XML or JSON document which may be signed and managed as required. The resources include both human readable and computer processable portions. In addition, the bundle may include CSS stylesheets , Provenance statements and a signature.

The composition resource is the foundation of the clinical document. It:

• provides identity and its purpose, and sets the context of the document

• carries key information such as the subject and author, and who attests to the document

• divides the document up into a series of sections, each with their own narrative

Resources referenced by the Composition as listed below SHALL be included in the bundle when the document is assembled:

• Composition.subject

• Composition.encounter

• Composition.author

• Composition.attester.party

• Composition.custodian

• Composition.event.detail

• Composition.section.author
• Composition.section.focus
• Composition.section.entry

Other resources that these referenced resources refer to may also be included in the bundle if the document construction system chooses to do so. Including these additional resources will make the document bigger but will save applications from needing to retrieve the linked resources if they need them while processing the document. Thus, whether these linked resources should be included or not depends on the implementation environment.

The document bundle SHALL include only:

• The Composition resource, and any resources directly or indirectly (e.g. recursively) referenced from it
• A Binary resource containing a stylesheet (as described below)
• Provenance Resources that have a target of Composition or another resource included in the document

There are two key identifiers in the document:

• The document identifier (mandatory). This is found in Bundle.identifier and is globally unique for this instance of the document, and is never re-used, including for other documents derived from the same composition
• The Composition identifier (optional). This is found in Composition.identifier, and is the same for all documents that are derived from this composition

The document has several dates in it:

• The document date (mandatory). This is found in Bundle.meta.lastUpdated and identifies when the document bundle was assembled from the underlying resources
• The Composition date (mandatory). This is found in Composition.date, which is when the author wrote the document logically
• The Attestation dates (optional). This is found in Composition.attester.time and is when the document was witnessed by the attesters. This would usually be at the same time as the composition date or afterwards
• The Composition last modified time (optional). This is found in Composition.meta.lastUpdated for the composition and is the last date of change of the composition. This must be >= the composition date

Document Bundles may be signed using digital signatures following the rules laid out in the digital signatures page. The signature SHOULD be provided by a listed attester of the document and the signature SHOULD contain a KeyInfo element that contains a KeyName element whose value is a URI that matches the fullUri for the matching attester resource.

Once assembled into a bundle, the document is immutable - its content can never be changed, and the document id can never be reused. Note that the document may be represented in either XML or JSON and interconverted between these or have its character encoding changed, all the while remaining the same document. However, the directly referenced content within the document and the presentation of the document cannot change substantially (such that it changes the clinical meaning of the content). Any additional documents derived from the same composition SHALL have a different document id.

3.3.1.1 Document Presentation

When the document is presented for human consumption, applications SHOULD present the collated narrative portions in order:

• The subject resource Narrative
• The Composition resource Narrative
• The section.text Narratives

The presentation of the document is called the 'attested content' of the document. Additional resources can be included in the bundle (e.g. resources referenced from the List that represent the section.content SHOULD be in the bundle, and other additional resources they reference can be included), but these (and any narrative) are not attested content. Specifically, the Composition.attester attests to the presented form of the document.

The Composition resource narrative should summarize the important parts of the document header that are required to establish clinical context for the document (other than the subject, which is displayed in its own right). To actually build the combined narrative, simply append all the narrative fragments together.

If the document is presented in a different order from that given above, it might not represent the original attested content. Implementation Guides may restrict document narrative and display behavior further.

The XML Tools reference implementation includes a XSLT transform that converts an XML document into browser-ready XHTML.

In addition to the basic style rules about Narratives, which must be followed, a document can reference or contain one or more stylesheets that contains additional styles that apply to the collated narrative. This is done by asserting stylesheet links on the feed:

```
<Bundle xmlns="http://hl7.org/fhir">
  <!-- metadata and type -->
  <link>
    <relation value="stylesheet"/>
    <url value="[uri]"/>
  </link>
</Bundle>
```

The url can be an absolute reference to a CSS stylesheet or a relative reference to a Binary resource that carries a CSS stylesheet. Stylesheet references can only refer to a CSS stylesheet - other forms of stylesheet are not acceptable.

Relative (internal) references SHOULD be used for stylesheets, because the viewer may be unable to resolve external content at the time of viewing, due to technical problems or local policy decisions.

Any stylesheet referenced or used SHALL NOT alter the presentation in such a way that it changes the clinical meaning of the content.

Unless otherwise agreed in local trading partner agreements, applications displaying the collated narrative SHOULD use the stylesheets specified by the document (see security note). Parties entering into a trading agreement to do otherwise should consider the implications this action will have on their long-term scope for document exchange very carefully. If the parties agree to use stylesheets that are not contained in the document, then it may be that they will never be able to share their documents safely in a more general context, such as a regional or national EHR or a global personal health record.

3.3.2 Document Profiles

Document profiles are used to describe documents for a particular purpose. Document profiles may make rules about:

• The content of the Composition resource, including

• The structure of the sections in the composition

• Which resources are to be included in the bundle along with the resources that are directly referenced in the Document resource

Applications should consider publishing Capability Statements that identify document types they support. Documents can identify a profile that they conform to by placing a profile identifier in the Bundle.meta.profile element - see Profile Tags for a discussion of the utility of this.

3.3.3 Document Handling Obligations

The authors/constructors and processors of Clinical Documents, whether human or software, have obligations that they must satisfy.

3.3.3.1 Author/Constructor Obligations

A document constructor is an application that creates a document. An author is a human, organization or device that uses the constructor to create a document. Between them, the constructor and the author may create new content resources and/or assemble already existing content resources while performing their tasks. They also have the following responsibilities:

• To assure that the document SHALL contain valid composition that conforms to the rules described here and that only links to other valid resources

• To assure that the content of document SHALL conform to any declared Profiles (see below).

• Ensure that the attesters are properly aware of the presentation of the document to which they are attesting

3.3.3.2 Processor Obligations

A document processor is an application and/or human user that receives documents and extracts data from them or makes decisions because of them. The documents may be received directly from a document constructor, accessed via a document management system or forwarded by a third party. The document processor is responsible for ensuring that received documents are processed and/or rendered in accordance to this specification. A document processor has the obligation to assure that the following rules are followed:

• When storing/transmitting a document, any method may be used as long as the bundled document can be (re-)assembled with sufficient integrity to validate a digital signature

• When presenting the narrative of the document, the rules described above SHALL be followed

• Resources or data from the document may be extracted for additional uses, but such data is no longer considered to be attested by the document author

• Wherever the data from the document is displayed to a user, there SHOULD always be a way for the user to access a presentation of the original document

In addition to these obligations, document receivers SHOULD carefully track the source of documents for new documents that supersede existing documents, particularly when the documents represent compositions that have been retracted. When documents have been replaced, they SHOULD either withdraw data extracted

from superseded documents or warn users when they view the document or data taken from it.

3.3.4 Document End-Points

There are several different RESTful end-points used when working with documents. The use of the various end-points can best be described by considering the consequences of posting to them:

End-Point | Type of Content | Description

[baseurl]/Bundle | Document Bundle | This works like a normal end-point for managing a type of resource, but it works with whole document bundles - i.e. a read operation returns a bundle, an update gets a bundle and a search returns a bundle of bundles. Note that if documents are POSTed using a create interaction the Bundle.id will change, but the Bundle.identifier will not. See Serving Bundles using the RESTful API for further comment

[baseurl]/Composition | Composition Resource | The normal end-point for managing composition resources. This can be used while building a document or after breaking a document up into its constituent resources or when using compositions separately from documents

[baseurl]/Binary | Document Bundle | Just store the entire document as a sequence of bytes and return exactly that sequence when requested. There is no way to find content in the /Binary end-point, so usually this would be associated with a Document Reference so that applications can find and process the document, though this is not required

[baseurl] (e.g. a transaction) | Document Bundle | Ignore the fact that the bundle is a document and process all of the resources that it contains as individual resources. Clients SHOULD not expect that a server that receives a document submitted using this method will be able to reassemble the document exactly. (Even if the server can reassemble the document (see below), the result cannot be expected to be in the same order, etc. Thus a document signature will very likely be invalid.). See Accepting other Bundle types for further details

Note: While these end-points are defined for use with document-related resources and document bundles, it is not necessary to use them. Documents may be transferred between systems using any method desired. In addition, servers and/or specifications may define additional operations for handling documents beyond the options described above.

3.3.4.1 Generating a Document

A client can ask a server to generate a fully bundled document from a composition resource. For details, see Generate Document Operation.

3.4 Messaging using FHIR Resources

URL: https://hl7.org/fhir/R4/messaging.html

FHIR Infrastructure Work Group | Maturity Level: 3 | Standards Status: Trial Use

FHIR Resources can be used in a traditional messaging context, much like HL7 v2 (see detailed comparison). Applications asserting conformance to this framework claim to be conformant to "FHIR messaging" (see Conformance).

In FHIR messaging, a "request message" is sent from a source application to a destination application when an event happens. Events mostly correspond to things that happen in the real world. The request message consists of a Bundle identified by the type "message", with the first resource in the bundle being a MessageHeader resource. The MessageHeader resource has a code - the message event - that identifies the nature of the request message, and it also carries additional request metadata. The other resources in the bundle depend on the type of the request.

The events supported in FHIR, along with the resources that are included in them, are defined below.

The destination application processes the request and returns one or more response messages which are also a bundle of resources identified by the type "message", with the first resource in each bundle being a MessageHeader resource with a response section that reports the outcome of processing the message and any additional response resources required.

- Example Request Message
- Example Response Message

3.4.1 Basic Messaging Assumptions

This specification assumes that content will be delivered from one application to another by some delivery mechanism, and then one or more responses will be returned to the source application. The exact mechanism of transfer is irrelevant to this specification, but may include file transfer, HTTP based transfer, MLLP (HL7 minimal lower layer protocol), MQ series messaging or anything else. The only requirement for the transfer layer is that requests are sent to a known location and responses are returned to the source of the request. This specification considers the source and destination applications as logical entities, and the mapping from logical source and destination to implementation specific addresses is outside the scope of this specification, though this specification does provide a direct delivery mechanism below.

The agreements around the content of the messages and the behavior of the two applications form the "contract" that describes the exchange. The contract will add regional and local agreements to the rules defined in this specification.

This specification ignores the existence of interface engines and message transfer agents that exist between the source and destination. Either they are transparent to the message/transaction content and irrelevant to

this specification, or they are actively involved in manipulating the message content (in particular, the source and destination headers are often changed). If these middleware agents are modifying the message content, then they become responsible for honoring the contract that applies (including applicable profiles) in both directions.

A key aspect of a message is the impact of its content:

3.4.1.1 The impact of the content of a message.

consequence | The message represents/requests a change that should not be processed more than once; e.g., making a booking for an appointment.
currency | The message represents a response to query for current information. Retrospective processing is wrong and/or wasteful.
notification | The content is not necessarily intended to be current, and it can be reprocessed, though there may be version issues created by processing old notifications.

Some Events defined by FHIR are assigned to one of these categories, but others are not able to be categorized in advance, and the category must be determined by the content, context, or use case.

3.4.1.1.1 Example: Elevation from Notification to Consequence

When it is necessary to receive an acknowledgement from multiple parties for a message of notification it becomes a message of consequence: The sender will have to send multiple messages, even if they have the same endpoint.

• Local protocol requires notification of all lab values

• Local protocol requires that critical values must be acknowledged by both the ordering and primary provider.

• Therefore a message of notification becomes a message of consequence.

• Two messages SHALL be sent, each with a unique identifier.

• One to the ordering provider

• One to the primary provider

• Each message SHALL have a unique acknowledgement

3.4.1.2 Implication to Receiver

Another key aspect of a message is the impact of its destination.receiver. In some circumstances it may be sufficient to direct a message to an endpoint, while in others it may be necessary to direct the message to a specific organization or person.

Consequence | A message of consequence SHOULD have one and only one receiver specified.
Currency | A message of currency MAY have one or more receivers specified.
Notification | A message of currency MAY have one or more receivers specified.

3.4.1.3 Message Exchange Patterns

Each FHIR request message has one or more response messages. There must be at least one response message so that the sender can know that the message was properly received. Multiple response messages SHALL NOT be returned for messages of consequence, and SHOULD not be returned for notifications.

In principle, source applications are not required to wait for a response to a transaction before issuing a new transaction. However, in many cases, the messages in a given stream are dependent on each other, and must be sent and processed in order. In addition, some transfer methods may require sequential delivery of messages.

3.4.1.3.1 Synchronous

For this reason, a synchronous exchange pattern - where the sender sends a message, and waits on the same channel for a single response, and then sends the next message - is the easiest to understand and manage:

• the sender sends a message to the receiver (the server)

• the server processes it, and then returns a response

• Usually (though not always) the sender waits for the response to the current message before sending the next message

This kind of messaging exchange is the most common because it's the simplest to understand.

3.4.1.3.2 Asynchronous

However synchronous message exchange does not cater for multiple response messages, which may arise when processing queries, and also imposes through-put limitations which may become relevant at high volumes. Additionally, it might not be practical or appropriate to wait for response messages. In these cases, the asynchronous message pattern should be used.

In Asynchronous messaging, the server acknowledges receipt of the message immediately, and responds to the sender separately. The server may respond more than once to any given message.

When a message is received, a receiver can determine from the content of the message header whether it's a new message to process, or a response to a message that has already been sent. Note that asynchronous messaging can be harder to implement due to the amount of complexity opposed to synchronous messaging; more can go wrong. This specification does not dictate any particular error handling protocols or responsibilities; these are left to trading partner agreements between implementers.

3.4.1.4 MessageHeader Identifiers & Timestamps

An incoming message contains two identifiers: the Bundle.id and the MessageHeader.id. Each time a new message is created, it SHALL be assigned an identifier (MessageHeader.id) that is unique within that message stream. Note that since message streams are often merged with other streams, it is recommended that the identifier should be globally unique. This can be achieved by using a UUID or an OID. Each time a message is sent, the Bundle.id should be changed to a new value.

When a receiver receives and processes the message, it responds with a new message with a new identifier, wrapped in a bundle which also has a new id. The response message also quotes the request MessageHeader.id in MessageHeader.response.identifier so that the source system can relate the response to its request.

A message has 2 important timestamps:

• Bundle.timestamp: the time the message was sent

• Bundle.meta.lastUpdated: the last time the message was updated (either by storing, or by modification)

In addition, the message may have additional timestamps in additional resources in the message, either .meta.lastUpdated or others throughout the resources. The meaning of these will depend on the message event.

3.4.1.5 Absence of Reliable Messaging

Some of the message delivery mechanisms mentioned above are reliable delivery systems - the message is always delivered, or an appropriate error is returned to the source. However most implementations use methods which do not provide reliable messaging, and either the request or the response can get lost in transit. FHIR messaging describes a simple approach that receivers SHOULD conform to in order to handle the absence of reliable messaging that maintains predictable functionality.

If the sender of the message implements reliable messaging, it SHALL do the following when it receives no response to a message within a configured timeout period based on the value specified in the CapabilityStatement messaging.event.category for the event associated with the message:

Consequence | Resend the same message (with the same MessageHeader.id) with the same Bundle.id
Currency | Resend the same message (with the same MessageHeader.id) with a different Bundle.id
Notification | Resend the same message (with the same MessageHeader.id) with a different Bundle.id

When a receiver implements reliable messaging, it SHALL check the incoming Bundle.id and MessageHeader.id against a cache of previously received messages. The correct action to take depends on what is received:

Both the Bundle.id and MessageHeader.id have not been received | This is the normal case, and the message should be processed
Both envelope and message already received | The original response has been lost (failed to return to the request issuer), and the original response SHALL be resent
The MessageHeader.id has already been received, but the Bundle.id is new | A previously seen message has been resubmitted for processing again. The server may either reprocess the message, or reject the message
The Bundle.id has already been received, but the MessageHeader.id is new | This is an error - Bundle.id values should never be reused

The duration period for caching does generally not need to be very long. At a minimum, it could be 1 minute longer than the timeout of the sending system, though it may need to be longer depending on the re-sending policies of the sending system.

Applications that implement reliable messaging declare their reliable cache period in their Capability Statement.

3.4.1.5.1 Example: Consequence

In the first example, a Clinical EHR issues an order for a particular imaging examination to be performed on a patient. This is considered to be a message of Consequence: multiple orders should not be created (in practice there are usually human review processes that catch multiple orders, but repeat orders create entropy in the system that is harmful). The EHR sends a message where the Bundle.id is UUID 1 (72edc4e0-6708-42ab-9734-f56721882c10), with a MessageHeader.id of UUID 2 (dad53a57-dcb4-4f18-b066-7239eb4b5229).

The EHR system never receives a response to the message; it does not know whether the request message got lost, or the imaging management systems was unable to process the request, or whether it successfully processed the message and the response was lost. In this case, the EHR system resends the message with same two identifiers.

In this case, the imaging system successfully received the message, and processed it. Because it receives the resent order after 1 minute (which is within its 15 minute cache time), and the two UUIDs 1 and 2 match a

message it has already processed, it knows that it already processed the order, and simply returns the previous response. In the case of additional resent queries, the application keeps sending the original response, though it may also alert system administrators that the same original message keeps being resent, since lost messages should be a rare occurrence.

When the EHR system finally receives the message, it knows how the imaging management system responded; it can be sure because the message id from the original request is echoed in the response portion of the returned message.

3.4.1.5.2 Example: Currency

In this second example, a Clinical EHR needs to know what appointment slots are available for a particular imaging procedure. This is a message of Currency: available slots are ever disappearing, and ordering a slot that has become unavailable is a waste of time for the humans and systems involved. The EHR sends a message where the Bundle.id is UUID 3 (4c7f5cb2-5964-4d42-b719-e0227461818c), with a MessageHeader.id is UUID 4 (63ed7d68-b2cc-421d-ba1c-a6c7785581f2).

The EHR system never receives a response to the message; it does not know whether the request message got lost, or the imaging management systems was unable to process the request, or whether it successfully processed the message and the response was lost. In this case, the EHR system resends the message with same MessageHeader.id (UUID 4), but creates a new Bundle.id (c7c17fe4-9560-49c7-b2ae-42636476fb86).

In this case, the imaging system successfully received the message, and processed it. When it receives the resent order after 1 minute (which is within its 15 minute cache time), it sees that although the message id is the same, the Bundle.id has changed, and it reprocesses the message again, and sends a new response.

When the EHR system finally receives the message, it knows the current slot availability on the imaging management system responded.

Note that the existence of active intermediaries (or "middleware") creates the need for this protocol - the original sender matches the response to the request based on the MessageHeader.id, and so an active intermediary that choose the re-initiate a query that it previously relayed cannot change the MessageHeader.id. This protocol avoids the need for the MessageHeader.id to change, and only requires change to the Bundle.id which is never the basis for context linking outside the immediate message exchange protocol described here.

3.4.2 Capability Statement

Applications may only assert conformance to "FHIR messaging" if they publish a Capability statement so the claim may be verified. A Capability statement lists all the message events supported (either as sender or receiver) and for each event, a profile that states which resources are bundled (sender), or are required to be bundled (receiver), and any rules about the information content of the individual resources.

3.4.3 $process-message

The simplest way to handle messages where there are also RESTful interactions occurring is to use the $process-message. This operation accepts a message, processes it according to the definition of the event in the message header, and returns a one or more response messages. See the opreation definition for further details.

3.4.4 Relationship between Messaging and REST

As well as this messaging framework documented here, FHIR also defines a RESTful API. The messaging and RESTful frameworks are related in that both share the same set of resources on which they operate. In fact, the basic MessageHeader resource that the messaging framework is implemented is itself a resource that can treated in a RESTful approach.

The kinds of functionality that the RESTful API and the messaging framework offer are very similar; their primary difference is architectural in nature.

For instance, the messaging framework defines an event for notifying that an administration resource has been created or updated; the REST API offers similar services (history and Subscription). On the other hand, there are differences in the capabilities offered - while a patient merge can be implemented as a series of RESTful operations performed by the client that update all resources linked to the patient, when a message command to merge patient records is processed, the server will do all the work, and is also able to merge in areas not exposed on the RESTful API. The REST API, however, provides a set of basic operations on all resources that would need special definitions in the messaging framework - definitions that are not provided.

There is no expectation that RESTful systems will need to offer messaging support, or vice versa, though systems may find it useful to support both sets of functionality in order to satisfy a wider range of implementers.

As a resource that can be used with the RESTful framework, the MessageHeader resource has the normal resource end-point (/MessageHeader), which is used to manage a set of static message resources. This could be used to make an archive of past messages available. Creating or updating MessageHeader resources in this fashion does not represent the actual occurrence of any event, nor can it trigger any logic associated with

the actual event. It is just for managing a set of message header resources.

3.4.4.1 Asynchronous Messaging using the RESTful API

It is possible to exchange messages using the RESTful end-point as a central point of exchange. This is not particularly efficient compared to other methods, but is useful for low-volume asynchronous exchange.

To send a message, a sender posts the message bundle to the /Bundle end-point, with a uri that identifies the receiver at MessageHeader.destination.endpoint. The RESTful server accepts the bundle, stores it as a single bundle, and indexes it on the MessageHeader.

To receive messages, a receiver searches for all messages destined for itself, since its last check:

```
GET [base]/Bundle?message.destination-uri=[rcv]&_lastUpdated=>2015-03-01T02:00:02+01:00
```

The receiver works through the response, processing each message. As each message is processed, the receiver creates a response message, reversing the source and destination, and posts it back to the server.

To check for responses, the original sender searches for response messages destined for itself, since its last check:

```
GET [base]/Bundle?message.destination-uri=[snd]&message.response-id:missing=false
     &_lastUpdated=>2015-03-03T06:03:522+01:00
```

This lightweight protocol needs ongoing administration to ensure that multiple parties do not interfere with each other by re-using the same system identifier (and against malicious attack).

3.4.5 Defining Events

The message.code element carries a Coding that identifies the event that the message conveys.

Events may be defined using the MessageDefinition resource. This specification does not define any events, but may do so in the future if implementers find this useful.

3.4.6 Invoking Operations via Messages

A message can be used to invoke an operation as defined for a RESTful interface using an operation definition. To invoke an operation using a message:

• The requester sends a message (a bundle with type = message, and a message header resource)

• The message header has an event.system of urn:ietf:rfc:3986

• The event.code is the URL from the operation definition OperationDefinition.url

• The MessageHeader.data refers to a Parameters resource

• The parameters resource is populated appropriately as specified by the nominated operation definition

The recipient executes the operation as specified, and then:

• The receiver sends a message (a bundle with type = message, and a message header resource)

• The message header has the same event as the original message

• The MessageHeader contains a response that refers to the original request message, and a code for the outcome, with details if the operation failed

• The MessageHeader.data refers to a Parameters resource

• The parameters resource is populated appropriately as specified for the response by the nominated operation definition

• If the operation definition specifies a single return, then this is returned as the target of the MesssageHeader.data directly

Here's an example:

```
<Bundle xmlns="http://hl7.org/fhir">
  <id value="urn:uuid:77831928-2a35-4c08-9496-8232323bf48c"/>
  <!-- normal bundle stuff -->
  <entry>
    <fullUrl value="urn:uuid:6080d4a7-5e05-45dc-96d5-f75329564d1f"/>
    <resource>
      <MessageHeader>
        <id value="cac8143e-6138-4f45-b086-bb8ebf976aae">
        <!-- normal message header stuff -->
        <event>
          <system value="urn:ietf:rfc:3986"/>
          <!-- value set expansion -->
          <code value="http://hl7.org/fhir/OperationDefinition/ValueSet-expand"/>
        </event>
        <!-- more normal message header stuff -->
        <data>
          <reference value="urn:uuid:00213637-dc7c-40d2-a7de-f4ef1eea5685"/>
        </data>
      </MessageHeader>
    </resource>
  </entry>
  <entry>
    <fullUrl value="urn:uuid:00213637-dc7c-40d2-a7de-f4ef1eea5685"/>
```

```
    <resource>
      <Parameters>
        <parameter>
          <name value="identifier"/>
          <valueUri value="http://hl7.org/fhir/ValueSet/identifier-type"/>
        </parameter>
      </Parameters>
    </resource>
  </entry>
</Bundle>
```

Note that there's no way to anchor the execution of the operation against a URL. The only operations that can be executed in this way are defined to be executed at the System or Resource level for a particular resource.

3.4.6.1 Invoking Search via Messages

In the same way that a defined operation can be invoked, a regular search operation can be invoked. This also uses the Parameters resource, with the following rules:

• The event code is "search-type" or "search-system" in the system http://hl7.org/fhir/restful-interaction

• If the event type is "search-type" there SHALL be a parameter "resourceType" with specifies the type of resource being searched

• The search parameters are converted to FHIR data types according to the following table

Search Parameter Type | Data Type
number | integer
date | dateTime
string | string
token | string or Coding (split the system and code apart)
reference | uri
composite | string
quantity | string or Quantity (split the syntax out)
uri | uri

Here's an example:

```
<Bundle xmlns="http://hl7.org/fhir">
  <id value="urn:uuid:77831928-2a35-4c08-9496-8232323bf48c"/>
  <!-- normal bundle stuff -->
  <entry>
    <fullUrl value="urn:uuid:c466754c-09c0-4f59-9f76-a48bd0ea27c9"/>
    <resource>
      <MessageHeader>
        <!-- normal message header stuff -->
        <event>
          <system value="http://hl7.org/fhir/restful-interaction"/>
          <!-- Search against Patient -->
          <code value="search-type"/>
        </event>
        <!-- more normal message header stuff -->
        <data>
          <reference value="urn:uuid:59a17a19-46eb-42d9-821a-f93a0c530cac"/>
        </data>
      </MessageHeader>
    </resource>
  </entry>
  <entry>
    <fullUrl value="urn:uuid:59a17a19-46eb-42d9-821a-f93a0c530cac"/>
    <resource>
      <Parameters>
        <parameter>
          <name value="resourceType"/>
          <valueString value="Patient"/>
        </parameter>
        <parameter>
          <name value="gender"/>
          <valueString value="m"/>
        </parameter>
      </Parameters>
    </resource>
  </entry>
</Bundle>
```

3.5 Using Resources with Services and Service-oriented Architecture

URL: https://hl7.org/fhir/R4/services.html

3.5 Using Resources with Services and Service-oriented Architecture

FHIR Infrastructure Work Group | Maturity Level: 2 | Standards Status: Trial Use

Though FHIR resources are primarily designed for the RESTful HTTP-based implementation, it is not necessary to use a RESTful interface when exchanging the resources; one way that they can be exchanged is in the

context of services. Service Oriented Architecture (SOA) is an architecture pattern using services to encapsulate and provide discrete pieces of application functionality to each other. Services communicate by invoking public interfaces and exchanging information (as parameters and outputs) in accordance with a well-defined service contract.

Services are cohesive sets of functions that maintain responsibility for both data and "state" for the scope of their responsibility. Services have a unity of function, such as Terminology Management, Identity Management, or so on, working with other services in collaboration as part of an orchestrated workflow. Data within a service typically follows a "black box" model and is not classically exposed, instead using pre-defined interfaces to make specific behavioral requests for the service to fill.

To relate this to FHIR, resources in a service-oriented implementation can serve two discrete roles: either as a payload parameter, specifying information flowing into or out of a service; or in a behavioral sense, using FHIR APIs as the invocation mechanism for engaging a service. Service APIs are often realized using either RESTful or SOAP-based interfaces.

Of note, there are potential benefits in considering usage of FHIR and SOA together. FHIR allows for ease of implementation and ready access to data payload in an open way. SOA has maturity around transactional integrity across distributed systems instances, providing a framework for loose-coupling and addressing pre-conditions, exception handling, and other implementation considerations of complex distributed systems.

This section illustrates different contexts to allow implementers to make informed decisions at implementation time, considering potential impacts and solution patterns to benefit specific situations. In doing so, this section will clarify what is meant to apply SOA principles and specifications within a FHIR environment/implementation, and the converse. It will identify the situational factors to consider, and the elements of the SOA discipline that can be applied for more effective FHIR implementation, providing a roadmap on how to optimize your implementation toolkit to make you and your team most effective.

This section is being included as FHIR offers a highly flexible framework for interoperability that can be deployed that can be used in a variety of different ways, and in ways that have the potential to hinder interoperability if not done thoughtfully or consistently. Use of SOA patterns and approaches can mitigate this risk, limiting the potential liability resulting from the extreme flexibility of FHIR and fostering an overall framework that leads to deployment and consistency. SOA provides guidance for how components interact, how to partition responsibilities, and how to manage workflows among different parts of systems, all of which have potential utility in FHIR implementation settings.

3.5.1 Service References and Patterns

The definition of specific services is a domain or context specific task. It is anticipated that this would be done as separate specifications that make use of the underlying facilities defined in this specification. HL7 has already defined a portfolio of services that take advantage of REST- and SOAP-based interfaces, but the parameters passed to and from those services are not at present resource-based. These are potential extension points that may lend well for FHIR implementation. Services defined like this are able to build on the common underlying platform features defined in this specification such as REST or messaging, and add specific interactions where appropriate. Alternatively, the service interfaces can build an entirely separate implementation.

Note that for each of the items below, there are paired sets of standards, an HL7 Standard defining the functional specification, and an OMG Technical Standard containing SOAP bindings (for all the items below) and REST bindings (for about half of the items below).

The portfolio of healthcare SOA service specifications includes:

• IXS (Identification and Cross-Reference Services) that identifies the characteristic of a generic identification management services (www.omg.org/spec/IXS)

• RLUS (Retrieve, Locate, Update Service) that supports the generic management resource in a SOA environment. (www.omg.org/spec/RLUS)

• ServD (Service directory) that supports the registration and discovery of health and human-related services (www.omg.org/spec/ServD)

• CDSS (Clinical Decision Support Service) that defines a standard access interface to a clinical decision support system (www.omg.org/spec/CDSS)

• CTS2 (Common Terminology Service V 2) that defines the capabilities, responsibilities, inputs, outputs, and expected behavior to support the management, maintenance, and interaction with ontologies and medical vocabulary systems (www.omg.org/spec/CTS2)

• CCS (Care Coordination Service) that provides the capabilities to support the coordination of patient care across the care continuum spanning multiple organization [emerging specification]

• OS (Order Service) that provides functional capabilities for ordering pharmacy, laboratory, radiology, consult and nutritional services individually or part of an order set [emerging specification].

• EPSS (Event Publication & Subscription Service) that provides a Service Functional Model (SFM) for services, components and systems to subscribe to clinical events of interest and receive notice when new data are available [emerging specification].

• UCS (Unified Communications Service) facilitates the bi-directional communication between participants (both SOA services and human participants) that is so essential for the efficient delivery and coordination of patient care [emerging specification].

Note: FHIR-enablement of the services above is already occurring on a limited basis. The Clinical Quality Improvement Framework (CQIF) portion of the FHIR specification is an example of a technical interface and resources that have been influenced by prior SOA work. Similarly, this specification defines a Terminology Service which is tightly integrated with the RESTful API (note: this terminology service has different goals than the CTS2 service, so they are not functionally equivalent). FHIR itself is a framework for the instantiation of an RLUS service. Definition of other additional services based on the service interface definitions provided by the HL7 SOA work group will be considered if there is enough interest in this.

3.5.2 Implementation Approaches

Given that there are a set of available services that can be applied directly or used as a pattern in conjunction with FHIR, it merits consideration that there are alternative ways to bring together FHIR and SOA, each of which has benefits and drawbacks. Three specific styles of implementation have been identified:

• FHIR + REST ("RESTful FHIR"), the predominant approach for implementing FHIR today

• FHIR + WS* represents FHIR implementations using the web services stack as the communication protocol instead of REST. This would include use of FHIR resources as payload parameters in SOAP calls, for example

• FHIR + SOA Pattern illustrates the impact of applying interaction patterns, exception handing and role definition, guided by SOA practices, atop an implementation technology (SOAP, REST, or others)

Determining which of the above implementation styles is best suited to a situation is context-dependent. The following table presents these alternatives, deliberately using qualitative measures to help navigate to the best-fit based upon influencing factors. "Harvey Balls" in the table below indicate relative strengths and weaknesses of the implementation alternatives.

Solution Quality | Short Description | #1:FHIR + REST (RESTful FHIR) | #2 FHIR + WS* | #3 FHIR + SOA Pattern | Comments
Support of transactional integrity | Transactions may involve multiple steps composed together into a single unit | ◒ | ◒ | ● | Transactional integrity is not assured in #1 or #2 but can be hand-coded. #3 assures that this is addressed
Support of stateless transactions | Transactions are independent and do not require supporting context | ● | ● | ● | Each of the implementation approaches can support stateless processing
Support of "loose coupling" | Minimum of dependencies between the calling and responding system; reduces burden on client to manage context of interaction with service | ◒ | ◒ | ● | The notion of "loose coupling" is fundamental to SOA and inherent in that architecture. Loose coupling can be supported by FHIR, but it does not innately do so
Use of FHIR Resources as Data Payload | Representation of FHIR resources for use as query or return parameters | ● | ● | ● | Data payload includes input data as well as data returned
Support of Dynamic Service Discovery | Provides for service registration, search, discovery, and late binding | ◒ | ◒ | ● | This is part of SOA specification and customary in implementations. For the others, it is up to the approach
Provides resource-oriented operations | Supports ability to create, read, update, delete resources; | ● | ◒ | ◒ | Direct access to resources/data revisions in a strength of REST and FHIR. Direct access to fine-grained transactional operations are generally not supported within SOA
Suitability for atomic transactions | Fitness to support fine-grained transactions, such as data access or targeted update | ● | ◒ | ◒ | REST is ideally suited for point access or updates of specific data elements/resources - functions discouraged within SOA and WS*
Suitability for composite transactions | Fitness to support complex transactions: context-sensitivity, multi-step workflows, etc. | ◒ | ● | ● | SOA provides for complex event processing, multi-step sequencing, orchestration

3.5.3 Architectural Considerations

FHIR resources are designed to be used in a wide variety of contexts. In particular, FHIR resources are required to be suitable for use in a REST environment. This means that there are number of design requirements and choices that impact how suitable resources are for use with services.

If service use was the only consideration, different decisions would be made, and resources would be more suitable for use with services. However, this would curtail their usefulness and reusability in other contexts.

API/Interface Design. Services' API design is explicit and exposes service capabilities which are a part of a service contract to consuming applications. This contrasts with what is the default interface into a FHIR server of a REST-based interface exposing CRUD operations and stateless access. Good SOA design practice indicates loosely-coupled access (e.g., external clients do not have visibility into the inner workings of a service) improves implementation flexibility over time, permitting design and performance improvements with the maturation of the implementation. As such, interfaces are typically exposed and aligned with business functionality (such as "Register Patient" or "Validate Identity").

Bridging what is not necessarily a natural-fit between SOA operations and RESTful operations are key to interface design. Implementers may elect to apply finer grained FHIR interactions to be encapsulated in coarse-grained functions, delegating those responsibilities to either implementations within a FHIR instance or to external code not visible to the service consumer.

Principal within the design objective is to minimize or eliminate responsibilities on behalf of the service consumer to understand or apply knowledge of the service implementation. This frees up the service provider to be able to evolve and improve the inner workings of the service without a "ripple effect" adversely impacting service consumers. In doing so, technical dependencies are minimized. Several good sources of SOA Design Patterns are available from HL7 and other public sources.

Data Storage and Coherence. The approach to and complexities associated with data storage and overall data coherence will significantly depend upon whether an implementation is subject to enterprise policies and whether data consistency is the responsibility of the service provider or consumer. In Enterprise settings, these issues are typically governed by organizational policy. FHIR servers and their data persistence would need to fit within that policy, with data access following that policy and services interacting with "policy enforcement points" to assure that appropriate permissions are in place.

Where data consistency is the responsibility of the service (or server) use of SOA-friendly interface protocols, such as SOAP and potentially REST, create the access channel for integration. This may necessitate defining minimum sets of data that are required to comply with data coherence expectations (in other words, a service contract that specifies the collection of resources that must be updated in tandem, and the scope of what constitutes a 'single transaction' - more on this in the Transactional Integrity section).

In alternative implementations data storage is managed within the FHIR server, consistent with non-SOA FHIR implementations. Data consistency is the responsibility of the Server, but responsibility may carry to consuming applications (if they are provided with sufficient update rights so as to make that a concern). Data validation also has a shared responsibility, first with the FHIR Server, and additionally with the authorized updating applications. Data visibility is a product of the REST interface, typically allowing CRUD operations to the data element level.

Transactional Integrity. When encapsulating functionality within a SOA-type service, data and transactional integrity are essential in making sure that data received and stored is accurate and consistent with what was received. An issue particularly with update functions, the ability to update multiple different resources and maintain harmony among them as part of a broader transaction is how this is typically realized.

Should one part of the update 'fail', one would expect all the other updates to be "rolled back" to a prior, stable state. Rollback is not expressly specified as part of the FHIR. Data update and integrity becomes the responsibility of each service. The FHIR specification does not inherently provide for concurrency management (e.g., managing of "deadlock" conditions where two resources each await update permission from the other). As a result, within the FHIR server itself it is assumed that fault tolerance and server availability is being managed (beyond the scope of the specification).

Applied use of SOA patterns have benefits in terms of distributed transactional integrity, as SOA has substantially removed burdens from calling (client) applications, instead levying these responsibilities upon the service provider. This has relevance because it is customary in larger scale implementations (such as large enterprises) that service consumers are varied and not necessarily know in advance to the service. As a result, these implementation patterns make for more robust infrastructure, mitigating potential implementation risk and ultimately improving data quality. We note that these elements can be manually accommodated in FHIR implementations, absent use of SOA patterns, though that can lead to inconsistent interpretation, delegated responsibility to client application(s), and the potential for incompatible implementations.

Modularity. The modularity of resources - which resources are defined, and why - is driven by a wide set of considerations around how they are used. Resources are a platform on which a set of business/clinical services of various kinds are provided. Accordingly, the resources are defined for general use, and they can be expected to be less suitable for a particular service than custom defined structures. The pay back is wider re-use of the information that the service deals with.

Effective "service-orientation" establishes modularity at both a macro and micro-level. At the macro-level, services themselves are modular, componentized, and composable. Since each service has a singularity of purpose with a well-defined interface, they are naturally suited to working in tandem with other services and consistent with modular design. At a more fine-grained level, the interfaces into and out of services also allow for modularity, and the ability to leverage the structure of FHIR resources as a payload descriptor for carrying data structures and corresponding semantics across APIs is modularly based.

Explicit State. All resources represent the various states of the record and real-world entities involved in a transaction explicitly. This is necessary for use in RESTful and document contexts, where there is no explicit transaction. Service interactions are typically associated with implicit semantics, such as a request to change the status of a resource to something else, for example. The fact that resources that carry this state explicitly as well as the transaction fixing state implicitly creates duplication between the two, and this will need to be managed.

Error Handling. SOA provides an implementation approach providing consistency in error handling, escalation, and error management. These tools can be leveraged to identify likely error conditions and exceptions based upon prior SOA work, helping to put into place infrastructure within FHIR to manage those exceptions. Moreover, this alignment creates future opportunity to more effectively integrate FHIR resources into an enterprise SOA fabric should that become a need. It is not anticipated that there is a need to change error management for FHIR, except to address payload-specific errors that may arise.

The FHIR specification does not specifically address error handling between and across servers and clients. When implementing within larger or more complex environments, particularly in situations where multiple FHIR servers may be involved (especially if they are provided by different vendors/implementations), error handing and management can quickly become complex and untenable.

Resource References. The most obvious impact is that resources refer to each other using full URL based references, and there are a number of rules around how these references are resolved. In the context of a service, this means that the references between modules carry this extra weight of choice and obligation, even when it might not be necessary.

FHIR resources or bundles may be used as the parameters or outputs of service interfaces.

3.5.4 Other Considerations

Orchestration. Orchestration is a term typically used within SOA to describe the steps, sequencing, and dynamic adjustment of workflow to meet a process need. Orchestrations may be entirely automated and fulfilled in a short timeline, or longer and multistep in fulfillment of a business process that may involve manual steps and human intervention. In the context of FHIR implementation, orchestration would refer to the sequencing of collections of FHIR (or service) calls that are used in tandem to fulfil a specific need.

Orchestration is neither natively supported nor unsupported in FHIR, save the availability of the FHIR Batch mode which allows for some degree of compounding of operations. In a SOA environment, orchestration is typically realized by documenting in some formalism, such as Business Process Modeling Notation (BPMN), a sequence of steps and flows, and inherent logic or decision-points affecting that flow. An engine is then capable of executing the process flow, receiving inputs during execution to adjust or adapt those processes based upon situational need as part of delivery fulfillment.

There are many examples where this approach is advantageous. For example, in a Care Management situation, test results, current problems, chief complaint, and potentially even resource availability might affect subsequent steps in fulfilling a care plan. Based upon changes to any of these factors, the sequence of calls and ultimately the systems or FHIR servers involved would vary. The role of automated clinical decision support is another example that naturally ties to workflow orchestration, adjusting care pathways based upon patient evaluation and affecting process flows within a health system.

At present, to support this complexity within a FHIR setting without the use of SOA tools, these flows would need to be manually coded, either by having a directed sequence of calls, or by creating an independent capability effectively acting as an orchestrator. This can be supported within FHIR exclusively, but all the steps and corresponding state management would need to be done by hand. FHIR Batch provides some of this capability, allowing for aggregation of multiple steps, but does not necessarily support the event processing needed in complex.

Security. Security is inherently a dimension of any enterprise SOA architecture, meaning that the responsibility within a service implementation is to provide the "hooks" to interact with that architecture. In other words, the service does not need to create or enforce security, it needs to interact with those enterprise components that have that responsibility. This includes topics such as identity management, access control, or other dimensions of a secure solution. SOA guidance can foster an effective design involving FHIR in support of authentication, policy enforcement, role-based access controls, and a host of other provisions and protections. It is important to note that within SOA, security is an established, mainstream, and mature offering. Reuse of these concepts can help prevent inadequate, incomplete, or ineffective security measures within FHIR, and eliminate the need to re-invent solutions exclusively for FHIR.

Enterprises typically have security architecture into which a SOA environment will have already integrated. Service implementations would rely upon this existing infrastructure as part of authentication, access control, etc. Policy enforcement is a function of the architecture and not a specific service itself. Services would rely upon policy enforcement points to govern access to information.

While technical dimensions of security would be addressed by the security architecture directly, FHIR implementers should anticipate that data-specific access control policies would need to be captured and formalized so that they are enforceable within the architecture. For example, if a new service is handling protected health information, the policy enforcement point within the architecture would need to know that the service has that nature of data.

Resource Identity. All resources have a single identity (the full URL), and a logical id which may be maintained as the resource moves from server to server (see Managing Resource Identity for further discussion). Since this identity is used by any reference to the resource, it must be maintained when the resource is exchanged so that references from other resources to the one being exchanged can still be resolved. Any use of resources in a service environment needs to address how these references can be resolved. This can be achieved by delegating the reference resolution to a RESTful framework, by ensuring that all the relevant resources are contained in the service call, or by making some service-based arrangement by which additional resources can be retrieved. Services that exchange resources SHOULD maintain resource identity. Further, as explicit version tracking is not a guaranteed capability of a service call, services that exchange resources should include version information associated with resources being exchanged.

Capability Statement. When using RESTful exchange, messaging, and document based exchange, the Capability statement allows authoring and reading applications to describe how they use a resource. The Capability Statement supports trading partner negotiation from specification time through to run-time discovery. The Capability statement doesn't provide any equivalent way to make declarations about services, though this might be added in the future if common requirements emerge. Services are expected to make appropriate arrangements around discovery and compatibility, though it is expected that these will vary

considerably.

Services that exchange resources MAY choose to provide support for describing and changing descriptions of service implementations.

3.5.5 Relating SOA and FHIR: Rationale

At its core, FHIR is based upon the elemental capabilities of Create, Read, Update, Delete (CRUD), allowing access to resources where interaction is primarily based upon these operations. Of note, multi-step or complex processing levies requirements upon the calling, client application. Sequencing of events, particularly when those events involve orchestration of complex or dynamic processes, can benefit from the application of SOA techniques and patterns. This is common where multi-step processing is involved, and particularly important when adjustments to a process flow can be made based upon situational context.

As these areas are both commonplace within health settings and not inherent strengths of FHIR, the marriage of FHIR and SOA techniques provides a viable and beneficial path forward to improving implementation and mitigating risks by leveraging proven and established industry best-practices.

SOA design patterns are frequently used to govern interactions for dynamic processes, non-sequential workflows, or dynamic workflows involving human intervention and/or consideration of external factors as part of processing. The ability to model interaction patterns using languages such as SOAml, and to define roles and role interactions, are tools that may be useful to FHIR development in these circumstances - and would be indicators signaling that SOA has the potential to add value to a FHIR implementation.

SOA design principles can provide guidance to a FHIR implementer resulting in reduction of co-dependencies between components, promoting "loose coupling" and minimizing potential impacts resulting from changes to inner workings of one component adversely affecting others. (For those familiar with the term, this encourages "black box" implementation).

SOA provides guidance around data persistence, durability, and expectations in support of the data lifecycle.

3.5.6 Summary

FHIR has proven itself as beneficial in that it provides an easy-to-implement and coherent approach to accessing healthcare resources using a modern protocol stack. SOA has proven itself over time in multiple vertical market segments to divide responsibilities and authoritatively manage information across distributed systems. Marrying the discipline of SOA with the implementation ease of FHIR is a winning combination when the situation warrants. Implementers should consider evaluation of the contextual landscape to determine where FHIR and SOA can benefit from complementary implementation leverage points.

Recognizing that there are situations where the coming-together of FHIR implementation and SOA techniques are advantageous, there are several implementation approaches available. Note that the selection of which approach is advantageous for any given situation will depend upon a variety of factors: existing legacy implementation and available APIs (particularly for interfacing applications), the strategic direction/IT roadmap of the organization, and so on. Noting that there are different styles of realizing FHIR and SOA together, a recap of the principal alternatives would include:

• "Full Support in FHIR" Approach. This alternative involves simplification of the exposed behaviors of the service/server so as to be supported within the REST protocol, thus allowing full support within FHIR. Behind a FHIR technology stack, SOA design principles such as loose coupling and "black box" API design would be coded to create service autonomy.

• "Classic SOA with FHIR Payload" Approach. This alternative is attractive particularly in enterprises that have or are making investments in SOA infrastructure. This alternative relies upon traditional SOA service definition, leveraging use of FHIR Resources as the parameters into and out of the service.

It is important to note that electing to apply SOA patterns to FHIR implementation does not fundamentally change the nature of what is being implemented. Applied correctly, it results in minor adaptation to coding and interface design resulting in enhanced consistency across implementations as well as enhanced interoperability and robustness.

These considerations are summarized in this table:

Capability	#1:FHIR + REST (RESTful FHIR)	#2 FHIR + WS*	#3 FHIR + SOA Pattern	Comments
Supports dynamic adjustment in workflows	○	◒	●	In a native FHIR environment, this must be done by hand. The WS* stack provides for some interaction patterns, but limited. Use of formal notation (BPMN) and service orchestration is most robust option
Supports ability to batch multiple operations	◒	◒	●	FHIR Batch allows for concatenation of multiple process steps. SOA Orchestration allows for fine grained control, grouping and sequencing of multiple operations
Provides for management of data coherence (e.g. Deadlocks, transactions)	○	○	●	SOA principles define data governance, assigning responsibility for all data management within a service contract
Provide support for orchestration languages (BPMN , SOAml)	○	◒	●	SOA applies a systematic framework for orchestration using industry accepted formalisms, avoiding costs and complexities associated with an ad-hoc approach

SOA in a FHIR Environment considers the benefits of applying SOA patterns and best-practices to a FHIR implementation community. In these cases, developers are faced with implementation decisions that have the potential to benefit from existing case studies, design patterns, or guidance that may either help provide consistency among FHIR implementations, or which may address gaps - either known or unidentified - resulting from implicit assumptions around a FHIR implementation.

FHIR in a SOA Environment considers the implications of using FHIR in a large, established enterprise that has or is making investments in SOA infrastructure. In these cases, shared services, enterprise policies, and existing infrastructure is common, and FHIR implementations would need to fit within the fabric of that environment.

Finally, it merits mention that not every FHIR implementation benefits from SOA, and vice-versa. The goal of this section is to help implementers navigate based upon their specific considerations to help determine if and to what extent these approaches provide benefits.

3.6 Using FHIR in persistent stores

URL: https://hl7.org/fhir/R4/storage.html

3.6 Using FHIR in persistent stores

FHIR Infrastructure Work Group | Maturity Level: n/a | Standards Status: Informative

Applications can use the resources defined by FHIR by storing them natively in a database or persistent store, where different applications or modules write and read the resources as part of their implementation. This page describes implementation issues encountered when storing resources natively in persistent stores.

Note that almost all applications store the information found in resources in some persistent store - usually a database. Most applications store the information found in the resources in some internal format, and only use resources natively when exchanging with other applications. This page concerns applications that store the FHIR resources natively.

3.6.1 Applicability of storing resources natively

In principle, resources are designed for exchange between systems, rather than as a database storage format. One practical consequence of this is that in some way, resources are highly denormalised, so that granular exchanges are fairly stand alone. (Note that in some ways, resources are also highly normalised; e.g. Patient demographic details are only found in the Patient resource).

To illustrate this, consider using an RxNorm code as a medication code. In a prescription resource (MedicationRequest), that will look like this:

```
<MedicationRequest xmlns="http://hl7.org/fhir">
  <medicationCodeableConcept>
   <coding>
     <system value="http://www.nlm.nih.gov/research/umls/rxnorm"/>
     <code value=""/>
   </coding>
  </medicationCodeableConcept>
</MedicationRequest>
```

If the resource was designed to prioritize storage efficiency, it would be normalized so that the code was just some primary key reference:

```
<MedicationRequest xmlns="http://hl7.org/fhir">
  <code key="123431232"/>
</MedicationRequest>
```

or something similar. Note: Whether to normalise or not is an engineering design question that has multiple different considerations, and very often product designers change their decision about this for engineering reasons with no associated change in functionality or requirements.

Resources, on the other hand, are designed for robustness and stability over efficiency. Using a key reference like the second example would require all the participants in an exchange to synchronize the key for the code, which is often not possible, and rarely stable.

In spite of this design criteria, it can still be useful and appropriate to store resources directly in a persistent store or database. Whether to do so depends on an application's requirements.

If an application's information requirements are nailed down - that is, well understood, clearly expressed, and not subject to uncontrolled change - then it is easy to design a data storage schema that is completely fit for purpose and highly efficient compared to storing FHIR resources. Classic Enterprise Information Systems (EHRs) tend to behave like this.

If, on the other hand, the application's information requirements are not at all nailed down, and implementers have to deal with whatever data comes in an ongoing basis, then resources are actually a very sound way to store an application's data. The extensibility approach makes FHIR a particularly robust choice as the primary data store. Clinical Data Repositories tend to behave like this.

Most information systems fall somewhere between these 2 extremes, and determining whether to store resources directly is not straight forward. Some applications take a hybrid approach - storing FHIR resources natively, and storing a well-controlled subset of information in some expressly designed persistent scheme. The balance between these is driven by the stability of information requirements for different parts of the system.

3.6.2 Dealing with Multiple FHIR Versions

Very often, applications storing FHIR resources natively must deal with multiple different FHIR versions, since they are receiving information from different systems with different timelines for adopting ongoing FHIR releases.

Typically, such systems have functional requirements that include some variation of these:

• The system must be able to make resources available in their original form for integrity/audit reasons

• The system must present some unified view/analysis/process based on a coherent view of the information from resources with different FHIR versions

There is no single pattern to resolve these somewhat inconsistent requirements, but application designers can consider several different techniques for dealing with these kinds of requirements:

• When storing resources, store the version of the resource alongside the resource

• Maintain forward (and possibly backwards) conversions for the parts of resources that matter to the application (and see the FHIR maintained version transforms)

• Store resources in both received and transformed state

• Host multiple different end-points for different purposes

3.6.3 Technology Choices

A wide variety of technologies exist for storing resources natively, including:

• Classic SQL servers with JSON support (built in, or added on top by the application)

• NoSQL servers (e.g. Mongodb , Couch , Hadoop , or Big Query)

• Some RDF based store using the RDF in the turtle format, or in some native triple story (e.g. Jena)

This specification does not recommend any particular approach. The rest of this page describes general information management considerations that apply across all technologies.

3.6.4 Managing Joins

Most applications that store resources natively find the way references work in FHIR deserves particular focus. References can be absolute or relative URLs, they can be version specific or not, they might or might not follow FHIR's well-defined RESTful interface pattern, and they might or might not resolve in the local system. In addition, given the requirements consideration above, it's not always possible to enforce referential integrity when storing resources directly.

All the flexibility is required in various exchange scenarios, but it can present challenges for application design when building a coherent data store with resources.

Applications that store resources natively may want to extend the storage format with an element to capture the resolved link in addition to the existing reference/identifier in the Reference and canonical data types.

The RDF format does this explicitly for references and codings (fhir:reference and fhir:concept).

Applications that store JSON, will have to remove the extension for exchange, unless a native FHIR extension is used, which might not be the most efficient way to store the resolved link.

Note to balloters: Is there enough interest to standardize an extension? Ballot comments are welcome.

3.7 FHIR Status Codes Grid

URL: https://hl7.org/fhir/R4/sc.html
FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative

This is an internal QA page. The intent of this table is to help reviewers compare the mappings between the general resource status codes and the status codes on particular resources.

Note: This page is not proscriptive; it's descriptive and comparative, to assist with review. The FHIR methodology says that if two domains use different terms for the same concept (e.g. 'withdrawn' and 'cancelled') then the domain concept name should be used. Conversely, where there is no explicit reason for the codes to differ based on domain usage, they should not.

All status codes must be mapped to one of the general status codes, which are found in the Resource Status code system.

Note: Bold indicates that the element is marked as a modifier.

code	error	proposed	planned	draft	requested	received	declined	accepted	arrived	active	suspended	failed	replaced	complete	inactive	abandoned	unknown	unconfirmed	confirmed	resolved	refuted	differential	partial	busy-unavailable	free	on-target	ahead-of-target	behind-target	not-ready	transduc-discon	hw-discon
stated codes	entered-in-error, marked deleted	proposed	planned	tentative, draft	requested	received	refused, declined, rejected	accepted, scheduled, booked, reserved	arrived	active, in-progress, published	on hold, onleave suspended	stopped, aborted, failed	Amended, replaced	completed, final, finished, achieved, done, dispensed, delivered	superseded, retired, not in use, inactive, off, refuted	cancelled, withdrawn, not done, abandoned	unknown	unconfirmed	confirmed	resolved	refuted	differential	partial	busy-unavailable	free	on-target	ahead-of-target	behind-target	not-ready	transduc-disconnected	hardware-disconnected
actual codes	entered-in-error	offered, proposed, proposed,waitlist	draft, intended, planned, preparation	attested, candidate, screening, eligible, potential-candidate, data-requested, draft, in-review, negotiable,policy, not-started, partial,preliminary,																											

pending, preliminary, tentative, test, waitlist | requested | received, registered | data-required, declined, disapproved, ineligible, not-billable, rejected | accepted, approved, booked, busy, reserved, scheduled, triaged, validated | active, arrived | active, active, closed-to-accrual, closed-to-accrual-and-intervention, available, billable, current, executable,resolved, in-process, in-progress, in-progress, sustaining, improving, worsening, no-change, issued, on, on-study, on-study-intervention, on-study-observation, pending, successful | disputed, noshow, on-hold, onhold, onleave, pending-on-study, standby, suspended, temporarily-closed-to-accrual, temporarily-closed-to-accrual-and-intervention, waiting | aborted, cancelled, error, failed, failure, needs-action, revoked, stopped, unsatisfactory, val-fail, reval-fail, withdrawn | amended, amended,appended, amended,appended,renewed, corrected, amended, corrected, amended, appended, follow-up, req-revalid | achieved, balanced, billed, complete, completed, completed, administratively-completed, executed, final, finished, fulfilled, off-study, retired, success | inactive, off, retired, superseded, unavailable | abandoned, cancelled, cancelled,rejected,terminated, not-attainable, not-done, not-registered, unknown, withdrawn | health-unknown, needs-action, unknown | busy-tentative, unconfirmed, unconfirmed, provisional | confirmed | resolved | refuted | differential | checked-in, partial | busy-unavailable, unavailable | free | ready
Issues? | | | | check test | in the Event pattern | | check data required | check busy | | check on, check Goal | check no-show | check needs-action | | | | | check needs action | check busy-tentative
Account.status | entered-in-error | | | | | | | | active | on-hold | | | | inactive | | unknown
ActivityDefinition.status | | | | draft | | | | | active | | | | | retired | | unknown
AllergyIntolerance.clinicalStatus | | | | | | | | | | active | | | | | inactive | | | | | resolved
AllergyIntolerance.verificationStatus | entered-in-error | | | | | | | | | | | | | | | | | | unconfirmed | confirmed | | refuted
Appointment.participant.status | entered-in-error | | | tentative | | | declined | accepted | | | | needs-action
Appointment.status | entered-in-error | proposed,waitlist | | pending | | | | booked | arrived | | noshow | | | fulfilled | | cancelled | | | | | | checked-in
AppointmentResponse.participantStatus | entered-in-error | | | tentative | | | declined | accepted | | in-process | | | | completed | | | needs-action
BiologicallyDerivedProduct.status | | | | | | | | | | available | | | | | | | | | | | | | | unavailable
CapabilityStatement.status | | | | draft | | | | | | active | | | | | retired | | unknown
CarePlan.activity.detail.status | entered-in-error | | | not-started | | | | scheduled | | in-progress | on-hold | stopped | | completed | | cancelled | unknown
CarePlan.status | entered-in-error | | | draft | | | | | | active | on-hold | revoked | | completed | | | unknown
CareTeam.status | entered-in-error | proposed | | | | | | | | active | suspended | | | | inactive
CatalogEntry.status | entered-in-error | | | draft | | | | | | active | | | | | retired | | unknown
ChargeItem.status | entered-in-error | | planned | | | | not-billable | | | billable | | aborted | | billed | | | unknown
ChargeItemDefinition.status | | | | draft | | | | | | active | | | | | retired | | unknown
Claim.status | entered-in-error | | | draft | | | | | | active | | | | | | cancelled
ClaimResponse.status | entered-in-error | | | draft | | | | | | active | | | | | | cancelled
ClinicalImpression.status | entered-in-error | | preparation | | | | | | | in-progress | on-hold | stopped | | completed | | not-done | unknown
CodeSystem.status | | | | draft | | | | | | active | | | | | retired | | unknown
Communication.status | entered-in-error | | preparation | | | | | | | in-progress | on-hold | stopped | | completed | | not-done | unknown
CommunicationRequest.status | entered-in-error | | | draft | | | | | | active | on-hold | revoked | | completed | | | unknown
CompartmentDefinition.status | | | | draft | | | | | | active | | | | | retired | | unknown
Composition.status | entered-in-error | | | preliminary | | | | | | | | | amended,appended | final
ConceptMap.status | | | | draft | | | | | | active | | | | | retired | | unknown
Condition.verificationStatus | entered-in-error | | | | | | | | | | | | | | | | | | | | unknown | unconfirmed, provisional | confirmed | | refuted | differential
Consent.status | entered-in-error | proposed | | draft | | | rejected | | | active | | | | | inactive
Contract.contentDefinition.publicationStatus | entered-in-error | offered | | negotiable,policy | | | | | | executable,resolved | disputed | revoked | amended,appended,renewed | executed | | cancelled,rejected,terminated
Contract.status | entered-in-error | offered | | negotiable,policy | | | | | | executable,resolved | disputed | revoked | amended,appended,renewed | executed | | cancelled,rejected,terminated
Coverage.status | entered-in-error | | | draft | | | | | | active | | | | | | cancelled
CoverageEligibilityRequest.status | entered-in-error | | | draft | | | | | | active | | | | | | cancelled
CoverageEligibilityResponse.status | entered-in-error | | | draft | | | | | | active | | | | | | cancelled
DetectedIssue.status | entered-in-error | | | preliminary | | registered | | | | | | | corrected, amended | final | | cancelled | unknown
Device.status | entered-in-error | | | | | | | | active | | | | | inactive | | unknown
DeviceMetric.operationalStatus | entered-in-error | | | | | | | | | on | standby | | | | off
DeviceRequest.status | entered-in-error | | | draft | | | | | | active | on-hold | revoked | | completed | | | unknown
DeviceUseStatement.status | entered-in-error | | intended | | | | | | | active | on-hold | stopped | | completed
DiagnosticReport.status | entered-in-error | | | partial,preliminary | | registered | | | | | | | corrected, amended, appended | final | | cancelled | unknown
DocumentManifest.status | entered-in-error | | | | | | | | | current | | | | | superseded
DocumentReference.docStatus | entered-in-error | | | preliminary | | | | | | | | | amended | final
DocumentReference.status | entered-in-error | | | | | | | | | current | | | | | superseded
EffectEvidenceSynthesis.status | | | | draft | | | | | | active | | | | | retired | | unknown
Encounter.location.status | | | planned | | | | | reserved | | | active | | | | completed
Encounter.status | entered-in-error | | planned | | | | | triaged | arrived | in-progress | onleave | | | finished | | cancelled | unknown
Encounter.statusHistory.status | entered-in-error | | planned | | | | | triaged | arrived | in-progress | onleave | | | finished | | cancelled | unknown
Endpoint.status | entered-in-error | | | test | | | | | | active | suspended | error | | | off
EnrollmentRequest.status | entered-in-error | | | draft | | | | | | active | | | | | | cancelled
EnrollmentResponse.status | entered-in-error | | | draft | | | | | | active | | | | | | cancelled
EpisodeOfCare.status | entered-in-error | | planned | waitlist | | | | | | active | onhold | | | finished | | cancelled
EpisodeOfCare.statusHistory.status | entered-in-error | | planned | waitlist | | | | | | active | onhold | | | finished | | cancelled
EventDefinition.status | | | | draft | | | | | | active | | | | | retired | | unknown

Evidence.status | | | | draft | | | | | | active | | | | | retired | | unknown
EvidenceVariable.status | | | | draft | | | | | | active | | | | | retired | | unknown
ExampleScenario.status | | proposed | planned | draft | | | | | | active | | | | finished | retired | | unknown
ExplanationOfBenefit.status | entered-in-error | | | draft | | | | | | active | | | | | | cancelled
FamilyMemberHistory.status | entered-in-error | | | | | | | | | | | | | completed | | | health-unknown | | | | | | partial
Flag.status | entered-in-error | | | | | | | | | active | | | | | inactive
Goal.achievementStatus | | | | | | | | | | in-progress, sustaining, improving, worsening, no-change | | | | achieved | | not-attainable
Goal.lifecycleStatus | entered-in-error | proposed | planned | | | | rejected | accepted | | active | on-hold | | | completed | | cancelled
GraphDefinition.status | | | | draft | | | | | | active | | | | | retired | | unknown
GuidanceResponse.status | entered-in-error | | | data-requested | | | data-required | | | in-progress | | failure | | success
ImagingStudy.status | entered-in-error | | | | | registered | | | | available | | | | | | cancelled | unknown
Immunization.status | entered-in-error | | | | | | | | | | | | | completed | | not-done
ImmunizationEvaluation.status | entered-in-error | | | | | | | | | | | | | completed
ImplementationGuide.status | | | | draft | | | | | | active | | | | | retired | | unknown
InsurancePlan.status | entered-in-error | | | draft | | | | | | active | | | | | retired | | unknown
Invoice.status | entered-in-error | | | draft | | | | | | issued | | cancelled | | balanced
Library.status | | | | draft | | | | | | active | | | | | retired | | unknown
List.status | entered-in-error | | | | | | | | | current | | | | | retired
Location.status | | | | | | | | | | active | suspended | | | | inactive
Measure.status | | | | draft | | | | | | active | | | | | retired | | unknown
MeasureReport.status | | | | | | | | | | pending | | error | | complete
Media.status | entered-in-error | | preparation | | | | | | | in-progress | on-hold | stopped | | completed | inactive | | unknown
Medication.status | entered-in-error | | | | | | | | | active | | | | | inactive
MedicationAdministration.status | entered-in-error | | | | | | | | | in-progress | on-hold | stopped | | completed | | | unknown
MedicationDispense.status | entered-in-error | | preparation | | | | | | | in-progress | on-hold | stopped | | completed | | | unknown
MedicationKnowledge.status | entered-in-error | | | | | | | | | active | | | | | inactive
MedicationRequest.status | entered-in-error | | | draft | | | | | | active | on-hold | stopped | | completed | | cancelled | unknown
MedicationStatement.status | entered-in-error | | intended | | | | | | | active | on-hold | stopped | | completed | | | unknown
MessageDefinition.status | | | | draft | | | | | | active | | | | | retired | | unknown
NamingSystem.status | | | | draft | | | | | | active | | | | | retired | | unknown
NutritionOrder.status | entered-in-error | | | draft | | | | | | active | on-hold | revoked | | completed | | | unknown
Observation.status | entered-in-error | | | preliminary | | registered | | | | | | | corrected, amended | final | | cancelled | unknown
OperationDefinition.status | | | | draft | | | | | | active | | | | | retired | | unknown
PaymentNotice.status | entered-in-error | | | draft | | | | | | active | | | | | | cancelled
PaymentReconciliation.status | entered-in-error | | | draft | | | | | | active | | | | | | cancelled
PlanDefinition.status | | | | draft | | | | | | active | | | | | retired | | unknown
Procedure.status | entered-in-error | | preparation | | | | | | | in-progress | on-hold | stopped | | completed | | not-done | unknown
Questionnaire.status | | | | draft | | | | | | active | | | | | retired | | unknown
QuestionnaireResponse.status | entered-in-error | | | | | | | | | in-progress | | stopped | amended | completed
RequestGroup.status | entered-in-error | | | draft | | | | | | active | on-hold | revoked | | completed | | | unknown
ResearchDefinition.status | | | | draft | | | | | | active | | | | | retired | | unknown
ResearchElementDefinition.status | | | | draft | | | | | | active | | | | | retired | | unknown
ResearchStudy.status | | | | in-review | | | disapproved | approved | | active, closed-to-accrual, closed-to-accrual-and-intervention | temporarily-closed-to-accrual, temporarily-closed-to-accrual-and-intervention | | | completed, administratively-completed | | withdrawn
ResearchSubject.status | | | | candidate, screening, eligible, potential-candidate | | | ineligible | | | on-study, on-study-intervention, on-study-observation | pending-on-study | withdrawn | follow-up | off-study | | not-registered
RiskAssessment.status | entered-in-error | | | preliminary | | registered | | | | | | | corrected, amended | final | | cancelled | unknown
RiskEvidenceSynthesis.status | | | | draft | | | | | | active | | | | | retired | | unknown
SearchParameter.status | | | | draft | | | | | | active | | | | | retired | | unknown
ServiceRequest.status | entered-in-error | | | draft | | | | | | active | on-hold | revoked | | completed | | | unknown
Slot.status | entered-in-error | | | | | | | busy | | | | | | | | | busy-tentative | | | | | | busy-unavailable | free
Specimen.status | entered-in-error | | | | | | | | | available | | unsatisfactory | | | unavailable
StructureDefinition.status | | | | draft | | | | | | active | | | | | retired | | unknown
StructureMap.status | | | | draft | | | | | | active | | | | | retired | | unknown
Subscription.status | entered-in-error | | | | requested | | | | | active | | error | | | off
Substance.status | entered-in-error | | | | | | | | | active | | | | | inactive
SupplyDelivery.status | entered-in-error | | | | | | | | | in-progress | | | | completed | | abandoned
SupplyRequest.status | entered-in-error | | | draft | | | | | | active | suspended | | | completed | | cancelled | unknown
Task.status | entered-in-error | | | draft | requested | received | rejected | accepted | | in-progress | on-hold | failed | | completed | | cancelled | | | | | | | | | | ready
TerminologyCapabilities.status | | | draft | | | | | | active | | | | | retired | | unknown
TestReport.status | entered-in-error | | | | | | | | | in-progress | waiting | stopped | | completed
TestScript.status | | | | draft | | | | | | active | | | | | retired | | unknown
ValueSet.status | | | | draft | | | | | | active | | | | | retired | | unknown
VerificationResult.primarySource.validationStatus | | | | | | | | | | successful | | failed | | | | unknown
VerificationResult.status | | | | attested | | | | validated | | in-process | | val-fail, reval-fail | req-revalid
VisionPrescription.status | entered-in-error | | | draft | | | | | | active | | | | | | cancelled

3.7.0.0.1 error

The resource was created in error, and should not be treated as valid (note: in many cases, for various data integrity related reasons, the information cannot be removed from the record)

Definitions for matching codes:

• Care plan activity was entered in error and voided.

• Contract was created in error. No Precedence Order. Status may be applied to a Contract with any status.

• Some of the actions that are implied by the medication request may have occurred. For example, the medication may have been dispensed and the patient may have taken some of the medication. Clinical decision support systems should take this status into account

• Some of the actions that are implied by the medication statement may have occurred. For example, the patient may have taken some of the medication. Clinical decision support systems should take this status into account.

• The DeviceMetric was entered in error.

• The administration was entered in error and therefore nullified.

• The care team should have never existed.

• The charge item has been entered in error and should not be processed for billing.

• The composition or document was originally created/issued in error, and this is an amendment that marks that the entire series should not be considered as valid.

• The consent was created wrongly (e.g. wrong patient) and should be ignored.

• The device was entered in error and voided.

• The dispense was entered in error and therefore nullified.

• The flag was added in error and should no longer be displayed.

• The goal was entered in error and voided.

• The imaging study has been withdrawn following a previous final release. This electronic record should never have existed, though it is possible that real-world decisions were based on it. (If real-world activity has occurred, the status should be "cancelled" rather than "entered-in-error".).

• The instance was entered in error.

• The list was never accurate. It is retained for medico-legal purposes only.

• The medication was entered in error.

• The observation has been withdrawn following previous final release. This electronic record should never have existed, though it is possible that real-world decisions were based on it. (If real-world activity has occurred, the status should be "cancelled" rather than "entered-in-error".).

• The report has been withdrawn following a previous final release. This electronic record should never have existed, though it is possible that real-world decisions were based on it. (If real-world activity has occurred, the status should be "cancelled" rather than "entered-in-error".).

• The resource instance was entered in error.

• The response was entered in error.

• The specimen was entered in error and therefore nullified.

• The statement was entered in error and is not valid.

• The statement was recorded incorrectly.

• The substance was entered in error.

• The task should never have existed and is retained only because of the possibility it may have used.

• This QuestionnaireResponse was entered in error and voided.

• This electronic record should never have existed, though it is possible that real-world decisions were based on it. (If real-world activity has occurred, the status should be "cancelled" rather than "entered-in-error".).

• This electronic record should never have existed, though it is possible that real-world decisions were based on it. (If real-world activity has occurred, the status should be "stopped" rather than "entered-in-error".).

• This electronic record should never have existed, though it is possible that real-world decisions were based on it. (If real-world activity has occurred, the status should be "abandoned" rather than "entered-in-error".).

• This instance should not have been part of this patient's medical record.

• This reference was created in error.

• This request should never have existed and should be considered 'void'. (It is possible that real-world decisions were based on it. If real-world activity has occurred, the status should be "revoked" rather than "entered-in-error".).

• This test report was entered or created in error.

• the invoice was determined as entered in error before it was issued.

3.7.0.0.2 proposed

The resource describes an action or plan that is proposed, and not yet approved by the participants

Definitions for matching codes:

• A goal is proposed for this patient.

• Contract is a proposal by either the Grantor or the Grantee. Aka - A Contract hard copy or electronic 'template', 'form' or 'application'. E.g., health insurance application; consent directive form. Usage: Beginning of contract negotiation, which may have been completed as a precondition because used for 0..* contracts. Precedence Order = 2. Comparable FHIR and v.3 status codes: requested; new.

• The care team has been drafted and proposed, but not yet participating in the coordination and delivery of patient care.

• The consent has been proposed but not yet agreed to by all parties. The negotiation stage.

3.7.0.0.3 planned

The resource describes a course of action that is planned and agreed/approved, but at the time of recording was still future

Definitions for matching codes:

• A goal is planned for this patient.

• The Encounter has not yet started.

• The charge item has been entered, but the charged service is not yet complete, so it shall not be billed yet but might be used in the context of pre-authorization.

• The core event has not started yet, but some staging activities have begun (e.g. initial compounding or packaging of medication). Preparation stages may be tracked for billing purposes.

• The core event has not started yet, but some staging activities have begun (e.g. surgical suite preparation). Preparation stages may be tracked for billing purposes.

• The device may be used at some time in the future.

• The medication may be taken at some time in the future.

• The patient is planned to be moved to this location at some point in the future.

• This episode of care is planned to start at the date specified in the period.start. During this status, an organization may perform assessments to determine if the patient is eligible to receive services, or be organizing to make resources available to provide care services.

• This resource is still under development and is not yet considered to be ready for normal use.

3.7.0.0.4 draft

The information in the resource is still being prepared and edited

Definitions for matching codes:

• ***TODO***

• A new instance the contents of which is not complete.

• A new resource instance the contents of which is not complete.

• Care plan activity is planned but no action has yet been taken.

• Protocol is submitted to the review board for approval.

• Some or all of the participant(s) have not finalized their acceptance of the appointment request.

• The consent is in development or awaiting use but is not yet intended to be acted upon.

• The participant has tentatively accepted the appointment. This could be automatically created by a system and requires further processing before it can be accepted. There is no commitment that attendance will occur.

• The prescription is not yet 'actionable', e.g. it is a work in progress, requires sign-off, verification or needs to be run through decision support process.

• The request has been created but is not yet complete or ready for action.

• The request was processed successfully, but more data may result in a more complete evaluation.

• The task is not yet ready to be acted upon.

• This endpoint is not intended for production usage.

• This episode has been placed on a waitlist, pending the episode being made active (or cancelled).

• This is a preliminary composition or document (also known as initial or interim). The content may be incomplete or unverified.

• This is an initial or interim observation: data may be incomplete or unverified.

• This resource is still under development and is not yet considered to be ready for normal use.

• the invoice has been prepared but not yet finalized.

3.7.0.0.5 requested

A fulfiller has been asked to perform this action, but it has not yet occurred

Definitions for matching codes:

• The client has requested the subscription, and the server has not yet set it up.

• The task is ready to be acted upon and action is sought.

3.7.0.0.6 received

The fulfiller has received the request, but not yet agreed to carry out the action

Definitions for matching codes:

• A potential performer has claimed ownership of the task and is evaluating whether to perform it.

• The existence of the imaging study is registered, but there is nothing yet available.

• The existence of the observation is registered, but there is no result yet available.

• The existence of the report is registered, but there is nothing yet available.

3.7.0.0.7 declined

The fulfiller chose not to perform the action

Definitions for matching codes:

• A person who did not meet one or more criteria required for participation in a study is considered to have failed screening or is ineligible for the study.

• A proposed goal was rejected.

• Protocol was disapproved by the review board.

• The charge item has been determined to be not billable (e.g. due to rules associated with the billing code).

• The consent has been rejected by one or more of the parties.

• The participant has declined the appointment and will not participate in the appointment.

• The potential performer who claimed ownership of the task has decided not to execute it prior to performing any action.

• The request was processed, but more data is required to complete the evaluation.

3.7.0.0.8 accepted

The fulfiller has decided to perform the action, and plans are in train to do this in the future

Definitions for matching codes:

• ***TODO***

• A proposed goal was accepted or acknowledged.

• All participant(s) have been considered and the appointment is confirmed to go ahead at the date/times specified.

• Appointment or other booking has occurred but activity has not yet begun.

• Indicates that the time interval is busy because one or more events have been scheduled for that interval.

• Protocol is approved by the review board.

• The participant has accepted the appointment.

• The patient has been assessed for the priority of their treatment based on the severity of their condition.

• The potential performer has agreed to execute the task but has not yet started work.

• This location is held empty for this patient.

3.7.0.0.9 arrived

The pre-conditions for the action are all fulfilled, and it is imminent

Definitions for matching codes:

• The Patient is present for the encounter, however is not currently meeting with a practitioner.

• The patient/patients has/have arrived and is/are waiting to be seen.

• This resource is ready for normal use.

3.7.0.0.10 active

The resource describes information that is currently valid or a process that is presently occuring

Definitions for matching codes:

• ***TODO***

• A current flag that should be displayed to a user. A system may use the category to determine which user roles should view the flag.

• A test operations is currently executing.

• At least one instance has been associated with this imaging study.

• Care plan activity has been started but is not yet complete.

• Product is currently available for use.
• Supply has been requested, but not delivered.
• The DeviceMetric is operating and will generate DeviceObservations.
• The Encounter has begun and the patient is present / the practitioner and the patient are meeting.
• The administration has started but has not yet completed.
• The care team is currently participating in the coordination and delivery of care.
• The charge item is ready for billing.
• The consent is to be followed and enforced.
• The device is available for use. Note: For *implanted devices* this means that the device is implanted in the patient.
• The device is still being used.
• The dispensed product is ready for pickup.
• The event is currently occurring.
• The goal is being sought actively.
• The instance is currently in-force.
• The list is considered to be an active part of the patient's record.
• The location is operational.
• The medication is available for use.
• The medication is still being taken.
• The patient is currently at this location, or was between the period specified. A system may update these records when the patient leaves the location to either reserved, or completed.
• The physical specimen is present and in good condition.
• The prescription is 'actionable', but not all actions that are implied by it have occurred yet.
• The report is currently being generated.
• The request is currently being processed.
• The request is in force and ready to be acted upon.
• The request is ready to be acted upon.
• The resource instance is currently in-force.
• The subject is currently experiencing, or is at risk of, a reaction to the identified substance.
• The subscription is active.
• The substance is considered for use or reference.
• The task has been started but is not yet complete.
• This QuestionnaireResponse has been partially filled out with answers but changes or additions are still expected to be made to it.
• This account is active and may be used.
• This endpoint is expected to be active and can be used.
• This episode of care is current.
• This is the current reference for this document.
• This resource is ready for normal use.
• the invoice has been finalized and sent to the recipient.

3.7.0.0.11 suspended

The process described/requested in this resource has been halted for some reason

Definitions for matching codes:

• A person is pre-registered for a study.
• A test operation is waiting for an external client request.
• Actions implied by the administration have been temporarily halted, but are expected to continue later. May also be called 'suspended'.
• Actions implied by the prescription are to be temporarily halted, but are expected to continue later. May also be called 'suspended'.
• Actions implied by the statement have been temporarily halted, but are expected to continue later. May also be called "suspended".
• Actions implied by the statement have been temporarily halted, but are expected to continue later. May also be called 'suspended'.

• Care plan activity was started but has temporarily ceased with an expectation of resumption at a future time.

• Contract is pended to rectify failure of the Grantor or the Grantee to fulfil contract provision(s). E.g., Grantee complaint about Grantor's failure to comply with contract provisions. Usage: Contract pended. Precedence Order = 7. Comparable FHIR and v.3 status codes: on hold; pended; suspended.

• Some or all of the participant(s) have not/did not appear for the appointment (usually the patient).

• The DeviceMetric is operating, but will not generate any DeviceObservations.

• The Encounter has begun, but the patient is temporarily on leave.

• The authorization/request to act has been temporarily withdrawn but is expected to resume in the future.

• The care team is temporarily on hold or suspended and not participating in the coordination and delivery of care.

• The dispense process is paused while waiting for an external event to reactivate the dispense. For example, new stock has arrived or the prescriber has called.

• The event has been temporarily stopped but is expected to resume in the future.

• The goal remains a long term objective but is no longer being actively pursued for a temporary period of time.

• The location is temporarily closed.

• The request (and any implicit authorization to act) has been temporarily withdrawn but is expected to resume in the future.

• The task has been started but work has been paused.

• This account is on hold.

• This endpoint is temporarily unavailable.

• This episode of care is on hold; the organization has limited responsibility for the patient (such as while on respite).

3.7.0.0.12 failed

The process described/requested in the resource could not be completed, and no further action is planned

Definitions for matching codes:

• A Contract that is rescinded. May be required prior to replacing with an updated Contract. Comparable FHIR and v.3 status codes: nullified.

• Actions implied by the administration have been permanently halted, before all of them occurred.

• Actions implied by the dispense have been permanently halted, before all of them occurred.

• Actions implied by the prescription are to be permanently halted, before all of the administrations occurred. This should not be used if the original order was entered in error

• Actions implied by the statement have been permanently halted, before all of them occurred.

• Actions implied by the statement have been permanently halted, before all of them occurred. This should not be used if the statement was entered in error.

• An error occurred attempting to generate the report.

• The event was terminated prior to the full completion of the intended activity but after at least some of the 'main' activity (beyond preparation) has occurred.

• The participant needs to indicate if they accept the appointment by changing this status to one of the other statuses.

• The person has withdrawn their participation in the study before registration.

• The planned care plan activity has been ended prior to completion after the activity was started.

• The processing of the charge was aborted.

• The request (and any implicit authorization to act) has been terminated prior to the known full completion of the intended actions. No further activity should occur.

• The request was not processed successfully.

• The server has an error executing the notification.

• The specimen cannot be used because of a quality issue such as a broken container, contamination, or too old.

• The task was attempted but could not be completed due to some error.

• The test script execution was manually stopped.

• This QuestionnaireResponse has been partially filled out with answers but has been abandoned. It is unknown whether changes or additions are expected to be made to it.

• This endpoint has exceeded connectivity thresholds and is considered in an error state and should no longer be attempted to connect to until corrective action is taken.

• the invoice was cancelled.

3.7.0.0.13 replaced

The information in this resource has been replaced by information in another resource

Definitions for matching codes:

• ***TODO***

• A person is no longer receiving study intervention and/or being evaluated with tests and procedures according to the protocol, but they are being monitored on a protocol-prescribed schedule.

• The composition content or the referenced resources have been modified (edited or added to) subsequent to being released as "final" and the composition is complete and verified by an authorized person.

• This QuestionnaireResponse has been filled out with answers, then marked as complete, yet changes or additions have been made to it afterwards.

3.7.0.0.14 complete

The process described/requested in the resource has been completed, and no further action is planned

Definitions for matching codes:

• A person that has ended their participation on a study either because their treatment/observation is complete or through not responding, withdrawal, non-compliance and/or adverse event.

• Activity against the request has been sufficiently completed to the satisfaction of the requester.

• All actions that are implied by the administration have occurred.

• All actions that are implied by the prescription have occurred.

• All available related health information is captured as of the date (and possibly time) when the family member history was taken.

• All test operations have completed.

• Care plan activity has been completed (more or less) as planned.

• Contract is activated for period stipulated when both the Grantor and Grantee have signed it. Usage: Required state for normal completion of contracting activity. Precedence Order = 6. Comparable FHIR and v.3 status codes: accepted; completed.

• Supply has been delivered ("completed").

• The Encounter has ended.

• The activity described by the request has been fully performed. No further activity will occur.

• The charge item has been billed (e.g. a billing engine has generated financial transactions by applying the associated ruled for the charge item to the context of the Encounter, and placed them into Claims/Invoices.

• The device is no longer being used.

• The dispensed product has been picked up.

• The event has now concluded.

• The goal has been met.

• The goal is no longer being sought.

• The medication is no longer being taken.

• The observation is complete and there are no further actions needed. Additional information such "released", "signed", etc would be represented using [Provenance](provenance.html) which provides not only the act but also the actors and dates and other related data. These act states would be associated with an observation status of `preliminary` until they are all completed and then a status of `final` would be applied.

• The patient was at this location during the period specified. Not to be used when the patient is currently at the location.

• The planning stages of the appointment are now complete, the encounter resource will exist and will track further status changes. Note that an encounter may exist before the appointment status is fulfilled for many reasons.

• The report is complete and ready for use.

• The report is complete and verified by an authorized person.

• The request was processed successfully.

• The task has been completed.

• This QuestionnaireResponse has been filled out with answers and the current content is regarded as definitive.

• This episode of care is finished and the organization is not expecting to be providing further care to the patient. Can also be known as "closed", "completed" or other similar terms.

• This resource has been withdrawn or superseded and should no longer be used.

• This version of the composition is complete and verified by an appropriate person and no further work is planned. Any subsequent updates would be on a new version of the composition.

• the invoice has been balaced / completely paid.

3.7.0.0.15 inactive

The resource describes information that is no longer valid or a process that is stopped occurring

Definitions for matching codes:

• The DeviceMetric is not operating.

• The care team was, but is no longer, participating in the coordination and delivery of care.

• The consent is terminated or replaced.

• The device is no longer available for use (e.g. lost, expired, damaged). Note: For *implanted devices* this means that the device has been removed from the patient.

• The flag no longer needs to be displayed.

• The list is "old" and should no longer be considered accurate or relevant.

• The location is no longer used.

• The medication is not available for use.

• The subject is no longer at risk of a reaction to the identified substance.

• The substance is considered for reference, but not for use.

• There is no physical specimen because it is either lost, destroyed or consumed.

• This account is inactive and should not be used to track financial information.

• This endpoint is no longer to be used.

• This reference has been superseded by another reference.

• This resource has been withdrawn or superseded and should no longer be used.

• Too many errors have occurred or the subscription has expired.

3.7.0.0.16 abandoned

The process described/requested in the resource did not complete - usually due to some workflow error, and no further action is planned

Definitions for matching codes:

• A person for whom registration was not completed.

• Delivery was not completed.

• Protocol was withdrawn by the lead organization.

• The Encounter has ended before it has begun.

• The appointment has been cancelled.

• The authoring system does not know which of the status values currently applies for this resource. Note: This concept is not to be used for "other" - one of the listed statuses is presumed to apply, it's just not known which one.

• The authorization/request to act has been terminated prior to the full completion of the intended actions. No further activity should occur.

• The episode of care was cancelled, or withdrawn from service, often selected during the planned stage as the patient may have gone elsewhere, or the circumstances have changed and the organization is unable to provide the care. It indicates that services terminated outside the planned/expected workflow.

• The event was terminated prior to any activity beyond preparation. I.e. The 'main' activity has not yet begun. The boundary between preparatory and the 'main' activity is context-specific.

• The goal has been abandoned.

• The goal is not possible to be met.

• The imaging study is unavailable because the imaging study was not started or not completed (also sometimes called "aborted").

• The instance is withdrawn, rescinded or reversed.

• The observation is unavailable because the measurement was not started or not completed (also sometimes called "aborted").

• The planned care plan activity has been withdrawn.

• The prescription has been withdrawn before any administrations have occurred

• The report is unavailable because the measurement was not started or not completed (also sometimes called "aborted").

• The resource instance is withdrawn, rescinded or reversed.

• The task was not completed.

3.7.0.0.17 unknown

Authoring system does not know the status

Definitions for matching codes:

• Health information for this family member is unavailable/unknown.

• The account status is unknown.

• The authoring system does not know which of the status values applies for this medication dispense. Note: this concept is not to be used for other - one of the listed statuses is presumed to apply, it's just now known which one.

• The authoring system does not know which of the status values currently applies for this charge item Note: This concept is not to be used for "other" - one of the listed statuses is presumed to apply, it's just not known which one.

• The authoring system does not know which of the status values currently applies for this request. Note: This concept is not to be used for 'other' - one of the listed statuses is presumed to apply, it's just not known which one.

• The authoring system does not know which of the status values currently applies for this resource. Note: This concept is not to be used for "other" - one of the listed statuses is presumed to apply, it's just not known which one.

• The authoring/source system does not know which of the status values currently applies for this event. Note: This concept is not to be used for "other" - one of the listed statuses is presumed to apply, but the authoring/source system does not know which.

• The authoring/source system does not know which of the status values currently applies for this observation. Note: This concept is not to be used for "other" - one of the listed statuses is presumed to apply, but the authoring/source system does not know which.

• The authoring/source system does not know which of the status values currently applies for this observation. Note: This concept is not to be used for 'other' - one of the listed statuses is presumed to apply, but the authoring/source system does not know which.

• The authoring/source system does not know which of the status values currently applies for this request. Note: This concept is not to be used for "other" - one of the listed statuses is presumed to apply, but the authoring/source system does not know which.

• The current state of the care plan activity is not known. Note: This concept is not to be used for "other" - one of the listed statuses is presumed to apply, but the authoring/source system does not know which one.

• The encounter status is unknown. Note that "unknown" is a value of last resort and every attempt should be made to provide a meaningful value other than "unknown".

• The participant needs to indicate if they accept the appointment by changing this status to one of the other statuses.

• The state of the medication use is not currently known.

• The status of the device has not been determined.

• The system does not know which of the status values currently applies for this request. Note: This concept is not to be used for "other" - one of the listed statuses is presumed to apply, it's just not known which one.

• The validations status has not been determined yet

3.7.0.0.18 unconfirmed

The information in this resource is not yet approved

Definitions for matching codes:

• A low level of certainty about the propensity for a reaction to the identified substance.

• Indicates that the time interval is busy because one or more events have been tentatively scheduled for that interval.

3.7.0.0.19 confirmed

The information in this resource is approved

Definitions for matching codes:

• A high level of certainty about the propensity for a reaction to the identified substance, which may include clinical evidence by testing or rechallenge.

• There is sufficient diagnostic and/or clinical evidence to treat this as a confirmed condition.

3.7.0.0.20 resolved

The issue identified by this resource is no longer of concern

Definitions for matching codes:

• A reaction to the identified substance has been clinically reassessed by testing or re-exposure and is considered no longer to be present. Re-exposure could be accidental, unplanned, or outside of any clinical setting.

3.7.0.0.21 refuted

This information has been ruled out by testing and evaluation

Definitions for matching codes:

• A propensity for a reaction to the identified substance has been disputed or disproven with a sufficient level of clinical certainty to justify invalidating the assertion. This might or might not include testing or rechallenge.

• This condition has been ruled out by diagnostic and clinical evidence.

3.7.0.0.22 differential

Potentially true?

Definitions for matching codes:

• One of a set of potential (and typically mutually exclusive) diagnoses asserted to further guide the diagnostic process and preliminary treatment.

3.7.0.0.23 partial

This information is still being assembled

Definitions for matching codes:

• Some health information is known and captured, but not complete - see notes for details.

• When checked in, all pre-encounter administrative work is complete, and the encounter may begin. (where multiple patients are involved, they are all present).

3.7.0.0.24 busy-unavailable

not available at this time/location

Definitions for matching codes:

• Indicates that the time interval is busy and that the interval cannot be scheduled.

• Product is not currently available for use.

3.7.0.0.25 free

Free for scheduling

Definitions for matching codes:

• Indicates that the time interval is free for scheduling.

3.7.0.0.26 on-target

Ready to act

Definitions for matching codes:

• The task is ready to be performed, but no action has yet been taken. Used in place of requested/received/accepted/rejected when request assignment and acceptance is a given.

3.7.0.0.27 ahead-of-target

Ahead of the planned timelines

Definitions for matching codes:

3.7.0.0.28 behind-target

Definitions for matching codes:

3.7.0.0.29 not-ready

Behind the planned timelines

Definitions for matching codes:

3.7.0.0.30 transduc-discon

The device transducer is disconnected

Definitions for matching codes:

3.7.0.0.31 hw-discon

The hardware is disconnected

Definitions for matching codes:

3.8 FHIR W5 Report

URL: https://hl7.org/fhir/R4/w5.html
FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative

This page is deprecated and will be a removed in a future version of this publication. Instead, see the FiveWs pattern.

Concise

Resource | id | id.version | status | class | grade | what | who.focus | context | when.init | when.planned | when.done | when.recorded | who.author | who.source | who.actor | who.cause | who.witness | who | where | why
Clinical Resources
ActivityDefinition (2) | url identifier* | version | status | experimental | | | | | | | | date | | | | | publisher | | | purpose
AdverseEvent (0) | | | | actuality category* | | event | | encounter | date | | | recordedDate | recorder | | | | | subject
AllergyIntolerance (3) | identifier* | | clinicalStatus verificationStatus | type category* | criticality | code | patient | encounter | onset[x] | | | recordedDate | recorder | asserter
BiologicallyDerivedProduct (0) | identifier*
BodyStructure (1) | identifier* | | active | | | morphology location locationQualifier* description image* | patient
CarePlan (2) | identifier* | | status | category* | | description | subject | encounter | | period | | created | author | | careTeam* | | | | | addresses*
CareTeam (2) | identifier* | | status | category* | | | subject | encounter | period | | | | | | member | | | | | reasonCode* reasonReference*
ClinicalImpression (0) | identifier* | | status | | | code | subject | encounter | | | effective[x] | date | assessor | | | | | | | problem*
Condition (3) | identifier* | | clinicalStatus verificationStatus | category* | severity | code | subject | encounter | onset[x] | | abatement[x] | recordedDate | recorder | asserter | | | | | | code* detail*
DetectedIssue (1) | identifier* | | status | code | severity | | patient | | | | | identified[x] | author | | | | | | evidence* code* detail*
DiagnosticReport (3) | identifier* | | status | category* | | code | subject | encounter | | | effective[x] | issued | | | performer* resultsInterpreter*
EffectEvidenceSynthesis (0) | url identifier* | version | status | | | | | | | | | date | | | | | publisher
EventDefinition (0) | url identifier* | version | status | experimental | | | | | | | | date | | | | | publisher | | | purpose
Evidence (0) | url identifier* | version | status | | | | | | | | | date | | | | | publisher
EvidenceVariable (0) | url identifier* | version | status | | | | | | | | | date | | | | | publisher
FamilyMemberHistory (2) | identifier* | | status | | | | patient | | | | | date | | | | | | | | reasonCode* reasonReference*
Goal (2) | identifier* | | lifecycleStatus | category* | priority | description | subject | | | start[x] | due[x] | statusDate | | expressedBy | | | | | | addresses*
GuidanceResponse (2) | identifier* | | status | | | | subject | encounter | | | occurrenceDateTime | | | | performer | | | | | reasonCode* reasonReference*
ImagingStudy (3) | identifier* | | status | modality* | | | subject | encounter | started | | | | | | actor | basedOn* referrer | interpreter* | | location | reasonCode* reasonReference*
Immunization (3) | identifier* | | status | | | vaccineCode | patient | encounter | | | occurrence[x] | recorded | | primarySource reportOrigin | actor | | | | location
ImmunizationEvaluation (0) | identifier* | | status | | | | patient | | date
ImmunizationRecommendation (1) | identifier* | | | | | | patient
Library (2) | url identifier* | version | status | experimental | | type | | | | | | date | | | | | publisher | | | purpose
Measure (2) | url identifier* | version | status | experimental | | | | | | | | date | | | | | publisher | | | purpose
MeasureReport (2) | identifier* | | status | type | | measure | subject | | | | | date | | reporter
Medication (3) | identifier* | | | code | | | | | | | | | | | manufacturer
MedicationAdministration (2) | identifier* | | status | | | medication[x] | subject | context supportingInformation* | | | effective[x] | | | | performer* | | | | | reasonReference*
MedicationDispense (2) | identifier* | | status | | | medication[x] | subject | supportingInformation*
MedicationKnowledge (0) | | | | code | | | | | | | | | | | manufacturer
MedicationRequest (3) | identifier* | | status | intent category* | priority | medication[x] | subject | encounter supportingInformation* | | | | authoredOn | requester | | performer | | | recorder performer | | reasonCode* reasonReference*
MedicationStatement (3) | identifier* | | status | category | | medication[x] | subject | | | | effective[x] | dateAsserted | | informationSource | | | | | | reasonCode* reasonReference*
MedicinalProduct (0)
MedicinalProductAuthorization (0)
MedicinalProductContraindication (0)
MedicinalProductIndication (0)
MedicinalProductIngredient (0)
MedicinalProductInteraction (0)
MedicinalProductManufactured (0)
MedicinalProductPackaged (0)
MedicinalProductPharmaceutical (0)
MedicinalProductUndesirableEffect (0)
MolecularSequence (1) | identifier*
NutritionOrder (2) | identifier* | | status | intent | | type* modifier type baseFormulaType additiveType | patient | encounter | | | | dateTime | orderer
Observation (5) | identifier* | | status | category* | | code code | subject focus* | encounter | | | effective[x] | issued | | | performer*
ObservationDefinition (0) | | | | category* | | code
PlanDefinition (2) | url identifier* | version | status | experimental | | | | | | | | date | | | | | publisher | | | purpose

Procedure (3) | identifier* | | status | category | | code | subject | encounter | | | performed[x] | | recorder | asserter | actor | | | | location | reasonCode* reasonReference*
RequestGroup (2) | identifier* | | status | intent | priority | code | subject | encounter | | | | authoredOn | author | | participant* | | | | | reasonCode* reasonReference*
ResearchDefinition (0) | url identifier* | version | status | experimental | | | | | | | | date | | | | | publisher | | | purpose
ResearchElementDefinition (0) | url identifier* | version | status | experimental | | | | | | | | date | | | | | publisher | | | purpose
ResearchStudy (1) | identifier* | | status | | | focus* condition* | enrollment* | | | | period | | | | sponsor principalInvestigator | | | | site* | reasonStopped
ResearchSubject (1) | identifier* | | status | | | | | | | period
RiskAssessment (1) | identifier* | | status | method | | code | subject | encounter | | | occurrence[x] | | | | performer | | | | | reasonCode* reasonReference*
RiskEvidenceSynthesis (0) | url identifier* | version | status | | | | | | | | | date | | | | | publisher
ServiceRequest (2) | identifier* | | status | intent category* | priority | code | subject | encounter | | | occurrence[x] | | authoredOn | requester | | performerType performer* locationCode* locationReference* | | | | | reasonCode* reasonReference*
Specimen (2) | identifier* accessionIdentifier | | status | | | type | subject | | collected[x] duration | | receivedTime | | | | collector | | | | | request*
SpecimenDefinition (0)
SubstanceNucleicAcid (0)
SubstancePolymer (0)
SubstanceProtein (0)
SubstanceReferenceInformation (0)
SubstanceSourceMaterial (0)
SubstanceSpecification (0)
VisionPrescription (2) | identifier* | | status | | | | patient | encounter | | | | dateWritten | prescriber
Administrative Resources
Device (2) | identifier* udiCarrier* | | status statusReason* | | | deviceIdentifier distinctIdentifier manufacturer manufactureDate expirationDate lotNumber serialNumber type modelNumber partNumber version component | patient | | | | | | owner contact* | | | | | location url
DeviceDefinition (0) | identifier* | | | | | manufacturer[x] type modelNumber version | | | | | | | | owner contact* | | | | | url
DeviceMetric (1) | identifier* | | | type category
Group (1) | identifier* | | active | type actual | | code
HealthcareService (2) | identifier* | | active | category* | | | | | | | | | | | | | | location*
InsurancePlan (0) | identifier* identifier* | | status | type*
Location (3) | identifier* | | status operationalStatus | mode type* physicalType
Organization (3) | identifier* | | active | type*
OrganizationAffiliation (0) | identifier* | | active | | | | | | | | period | | | | | | | | location*
Patient (5) | identifier* | | active
Person (2) | identifier* | | active
Practitioner (3) | identifier* | | active
PractitionerRole (2) | identifier* | | active | | | | | | | | period | | | | | | | | location*
RelatedPerson (2) | identifier* | | active | relationship* | | | | | | | period
Substance (2) | identifier* | | status | category* | | code
VerificationResult (0) | | | | | | | | | | | | | | | | | | | | who
Workflow Resources
Appointment (3) | identifier* | | status | specialty* priority | | | | supportingInformation* | start | | end | | | | | | | actor
AppointmentResponse (3) | identifier* | | | | | | | | start | | end | | | | | | | actor
Communication (2) | identifier* | | status | category* | priority | | subject | topic about* encounter | | | sent received | | | | recipient* sender | | | | | reasonCode* reasonReference*
CommunicationRequest (2) | identifier* | | status | category* | priority | | subject | about* encounter | | | occurrence[x] | | authoredOn | requester | | recipient* sender | | | | | reasonCode* reasonReference*
DeviceRequest (1) | identifier* | | status | intent | priority | code[x] parameter* code value[x] | subject | encounter | | | occurrence[x] | | authoredOn | requester | | performerType performer | | | | | reasonCode* reasonReference*
DeviceUseStatement (0) | identifier* | | status | | | | subject | | | | timing[x] | recordedOn | | | source device | | | | | reasonCode* reasonReference*
Encounter (2) | identifier* | | status | class type* | priority | | subject | episodeOfCare* | | | period | | | | | | | individual | location | reasonCode* reasonReference* condition
EpisodeOfCare (2) | identifier* | | status | type* | | condition | patient | | period
Flag (1) | identifier* | | status | category* | | code | subject | period encounter | | | | | author
Schedule (3) | identifier* | | active | serviceCategory* serviceType* specialty* | | | actor* | | | planningHorizon
Slot (3) | identifier* | | | serviceCategory* serviceType* specialty* appointmentType | | | | | | start end
SupplyDelivery (1) | | | | | | | | | | | | occurrence[x]
SupplyRequest (1) | identifier* | | status | category | priority | parameter* code value[x] | | | | occurrence[x] | | authoredOn | requester | | supplier* | | | | | reasonCode* reasonReference*
Task (2) | identifier* | | status | intent | priority | code focus | for | encounter | | | executionPeriod | authoredOn | requester | | performerType* owner | | | | location | reasonCode reasonReference
Infrastructural Resources
AuditEvent (3) | | | | | | type subtype* action outcome outcomeDesc entity* what type | | role lifecycle securityLabel* name description query detail* type value[x] | | | period | recorded | | | | | source site observer type* | agent* type role* who altId name requestor | location media network address type | purposeOfEvent* policy* purposeOfUse*
Basic (1) | identifier* | | | | | code | subject | | | | | created | author
Binary (5)
Bundle (5) | identifier | | | type | | | | | timestamp
Composition (2) | identifier | | status | type category* mode | | | subject | encounter | | | date | | author* author* | | | | party
Consent (2) | identifier* | | status | category* | | | patient | | | | | dateTime | | | performer* | | organization*

DocumentManifest (2) | masterIdentifier identifier* | | status | type | | | subject | | | | created | | author* | | | recipient*
DocumentReference (3) | masterIdentifier identifier* | | status docStatus | type category* | | | subject | encounter* | | | | date | | | | | authenticator
Endpoint (2) | identifier* | | status | connectionType | | name | | | | | period
ExampleScenario (0) | url identifier* | version | status | experimental | | | | | | | | date | | | | | publisher | | | purpose
Linkage (0) | | | active | | | | | | | | | | author
List (1) | identifier* | | status | mode | | code | subject | encounter | | | | date | source
Media (1) | identifier* | | status | type | | modality | subject | encounter | | | created[x] | issued | | | operator | | | | | reasonCode*
MessageDefinition (1) | url identifier* | version | status | experimental | | | | | | | | date | | | | | publisher | | | purpose
MessageHeader (4) | | | | | | event[x] | destination* receiver sender | | | | | | enterer author | | source | | responsible | | | reason
OperationOutcome (5)
Provenance (3) | | | | | | target* | | | | | occurred[x] | recorded | | | who | | | agent* | location | reason* activity
Questionnaire (3) | url identifier* | version | status | experimental | | | | | | | | date | | | | | publisher | | | purpose
QuestionnaireResponse (3) | identifier | | status | | | | subject | encounter | | | | authored | author | source
Subscription (3) | | | status | | | | contact* | | | | end | | | | | | | | | reason
Conformance Resources
CapabilityStatement (5) | url | version | status | experimental | | | | | | | | date | | | | | publisher | | | purpose
CatalogEntry (0)
CodeSystem (5) | url identifier* | version | status | experimental | | | | | | | | date | | | | | publisher | | | purpose
CompartmentDefinition (1) | url | version | status | experimental | | | | | | | | date | | | | | publisher | | | purpose
ConceptMap (3) | url identifier | version | status | experimental | | | | | | | | date | | | | | publisher | | | purpose
GraphDefinition (1) | url | version | status | experimental | | | | | | | | date | | | | | publisher | | | purpose
ImplementationGuide (1) | url | version | status | experimental | | | | | | | | date | | | | | publisher
NamingSystem (1) | value | | status | kind | | | | | | | | date | | | | | publisher responsible
OperationDefinition (5) | url | version | status | kind experimental | | | | | | | | date | | | | | publisher | | | purpose
SearchParameter (3) | url | version | status | experimental | | | | | | | | date | | | | | publisher | | | purpose
StructureDefinition (5) | url identifier* | version | status | experimental | | | | | | | | date | | | | | publisher | | | purpose
StructureMap (2) | url identifier* | version | status | experimental | | | | | | | | date | | | | | publisher | | | purpose
TerminologyCapabilities (0) | url | version | status | experimental | | | | | | | | date | | | | | publisher | | | purpose
TestReport (0) | identifier | | status | | | | | | | | issued
TestScript (2) | url identifier | version | status | experimental | | | | | | | | date | | | | | publisher | | | purpose
ValueSet (5) | url identifier* | version | status | experimental | | | | | lockedDate | | | date | | | | | publisher | | | purpose
Financial Resources
Account (2) | identifier* | | status | type | | name | subject* | | | | servicePeriod
ChargeItem (0) | identifier* | | status | | | code | subject | context | | | occurrence[x] | | | | actor enterer | | | | | reason* service*
ChargeItemDefinition (0) | url identifier* | version | status | experimental | | | | | | | | date | | | | | publisher
Claim (2) | identifier* identifier | | status | type subType use | | | patient | | | | billablePeriod serviced[x] | created | enterer | provider | provider | referral | | | facility location[x]
ClaimResponse (2) | identifier* | | status | type subType use | | | patient | | | | serviced[x] | created | | requestor provider* | | | | | location[x] | request
Contract (1) | | | status | | | contentDerivative type subType* term* type subType type decision entity[x] quantity unitPrice factor points net group* friendly* content[x] legal* content[x] rule* content[x] | | | | | applies effectiveTime | issued | party signature* | | type | | | | | intent
Coverage (2) | identifier* | | status | type | | | policyHolder subscriber subscriberId beneficiary | | | | period | | payor*
CoverageEligibilityRequest (2) | identifier* | | status | priority purpose* | | | patient | | | | serviced[x] | created | enterer | provider provider | | | | insurer | facility
CoverageEligibilityResponse (2) | identifier* | | status | purpose* | | | patient | | | | serviced[x] | created | | requestor provider | | | | | | request
EnrollmentRequest (0) | | | | | | | | | | | | | | | | provider candidate
EnrollmentResponse (0) | | | | | | | | | | | | | | | | organization
ExplanationOfBenefit (2) | identifier* | | status | type subType use | | | patient | | | | billablePeriod serviced[x] serviced[x] | created | enterer insurer | provider provider* | provider | referral | | | facility location[x] location[x] | claim
Invoice (0) | identifier* | | status | | | type | subject | | | | date | | | | actor
PaymentNotice (2) | identifier* | | status | | | | request response | | | | | created | | provider
PaymentReconciliation (2) | identifier* | | status | | | | | | | | period | created | | paymentIssuer | | | | | request

Full

Resource | id | id.version | status | class | grade | what | who.focus | context | when.init | when.planned | when.done | when.recorded | who.author | who.source | who.actor | who.cause | who.witness | who | where | why
Clinical Resources
ActivityDefinition (2) | url : uri identifier* : Identifier | version : string | status : code | experimental : boolean | | | | | | | | date : dateTime | | | | | publisher : string | | | purpose : markdown
AdverseEvent (0) | | | | actuality : code category* : CodeableConcept | | event : CodeableConcept | | encounter : Encounter | date : dateTime | | | recordedDate : dateTime | recorder : Patient|Practitioner|PractitionerRole|RelatedPerson | | | | subject : Patient|Group|Practitioner|RelatedPerson
AllergyIntolerance (3) | identifier* : Identifier | | clinicalStatus : CodeableConcept verificationStatus : CodeableConcept | type : code category* : code | criticality : code | code : CodeableConcept | patient : Patient | encounter : Encounter | onset[x] : dateTime|Age|Period|Range|string | | | recordedDate : dateTime | recorder : Practitioner|PractitionerRole|Patient|RelatedPerson | asserter : Patient|RelatedPerson|Practitioner|PractitionerRole
BiologicallyDerivedProduct (0) | identifier* : Identifier
BodyStructure (1) | identifier* : Identifier | | active : boolean | | | morphology : CodeableConcept location : CodeableConcept locationQualifier* : CodeableConcept description : string image* : Attachment | patient : Patient
CarePlan (2) | identifier* : Identifier | | status : code | category* : CodeableConcept | | description : string | subject : Patient|Group | encounter : Encounter | | period : Period | | created : dateTime | author :

Patient|Practitioner|PractitionerRole|Device|RelatedPerson|Organization|CareTeam | | careTeam* : CareTeam | | | | | addresses* : Condition

CareTeam (2) | identifier* : Identifier | | status : code | category* : CodeableConcept | | | subject : Patient|Group | encounter : Encounter | period : Period | | | | | | member : Practitioner|PractitionerRole|RelatedPerson|Patient|Organization|CareTeam | | | | | reasonCode* : CodeableConcept reasonReference* : Condition

ClinicalImpression (0) | identifier* : Identifier | | status : code | | | code : CodeableConcept | subject : Patient|Group | encounter : Encounter | | | effective[x] : dateTime|Period | date : dateTime | assessor : Practitioner|PractitionerRole | | | | | | | problem* : Condition|AllergyIntolerance

Condition (3) | identifier* : Identifier | | clinicalStatus : CodeableConcept verificationStatus : CodeableConcept | category* : CodeableConcept | severity : CodeableConcept | code : CodeableConcept | subject : Patient|Group | encounter : Encounter | onset[x] : dateTime|Age|Period|Range|string | | abatement[x] : dateTime|Age|Period|Range|string | recordedDate : dateTime | recorder : Practitioner|PractitionerRole|Patient|RelatedPerson | asserter : Practitioner|PractitionerRole|Patient|RelatedPerson | | | | | code* : CodeableConcept detail* : Any

DetectedIssue (1) | identifier* : Identifier | | status : code | code : CodeableConcept | severity : code | | patient : Patient | | | | | identified[x] : dateTime|Period | author : Practitioner|PractitionerRole|Device | | | | | | evidence* : code* : CodeableConcept detail* : Any

DiagnosticReport (3) | identifier* : Identifier | | status : code | category* : CodeableConcept | | code : CodeableConcept | subject : Patient|Group|Device|Location | encounter : Encounter | | | effective[x] : dateTime|Period | issued : instant | | | performer* : Practitioner|PractitionerRole|Organization|CareTeam resultsInterpreter* : Practitioner|PractitionerRole|Organization|CareTeam

EffectEvidenceSynthesis (0) | url : uri identifier* : Identifier | version : string | status : code | | | | | | | | date : dateTime | | | | | publisher : string

EventDefinition (0) | url : uri identifier* : Identifier | version : string | status : code | experimental : boolean | | | | | | | date : dateTime | | | | | publisher : string | | | purpose : markdown

Evidence (0) | url : uri identifier* : Identifier | version : string | status : code | | | | | | | | date : dateTime | | | | | publisher : string

EvidenceVariable (0) | url : uri identifier* : Identifier | version : string | status : code | | | | | | | | date : dateTime | | | | | publisher : string

FamilyMemberHistory (2) | identifier* : Identifier | | status : code | | | | patient : Patient | | | | | date : dateTime | | | | | | | | reasonCode* : CodeableConcept reasonReference* : Condition|Observation|AllergyIntolerance|QuestionnaireResponse|DiagnosticReport|DocumentReference

Goal (2) | identifier* : Identifier | | lifecycleStatus : code | category* : CodeableConcept | priority : CodeableConcept | description : CodeableConcept | subject : Patient|Group|Organization | | | start[x] : date|CodeableConcept | due[x] : date|Duration | statusDate : date | | expressedBy : Patient|Practitioner|PractitionerRole|RelatedPerson | | | | | addresses* : Condition|Observation|MedicationStatement|NutritionOrder|ServiceRequest|RiskAssessment

GuidanceResponse (2) | identifier* : Identifier | | status : code | | | | subject : Patient|Group | encounter : Encounter | | | occurrenceDateTime : dateTime | | | performer : Device | | | | | reasonCode* : CodeableConcept reasonReference* : Condition|Observation|DiagnosticReport|DocumentReference

ImagingStudy (3) | identifier* : Identifier | | status : code | modality* : Coding | | | subject : Patient|Device|Group | encounter : Encounter | started : dateTime | | | | | | actor : Practitioner|PractitionerRole|Organization|CareTeam|Patient|Device|RelatedPerson | basedOn* : CarePlan|ServiceRequest|Appointment|AppointmentResponse|Task referrer : Practitioner|PractitionerRole | interpreter* : Practitioner|PractitionerRole | | location : Location | reasonCode* : CodeableConcept reasonReference* : Condition|Observation|Media|DiagnosticReport|DocumentReference

Immunization (3) | identifier* : Identifier | | status : code | | | vaccineCode : CodeableConcept | patient : Patient | encounter : Encounter | | | occurrence[x] : dateTime|string | recorded : dateTime | | primarySource : boolean reportOrigin : CodeableConcept | actor : Practitioner|PractitionerRole|Organization | | | | location : Location

ImmunizationEvaluation (0) | identifier* : Identifier | | status : code | | | | patient : Patient | | date : dateTime

ImmunizationRecommendation (1) | identifier* : Identifier | | | | | | patient : Patient

Library (2) | url : uri identifier* : Identifier | version : string | status : code | experimental : boolean | | type : CodeableConcept | | | | | date : dateTime | | | | | publisher : string | | | purpose : markdown

Measure (2) | url : uri identifier* : Identifier | version : string | status : code | experimental : boolean | | | | | | | date : dateTime | | | | | publisher : string | | | purpose : markdown

MeasureReport (2) | identifier* : Identifier | | status : code | type : code | | measure : canonical(Measure) | subject : Patient|Practitioner|PractitionerRole|Location|Device|RelatedPerson|Group | | | | | date : dateTime | | reporter : Practitioner|PractitionerRole|Location|Organization

Medication (3) | identifier* : Identifier | | | code : CodeableConcept | | | | | | | | | | | manufacturer : Organization

MedicationAdministration (2) | identifier* : Identifier | | status : code | | | medication[x] : CodeableConcept|Reference(Medication) | subject : Patient|Group | context : Encounter|EpisodeOfCare supportingInformation* : Any | | | effective[x] : dateTime|Period | | | | performer* : | | | | | reasonReference* : Condition|Observation|DiagnosticReport

MedicationDispense (2) | identifier* : Identifier | | status : code | | | medication[x] : CodeableConcept|Reference(Medication) | subject : Patient|Group | supportingInformation* : Any

MedicationKnowledge (0) | | | | code : CodeableConcept | | | | | | | | | | | manufacturer : Organization

MedicationRequest (3) | identifier* : Identifier | | status : code | intent : code category* : CodeableConcept | priority : code | medication[x] : CodeableConcept|Reference(Medication) | subject : Patient|Group | encounter : Encounter supportingInformation* : Any | | | | authoredOn : dateTime | requester : Practitioner|PractitionerRole|Organization|Patient|RelatedPerson|Device | | performer : Practitioner|PractitionerRole|Organization|Patient|Device|RelatedPerson|CareTeam | | | recorder : Practitioner|PractitionerRole performer : Organization | | reasonCode* : CodeableConcept reasonReference* : Condition|Observation

MedicationStatement (3) | identifier* : Identifier | | status : code | category : CodeableConcept | | medication[x] : CodeableConcept|Reference(Medication) | subject : Patient|Group | | | | effective[x] : dateTime|Period | dateAsserted : dateTime | | informationSource : Patient|Practitioner|PractitionerRole|RelatedPerson|Organization | | | | | | reasonCode* : CodeableConcept reasonReference* : Condition|Observation|DiagnosticReport

MedicinalProduct (0)

MedicinalProductAuthorization (0)

MedicinalProductContraindication (0)

MedicinalProductIndication (0)

MedicinalProductIngredient (0)

MedicinalProductInteraction (0)

MedicinalProductManufactured (0)
MedicinalProductPackaged (0)
MedicinalProductPharmaceutical (0)
MedicinalProductUndesirableEffect (0)
MolecularSequence (1) | identifier* : Identifier
NutritionOrder (2) | identifier* : Identifier | | status : code | intent : code | | type* : CodeableConcept modifier : CodeableConcept type : CodeableConcept baseFormulaType : CodeableConcept additiveType : CodeableConcept | patient : Patient | encounter : Encounter | | | | dateTime : dateTime | orderer : Practitioner|PractitionerRole
Observation (5) | identifier* : Identifier | | status : code | category* : CodeableConcept | | code : CodeableConcept code : CodeableConcept | subject : Patient|Group|Device|Location focus* : Any | encounter : Encounter | | | effective[x] : dateTime|Period|Timing|instant | issued : instant | | | performer* : Practitioner|PractitionerRole|Organization|CareTeam|Patient|RelatedPerson
ObservationDefinition (0) | | | | category* : CodeableConcept | | code : CodeableConcept
PlanDefinition (2) | url : uri identifier* : Identifier | version : string | status : code | experimental : boolean | | | | | | | | date : dateTime | | | | | publisher : string | | | purpose : markdown
Procedure (3) | identifier* : Identifier | | status : code | category : CodeableConcept | | code : CodeableConcept | subject : Patient|Group | encounter : Encounter | | | performed[x] : dateTime|Period|string|Age|Range | | recorder : Patient|RelatedPerson|Practitioner|PractitionerRole | asserter : Patient|RelatedPerson|Practitioner|PractitionerRole | actor : Practitioner|PractitionerRole|Organization|Patient|RelatedPerson|Device | | | | location : Location | reasonCode* : CodeableConcept reasonReference* : Condition|Observation|Procedure|DiagnosticReport|DocumentReference
RequestGroup (2) | identifier* : Identifier | | status : code | intent : code | priority : code | code : CodeableConcept | subject : Patient|Group | encounter : Encounter | | | | authoredOn : dateTime | author : Device|Practitioner|PractitionerRole | | participant* : Patient|Practitioner|PractitionerRole|RelatedPerson|Device | | | | | reasonCode* : CodeableConcept reasonReference* : Condition|Observation|DiagnosticReport|DocumentReference
ResearchDefinition (0) | url : uri identifier* : Identifier | version : string | status : code | experimental : boolean | | | | | | | | date : dateTime | | | | | publisher : string | | | purpose : markdown
ResearchElementDefinition (0) | url : uri identifier* : Identifier | version : string | status : code | experimental : boolean | | | | | | | | date : dateTime | | | | | publisher : string | | | purpose : markdown
ResearchStudy (1) | identifier* : Identifier | | status : code | | | focus* : CodeableConcept condition* : CodeableConcept | enrollment* : Group | | | | period : Period | | | | sponsor : Organization principalInvestigator : Practitioner|PractitionerRole | | | | site* : Location | reasonStopped : CodeableConcept
ResearchSubject (1) | identifier* : Identifier | | status : code | | | | | | | period : Period
RiskAssessment (1) | identifier* : Identifier | | status : code | method : CodeableConcept | | code : CodeableConcept | subject : Patient|Group | encounter : Encounter | | | occurrence[x] : dateTime|Period | | | | performer : Practitioner|PractitionerRole|Device | | | | | reasonCode* : CodeableConcept reasonReference* : Condition|Observation|DiagnosticReport|DocumentReference
RiskEvidenceSynthesis (0) | url : uri identifier* : Identifier | version : string | status : code | | | | | | | | date : dateTime | | | | | publisher : string
ServiceRequest (2) | identifier* : Identifier | | status : code | intent : code category* : CodeableConcept | priority : code | code : CodeableConcept | subject : Patient|Group|Location|Device | encounter : Encounter | | occurrence[x] : dateTime|Period|Timing | | authoredOn : dateTime | requester : Practitioner|PractitionerRole|Organization|Patient|RelatedPerson|Device | | performerType : CodeableConcept performer* : Practitioner|PractitionerRole|Organization|CareTeam|HealthcareService|Patient|Device|RelatedPerson locationCode* : CodeableConcept locationReference* : Location | | | | | reasonCode* : CodeableConcept reasonReference* : Condition|Observation|DiagnosticReport|DocumentReference
Specimen (2) | identifier* : Identifier accessionIdentifier : Identifier | | status : code | | | type : CodeableConcept | subject : Patient|Group|Device|Substance|Location | | collected[x] : dateTime|Period duration : Duration | | receivedTime : dateTime | | | | collector : Practitioner|PractitionerRole | | | | | request* : ServiceRequest
SpecimenDefinition (0)
SubstanceNucleicAcid (0)
SubstancePolymer (0)
SubstanceProtein (0)
SubstanceReferenceInformation (0)
SubstanceSourceMaterial (0)
SubstanceSpecification (0)
VisionPrescription (2) | identifier* : Identifier | | status : code | | | | patient : Patient | encounter : Encounter | | | | dateWritten : dateTime | prescriber : Practitioner|PractitionerRole
Administrative Resources
Device (2) | identifier* : Identifier udiCarrier* : | | status : code statusReason* : CodeableConcept | | | deviceIdentifier : string distinctIdentifier : string manufacturer : string manufactureDate : dateTime expirationDate : dateTime lotNumber : string serialNumber : string type : code modelNumber : string partNumber : string version : string component : Identifier | patient : Patient | | | | | | owner : Organization contact* : ContactPoint | | | | | location : Location url : uri
DeviceDefinition (0) | identifier* : Identifier | | | | | manufacturer[x] : string|Reference(Organization) type : code modelNumber : string version : string | | | | | | | owner : Organization contact* : ContactPoint | | | | url : uri
DeviceMetric (1) | identifier* : Identifier | | | type : CodeableConcept category : code
Group (1) | identifier* : Identifier | | active : boolean | type : code actual : boolean | | code : CodeableConcept
HealthcareService (2) | identifier* : Identifier | | active : boolean | category* : CodeableConcept | | | | | | | | | | | | | | | location* : Location
InsurancePlan (0) | identifier* : Identifier identifier* : Identifier | | status : code | type* : CodeableConcept
Location (3) | identifier* : Identifier | | status : code operationalStatus : Coding | mode : code type* : CodeableConcept physicalType : CodeableConcept
Organization (3) | identifier* : Identifier | | active : boolean | type* : CodeableConcept
OrganizationAffiliation (0) | identifier* : Identifier | | active : boolean | | | | | | | | period : Period | | | | | | | | location* : Location
Patient (5) | identifier* : Identifier | | active : boolean
Person (2) | identifier* : Identifier | | active : boolean
Practitioner (3) | identifier* : Identifier | | active : boolean

PractitionerRole (2) | identifier* : Identifier | | active : boolean | | | | | | | period : Period | | | | | | | location* : Location
RelatedPerson (2) | identifier* : Identifier | | active : boolean | relationship* : CodeableConcept | | | | | | period : Period
Substance (2) | identifier* : Identifier | | status : code | category* : CodeableConcept | | code : CodeableConcept
VerificationResult (0) | | | | | | | | | | | | | | | | | | | who : Organization|Practitioner|PractitionerRole

Workflow Resources

Appointment (3) | identifier* : Identifier | | status : code | specialty* : CodeableConcept priority : unsignedInt | | | | supportingInformation* : Any | start : instant | | end : instant | | | | | | actor : Patient|Practitioner|PractitionerRole|RelatedPerson|Device|HealthcareService|Location
AppointmentResponse (3) | identifier* : Identifier | | | | | | | start : instant | | end : instant | | | | | | actor : Patient|Practitioner|PractitionerRole|RelatedPerson|Device|HealthcareService|Location
Communication (2) | identifier* : Identifier | | status : code | category* : CodeableConcept | priority : code | | subject : Patient|Group | topic : CodeableConcept about* : Any encounter : Encounter | | | sent : dateTime received : dateTime | | | | recipient* : Device|Organization|Patient|Practitioner|PractitionerRole|RelatedPerson|Group|CareTeam|HealthcareService sender : Device|Organization|Patient|Practitioner|PractitionerRole|RelatedPerson|HealthcareService | | | | | reasonCode* : CodeableConcept reasonReference* : Condition|Observation|DiagnosticReport|DocumentReference
CommunicationRequest (2) | identifier* : Identifier | | status : code | category* : CodeableConcept | priority : code | | subject : Patient|Group | about* : Any encounter : Encounter | | occurrence[x] : dateTime|Period | | authoredOn : dateTime | requester : Practitioner|PractitionerRole|Organization|Patient|RelatedPerson|Device | | recipient* : Device|Organization|Patient|Practitioner|PractitionerRole|RelatedPerson|Group|CareTeam|HealthcareService sender : Device|Organization|Patient|Practitioner|PractitionerRole|RelatedPerson|HealthcareService | | | | | reasonCode* : CodeableConcept reasonReference* : Condition|Observation|DiagnosticReport|DocumentReference
DeviceRequest (1) | identifier* : Identifier | | status : code | intent : code | priority : code | code[x] : Device)|CodeableConcep parameter* : code : CodeableConcept value[x] : CodeableConcept|Quantity|Range|boolean | subject : Patient|Group|Location|Device | encounter : Encounter | | occurrence[x] : dateTime|Period|Timing | | authoredOn : dateTime | requester : Device|Practitioner|PractitionerRole|Organization | | performerType : CodeableConcept performer : Practitioner|PractitionerRole|Organization|CareTeam|HealthcareService|Patient|Device|RelatedPerson | | | | | reasonCode* : CodeableConcept reasonReference* : Condition|Observation|DiagnosticReport|DocumentReference
DeviceUseStatement (0) | identifier* : Identifier | | status : code | | | | subject : Patient|Group | | | | timing[x] : Timing|Period|dateTime | recordedOn : dateTime | | | source : Patient|Practitioner|PractitionerRole|RelatedPerson device : Device | | | | | reasonCode* : CodeableConcept reasonReference* : Condition|Observation|DiagnosticReport|DocumentReference|Media
Encounter (2) | identifier* : Identifier | | status : code | class : Coding type* : CodeableConcept | priority : CodeableConcept | | subject : Patient|Group | episodeOfCare* : EpisodeOfCare | | | period : Period | | | | | | individual : Practitioner|PractitionerRole|RelatedPerson | location : Location | reasonCode* : CodeableConcept reasonReference* : Condition|Procedure|Observation|ImmunizationRecommendation condition : Condition|Procedure
EpisodeOfCare (2) | identifier* : Identifier | | status : code | type* : CodeableConcept | | condition : Condition | patient : Patient | | period : Period
Flag (1) | identifier* : Identifier | | status : code | category* : CodeableConcept | | code : CodeableConcept | subject : Patient|Location|Group|Organization|Practitioner|PlanDefinition|Medication|Procedure | period : Period encounter : Encounter | | | | | author : Device|Organization|Patient|Practitioner|PractitionerRole
Schedule (3) | identifier* : Identifier | | active : boolean | serviceCategory* : CodeableConcept serviceType* : CodeableConcept specialty* : CodeableConcept | | | actor* : Patient|Practitioner|PractitionerRole|RelatedPerson|Device|HealthcareService|Location | | | planningHorizon : Period
Slot (3) | identifier* : Identifier | | | serviceCategory* : CodeableConcept serviceType* : CodeableConcept specialty* : CodeableConcept appointmentType : CodeableConcept | | | | | | start : instant end : instant
SupplyDelivery (1) | | | | | | | | | | | occurrence[x] : dateTime|Period|Timing
SupplyRequest (1) | identifier* : Identifier | | status : code | category : CodeableConcept | priority : code | parameter* : code : CodeableConcept value[x] : CodeableConcept|Quantity|Range|boolean | | | | occurrence[x] : dateTime|Period|Timing | | authoredOn : dateTime | requester : Practitioner|PractitionerRole|Organization|Patient|RelatedPerson|Device | | supplier* : Organization|HealthcareService | | | | | reasonCode* : CodeableConcept reasonReference* : Condition|Observation|DiagnosticReport|DocumentReference
Task (2) | identifier* : Identifier | | status : code | intent : code | priority : code | code : CodeableConcept focus : Any | for : Any | encounter : Encounter | | | executionPeriod : Period | authoredOn : dateTime | requester : Device|Organization|Patient|Practitioner|PractitionerRole|RelatedPerson | | performerType* : CodeableConcept owner : Practitioner|PractitionerRole|Organization|CareTeam|HealthcareService|Patient|Device|RelatedPerson | | | | location : Location | reasonCode : CodeableConcept reasonReference : Any

Infrastructural Resources

AuditEvent (3) | | | | | | type : Coding subtype* : Coding action : code outcome : code outcomeDesc : string entity* : what : Any type : Coding | | role : Coding lifecycle : Coding securityLabel* : Coding name : string description : string query : base64Binary detail* : type : string value[x] : string|base64Binary | | | period : Period | recorded : instant | | | | | source : site : string observer : PractitionerRole|Practitioner|Organization|Device|Patient|RelatedPerson type* : Coding | agent* : type : CodeableConcept role* : CodeableConcept who : PractitionerRole|Practitioner|Organization|Device|Patient|RelatedPerson altId : string name : string requestor : boolean | location : Location media : Coding network : address : string type : code | purposeOfEvent* : CodeableConcept policy* : uri purposeOfUse* : CodeableConcept
Basic (1) | identifier* : Identifier | | | | | code : CodeableConcept | subject : Any | | | | | created : date | author : Practitioner|PractitionerRole|Patient|RelatedPerson|Organization
Binary (5)
Bundle (5) | identifier : Identifier | | | type : code | | | | | timestamp : instant
Composition (2) | identifier : Identifier | | status : code | type : CodeableConcept category* : CodeableConcept mode : code | | | subject : Any | encounter : Encounter | | | date : dateTime | | author* : Practitioner|PractitionerRole|Device|Patient|RelatedPerson|Organization author* : Practitioner|PractitionerRole|Device|Patient|RelatedPerson|Organization | | | | party : Patient|RelatedPerson|Practitioner|PractitionerRole|Organization
Consent (2) | identifier* : Identifier | | status : code | category* : CodeableConcept | | | patient : Patient | | | | | dateTime : dateTime | | | performer* : Organization|Patient|Practitioner|RelatedPerson|PractitionerRole | | organization* : Organization
DocumentManifest (2) | masterIdentifier : Identifier identifier* : Identifier | | status : code | type : CodeableConcept | | | subject : Patient|Practitioner|Group|Device | | | | created : dateTime | | author* : Practitioner|PractitionerRole|Organization|Device|Patient|RelatedPerson | | | recipient* :

Patient|Practitioner|PractitionerRole|RelatedPerson|Organization

DocumentReference (3) | masterIdentifier : Identifier identifier* : Identifier | | status : code docStatus : code | type : CodeableConcept category* : CodeableConcept | | | subject : Patient|Practitioner|Group|Device | encounter* : Encounter|EpisodeOfCare | | | | date : instant | | | | | authenticator : Practitioner|PractitionerRole|Organization

Endpoint (2) | identifier* : Identifier | | status : code | connectionType : Coding | | name : string | | | | | period : Period

ExampleScenario (0) | url : uri identifier* : Identifier | version : string | status : code | experimental : boolean | | | | | | | | date : dateTime | | | | | publisher : string | | | purpose : markdown

Linkage (0) | | | active : boolean | | | | | | | | | | author : Practitioner|PractitionerRole|Organization

List (1) | identifier* : Identifier | | status : code | mode : code | | code : CodeableConcept | subject : Patient|Group|Device|Location | encounter : Encounter | | | | date : dateTime | source : Practitioner|PractitionerRole|Patient|Device

Media (1) | identifier* : Identifier | | status : code | type : CodeableConcept | | modality : CodeableConcept | subject : Patient|Practitioner|PractitionerRole|Group|Device|Specimen|Location | encounter : Encounter | | | created[x] : dateTime|Period | issued : instant | | | operator : Practitioner|PractitionerRole|Organization|CareTeam|Patient|Device|RelatedPerson | | | | | reasonCode* : CodeableConcept

MessageDefinition (1) | url : uri identifier* : Identifier | version : string | status : code | experimental : boolean | | | | | | | | date : dateTime | | | | | publisher : string | | | purpose : markdown

MessageHeader (4) | | | | | | event[x] : Coding|uri | destination* : receiver : Practitioner|PractitionerRole|Organization sender : Practitioner|PractitionerRole|Organization | | | | | | enterer : Practitioner|PractitionerRole author : Practitioner|PractitionerRole | | source : | | responsible : Practitioner|PractitionerRole|Organization | | | reason : CodeableConcept

OperationOutcome (5)

Provenance (3) | | | | | | target* : Any | | | | | occurred[x] : Period|dateTime | recorded : instant | | | who : Practitioner|PractitionerRole|RelatedPerson|Patient|Device|Organization | | | agent* : | location : Location | reason* : CodeableConcept activity : CodeableConcept

Questionnaire (3) | url : uri identifier* : Identifier | version : string | status : code | experimental : boolean | | | | | | | | date : dateTime | | | | | publisher : string | | | purpose : markdown

QuestionnaireResponse (3) | identifier : Identifier | | status : code | | | | subject : Any | encounter : Encounter | | | | authored : dateTime | author : Device|Practitioner|PractitionerRole|Patient|RelatedPerson|Organization | source : Patient|Practitioner|PractitionerRole|RelatedPerson

Subscription (3) | | | status : code | | | | contact* : ContactPoint | | | | end : instant | | | | | | | | | reason : string

Conformance Resources

CapabilityStatement (5) | url : uri | version : string | status : code | experimental : boolean | | | | | | | | date : dateTime | | | | | publisher : string | | | purpose : markdown

CatalogEntry (0)

CodeSystem (5) | url : uri identifier* : Identifier | version : string | status : code | experimental : boolean | | | | | | | | date : dateTime | | | | | publisher : string | | | purpose : markdown

CompartmentDefinition (1) | url : uri | version : string | status : code | experimental : boolean | | | | | | | | date : dateTime | | | | | publisher : string | | | purpose : markdown

ConceptMap (3) | url : uri identifier : Identifier | version : string | status : code | experimental : boolean | | | | | | | | date : dateTime | | | | | publisher : string | | | purpose : markdown

GraphDefinition (1) | url : uri | version : string | status : code | experimental : boolean | | | | | | | | date : dateTime | | | | | publisher : string | | | purpose : markdown

ImplementationGuide (1) | url : uri | version : string | status : code | experimental : boolean | | | | | | | | date : dateTime | | | | | publisher : string

NamingSystem (1) | value : string | | status : code | kind : code | | | | | | | | date : dateTime | | | | | publisher : string responsible : string

OperationDefinition (5) | url : uri | version : string | status : code | kind : code experimental : boolean | | | | | | | | date : dateTime | | | | | | publisher : string | | | purpose : markdown

SearchParameter (3) | url : uri | version : string | status : code | experimental : boolean | | | | | | | | date : dateTime | | | | | publisher : string | | | purpose : markdown

StructureDefinition (5) | url : uri identifier* : Identifier | version : string | status : code | experimental : boolean | | | | | | | | date : dateTime | | | | | publisher : string | | | purpose : markdown

StructureMap (2) | url : uri identifier* : Identifier | version : string | status : code | experimental : boolean | | | | | | | | date : dateTime | | | | | publisher : string | | | purpose : markdown

TerminologyCapabilities (0) | url : uri | version : string | status : code | experimental : boolean | | | | | | | | date : dateTime | | | | | publisher : string | | | purpose : markdown

TestReport (0) | identifier : Identifier | | status : code | | | | | | | | | issued : dateTime

TestScript (2) | url : uri identifier : Identifier | version : string | status : code | experimental : boolean | | | | | | | | date : dateTime | | | | | | publisher : string | | | purpose : markdown

ValueSet (5) | url : uri identifier* : Identifier | version : string | status : code | experimental : boolean | | | | | lockedDate : date | | | date : dateTime | | | | | publisher : string | | | purpose : markdown

Financial Resources

Account (2) | identifier* : Identifier | | status : code | type : CodeableConcept | | name : string | subject* : Patient|Device|Practitioner|PractitionerRole|Location|HealthcareService|Organization | | | | servicePeriod : Period

ChargeItem (0) | identifier* : Identifier | | status : code | | | code : CodeableConcept | subject : Patient|Group | context : Encounter|EpisodeOfCare | | | occurrence[x] : dateTime|Period|Timing | | | | actor : Practitioner|PractitionerRole|Organization|CareTeam|Patient|Device|RelatedPerson enterer : Practitioner|PractitionerRole|Organization|Patient|Device|RelatedPerson | | | | | reason* : CodeableConcept service* : DiagnosticReport|ImagingStudy|Immunization|MedicationAdministration|MedicationDispense|Observation|Procedure|SupplyDelivery

ChargeItemDefinition (0) | url : uri identifier* : Identifier | version : string | status : code | experimental : boolean | | | | | | | | date : dateTime | | | | | publisher : string

Claim (2) | identifier* : Identifier identifier : Identifier | | status : code | type : CodeableConcept subType : CodeableConcept use : code | | | patient : Patient | | | | billablePeriod : Period serviced[x] : date|Period | created : dateTime | enterer : Practitioner|PractitionerRole | provider : Practitioner|PractitionerRole|Organization | provider : Practitioner|PractitionerRole|Organization | referral : ServiceRequest | | | facility : Location location[x] : CodeableConcept|Address|Reference(Location)

ClaimResponse (2) | identifier* : Identifier | | status : code | type : CodeableConcept subType : CodeableConcept use : code | | | patient : Patient | | | | serviced[x] : date|Period | created : dateTime | | requestor : Practitioner|PractitionerRole|Organization provider* : Practitioner|PractitionerRole|Organization | | | | | location[x] : CodeableConcept|Address|Reference(Location) | request : Claim

Contract (1) | | | status : code | | | contentDerivative : CodeableConcept type : CodeableConcept subType* : CodeableConcept term* : type : CodeableConcept subType : CodeableConcept type : CodeableConcept decision : CodeableConcept entity[x] : CodeableConcept|Reference(Any) quantity : SimpleQuantity unitPrice : Money factor : decimal points : decimal net : Money group* : @Contract.term friendly* : content[x] : Attachment|Reference(Composition|DocumentReference|QuestionnaireResponse) legal* : content[x] : Attachment|Reference(Composition|DocumentReference|QuestionnaireResponse) rule* : content[x] : Attachment|Reference(DocumentReference) | | | | | applies : Period effectiveTime : dateTime | issued : dateTime | party : Organization|Patient|Practitioner|PractitionerRole|RelatedPerson signature* : Signature | | type : Coding | | | | | intent : CodeableConcept

Coverage (2) | identifier* : Identifier | | status : code | type : CodeableConcept | | | policyHolder : Patient|RelatedPerson|Organization subscriber : Patient|RelatedPerson subscriberId : string beneficiary : Patient | | | | period : Period | | payor* : Organization|Patient|RelatedPerson

CoverageEligibilityRequest (2) | identifier* : Identifier | | status : code | priority : CodeableConcept purpose* : code | | | patient : Patient | | | | serviced[x] : date|Period | created : dateTime | enterer : Practitioner|PractitionerRole | provider : Practitioner|PractitionerRole|Organization provider : Practitioner|PractitionerRole | | | | insurer : Organization | facility : Location

CoverageEligibilityResponse (2) | identifier* : Identifier | | status : code | purpose* : code | | | patient : Patient | | | | serviced[x] : date|Period | created : dateTime | | requestor : Practitioner|PractitionerRole|Organization provider : Practitioner|PractitionerRole | | | | | | request : CoverageEligibilityRequest

EnrollmentRequest (0) | | | | | | | | | | | | | | | provider : Practitioner|PractitionerRole|Organization candidate : Patient

EnrollmentResponse (0) | | | | | | | | | | | | | | | organization : Organization

ExplanationOfBenefit (2) | identifier* : Identifier | | status : code | type : CodeableConcept subType : CodeableConcept use : code | | | patient : Patient | | | | billablePeriod : Period serviced[x] : date|Period serviced[x] : date|Period | created : dateTime | enterer : Practitioner|PractitionerRole insurer : Organization | provider : Practitioner|PractitionerRole|Organization provider* : Practitioner|PractitionerRole|Organization | provider : Practitioner|PractitionerRole|Organization | referral : ServiceRequest | | | facility : Location location[x] : CodeableConcept|Address|Reference(Location) location[x] : CodeableConcept|Address|Reference(Location) | claim : Claim

Invoice (0) | identifier* : Identifier | | status : code | | | type : CodeableConcept | subject : Patient|Group | | | | date : dateTime | | | | actor : Practitioner|Organization|Patient|PractitionerRole|Device|RelatedPerson

PaymentNotice (2) | identifier* : Identifier | | status : code | | | | request : Any response : Any | | | | | created : dateTime | | provider : Practitioner|PractitionerRole|Organization

PaymentReconciliation (2) | identifier* : Identifier | | status : code | | | | | | | | period : Period | created : dateTime | | paymentIssuer : Organization | | | | | request : Task